



Computer Science 9618

Paper 1 · Theory Fundamentals

Part-level Topical Workbook

with Mark Scheme

- Every question part filed under its own syllabus topic (units 1 to 8 of the 2027-29 syllabus)
- Each item carries the official context it needs: stem, code, tables, diagrams, earlier parts
- Answer lines removed; tables, trace tables and code to complete kept
- Newest papers first; mark scheme in an Answers Section after each unit
- Index of every part with its unit, marks and page

Coverage

37 papers: M/J 21/P11 to M/J 26/P23 (newest first in each unit)

326 items, 2267 marks across 8 units

Syllabus codes: 9618

8 items come from Paper 2 papers (filed here by topic)

Contents

UNIT 1	Information representation	3
	Answers Section	49
UNIT 2	Communication	84
	Answers Section	108
UNIT 3	Hardware	132
	Answers Section	182
UNIT 4	Processor Fundamentals	217
	Answers Section	300
UNIT 5	System Software	335
	Answers Section	361
UNIT 6	Security, privacy and data integrity	384
	Answers Section	408
UNIT 7	Ethics and Ownership	424
	Answers Section	436
UNIT 8	Databases	450
	Answers Section	489
INDEX	Topic index	524

Unit 1

INFORMATION REPRESENTATION

59 items · 399 marks

Syllabus sections in this unit

- 1.1 Data Representation
- 1.2 Multimedia
- 1.3 Compression

Unit 1: Information representation

1. M/J 26/P11/Q1/a,b,c,d

1 A computer can store numbers and text in binary.

(a) The computer uses the UNICODE character set.

(i) Complete the table by calculating the missing values.

character	denary	binary	hexadecimal
⋮		0001 1110 1001	1E9
@	64		40
Ω	911	0011 1000 1111	
Ψ		0001 0101 1101 0110	15D6

[4]

working space

(ii) Identify **one** other character set and explain how it is different from UNICODE.

character set

differences

[3]

(b) The computer also uses two's complement.

Convert the 8-bit two's complement binary integer 01101011 into denary.

[1]

(c) Show how the denary number 30 is subtracted from the binary number 01101011 using binary subtraction.

Show your working.

[2]

(d) Identify **two** differences between binary prefixes and decimal prefixes.

1

2

[2]

2. M/J 26/P11/Q5/a,b,c

- 5** A school needs to create a new logo. The logo can be created as a bitmap image or a vector graphic.

(a) Explain how a bitmap image is encoded in a computer.

[3]

(b) Complete the description of a vector graphic by writing the missing terms.

A is an individual element of an image, for example: a line, a square.

A is an attribute of one part of the image, for example: a line colour, the width of a line.

A is all of the elements that are included in the vector graphic.

[3]

(c) Tick (✓) **one** box to identify whether a bitmap image or a vector graphic will be most appropriate for the school logo.

Justify your choice.

	tick (✓)
bitmap image	
vector graphic	

justification

[3]

3. M/J 26/P12/Q1/a,b,c,d

- 1 (a) Tick (✓) **one** box to identify the correct definition of a pixel in a bitmap image.

- | | |
|--------------------------|---|
| ☐ | a colour in an image |
| ☐ | a row of consecutive identical colours in an image |
| ☐ | the binary code for one colour in an image |
| ☐ | the smallest element of an image that can be one colour |

[1]

- (b) Describe the difference between the terms image resolution and screen resolution.

[2]

- (c) Two items in a bitmap image file header are the resolution and colour depth.

Identify **one** other item included in the file header of a bitmap image.

[1]

- (d) The following bitmap image is created. Each letter represents a colour: R is red, W is white, B is blue.

R	R	W	W	W	W
B	B	R	R	W	W
W	W	W	R	R	R
W	W	W	W	R	R
W	W	R	R	R	R
R	R	R	R	R	B
B	B	B	B	B	B

- (i) Identify the smallest bit depth that can be used for the image shown.

[1]

- (ii) Explain how the image shown can be compressed using lossless compression.

[3]

4. M/J 26/P12/Q4/a,b

4 A sound file has a sampling rate of 44.1 kHz.

(a) Describe what is meant by a sampling rate of 44.1 kHz.

[1]

(b) A student creates a recording of a sound with a sampling resolution of 2 bytes.

The student creates a second recording of the same sound but with a sampling resolution of 8 bits.

Explain how the two sound recordings **and** the two sound files will be affected by the different sampling resolutions.

[3]

5. M/J 26/P12/Q5/a,b

5 (a) Complete the table by calculating the missing values.

denary	unsigned binary	hexadecimal	Binary Coded Decimal (BCD)
160	10100000	A0	
	100100001	121	001010001001
2486		9B6	0010010010000110
35123	1000100100110011		00110101000100100011

[4]

working space

(b) Complete the table by writing the answer to each statement.

statement	answer
The smallest 12-bit two's complement binary number	
The denary value represented by the 8-bit unsigned binary number 01100010 after a logical right shift of 2 places has been applied.	
The binary value after a cyclic left shift of 4 places has been applied to the binary number 10010101	

[3]

6. M/J 26/P13/Q3/a,b,c(i)

3 An animation is a sequence of bitmap images with a sound recording played at the same time.

(a) The sound is recorded with a sampling resolution of 2 bytes.

State what is meant by a sampling resolution of 2 bytes.

[1]

(b) (i) Explain how the data for the bitmapped images are encoded.

[3]

(ii) The bitmap images have a bit depth of 2 bytes. Each image is 2048 pixels wide by 1536 pixels high. One second of video contains eight individual bitmap images.

There is no sound recording in this part of the video.

Calculate an estimate for the file size of one second of the video in mebibytes (MiB).
Show all of your working.

working:

answer MiB

[2]

(c) The sound and images are compressed independently before being transmitted over the internet to public cloud storage.

(i) Explain the different ways that lossless compression can be used to compress the images and the sound file.

image file

sound file

[4]

7. M/J 26/P13/Q6/a,b,c,d

- 6 (a) Draw **one or more** lines to match each denary number with its equivalent unsigned binary, hexadecimal or Binary Coded Decimal (BCD) value(s).

Denary number	Equivalent unsigned value
	Hexadecimal: 121
3120	BCD: 00010101100001010001
289	Binary: 110000110000
22516	Hexadecimal: 57F4
15851	Binary: 11110111101011
	BCD: 0011000100100000

[4]

- (b) Complete the binary addition. Show all of your working.

$$\begin{array}{r}
 110101001111 \\
 + 111111001111 \\
 \hline
 \end{array}$$

[2]

- (c) Convert the two's complement integer 011010110001 into denary.

[1]

- (d) Tick (✓) **one** box to identify the smallest integer that can be represented by a 16-bit two's complement integer.

- ☐ 0000000000000000
☐ 1111111111111111
☐ 0111111111111110
☐ 1000000000000000

[1]

8. O/N 25/P11/Q1/a,b

- 1 (a) (i) Convert the binary number into hexadecimal.

101100111010

[1]

- (ii) Convert the denary number into Binary Coded Decimal (BCD).

108

[1]

- (iii) Convert the 12-bit two's complement binary integer into denary.

Show your working.

111110111100

Working

Denary value

[2]

- (b) (i) The following binary addition is performed using 8-bit registers.

Complete the calculation using binary addition.

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0\ 0\ 1\ 1 \\ +\ 0\ 1\ 1\ 1\ 1\ 0\ 0\ 0 \\ \hline \end{array}$$

[1]

- (ii) Name and describe the error that can occur when binary addition is performed.

[2]

9. O/N 25/P11/Q7/d

- 7 Web browsers use Internet Protocol (IP) addresses.

- (d) A bitmap file is downloaded. The image has a maximum of 256 colours and measures 512 pixels wide by 2048 pixels high.

- (i) Calculate an estimate of the file size of the bitmap in kibibytes.

Show your working.

Working

File size of bitmap kibibytes

[2]

- (ii) Explain why the actual file size may be larger than the one calculated in (d)(i).

[2]

- (iii) Explain how a bitmap image is compressed using run-length encoding (RLE).

[2]

10. O/N 25/P12/Q6/a,b,d

- 6 (a)** A sound file is compressed by reducing the sampling rate.

State whether this is lossless or lossy compression. Justify your choice.

Type of compression

Justification

[1]

- (b)** The following table shows some words and corresponding denary values.

Word	Denary value
Computing	55
Science	56
Computers	57
are	58
Brilliant!	59
is	60
Fun!	61
Amazing!	62

The following table shows three bytes of data that have been received.

Use the table to find the corresponding words from the binary values received.

Binary value	00111000	00111100	00111110
Word			

Working

[1]

- (d)** A bitmap image has a resolution of 1000 pixels wide by 2000 pixels high. The colour depth is 16 bits.

Calculate an estimate of the file size in megabytes.

Show your working.

File size megabytes

[2]

11. O/N 25/P12/Q7/a,b,c

- 7 (a) The following binary addition is performed using 8-bit registers.

Complete the calculation using binary addition.

$$\begin{array}{r} 1\ 0\ 0\ 0\ 0\ 1\ 1\ 1 \\ +\ 0\ 0\ 1\ 1\ 1\ 0\ 0\ 1 \\ \hline \end{array}$$

[1]

- (b) A computer uses the Unicode character set.

State the number of bits used to store **one** character from the Unicode character set.

[1]

- (c) ASCII is another character set. The ASCII value for the character 'h' has the denary value 104.

(i) Write the Binary Coded Decimal (BCD) value for the ASCII character 'h'.

[1]

(ii) Write the hexadecimal value for the ASCII character 'h'.

[1]

12. O/N 25/P13/Q1/a,b,c

- 1 (a) A student creates sound recordings.

Making changes to a sound recording may impact the accuracy of the resulting sound file or the size of the file.

Draw **two** lines from each change to the impacts it has on the sound file.

Change	Impact
increase the duration of the recording	the file size gets bigger
	no change in the file size
increase the sampling rate	the file size gets smaller
	the accuracy of the sound file improves
decrease the sampling resolution	no change to the accuracy of the sound file
	the accuracy of the sound file worsens

[3]

- (b) The file names of the sound recordings are stored using the ASCII character set.

Explain how text is represented by the ASCII character set.

[2]

- (c) Give **two** differences between the ASCII and UNICODE character sets.

1

2

[2]

13. O/N 25/P13/Q2/a,b,d

- 2 (a)** Data in a computer system is represented in binary.

Put **one** tick (✓) in each row to identify the minimum number of bits used to store each example of data.

Example of data	Number of bits						
	4	8	16	24	32	64	128
the hexadecimal value F139							
16 000 000 unique amplitude values							
an IPv4 address							
256 unique colours							
an IPv6 address							
the denary value 65 000							

[3]

- (b)** Convert the denary number –108 into a 12-bit two's complement binary number.

[1]

- (d)** Convert the following positive binary integer into hexadecimal.

1110001100111011

[1]

14. O/N 25/P13/Q7/b

- 7** A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

- (b)** A video is made of many bitmap images called frames. 30 frames are recorded every second.

Each frame is 4000 pixels wide by 3000 pixels high. The video records using 16-bit colour depth.

Calculate an estimate for the file size for **one** second of the video in gigabytes.

Show your working.

File size gigabytes

[2]

15. M/J 25/P11/Q2/b

- 2** Programmers in a software development company take part in live video conferences to discuss their work.

The live video conferences take place using real-time bit streaming. The video is compressed before it is transmitted.

- (b) (i)** Explain the reasons why a video is compressed before it is transmitted using real-time bit streaming.

[4]

- (ii)** Identify whether the lossy or lossless compression method is more appropriate for real-time bit streaming. Justify your answer.

Compression method

Justification

[3]

16. M/J 25/P11/Q3/a,b

3 A computer stores images and text files.

(a) One of the images is a bitmapped image.

Complete the table by writing the answer for each statement.

Statement	Answer
the term for the smallest element that makes up an image	
the largest number of different colours that can be represented with a bit depth of 8 bits	
the term for the dots per inch (dpi) when an image is displayed	

[3]

(b) The text in the files is stored using the Unicode character set.

(i) Give **two** advantages of using the Unicode character set instead of using the ASCII character set.

1

2

[2]

(ii) The Unicode character 'Ƨ' has the binary value: 0010 0111 0110 1110

Convert the binary value for the character 'Ƨ' into denary.

[1]

(iii) The Unicode character 'Σ' has the hexadecimal value 2140

Convert the hexadecimal code for the character 'Σ' into denary.

[1]

17. M/J 25/P12/Q2/a,b,c

- 2 (a) Convert the denary integer 558 into 12-bit binary and hexadecimal.

Binary

Hexadecimal

[2]

- (b) (i) Convert the two's complement binary integer 11100010 into denary.

[1]

- (ii) Write the smallest and the largest two's complement binary integers that can be represented in 8 bits.

Smallest

Largest

[2]

- (c) Give **one** application where Binary Coded Decimal (BCD) is used and justify its use.

Application

Justification

[2]

18. M/J 25/P13/Q1/a,b,c

- 1 (a) A student paints a picture in an art class.

The student takes a photograph of the picture using a digital camera. The digital camera creates an image with a resolution of 2 million pixels and uses a bit depth of 16 bits.

- (i) Calculate the file size of the image created by the digital camera in megabytes (MB).

Show your working.

Working space

Answer MB

[2]

- (ii) The student takes a second photograph with a lower bit depth.

Explain the effect of decreasing the bit depth on the image and on the image file.

Effect on image

Effect on image file

[4]

- (b) A teacher records an audio file of the student playing the piano during a music lesson.

Complete the table by giving the term for each description about sound representation on a computer.

Description	Term
the number of times the amplitude is measured per time interval	
the number of bits used to store each amplitude measurement	
the type of sound wave before it is recorded by a computer	

[3]

(c) The student types a report on a computer in a history lesson.

(i) The computer uses the Unicode character set.

Give **two** characteristics of the Unicode character set.

1

2

[2]

(ii) Some of the school's computers use the extended ASCII character set.

The table gives some characters from the extended ASCII character set, their denary, 8-bit binary and hexadecimal numbers.

Complete the table by filling in the missing numbers.

Character	Denary	8-bit Binary	Hexadecimal
!	33		21
L		01001100	4C
ü	252	11111100	

[3]

19. M/J 25/P13/Q7/b,c

(b) Complete the following binary addition. Show your working. Include any overflow bit(s).

$$\begin{array}{r}
 1 \ 1 \ 1 \ 1 \ 0 \ 1 \ 0 \ 1 \\
 + \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 1 \\
 \hline
 \end{array}$$

[1]

(c) Subtract the binary number 00011110 from the binary number 01100100 using binary subtraction. Show your working.

[2]

20. O/N 24/P11/Q1/a,b,c,d

- 1 (a) State **one** difference between a tebibyte and a gigabyte.

[1]

- (b) (i) Convert the unsigned binary integer into hexadecimal.

110001100111

Answer

[1]

- (ii) Convert the two's complement binary number into denary.

100110010111

Answer

[1]

- (iii) Convert the Binary Coded Decimal (BCD) into denary.

010101110011

Answer

[1]

- (c) Subtract the denary number 23 from the two's complement binary number 01001010

Perform this calculation using binary subtraction.

Show your working.

Working

Answer

[2]

- (d) State **one** reason why binary addition and subtraction can result in overflow.

[1]

21. O/N 24/P11/Q4/b

- 4 A student uses a laptop to write a program that is saved as a text file.

- (b) The student compresses the file before it is emailed to their teacher as an attachment.

- (i) Explain the benefits to the teacher of the attachment being a compressed file.

[3]

- (ii) Describe **one** lossless method of compressing a text file.

[3]

22. O/N 24/P12/Q7/a,b

7 A student takes a photograph of a science experiment.

(a) The photograph is saved as a bitmapped image.

(i) Define the following bitmap terms.

Colour depth

File header

[2]

(ii) Explain why changing the image resolution will affect the image quality and file size.

Image quality

File size

[2]

(iii) Identify **one** lossless method of compressing an image.

[1]

(b) The student draws a picture on paper that is scanned into the computer and saved as a vector graphic.

Define the vector graphic terms property and drawing list.

Property

Drawing list

[2]

23. O/N 24/P13/Q6/a,b,c

6 A computer system stores text, images and sound.

(a) A character set is used to represent characters in a computer.

Identify **and** describe **one** character set.

Character set

Description

[2]

(b) The colour of each pixel in a bitmapped image is represented by 8 bits.

(i) State the largest number of different colours that can be represented by 8 bits.

[1]

(ii) State **one** drawback of increasing the number of bits that represents each pixel in the bitmap image.

[1]

(iii) A bitmap image can be compressed using lossy compression.

Explain the reasons why lossy compression is often suitable for a bitmap image.

[2]

(c) (i) Explain how an analogue sound wave is converted into digital data.

[2]

(ii) Describe **one** method of compressing a sound file using lossy compression.

[2]

24. O/N 24/P13/Q8/a,b

8 (a) Convert the hexadecimal number 1FAB into denary.

Working

Denary value

[1]

(b) Explain how to convert the two's complement integer 10011111 into denary. Give the denary value after conversion.

Explanation

Denary value

[3]

25. M/J 24/P11/Q2/d

- 2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:

- The digital camera in the doorbell starts recording a video.
- A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell's internal secondary storage device and overwritten when the secondary storage device is full.



- (d) The digital camera has a microphone which is used to record the sound for the video.

The user changes the sampling rate that the microphone uses from 44.1 kHz to 88.2 kHz.

Describe how this change in sampling rate will affect the performance of the video doorbell.

[3]

26. M/J 24/P11/Q7

- 7 Complete the binary addition. Show your working.

$$\begin{array}{r}
 10011110 \\
 01100001 \\
 + 00011001 \\
 \hline
 \end{array}$$

[3]

27. M/J 24/P12/Q2/d

- 2** A computer game is being designed that users will be able to play using a virtual reality (VR) headset.

(d) The computer can transmit a video made from bitmap images and vector graphic animations to the VR headset.

(i) Describe how the data for a bitmapped image is encoded.

[3]

(ii) Describe the contents of a vector graphic drawing list.

[2]

(iii) The bitmap video is **not** compressed before transmission to the VR headset.

Give **two** reasons why the video does **not** need to be compressed.

1

2

[2]

28. M/J 24/P12/Q7/a,b,c

- 7** A computer stores binary data.

(a) Tick (✓) **one** box only to identify the **largest** file size.

- | | |
|--------------------------|----------------|
| ☐ | 3300 kibibytes |
| ☐ | 0.3 megabytes |
| ☐ | 3 mebibytes |
| ☐ | 3300 kilobytes |

[1]

(b) Subtract the denary number 10 from the denary number 100 using binary subtraction.

Show your working.

Working

Answer

[3]

(c) Convert the hexadecimal number C0F into denary.

Show your working.

Working

Answer

[2]

29. M/J 24/P13/Q1/a,b,c,d

1 (a) Complete the following description.

A kibibyte has a prefix. Three kibibytes is the same as bytes.

A megabyte has a prefix. Two terabytes is the same as gigabytes.

[4]

(b) Convert the denary number 241 to hexadecimal.

[1]

(c) State what is meant by an **overflow in binary addition**.

[1]

(d) Computers use character sets when representing characters in binary.

(i) Complete the table by identifying the number of bits each of the character sets allocates to each character.

Character set	Number of bits
ASCII	
extended ASCII	
Unicode	

[1]

(ii) Explain how the word 'Clock' is represented by a character set.

[2]



30. M/J 24/P13/Q2/a,b,c

2 A photograph is stored as a bitmap image.

- (a) The photograph has a resolution of 4000 pixels wide by 3000 pixels high. The bit depth is 4 bytes.

Calculate an estimate for the file size of the photograph in megabytes.

Show your working.

Working

Answer megabytes

[2]

- (b) The photograph is compressed before being uploaded to a web server.

- (i) Give **three** benefits of this photograph being compressed using lossy compression instead of lossless compression.

1

2

3

[3]

- (ii) Explain how run-length encoding (RLE) will compress the photograph.

[2]

- (c) Identify **two** elements of a bitmap image that can be changed to reduce its file size.

1

2

[2]



31. O/N 23/P11/Q1/a,b,c

- 1 (a) Draw **one** line from each vector graphic term to its most appropriate description.

Term	Description
drawing list	a component created using a formula
drawing object	defines one characteristic of a component
property	data required to create all components in the graphic

[2]

- (b) State what is meant by the **bit depth** of a bitmap image **and** explain how changing the bit depth affects the image.

Definition

Explanation

[3]

- (c) Explain why a bitmap image is often compressed before it is attached to an email.

[2]

32. O/N 23/P12/Q3/a,b,c

- 3 (a) State **one** difference between a kibibyte and a megabyte.

[1]

- (b) (i) Convert the denary value into a 12-bit two's complement binary integer.

–196

Answer

[1]

- (ii) Convert the Binary Coded Decimal (BCD) into denary.

100001100101

Answer

[1]

- (iii) Convert the unsigned binary integer into denary.

000111010110

Answer

[1]

- (c) Identify **one** practical application of BCD **and** justify why BCD is used in this application.

Application

Justification

[2]

33. O/N 23/P12/Q6/a,b,c

- 6 (a) A real-time video of a music concert needs to be streamed to subscribers.

Tick (✓) **one** box to identify the most appropriate type of compression **and** justify your answer.

Lossy	Lossless
☐	☐

Justification

[3]

- (b) Explain the impact of changing the sampling resolution on the accuracy of a sound recording.

[3]

- (c) A bitmap image has a resolution of 2048 pixels wide and 1024 pixels high.

The image has a bit depth of 10 bits per pixel.

Estimate the file size of the bitmap image in mebibytes. Show your working.

Working

Estimated file size in mebibytes

[2]

34. O/N 23/P13/Q1/a,b

- 1 (a) State what is meant by **analogue data**.

[1]

- (b) Draw **one** line from each term to its most appropriate description.

Term	Description
sampling	the number of samples taken per second
sampling rate	taking measurements at regular intervals and storing the values
sampling resolution	the number of bits used to store each sample

[2]

35. O/N 23/P13/Q2/a,b

- 2 (a) Describe the impact of increasing the image resolution on the quality of a bitmap graphic.

[2]

- (b) Calculate the file size of a bitmap image using the following information:

- image resolution of 2048 pixels wide and 1024 pixels high
- bit depth of 16 bits.

Give your answer in kibibytes. Show your working.

Working

Answer in kibibytes

[2]

36. O/N 23/P13/Q5/b(i)

- 5 A programmer uses an Integrated Development Environment (IDE) to develop a program that monitors air quality.

- (b) The program is distributed by downloading the source code and its library files from a web server.

- (i) Explain the reasons for compressing the files.

[2]

37. M/J 23/P11/Q1/a,b,c

- 1 Images are being created to advertise holidays.

Some of the images are bitmap images and some are vector graphics.

(a) Complete the table by defining the image terms.

Term	Definition
Drawing list	
Pixel	
Colour depth	

[3]

(b) The bitmap images are photographs of the holiday locations.

- (i) Colour depth and image resolution are both included in the file header of a bitmap image.

Identify **two other** items that could be included in the file header of each photograph.

1

2

[2]

- (ii) One of the photographs has a bit depth of 8 bytes and an image resolution of 1500 pixels wide and 3000 pixels high.

Calculate the file size of the photograph in megabytes. Show your working.

Working

File size MB

[2]

(c) The photographs are compressed before they are uploaded to a web server. Customers download the photographs from this web server.

- (i) Explain the reasons why compressing the photographs will benefit the customers.

[3]

- (ii) An image can be compressed using run-length encoding (RLE).

Explain the reasons why RLE may **not** reduce the file size of a bitmap image.
Give **one** example in your answer.

[3]

38. M/J 23/P11/Q3/d(i,ii,iii,iv,v)

- 3** A computer has an Operating System (OS).

- (d) The computer stores data in binary form.

- (i) State the difference between a kibibyte and a kilobyte.

[1]

- (ii) Convert the denary number 964 into Binary Coded Decimal (BCD).

[1]

- (iii) Convert the positive binary integer 11110010 into hexadecimal.

[1]

- (iv) Give the smallest **and** largest two's complement binary number that can be represented using 8 bits.

Smallest

Largest

[2]

- (v) Add the following two binary integers using binary addition. Show your working.

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0\ 0\ 0\ 0 \\ +\ 0\ 0\ 0\ 1\ 1\ 0\ 1\ 1 \\ \hline \end{array}$$

[2]

39. M/J 23/P12/Q4/a,b,c

4 Data in a computer is stored in binary form.

(a) State the number of unique binary values that can be represented in 16 bits.

[1]

(b) Give the 8-bit one's complement representation of the denary number -120 .

Show your working.

Working

Answer

[2]

(c) Convert the hexadecimal number A04 into denary.

Show your working.

Working

Answer

[2]

40. M/J 23/P13/Q3/c

3 A mobile telephone is used to record a video.

(c) The video includes a sound recording.

(i) Describe how sound is represented in a computer.

[3]

(ii) A second video is recorded. The sound in the second video needs to be more precise.

Explain the reasons why increasing the sampling rate and the sampling resolution will improve the precision of the second recording.

Sampling rate

Sampling resolution

[4]

41. M/J 23/P13/Q7/a

7 A computer stores data in binary form.

(a) Draw **one** line from each description to its matching denary value.

Description	Denary value
	-127
The smallest integer that can be represented in 8-bit two's complement.	127
	-255
The largest integer that can be represented in 8-bit two's complement.	-128
	-256
The largest unsigned integer that can be represented in 8 bits.	256
	128
	255

[3]

42. O/N 22/P11/Q1/a,b,c,d

- 1 (a) (i) Convert the unsigned binary integer into denary.

00100111

Answer

[1]

- (ii) Convert the Binary Coded Decimal (BCD) into denary.

00100111

Answer

[1]

- (iii) Convert the 8-bit two's complement binary integer into denary.

11100111

Answer

[1]

- (b) Perform the following binary subtraction. Show your working.

$$\begin{array}{r} 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \\ - \ 0 \ 1 \ 1 \ 1 \ 0 \ 1 \ 0 \ 1 \\ \hline \end{array}$$

[2]

- (c) Give **one** similarity and **two** differences between the ASCII and Unicode character sets.

Similarity

Difference 1

Difference 2

[3]

- (d) Sound samples are recorded and saved in a file.

- (i) State what is meant by **sampling rate**.

[1]

- (ii) Explain the effect of increasing the **sampling resolution** on the sound file.

[2]

43. O/N 22/P11/Q7/c

- (c) An Operating System may include a utility program to compress text files.

Describe **one** appropriate method of compressing a text file.

[3]

44. O/N 22/P12/Q2/a,b

- 2 (a) (i) Convert the two's complement binary integer into denary.

10010110

Answer [1]

- (ii) Convert the unsigned binary integer into hexadecimal.

10010110

Answer [1]

- (iii) Convert the unsigned binary integer into Binary Coded Decimal (BCD). Show your working.

10010101

Working

Answer [2]

- (b) Perform the following binary addition.

$$\begin{array}{r} 1\ 0\ 0\ 0\ 1\ 1\ 0\ 0 \\ +\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \end{array}$$

[1]

45. O/N 22/P12/Q6/b

- 6 (a) A student uses a networked laptop computer to send an email to a colleague.

- (b) The student records a sound file.

- (i) Explain the effect of increasing the sampling rate on the accuracy of the sound recording.

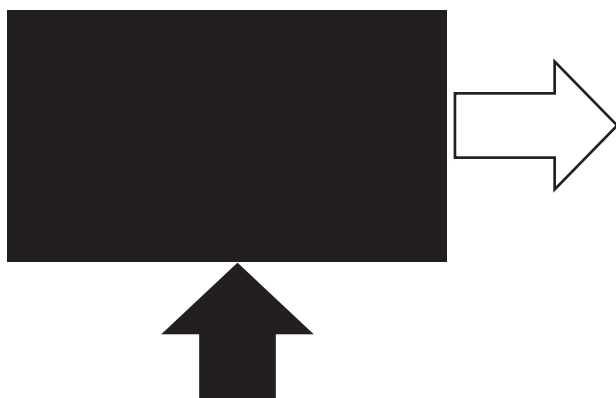
[2]

- (ii) Explain the effect of decreasing the sampling resolution on the file size of the sound recording.

[2]

46. O/N 22/P12/Q8/a,b,c

- 8 The following bitmap image has a resolution of 4096×4096 pixels and a colour depth of 24 bits per pixel.



The image is displayed on a monitor that has a screen resolution of 1920×1080 pixels.

(a) Tick (✓) **one** box in each row to identify the effect of each action on the image file size.

Action	Increases the file size	Decreases the file size	No change to the file size
Change the colour depth of the image file to 16 bits per pixel.			
Change the screen resolution to 1366×768 pixels.			
Change the colour of the rectangle from black to red.			

[2]

(b) State **two** benefits of creating a vector graphic instead of a bitmap image.

1

2

[2]

(c) A second bitmap image is stored using a colour depth of 8 bits per pixel.

The file is compressed using run-length encoding (RLE).

(i) The table shows the compressed and uncompressed values for parts of the image file.

Each colour of the pixel in the image is represented by a hexadecimal value.

Complete the table. The first row has been completed for you.

Uncompressed image	RLE compressed image
EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA
	2AB 2FF 11D 167
32 32 80 81 81	

[2]

(ii) RLE is an example of lossless compression.

Explain why lossless compression is more appropriate than lossy compression for a text file.

[2]

47. O/N 22/P13/Q1/a,b

1 A digital audio message needs to be recorded.

(a) Tick (✓) **one** box in each row to identify the effect of each action on the accuracy of the recording.

Action	Accuracy increases	Accuracy decreases	Accuracy does not change
Change the sampling rate from 40 kHz to 60 kHz.			
Change the duration of the recording from 20 minutes to 40 minutes.			
Change the sampling resolution from 24 bits to 16 bits.			

[2]

(b) The audio message is recorded with a sampling rate of 50 kHz and a sampling resolution of 16 bits.

The recording is 20 minutes in length.

Calculate the file size of the recording.

Give your answer in megabytes **and** show your working.

Working

Answer megabytes

[2]

48. O/N 22/P13/Q2/e

- 2 The relational database `ASTRONOMY` is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

`COMPANY (TelephoneNumber, CompanyID, CompanyName)`

`PHOTOGRAPH (PhotoID, TelescopeID, DateTaken, TimeTaken, Elevation)`

`TELESCOPE (TelescopeID, CompanyID, SerialNumber)`

- (e) The photographs are stored as bitmap images.

Complete the statements about bitmap images by writing the missing words.

The of a bitmap image is the number of bits that are used to store each pixel.

Metadata about the image is stored in the of the file.

[2]

49. O/N 22/P13/Q9/a,b

- 9 (a) (i) Convert the unsigned binary value into hexadecimal.

10010011

Answer

[1]

- (ii) Convert the unsigned binary value into denary.

10010011

Answer

[1]

- (b) State **two** benefits of using Binary Coded Decimal (BCD) to represent values.

Benefit 1

Benefit 2

[2]

50. O/N 21/P11/Q1/a,b,c

- 1 (a) Draw **one** line from each binary value to its equivalent (same) value on the right.

Binary value

8 bits

8000 bits

1000 kilobytes

1024 mebibytes

8192 bits

1 kibibyte

1 gigabyte

1 byte

1 kilobyte

1 gibibyte

1 megabyte

1 mebibyte

[5]

- (b) (i) Perform the following binary addition. Show your working.

$$\begin{array}{r} 10101010 \\ + 00110111 \\ \hline \end{array}$$

[2]

- (ii) State how an overflow can occur when adding two binary integers.

[1]

- (c) Convert the hexadecimal value F0 into denary.

[1]

51. O/N 21/P11/Q7/a,b

7 Bobby is recording a sound file for his school project.

(a) He repeats the recording of the sound several times, with a different sample rate each time.

(i) Describe the reasons why the sound is closer to the original when a higher sample rate is used.

[2]

(ii) Describe the reasons why the sound file size increases when a higher sample rate is used.

[2]

(b) Bobby wants to email the sound file to his school email address. He compresses the file before sending the email.

(i) Explain the reasons why Bobby compresses the sound file.

[2]

(ii) Bobby uses lossless compression.

Describe how lossless compression can compress the sound file.

[2]

52. O/N 21/P12/Q4/a,b,c,d,e

4 A register stores the following binary number:

1	1	0	0	1	1	0	1
---	---	---	---	---	---	---	---

(a) The binary value in the register represents an unsigned binary integer.

Convert the unsigned binary integer into denary.

[1]

(b) The binary value in the register represents a two's complement binary integer.

Convert the two's complement binary integer into denary.

[1]

(c) The binary value in the register represents a hexadecimal number.

Convert the binary number into hexadecimal.

[1]

(d) State why the value in the register cannot be interpreted as a Binary Coded Decimal (BCD).

[1]

(e) The binary contents of **two** registers are:

Register 1	0	0	1	1	1	1	0	1
Register 2	0	0	1	0	1	1	0	1

(i) Add the contents of **Register 1** and **Register 2**. Show your working.

Answer

[2]

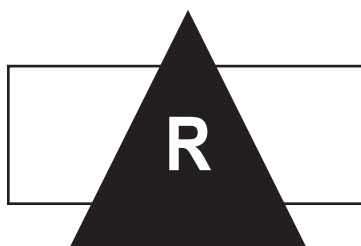
(ii) Subtract the contents of **Register 2** from the contents of **Register 1**. Show your working.

Answer

[2]

53. O/N 21/P12/Q5/a,b

5 Riya has created the following logo as a vector graphic.



- (a) Complete the table by writing a description of each vector graphic term **and** give an example for this logo.

Term	Description	Example from logo
Property		
Drawing list		

[4]

- (b) Riya takes a photograph using a digital camera. The photograph is stored as a bitmap image.

- (i) Describe **two** differences between a vector graphic and a bitmap image.

1

2

[4]

- (ii) Riya needs to email the photograph. She compresses the photograph before sending it using an email.

Describe **two** lossy methods that Riya can use to compress the image.

Method 1

Method 2

[4]

54. O/N 21/P13/Q1/a,b,c

- 1 (a) Draw **one** line from each binary value to its equivalent (same) value on the right.

Binary value

8 bits

8000 bits

1000 kilobytes

1024 mebibytes

8192 bits

1 kibibyte

1 gigabyte

1 byte

1 kilobyte

1 gibibyte

1 megabyte

1 mebibyte

[5]

- (b) (i) Perform the following binary addition. Show your working.

$$\begin{array}{r} 10101010 \\ + 00110111 \\ \hline \end{array}$$

[2]

- (ii) State how an overflow can occur when adding two binary integers.

[1]

- (c) Convert the hexadecimal value F0 into denary.

[1]

55. O/N 21/P13/Q7/a,b

7 Bobby is recording a sound file for his school project.

(a) He repeats the recording of the sound several times, with a different sample rate each time.

(i) Describe the reasons why the sound is closer to the original when a higher sample rate is used.

[2]

(ii) Describe the reasons why the sound file size increases when a higher sample rate is used.

[2]

(b) Bobby wants to email the sound file to his school email address. He compresses the file before sending the email.

(i) Explain the reasons why Bobby compresses the sound file.

[2]

(ii) Bobby uses lossless compression.

Describe how lossless compression can compress the sound file.

[2]

56. M/J 21/P11/Q1/a,b,c

1 Anya scans an image into her computer for a school project.

(a) The scanned image is a bitmapped image.

(i) Complete the following table to describe the two terms about graphics.

Term	Description
Pixel	
File header	

[2]

(ii) The image is scanned with an image resolution of 1024×512 pixels, and a colour depth of 8 bits per pixel.

Calculate an estimate for the file size, giving your answer in mebibytes. Show your working.

Working

Answer mebibytes

[3]

- (b) The image is compressed using lossless compression.

Identify **one** method of lossless compression that can be used to compress the image **and** describe how the method will reduce the file size.

Lossless compression method

Description

[3]

- (c) One of the colours used in the image has the hexadecimal colour code:

#FC238A

FC is the amount of red, 23 is the amount of green and 8A is the amount of blue in the colour.

- (i) Convert the hexadecimal code FC into denary.

[1]

- (ii) The amount of green in binary is 00100011. This has the denary number 15 added to it to create a second colour.

Add the denary number 15 to the binary number 00100011 and give your answer in binary.

Perform the addition in binary. Show your working.

Working

Answer (in binary)

[3]

- (iii) Hexadecimal 23 in two's complement representation is 00100011. The denary number 10 needs to be subtracted from this value.

Subtract the denary number 10 from the two's complement representation 00100011.

Give your answer in binary. Show your working.

Working

Answer (in binary)

[3]

57. M/J 21/P12/Q4/b(i)

- 4 The table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
CMP	<address>	Compare the contents of ACC with the contents of <address>
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
JMP	<address>	Jump to the given address
IN		Key in a character and store its ASCII value in ACC
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
# denotes a denary number, e.g. #123		

The current contents of the main memory and selected values from the ASCII character set are:

Address	Instruction
70	IN
71	CMP 100
72	JPE 80
73	CMP 101
74	JPE 76
75	JMP 80
76	LDD 102
77	INC ACC
78	STO 102
79	JMP 70
80	LDD 102
81	DEC ACC
82	STO 102
83	JMP 70
...	
100	68
101	65
102	100

ASCII code table (selected codes only)

ASCII code	Character
65	A
66	B
67	C
68	D

(b) Some bit manipulation instructions are shown in the table:

Instruction		Explanation
Opcode	Operand	
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
<address> can be an absolute address or a symbolic address # denotes a denary number, e.g. #123		

The contents of the memory address 300 are shown:

Bit Number	7	6	5	4	3	2	1	0
300	0	1	1	0	0	1	1	0

(i) The contents of memory address 300 represent an unsigned binary integer.

Write the denary value of the unsigned binary integer in memory address 300.

[1]

58. M/J 21/P12/Q6/a,b,c

6 A computer uses the ASCII character set.

(a) State the number of characters that can be represented by the ASCII character set and the extended ASCII character set.

ASCII

Extended ASCII

[2]

(b) Explain how a word such as 'HOUSE' is represented by the ASCII character set.

[2]

(c) Unicode is a different character set.

The Unicode value for the character '1' is denary value 49.

(i) Write the hexadecimal value for the Unicode character '1'.

[1]

(ii) Write the denary value for the Unicode character '5'.

[1]

59. M/J 21/P13/Q1/a,b,c

1 Anya scans an image into her computer for a school project.

(a) The scanned image is a bitmapped image.

(i) Complete the following table to describe the two terms about graphics.

Term	Description
Pixel	
File header	

[2]

(ii) The image is scanned with an image resolution of 1024×512 pixels, and a colour depth of 8 bits per pixel.

Calculate an estimate for the file size, giving your answer in mebibytes. Show your working.

Working

Answer mebibytes

[3]

(b) The image is compressed using lossless compression.

Identify **one** method of lossless compression that can be used to compress the image **and** describe how the method will reduce the file size.

Lossless compression method

Description

[3]

(c) One of the colours used in the image has the hexadecimal colour code:

#FC238A

FC is the amount of red, 23 is the amount of green and 8A is the amount of blue in the colour.

(i) Convert the hexadecimal code FC into denary.

[1]

(ii) The amount of green in binary is 00100011. This has the denary number 15 added to it to create a second colour.

Add the denary number 15 to the binary number 00100011 and give your answer in binary.

Perform the addition in binary. Show your working.

Working

Answer (in binary)

[3]

- (iii) Hexadecimal 23 in two's complement representation is 00100011. The denary number 10 needs to be subtracted from this value.

Subtract the denary number 10 from the two's complement representation 00100011.

Give your answer in binary. Show your working.

Working

Answer (in binary)

[3]

Answers Section

1. M/J 26/P11/Q1/a,b,c,d

1(a)(i)	1 mark for each correct value <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <thead> <tr> <th>Character</th><th>Denary</th><th>Binary</th><th>Hexadecimal</th></tr> </thead> <tbody> <tr> <td>☼</td><td>489</td><td>0001 1110 1001</td><td>1E9</td></tr> <tr> <td>@</td><td>64</td><td>0100 0000</td><td>40</td></tr> <tr> <td>Ω</td><td>911</td><td>0011 1000 1111</td><td>38F</td></tr> <tr> <td>Ψ</td><td>5590</td><td>0001 0101 1101 0110</td><td>15D6</td></tr> </tbody> </table>	Character	Denary	Binary	Hexadecimal	☼	489	0001 1110 1001	1E9	@	64	0100 0000	40	Ω	911	0011 1000 1111	38F	Ψ	5590	0001 0101 1101 0110	15D6	4	Correct answers only
Character	Denary	Binary	Hexadecimal																				
☼	489	0001 1110 1001	1E9																				
@	64	0100 0000	40																				
Ω	911	0011 1000 1111	38F																				
Ψ	5590	0001 0101 1101 0110	15D6																				
1(a)(ii)	1 mark for the identification - ASCII 1 mark for each difference to max 2 - There are fewer bits per character // ASCII has 7/8 bits per character, UNICODE has many more - Fewer different characters can be represented // ASCII only represents the Latin alphabet - ASCII Cannot be expanded indefinitely	3	Accept extended ASCII, or any other valid character set. English Language BOD																				
1(b)	107	1	Correct answer only																				
1(c)	1 mark for correct working - showing/working binary subtraction 1 mark for correct answer 0100 1101 Method 1: Change +30 to -30 and add 30 in binary: 0001 1110 Flip bits: 1110 0001 +0000 0001 Add the 1: 1110 0010 Add -30: 0110 1011 1110 0010 (1) 0100 1101 Method 2: Direct Subtraction 0110 1011 -0001 1110 0100 1101 11 1 Carries	2	No marks for converting to denary and back Method 1: Direct subtraction Needs to show the carries for the working mark Method 2: Change +30 to -30 and add Needs to show the conversion and the addition																				
1(d)	1 mark each - Binary is in powers of 2 whilst decimal is in powers of 10 - Any example of values e.g. Kibibyte is 1024 bytes whilst Kilobyte is 1000 bytes	2	Do not allow: Decimal uses the digits 0,1,2,3,4,5,6,7,8,9 binary only uses the digits 0 and 1 The question asks about binary and denary prefixes MP2: Binary is mebibyte, denary is megabyte is ok, binary is mebibyte, denary is kilobyte is not																				

2. M/J 26/P11/Q5/a,b,c

5(a)	1 mark for each point to max 3 - The image is made of pixels - Each pixel has one colour - Each colour has a unique binary code - The code for the colour of each pixel is stored in sequence / order	3	
------	---	---	--

5(b)	1 mark for each correct term - drawing object - property - drawing list A drawing object is an individual element of an image, for example: a line, a square. A property is an attribute of one part of the image, for example: a line colour, the width of a line. A drawing list is all of the elements that are included in the vector graphic.	3	
5(c)	No mark for the choice. 1 mark for each point for a matching justification to max 3 Bitmap image: - The logo might include something like a photograph - Bitmaps can be more detailed/precise/high quality - which may be more attractive/appealing on documents - Bitmaps can be edited more precisely - in case changes need to be made to the logo in future Vector graphic: - A logo should be made of simple shapes which are calculated mathematically - A vector graphic logo can be resized without pixelation - so, it could be used on different documents without losing quality - The file size is smaller than a bitmap image - so, the logo will not use excessive file storage if used in many documents	3	Justification must match the candidate's choice. No mix and match. Accept any sensible justification of their choice Accept other examples such as tee shirts for documents in MP3 and MP8

3. M/J 26/P12/Q1/a,b,c,d

1(a)	1 mark for a correct tick The smallest element of an image that can be one colour	1	Correct answer only
1(b)	1 mark for each correct answer - Image resolution is the number of pixels wide by pixels high that form the image and screen resolution is the number of pixels wide by pixels high that can be displayed on a screen. - If screen resolution is not as high as image resolution the quality of the image is worsened. // Image resolution can be changed, but screen resolution has a physical limit	2	Accept horizontal and/or vertical for wide and/or high Just number of pixels for either or both is NE Do not accept bits or characters for pixels
1(c)	1 mark for a correct answer For example: - File size - File type - Dimensions of the image // image size - Compression type - Data offset – bytes from the beginning of the file to the actual pixel data	1	Mark the first answer given Do not accept resolution or colour depth, they are given in the question Do not accept a description of a file header or metadata Accept height and/or width for dimensions BOD
1(d)(i)	1 mark for: Two // 2	1	Correct answer only
1(d)(ii)	1 mark for each correct point to max 2 - Run Length Encoding (RLE) can be used - A series of consecutive pixels of the same colour // a sequence of the same colour pixel is stored as a single copy of the colour and the number of consecutive occurrences 1 mark for a suitable reference to the given image - RLE: e.g. R2 // W4 // 2R4W // R2W4	3	Take care that the answers are compressing pixels not anything else Do not accept bits or characters or elements etc. for pixels If no reference to the diagram included in the answer, please check the diagram for annotations

4. M/J 26/P12/Q4/a,b

4(a)	1 mark for: The sound wave is measured 44100 times each second // 44100 samples are taken every second	1 Accept appropriate alternative words for measured including 'sampled' Accept 44.1 thousand instead of 44100
4(b)	1 mark each to max 3 - The resolution of 2 bytes (16 bits) will have more bits per sample ... which will result in a larger file size ... but the digital sound will be more accurate to original ... because a greater range/more exact amplitudes can be recorded ... with smaller quantisation errors	3 Must be clear which resolution is being described Allow reverse arguments e.g. The resolution of 8 bits will have fewer bits per sample ... which will result in a smaller file size etc.

5. M/J 26/P12/Q5/a,b

5(a)	1 mark for each correctly completed value <table><thead><tr><th>denary</th><th>unsigned binary</th><th>hexadecimal</th><th>Binary Coded Decimal (BCD)</th></tr></thead><tbody><tr><td>160</td><td>10100000</td><td>A0</td><td>0001 0110 0000</td></tr><tr><td>289</td><td>100100001</td><td>121</td><td>001010001001</td></tr><tr><td>2486</td><td>1001 1011 0110</td><td>9B6</td><td>0010010010000110</td></tr><tr><td>35123</td><td>1000100100110011</td><td>8933</td><td>00110101000100100011</td></tr></tbody></table>	denary	unsigned binary	hexadecimal	Binary Coded Decimal (BCD)	160	10100000	A0	0001 0110 0000	289	100100001	121	001010001001	2486	1001 1011 0110	9B6	0010010010000110	35123	1000100100110011	8933	00110101000100100011	4
denary	unsigned binary	hexadecimal	Binary Coded Decimal (BCD)																			
160	10100000	A0	0001 0110 0000																			
289	100100001	121	001010001001																			
2486	1001 1011 0110	9B6	0010010010000110																			
35123	1000100100110011	8933	00110101000100100011																			
5(b)	1 mark for each correct answer <table><thead><tr><th>statement</th><th>answer</th></tr></thead><tbody><tr><td>The smallest 12-bit two's complement binary number</td><td>1000 0000 0000</td></tr><tr><td>The denary value represented by the 8-bit unsigned binary number 01100010 after a logical right shift of 2 places has been applied.</td><td>24</td></tr><tr><td>The binary value after a cyclic left shift of 4 places has been applied to the binary number 10010101</td><td>0101 1001</td></tr></tbody></table>	statement	answer	The smallest 12-bit two's complement binary number	1000 0000 0000	The denary value represented by the 8-bit unsigned binary number 01100010 after a logical right shift of 2 places has been applied.	24	The binary value after a cyclic left shift of 4 places has been applied to the binary number 10010101	0101 1001	3 Note: The first answer must be 12 bits and a binary value Note: The second answer must be in denary.												
statement	answer																					
The smallest 12-bit two's complement binary number	1000 0000 0000																					
The denary value represented by the 8-bit unsigned binary number 01100010 after a logical right shift of 2 places has been applied.	24																					
The binary value after a cyclic left shift of 4 places has been applied to the binary number 10010101	0101 1001																					

6. M/J 26/P13/Q3/a,b,c(i)

3(a)	1 mark for: Each sample is stored as a 16-bit binary number // there are 2^{16} / 65536 different amplitudes that can be recorded	1 Accept 2 bytes Or any equivalent.
3(b)(i)	1 mark each to max 3 - Image is a series of pixels ... where each pixel has one colour - Each colour has a unique binary number - The binary number/code of the colour of each pixel is stored in sequence	3
3(b)(ii)	1 mark for working 1 mark for answer $2048 * 1536 * 2 * 8 / 1024 / 1024$ 48 MiB	2

3(c)(i)	1 mark each max 2 for image and max 2 for sound Image e.g. - Run-length encoding / RLE - Store each colour and the number of times it appears consecutively Sound e.g. - Reduce amplitude values (dynamic range) to only those sounds used in the actual recording reducing sampling resolution without impacting the sound - Record changes in amplitudes instead of each individual amplitudeeach change requires less bits than a new amplitude	4	Accept RLE for sound but do not allow for both parts
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7. M/J 26/P13/Q6/a,b,c,d

6(a)	1 mark for each set of lines from each denary number <table><thead><tr><th>Denary number</th><th>Equivalent unsigned value</th></tr></thead><tbody><tr><td>3120</td><td>Hexadecimal: 121</td></tr><tr><td>289</td><td>BCD: 00010101100001010001</td></tr><tr><td>22516</td><td>Binary: 110000110000</td></tr><tr><td>15851</td><td>Hexadecimal: 57F4</td></tr><tr><td></td><td>Binary: 11110111101011</td></tr><tr><td></td><td>BCD: 0011000100100000</td></tr></tbody></table>	Denary number	Equivalent unsigned value	3120	Hexadecimal: 121	289	BCD: 00010101100001010001	22516	Binary: 110000110000	15851	Hexadecimal: 57F4		Binary: 11110111101011		BCD: 0011000100100000	4	
Denary number	Equivalent unsigned value																
3120	Hexadecimal: 121																
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22516	Binary: 110000110000																
15851	Hexadecimal: 57F4																
	Binary: 11110111101011																
	BCD: 0011000100100000																
6(b)	1 mark for working 1 mark for answer. 1101 0100 1111 1111 1100 1111 1 1101 0001 1110 1111 1 1 111	2	No marks for answer if no evidence of using binary addition.														
6(c)	1 mark for correct answer 1713	1															
6(d)	1 mark for correct answer 1000000000000000	1	Last answer														

8. O/N 25/P11/Q1/a,b

1(a)(i)	B3A	1
1(a)(ii)	0001 0000 1000	1

1(a)(iii)	1 mark for the working 1 mark for the correct denary value Working: Method 1: <pre> 1111 1011 1100 Flip the bits 0000 0100 0011 Add 1 0000 0000 0001 + 0000 0100 0100 </pre> Method 2: -2048 + 1024 + 512 + 256 + 128 + 32 + 16 + 8 + 4 // -128 + 32 + 16 + 8 + 4 Denary value: -68	2
1(b)(i)	(1) 00101011	1
1(b)(ii)	1 mark per bullet point, max 2 marks - An overflow error occurs - The answer cannot be represented in the number of bits available // The answer is larger than the maximum positive number that can be stored in the register // The answer is smaller than the most negative number that can be stored in the register	2

9. O/N 25/P11/Q7/d

7(d)(i)	1 mark for the working 1 mark for the correct answer Working: $(512 * 2048 * 8) / (8 * 1024) // (512 * 2048) / 1024$ $// (2^9 * 2^{11} * 2^3) / (2^3 * 2^{10}) // (2^9 * 2^{11}) / 2^{10}$ Answer: 1024 kibibytes // 2^{10} kibibytes	2
7(d)(ii)	1 mark per bullet point, max 2 marks - The file will contain metadata - The file will have a header	2
7(d)(iii)	1 mark per bullet point, max 2 marks - Sequences of consecutive identical colours / pixels ... are stored as the colour value and the number of times it occurs consecutively	2

10. O/N 25/P12/Q6/a,b,d

6(a)	1 mark for correct answer Type of compression: lossy Justification: There will be fewer samples per second, so data will be permanently lost // There will be fewer samples per second, so the original sound cannot be re-created	1								
6(b)	1 mark for correct words in the correct order <table><tr><td>Binary Value</td><td>0011 1000</td><td>0011 1100</td><td>0011 1110</td></tr><tr><td>Word</td><td>Science</td><td>is</td><td>Amazing!</td></tr></table>	Binary Value	0011 1000	0011 1100	0011 1110	Word	Science	is	Amazing!	1
Binary Value	0011 1000	0011 1100	0011 1110							
Word	Science	is	Amazing!							
6(d)	1 mark for the working 1 mark for the correct answer Working: (1000 * 2000 * 16) / (8 * 1000 * 1000) // (1000 * 2000 * 2 / (1000 * 1000) Answer: 4 megabytes	2								

11. O/N 25/P12/Q7/a,b,c

7(a)	1 mark for the correct answer 1100 0000	1
7(b)	1 mark for the correct answer 8 // 16 // 32 // 64	1
7(c)(i)	1 mark for the correct answer 0001 0000 0100	1
7(c)(ii)	1 mark for the correct answer 68	1

12. O/N 25/P13/Q1/a,b,c

1(a)	1 mark for each correctly connected change box, max 3 marks	3
1(b)	1 mark per bullet point, max 2 marks - Each character has a unique code - Each character in the text is replaced sequentially / in order by its code	2
1(c)	1 mark per bullet point, max 2 marks - ASCII uses 7 / 8 bits Unicode can use many more / up to 32 bits - Unicode can represent a wider range of characters including different languages	2

13. O/N 25/P13/Q2/a,b,d

2(a)	3 marks for 6 correct ticks 2 mark for 4 or 5 correct ticks 1 mark for 2 or 3 correct ticks	3																																																															
	<table><tr><th rowspan="2">Example of data</th><th colspan="7">Number of bits</th></tr><tr><th>4</th><th>8</th><th>16</th><th>24</th><th>32</th><th>64</th><th>128</th></tr><tr><td>the hexadecimal value F139</td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td></tr><tr><td>16 000 000 unique amplitude values</td><td></td><td></td><td></td><td>✓</td><td></td><td></td><td></td></tr><tr><td>an IPv4 address</td><td></td><td></td><td></td><td></td><td>✓</td><td></td><td></td></tr><tr><td>256 unique colours</td><td></td><td>✓</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>an IPv6 address</td><td></td><td></td><td></td><td></td><td></td><td></td><td>✓</td></tr><tr><td>the denary value 65 000</td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td></tr></table>	Example of data	Number of bits							4	8	16	24	32	64	128	the hexadecimal value F139			✓					16 000 000 unique amplitude values				✓				an IPv4 address					✓			256 unique colours		✓						an IPv6 address							✓	the denary value 65 000			✓					
Example of data	Number of bits																																																																
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the denary value 65 000			✓																																																														
2(b)	1 mark 1111 1001 0100	1																																																															
2(d)	1 mark E33B	1																																																															

14. O/N 25/P13/Q7/b

7(b)	1 mark for the working 1 mark for the correct answer Working: $(4000 \times 3000 \times 30 \times 16) / (8 \times 1000 \times 1000 \times 1000)$ $// (4000 \times 3000 \times 30 \times 2) / (1000 \times 1000 \times 1000)$ Answer: 0.72 gigabytes	2
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15. M/J 25/P11/Q2/b

2(b)(i)	1 mark each to max 4 e.g. • Video is data-intensive • The file size needs reducing in order to • reduce the amount of bandwidth used • and reduce buffering • This means people are not behind in the conversation • and people with lower bandwidth can still take part	4
2(b)(ii)	No mark for choice but lossy is most appropriate 1 mark each to max 3 e.g. • Reduces file size more than lossless • so significantly less bandwidth / data is needed • so buffering is reduced even more than with lossless • Data can be removed which cannot be seen • reducing quality without impacting experience • for example, because resolution of video can be reduced // sample rate of audio can be reduced	3

16. M/J 25/P11/Q3/a,b

3(a)	1 mark for each correct answer <table><tr><th>Statement</th><th>Answer</th></tr><tr><td>The term for the smallest element that makes up an image.</td><td>pixel</td></tr><tr><td>The largest number of different colours that can be represented with a bit depth of 8 bits.</td><td>256 // 2⁸</td></tr><tr><td>The term for the dots per inch (dpi) when an image is displayed.</td><td>Screen resolution</td></tr></table>	Statement	Answer	The term for the smallest element that makes up an image.	pixel	The largest number of different colours that can be represented with a bit depth of 8 bits.	256 // 2⁸	The term for the dots per inch (dpi) when an image is displayed.	Screen resolution	3
Statement	Answer									
The term for the smallest element that makes up an image.	pixel									
The largest number of different colours that can be represented with a bit depth of 8 bits.	256 // 2⁸									
The term for the dots per inch (dpi) when an image is displayed.	Screen resolution									
3(b)(i)	1 mark each to max 2 • A wider range of characters can be represented • so characters from more languages can be represented • and symbols such as emojis can be used	2								
3(b)(ii)	1 mark for: 10,094	1								
3(b)(iii)	1 mark for: 8,512	1								

17. M/J 25/P12/Q2/a,b,c

2(a)	1 mark for: 0010 0010 1110 1 mark for: 22E	2
2(b)(i)	1 mark for: -30	1
2(b)(ii)	1 mark for smallest, 1 mark for largest Smallest: 1000 0000 Largest: 0111 1111	2
2(c)	1 mark for application and 1 mark for corresponding justification e.g. - an application that performs financial / banking calculations - because financial transactions use only two decimal places and must be accurate, no accumulating / rounding errors and it is difficult to represent decimal values exactly in normal binary. Or - electronic displays - because visual displays only need to show individual digits - because conversion between denary and BCD is straightforward Or - The storage of the date and time in the BIOS of a PC - because conversion between denary and BCD is more straightforward Or - Barcode systems - because conversion between denary and BCD can be accurately completed	2

18. M/J 25/P13/Q1/a,b,c

1(a)(i)	1 mark for correct working: e.g. $2000000 * 16 / (8 * 1000 * 1000)$ 1 mark for answer: 4MB	2
1(a)(ii)	1 mark each Image: - There will be fewer shades of colour available - so the image does not match the original as detail is lost Image file: - Fewer bits are used to store each pixel - so less data is stored, therefore the file size is reduced	4

1(b)	1 mark for each correct term:	3																
<table><tr><th>Description</th><th>Sound term</th></tr><tr><td>the number of times the amplitude is measured per time interval</td><td>Sampling rate</td></tr><tr><td>the number of bits used to store each amplitude measurement</td><td>Sampling resolution</td></tr><tr><td>the type of sound wave before it is recorded by a computer</td><td>Analogue</td></tr></table>		Description	Sound term	the number of times the amplitude is measured per time interval	Sampling rate	the number of bits used to store each amplitude measurement	Sampling resolution	the type of sound wave before it is recorded by a computer	Analogue									
Description	Sound term																	
the number of times the amplitude is measured per time interval	Sampling rate																	
the number of bits used to store each amplitude measurement	Sampling resolution																	
the type of sound wave before it is recorded by a computer	Analogue																	
1(c)(i)	1 mark each to max 2 8 / 16 / 32 / bits per character - Represents $2^8 / 2^{16}$ / etc. characters - Represents every language and other characters such as emojis	2																
1(c)(ii)	1 mark for each correctly completed space <table><tr><th>Character</th><th>Denary</th><th>8–bit Binary</th><th>Hexadecimal</th></tr><tr><td>!</td><td>33</td><td>0010 0001</td><td>21</td></tr><tr><td>L</td><td>76</td><td>0100 1100</td><td>4C</td></tr><tr><td>ü</td><td>252</td><td>1111 1100</td><td>FC</td></tr></table>	Character	Denary	8–bit Binary	Hexadecimal	!	33	0010 0001	21	L	76	0100 1100	4C	ü	252	1111 1100	FC	3
Character	Denary	8–bit Binary	Hexadecimal															
!	33	0010 0001	21															
L	76	0100 1100	4C															
ü	252	1111 1100	FC															

19. M/J 25/P13/Q7/b,c

7(b)	1 mark for $ \begin{array}{r} 1\ 1\ 1\ 1\ 0\ 1\ 0\ 1 \\ +_1 1_1\ 0_1\ 1_1\ 1\ 0\ 0\ 0_1\ 1 \\ \hline (1)\ 1\ 0\ 1\ 0\ 0\ 1\ 1\ 0 \end{array} $ Answer: (1) 1010 0110	1
7(c)	1 mark for showing binary subtraction (any method) Direct subtraction: $ \begin{array}{r} 0\ 1\ 1\ 10\ 10\ 11\ 10\ 0 \\ -0\ 0\ 0_1\ 1_1\ 1_1\ 1_1\ 1\ 0 \\ \hline 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \end{array} $ Adding the two's complement: $ \begin{array}{r} 0\ 1\ 1\ 0\ 0\ 1\ 0\ 0 \\ +_1 1_1\ 1_1\ 1\ 0\ 0\ 0\ 1\ 0 \\ \hline (1)\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \end{array} $ 1 mark for answer 0100 0110	2

20. O/N 24/P11/Q1/a,b,c,d

1(a)	1 mark for: - A tebibyte = 1024 gibibytes / 1 048 576 kibibytes / 2^{40} bytes whereas a gigabyte = 1000 megabytes / 1 000 000 kilobytes / 10^9 bytes - Tebi is binary prefix giga is denary prefix	1
1(b)(i)	1 mark for: C67	1
1(b)(ii)	1 mark for: -1641	1
1(b)(iii)	1 mark for: 573	1
1(c)	1 mark for working 1 mark for answer. Working Using the two's complement and adding: 23 in binary = 00010111 so minus 23 is 11101001 0100 1010 + <u>1110 1001</u> (1)0011 0011 Direct subtraction: 0100 1010 - <u>0001 0111</u> 0011 0011 Answer: 0011 0011	2
1(d)	1 mark from: - The answer cannot be represented in the number of bits available - The answer is larger than the maximum positive number that can be stored in the register - The answer is smaller than the most negative number that can be stored in the register	1

21. O/N 24/P11/Q4/b

4(b)(i)	1 mark for each bullet point (max 3) e.g. • Less of the teacher's storage space is used • ... so more student work can be stored • Transmission time is reduced • ... so the teacher does not have to wait as long for it to arrive • Bandwidth usage is reduced • ... so other transmissions are not adversely affected • Less data allowance is used on the teacher's email system	3
4(b)(ii)	1 mark for each bullet point (max 3) • Run-Length Encoding // RLE • Repeated sequences of the same characters are replaced by • ... a single copy of the character • ... and a count of the number of characters	3

22. O/N 24/P12/Q7/a,b

7(a)(i)	1 mark for each correct definition Colour depth: • the number of bits used to represent a colour // the number of colours that can be represented in an image File header: • stores data about the image file / metadata	2
7(a)(ii)	1 mark for each correct explanation Image quality: • Decreasing resolution means details within the image are lost because there are fewer pixels // Increasing resolution means the image is more detailed because there are more pixels File size: • Decreasing the resolution will decrease the file size because there are fewer pixels therefore less data // Increasing the resolution will increase the file size because there are more pixels therefore more data	2
7(a)(iii)	1 mark for correct method For example: • Run-Length Encoding	1

7(b)	1 mark for each correct definition Property: - an attribute of a drawing object // data about a shape // defines one aspect of the appearance of a drawing object Drawing list: - all the drawing objects/shapes in an image // stores the commands/descriptions / mathematical equations required to draw each object	2
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23. O/N 24/P13/Q6/a,b,c

6(a)	1 mark for identification 1 mark for matching description e.g. - ASCII 7/8 bits per character // represents 128/256 characters // represents all characters from Latin alphabet - UNICODE 8/16/32 bits per character // represents 256/65536+ characters // represents all characters in all languages	2
6(b)(i)	1 mark for: 256 // 2^8	1
6(b)(ii)	1 mark for: Increased file size	1
6(b)(iii)	1 mark for each bullet point (max 2) e.g. - The change may not be noticeable // Data removed is usually not noticed by the human eye ... for example, changes in shade/detail - It produces a larger decrease in file size compared to lossless // Lossy decreases file size considerably	2
6(c)(i)	1 mark for each bullet point (max 2) - Value/magnitude/size of the analogue sound wave is measured a set number of times each second/time / at set intervals - Each sample/reading/measurement is given the binary number and stored in sequence	2

6(c)(ii)	1 mark for each correct point and 1 mark for matching expansion e.g. • Decrease sample rate • ... fewer samples/readings/measurements stored per second // fewer bits per second stored • Decrease sample resolution • ... fewer bits per sample/reading/measurement // each sample has fewer bits • Sound outside of set/human hearing range is removed • ... fewer measurements are stored / decreases the number of possible binary values so fewer bits are stored	2
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24. O/N 24/P13/Q8/a,b

8(a)	1 mark for: Denary value: 8107	1
8(b)	1 mark for each bullet point for the method (max 2) e.g. • Flip each bit then add 1 ... • ... method of converting the new binary number into denary • Most significant 1 bit is treated as the corresponding negative denary value ... • ... add the other positive corresponding denary values 1 mark for correct conversion Denary value: -97	3

25. M/J 24/P11/Q2/d

2(d)	1 mark each to max 3 : • Data transmission to user's smartphone will take longer • ... because there is more data to transmit • The secondary storage device will fill faster • ... fewer videos will be able to be stored long-term // videos are overwritten more often	3
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26. M/J 24/P11/Q7

7	1 mark each: - Working – carried values clearly indicated - Correct answer 0001 1000 - Overflow clearly indicated as overflow Example: <pre> 1 0 0 1 1 1 1 0 0 1 1 0 0 0 0 1 + 0 0 0 1 1 0 0 1 ----- (1) 0 0 0 1 1 0 0 0 1 1 1 1 1 1 1(carries) </pre>	3
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27. M/J 24/P12/Q2/d

2(d)(i)	1 mark each: - Image is made of pixels and each pixel has one colour - Each colour has a unique binary code Code for the colour of each pixel is stored in sequence	3
2(d)(ii)	1 mark each to max 2: - List of objects in the drawing - A list that stores the command/description/equation required to draw each object - Properties of each object e.g. the fill colour, line weight/colour	2
2(d)(iii)	1 mark each to max 2: - Dedicated connection to the headset // not sharing bandwidth - Already fast connection that can transmit the data without slowing - Video may already be a small file size and does not need further reduction - Video is not saved so storage is not an issue in the headset	2

28. M/J 24/P12/Q7/a,b,c

7(a)	1 mark for: 3300 kibibytes	1
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7(b)	1 mark each: - Converting 100 to binary 0110 0100 and 10 to binary 0000 1010 - Subtraction method - converting 10 to – 10 and adding // direct subtraction correct answer 0101 1010 Method 1: Converting to -10 and adding: Binary for +10 is 0000 1010 Binary for –10 is 1111 0110 Binary for 100 is 0110 0100 100 + (-10): <pre> 0 1 1 0 0 1 0 0 +1 1 1 1 0 1 1 0 (1) 0 1 0 1 1 0 1 0 Carries: 1 1 0 0 1 0 0 0 </pre> Method 2: Direct Subtraction Borrows: <pre> 0 0 0 1 1 0 1 0 0 1 1 0 0 1 0 0 - 0 0 0 0 1 0 1 0 0 1 0 1 1 0 1 0 </pre>	3
7(c)	1 mark for working: 1100 0000 1111 // 2048 + 1024 + 8 + 4 + 2 + 1 // (12 * 16²) + 15 // (12 * 16 * 16) + 15 // 3072 + 15 1 mark for correct answer: 3087	2

29. M/J 24/P13/Q1/a,b,c,d

1(a)	1 mark for each correct answer: - binary 3072 - denary/decimal 2000 A kibibyte has a binary prefix. Three kibibytes is the same as 3072 bytes. A megabyte has a decimal/denary prefix. Two terabytes is the same as 2000 gigabytes.	4
1(b)	1 mark for correct answer: F1	1

1(c)	1 mark for a correct answer: The answer is too long to be represented in the same number of bits as the binary numbers being added	1								
1(d)(i)	1 mark for all 3 answers correct: <table><tr><th>Character set</th><th>Number of bits</th></tr><tr><td>ASCII</td><td>7</td></tr><tr><td>extended ASCII</td><td>8</td></tr><tr><td>Unicode</td><td>16/32</td></tr></table>	Character set	Number of bits	ASCII	7	extended ASCII	8	Unicode	16/32	1
Character set	Number of bits									
ASCII	7									
extended ASCII	8									
Unicode	16/32									
1(d)(ii)	1 mark each: - Each character has a unique binary code - The binary code for each character is stored in sequence	2								

30. M/J 24/P13/Q2/a,b,c

2(a)	1 mark for working: $4000 * 3000 * 4$ 1 mark for correct answer: 48MB	2
2(b)(i)	1 mark each to max 3 : - The file takes less storage space on the web server than if lossless compression was used - The file is faster to upload/download to/from the server than if lossless compression was used - The file uses less bandwidth to transmit than if lossless compression was used - The file consumes less data allowance than if lossless compression was used	3
2(b)(ii)	1 mark each: - Identifies consecutive repeating pixels of the same colour - Stores the colour /pattern and the number of times it repeats	2
2(c)	1 mark each to max 2 : - Colour/bit depth - Image resolution	2

31. O/N 23/P11/Q1/a,b,c

1(a)	2 marks for all 3 lines correct 1 mark for 1 line correct <table><thead><tr><th>Term</th><th>Description</th></tr></thead><tbody><tr><td>drawing list</td><td>a component created using a formula</td></tr><tr><td>drawing object</td><td>defines one characteristic of a component</td></tr><tr><td>property</td><td>data required to create all components in the graphic</td></tr></tbody></table>	Term	Description	drawing list	a component created using a formula	drawing object	defines one characteristic of a component	property	data required to create all components in the graphic	2
Term	Description									
drawing list	a component created using a formula									
drawing object	defines one characteristic of a component									
property	data required to create all components in the graphic									
1(b)	1 mark for the definition - The number of bits used to represent each colour 1 mark for each bullet point for the explanation - Increase in bit depth means the image has a greater range of colours // Decrease in bit depth means the image has a smaller range of colours - Increase in bit depth makes the image closer to the original / more realistic // Decrease in bit depth makes the image less like the original / less realistic	3								
1(c)	1 mark for each bullet point (max 2) - Reduced bandwidth usage when transmitting the message - Reduced transmission time from email client to email server - Reduced storage space on the email - Email accounts often have a maximum size for an attachment	2								

32. O/N 23/P12/Q3/a,b,c

3(a)	1 mark for: Either - kibibyte = 1,024 bytes // 2^{10} bytes and megabyte = 1000 kilobytes // 1 000 000 bytes // 10^3 kilobytes // 10^6 bytes Or - kibi is binary prefix and mega is denary prefix	1
3(b)(i)	1111 0011 1100	1
3(b)(ii)	865	1
3(b)(iii)	470	1

3(c)	1 mark for a correct application 1 mark for a corresponding justification • An application that performs financial / banking calculations • ... because it is difficult to represent decimal values exactly in normal binary and financial transactions use only two decimal places and must be accurate, no accumulating errors Or • Electronic displays, e.g. calculators, digital clocks • ... because visual displays only need to show individual digits • ... because conversion between denary and BCD is easier Or • The storage of the date and time in the BIOS of a PC • ... because conversion with denary is easier	2
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33. O/N 23/P12/Q6/a,b,c

6(a)	1 mark for each bullet point (max 3) Lossy compression (ticked) • Loss of quality will not be noticed • Needs to be viewed in real time so less bandwidth needed if file size smaller • Smaller file sizes will reduce buffering so the video will play more smoothly • Viewers may watch on different devices, so may not need high quality resolution Lossless compression (ticked) • Original recording may not have been made in high resolution • Could be streaming to high bandwidth devices • The reduction in the file size is sufficient for the receiving device • Viewers do not want any loss of quality	3
6(b)	1 mark for each bullet point (max 3) Increase sampling resolution • ... the number of bits used for each sample is increased • ... there will be more values available to represent each sample // more amplitudes can be represented • ... each binary amplitude/note in the digital recording is closer to the analogue amplitude/note • ... quantisation errors are reduced • ... the digital soundwave is closer to the original analogue soundwave Decrease sampling resolution • ... the number of bits used for each sample is decreased • ... there will be fewer values available to represent each sample // fewer amplitudes can be stored • ... each binary amplitude/note in the digital recording is further from the analogue amplitude/note • ... quantisation errors are increased • ... the digital soundwave is less like the original analogue soundwave	3

6(c)	1 mark for answer, 1 mark for working Answer: 2.5 mebibytes Working: $(2048 \times 1024 \times 10) / (8 \times 1024 \times 1024)$	2
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34. O/N 23/P13/Q1/a,b

1(a)	1 mark for: data values that are continuously changing // variable // any value	1
1(b)	1 mark for 1 correct line 2 marks for 3 correct lines Term sampling sampling rate sampling resolution Description the number of samples taken per second taking measurements at regular intervals and storing the values the number of bits used to store each sample	2

35. O/N 23/P13/Q2/a,b

2(a)	1 mark for each bullet point: - more pixels can be stored / are available - the image is sharper / less pixelated	2
2(b)	1 mark for answer, 1 mark for working Answer: 4096 kibibytes Working: File size = $(2048 \times 1024 \times 16) / (8 \times 1024)$	2

36. O/N 23/P13/Q5/b(i)

5(b)(i)	1 mark for each bullet point (max 2). - To reduce the time it takes to download the program files from the web server // to upload them to the server in the first place - To reduce the amount of storage space used on the web server // user's device	2
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37. M/J 23/P11/Q1/a,b,c

1(a)	1 mark for each correct definition <table><tr><th>Term</th><th>Definition</th></tr><tr><td>Drawing list</td><td>All the drawing objects in an image // a list that stores the commands required to draw each object</td></tr><tr><td>Pixel</td><td>The smallest part of the image // one square / dot of one colour</td></tr><tr><td>Colour depth</td><td>The number of bits per pixel // determines the number of colours that can be represented in the image</td></tr></table>	Term	Definition	Drawing list	All the drawing objects in an image // a list that stores the commands required to draw each object	Pixel	The smallest part of the image // one square / dot of one colour	Colour depth	The number of bits per pixel // determines the number of colours that can be represented in the image	3
Term	Definition									
Drawing list	All the drawing objects in an image // a list that stores the commands required to draw each object									
Pixel	The smallest part of the image // one square / dot of one colour									
Colour depth	The number of bits per pixel // determines the number of colours that can be represented in the image									
1(b)(i)	1 mark each Examples: • Confirmation that it is a bitmap // file type • Compression type • Location/offset of data within the file • Dimensions e.g.100 × 100 pixels	2								
1(b)(ii)	1 mark for working; 1 mark for answer • Working: (1500 * 3000 * 8) / 1000 / 1000 • Answer: 36 MB	2								
1(c)(i)	1 mark each Examples: • The customers will be able to download the photographs in less time • ...and they will take less of the customer's bandwidth • The photographs will take up less space on the customer's storage medium • ...therefore the customers can store more images • ...and will have more space for other files	3								

1(c)(ii)	1 mark each to max 2 for explanation; 1 mark for an image related example - RLE stores a colour and the number of times it occurs consecutively - An image may not have many sequences of the same colour - It would need to store each colour and then the count/number 1 which adds data Example: - Red-Green-Blue would become Red 1 Green 1 Blue 1	3
----------	---	----------

38. M/J 23/P11/Q3/d(i,ii,iii,iv,v)

3(d)(i)	1 mark from - Kibibyte is 1024 bytes and kilobyte is 1000 bytes - Kibibyte is binary prefix and kilobyte is denary prefix	1
3(d)(ii)	1001 0110 0100	1
3(d)(iii)	F2	1
3(d)(iv)	Smallest: 10000000 Largest: 01111111	2
3(d)(v)	1 mark for working 1 mark for answer <pre> 1 0 1 1 0 0 0 0 + 0 0 0 1 1 0 1 1 1 1 0 0 1 0 1 1 1 1 </pre>	2

39. M/J 23/P12/Q4/a,b,c

4(a)	2^{16} // 65536	1
4(b)	1 mark for working; 1 mark for answer - Working: $+120 = 0111\ 1000$ - Answer: 1000 0111	2
4(c)	1 mark for working; 1 mark for answer - Working: $A04 = (10 * 16^2) + 4$ // $A04 = (10 * 256) + 4$ // $A04 = 1010\ 0000\ 0100$ - Answer: 2564	2

40. M/J 23/P13/Q3/c

3(c)(i)	1 mark each • The amplitude is recorded a set number of times a second • Each (instance of an) amplitude is given a corresponding binary number • The binary number (of each amplitude) is saved in sequence	3
3(c)(ii)	1 mark each; max 2 for rate and max 2 for resolution Sampling rate • There are smaller 'gaps' in the sound wave // sound is recorded more often • Digital waveform is closer to the analogue waveform • The quantisation errors are smaller Sampling resolution • There are more bits per sample // a wider range of amplitudes can be stored • Each binary amplitude/note (in the digital recording) is closer to the analogue amplitude/note • Digital waveform is closer to the analogue waveform • The quantisation errors are smaller	4

41. M/J 23/P13/Q7/a

7(a)	1 mark for each correct line	3													
	<table><thead><tr><th>Description</th><th>Denary value</th></tr></thead><tbody><tr><td rowspan="2">The smallest integer that can be represented in 8-bit two's complement.</td><td>-127</td></tr><tr><td>127</td></tr><tr><td rowspan="3">The largest integer that can be represented in 8-bit two's complement.</td><td>-255</td></tr><tr><td>-128</td></tr><tr><td>-256</td></tr><tr><td rowspan="3">The largest unsigned integer that can be represented in 8 bits.</td><td>256</td></tr><tr><td>128</td></tr><tr><td>255</td></tr></tbody></table>	Description	Denary value	The smallest integer that can be represented in 8-bit two's complement.	-127	127	The largest integer that can be represented in 8-bit two's complement.	-255	-128	-256	The largest unsigned integer that can be represented in 8 bits.	256	128	255	
Description	Denary value														
The smallest integer that can be represented in 8-bit two's complement.	-127														
	127														
The largest integer that can be represented in 8-bit two's complement.	-255														
	-128														
	-256														
The largest unsigned integer that can be represented in 8 bits.	256														
	128														
	255														

42. O/N 22/P11/Q1/a,b,c,d

1(a)(i)	39	1
1(a)(ii)	27	1
1(a)(iii)	-25	1

1(b)	1 mark for working, 1 mark for answer (0011 1110) Working using borrowing: <pre> 1 10 10 1 0 10 0 0 10 10 1 0 1 1 0 0 1 1 0 1 1 1 0 1 0 1 - ----- 0 0 1 1 1 1 1 0 </pre> Working using two's complement: <pre> 1 0 1 1 0 0 1 1 Two's complement 1 0 0 0 1 0 1 1 + ----- (1) 0 0 1 1 1 1 1 0 1 1 </pre>	2
1(c)	1 mark for similarity, 2 marks for differences Similarity (max 1): • both can use 8 bits • both represent each character using a unique code • Unicode will contain all the characters that ASCII contains // ASCII is a subset of Unicode Differences (max 2): • Unicode can go up to 32 bits per character whereas ASCII is 7 or 8 bits Unicode can represent a wider range of characters than ASCII • different languages are represented using Unicode, ASCII is only for one language	3
1(d)(i)	the number of samples taken per unit time / per second	1
1(d)(ii)	1 mark for each bullet point (max 2): • increases the number of bits per sample // larger range of values • which means that the file size increases • makes the sound file more accurate //digital waveform closer to original (analogue) waveform • smaller quantisation errors	2

43. O/N 22/P11/Q7/c

7(c)	1 mark for each bullet point (max 3): • lossless compression • Run Length Encoding • repeated sequences of <u>characters</u> are replaced by • ... a single copy of the character • ... and a counter of the number of characters	3
------	--	----------

44. O/N 22/P12/Q2/a,b

2(a)(i)	−106	1
2(a)(ii)	96	1
2(a)(iii)	1 mark for each bullet point: • 149 decimal • 0001 0100 1001	2
2(b)	1101 0010	1

45. O/N 22/P12/Q6/b

6(b)(i)	1 mark for each bullet point (max 2): • improves the accuracy of the sound file • ... because (digital) waveform more closely resembles the analogue waveform • quantization errors are reduced • increases the amount of detail stored	2
6(b)(ii)	1 mark for each bullet point: • decreases the file size of the sound file • ... because fewer bits are used to store each sample	2

46. O/N 22/P12/Q8/a,b,c

8(a)	1 mark one or two correct row(s). 2 marks for all three correct rows.			2																
<table><tr><th>Action</th><th>Increases the file size</th><th>Decreases the file size</th><th>No change to the file size</th></tr><tr><td>Change the colour depth of the image file to 16 bits per pixel.</td><td></td><td>✓</td><td></td></tr><tr><td>Change the screen resolution to 1366 × 768 pixels.</td><td></td><td></td><td>✓</td></tr><tr><td>Change the colour of the rectangle from black to red.</td><td></td><td></td><td>✓</td></tr></table>					Action	Increases the file size	Decreases the file size	No change to the file size	Change the colour depth of the image file to 16 bits per pixel.		✓		Change the screen resolution to 1366 × 768 pixels.			✓	Change the colour of the rectangle from black to red.			✓
Action	Increases the file size	Decreases the file size	No change to the file size																	
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Change the screen resolution to 1366 × 768 pixels.			✓																	
Change the colour of the rectangle from black to red.			✓																	

8(b)	1 mark for each bullet point (max 2). For example: • can be enlarged without pixelation / loss of quality • individual components of the image can be edited • generally a smaller file size	2								
8(c)(i)	1 mark for each correct underlined part: <table><tr><td>Uncompressed sound</td><td>RLE compressed sound</td></tr><tr><td>EA F1 F1 F2 F2 F2 EA</td><td>1EA 2F1 3F2 1EA</td></tr><tr><td><u>AB AB FF FF 1D 67</u></td><td>2AB 2FF 11D 167</td></tr><tr><td>32 32 80 81 81</td><td><u>232 180 281</u></td></tr></table>	Uncompressed sound	RLE compressed sound	EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA	<u>AB AB FF FF 1D 67</u>	2AB 2FF 11D 167	32 32 80 81 81	<u>232 180 281</u>	2
Uncompressed sound	RLE compressed sound									
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<u>AB AB FF FF 1D 67</u>	2AB 2FF 11D 167									
32 32 80 81 81	<u>232 180 281</u>									
8(c)(ii)	1 mark for each bullet point: • all the data is required // no data can be lost • ... otherwise text file will be corrupted / not make sense	2								

47. O/N 22/P13/Q1/a,b

1(a)	1 mark for one or two correct ticks, 2 marks for three correct ticks.	2																
	<table><tr><th>Action</th><th>Accuracy increases</th><th>Accuracy decreases</th><th>Accuracy does not change</th></tr><tr><td>Change the sampling rate from 40 kHz to 60 kHz.</td><td>✓</td><td></td><td></td></tr><tr><td>Change the duration of the recording from 20 minutes to 40 minutes.</td><td></td><td></td><td>✓</td></tr><tr><td>Change the sampling resolution from 24 bits to 16 bits.</td><td></td><td>✓</td><td></td></tr></table>	Action	Accuracy increases	Accuracy decreases	Accuracy does not change	Change the sampling rate from 40 kHz to 60 kHz.	✓			Change the duration of the recording from 20 minutes to 40 minutes.			✓	Change the sampling resolution from 24 bits to 16 bits.		✓		
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Change the duration of the recording from 20 minutes to 40 minutes.			✓															
Change the sampling resolution from 24 bits to 16 bits.		✓																
1(b)	1 mark for answer; 1 mark for working. Working: Size = 50KHz * (20 × 60) * 16 bits = 50 000 * 1200 * 16 bits // 50 000 *1200 * 2 bytes = 960 000 000 bits = 120 000 000 bytes = 120 000 kilobytes = 120 megabytes Answer = 120 megabytes	2																

48. O/N 22/P13/Q2/e

2(e)	1 mark for each correctly completed term; The bit depth of a bitmap image is the number of bits that are used to store each pixel. Metadata about the image is stored in the header of the file.	2
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49. O/N 22/P13/Q9/a,b

9(a)(i)	93	1
9(a)(ii)	147	1
9(b)	1 mark for each correct benefit (max 2) Examples: • straightforward to convert to / from BCD and denary • ... so it is less complex to encode and decode for programmers • easier for digital equipment use BCD to display output information • can represent monetary values exactly	2

50. O/N 21/P11/Q1/a,b,c

1(a)	1 mark for each correct line 	5
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1(b)(i)	1 mark for answer 1 mark for working e.g. <pre> 1010 1010 0011 0111 <u>1110 0001</u> 1 1 1 1 1 1 </pre>	2
1(b)(ii)	The result is a larger number than can be stored in the given number of bits. // The result is greater than <u>255</u>	1
1(c)	240	1

51. O/N 21/P11/Q7/a,b

7(a)(i)	1 mark per bullet point • Smaller time gaps between the samples • Makes the digital sound wave more accurate • Smaller quantisation errors	2
7(a)(ii)	1 mark per bullet point • More samples/data are taken/recorded • ... so more bits are stored altogether	2
7(b)(i)	1 mark per bullet point • Reduces the file size • Faster to transmit/download • Original file is too large for email storage/attachment	2
7(b)(ii)	1 mark per bullet point to max 2 e.g. • Reduce amplitude to only the range used • ... limited amplitudes mean fewer bits per sample • Run-length-encoding • ... Where consecutive sounds are the same record the binary value of the sound and number of times it repeats • Record the changes instead of the actual sounds	2

52. O/N 21/P12/Q4/a,b,c,d,e

4(a)	205	1
4(b)	-51	1
4(c)	CD	1

4(d)	1 mark for: The denary value in each group of 4 bits is greater than 9 // the denary value in each nibble is greater than 9	1
4(e)(i)	1 mark for working, 1 mark for answer <pre> 0011 1101 +0010 1101 ----- 0110 1010 111 1 1 </pre>	2
4(e)(ii)	1 mark for working, 1 mark for answer <pre> 0011 1101 +1101 0011 (two's complement) ----- 0001 0000 1111 111 </pre>	2

53. O/N 21/P12/Q5/a,b

5(a)	1 mark for each description, 1 mark for each valid example <table border="1"> <thead> <tr> <th>Term</th><th>Description</th><th>Example from logo</th></tr> </thead> <tbody> <tr> <td>Property</td><td>data about the shapes // defines one aspect of the appearance of the drawing object</td><td>e.g. black line // white fill // black fill //solid (line) // font of letter // colour of triangle</td></tr> <tr> <td>Drawing list</td><td>the list of shapes involved in an image // a list that stores the command/description required to draw each object</td><td>e.g. triangle // capital letter R // rectangle // line</td></tr> </tbody> </table>	Term	Description	Example from logo	Property	data about the shapes // defines one aspect of the appearance of the drawing object	e.g. black line // white fill // black fill //solid (line) // font of letter // colour of triangle	Drawing list	the list of shapes involved in an image // a list that stores the command/description required to draw each object	e.g. triangle // capital letter R // rectangle // line	4
Term	Description	Example from logo									
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Drawing list	the list of shapes involved in an image // a list that stores the command/description required to draw each object	e.g. triangle // capital letter R // rectangle // line									
5(b)(i)	1 mark for each bullet point to max 2 for each difference • Bitmap made up of pixels // bitmap is made of colours stored for individual pixels • Vector graphic store a set of instructions about how to draw the shape • When bitmap is enlarged the pixels get bigger and it pixelates • When vector is enlarged it is recalculated and does not pixelate • Bitmap files are usually bigger than vector graphics files because of the need to store data about each pixel • Vector graphics have smaller file size because they contain just the instructions to create the shapes • Bitmap images can be compressed with significant reduction in file size • Vector graphic images do not compress well because of little redundant data	4									

5(b)(ii)	1 mark for each bullet point to max 2 for each method • Reduce bit depth • ... reduces the number of bits per colour / pixel which means each pixel has fewer bits • Reduce colour palette // reduce number of colours • ... fewer colours mean fewer bits needed to store each colour • Reduce image resolution • ... fewer pixels per unit measurement means less binary to store	4
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54. O/N 21/P13/Q1/a,b,c

1(a)	1 mark for each correct line	5
1(b)(i)	1 mark for answer 1 mark for working e.g. <pre> 1010 1010 0011 0111 <u>1110 0001</u> 1 1 1 1 1 1 </pre>	2

1(b)(ii)	The result is a larger number than can be stored in the given number of bits. // The result is greater than <u>255</u>	1
1(c)	240	1

55. O/N 21/P13/Q7/a,b

7(a)(i)	1 mark per bullet point - Smaller time gaps between the samples - Makes the digital sound wave more accurate - Smaller quantisation errors	2
7(a)(ii)	1 mark per bullet point - More samples/data are taken/recorded ... so more bits are stored altogether	2
7(b)(i)	1 mark per bullet point - Reduces the file size - Faster to transmit/download - Original file is too large for email storage/attachment	2
7(b)(ii)	1 mark per bullet point to max 2 e.g. - Reduce amplitude to only the range used ... limited amplitudes mean fewer bits per sample - Run-length-encoding ... Where consecutive sounds are the same record the binary value of the sound and number of times it repeats - Record the changes instead of the actual sounds	2

56. M/J 21/P11/Q1/a,b,c

1(a)(i)	1 mark for each description Pixel: - A single square of one colour - The smallest addressable element in an image File header: - Data about the bitmap image (e.g. number of colours)	2
1(a)(ii)	1 mark per bullet point for working, 1 mark for answer Working: $1024 \times 512 = 524\,288$ pixels/bytes $524288 / 1024 / 1024$ Answer: 0.50 mebibytes	3

1(b)	1 mark for naming method, 1 mark per description to max 2 • Run-length encoding • Replace sequences of the same colour pixel • ... with colour code and number of identical pixels	3
1(c)(i)	252	1
1(c)(ii)	1 mark per bullet point • Converting 15 to binary 0000 1111 • Method for addition • Final answer $\begin{array}{r} 0010 \ 0011 \\ + \ 0000 \ 1111 \\ \hline 0011 \ 0010 \\ 1 \ 111 \end{array}$	3
1(c)(iii)	1 mark per bullet point • Converting -10 to two's complement binary 1111 0110 • Adding values • Final answer 0001 1001 $\begin{array}{r} 10 = 0000 \ 1010 \\ -10 = 1111 \ 0110 \\ \hline 0010 \ 0011 \\ + \ 1111 \ 0110 \\ \hline 0001 \ 1001 \\ 11 \ 11 \end{array}$	3

57. M/J 21/P12/Q4/b(i)

4(b)(i)	102	1
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58. M/J 21/P12/Q6/a,b,c

6(a)	1 mark for each correct answer ASCII = $128 // 2^7$ Extended ASCII = $256 // 2^8$	2
6(b)	1 mark per bullet point to max 2 • Each character has its own unique code • Each character in the word is replaced by its code • The codes are stored in the order in the word	2
6(c)(i)	31	1
6(c)(ii)	53	1

59. M/J 21/P13/Q1/a,b,c

1(a)(i)	1 mark for each description Pixel: • A single square of one colour • The smallest addressable element in an image File header: • Data about the bitmap image (e.g. number of colours)	2
1(a)(ii)	1 mark per bullet point for working, 1 mark for answer Working: • $1024 \times 512 = 524\,288$ pixels/bytes • $524288 / 1024 / 1024$ Answer: 0.50 mebibytes	3
1(b)	1 mark for naming method, 1 mark per description to max 2 • Run-length encoding • Replace sequences of the same colour pixel • ... with colour code and number of identical pixels	3
1(c)(i)	252	1
1(c)(ii)	1 mark per bullet point • Converting 15 to binary 0000 1111 • Method for addition • Final answer <pre> 0010 0011 + 0000 1111 ----- 0011 0010 1 111 </pre>	3
1(c)(iii)	1 mark per bullet point • Converting -10 to two's complement binary 1111 0110 • Adding values • Final answer 0001 1001 <pre> 10 = 0000 1010 -10 = 1111 0110 0010 0011 + 1111 0110 ----- 0001 1001 11 11 </pre>	3

Unit 2

COMMUNICATION

31 items · 270 marks

Syllabus sections in this unit

2.1 Networks including the internet

Unit 2: Communication

1. M/J 26/P11/Q7/a,b,c

- 7 A networked computer game competition takes place in a school.

The game is hosted on a server on the school LAN (local area network). Each competitor logs on using their own computer.

- (a) The game runs as a thin-client.

Give **two** characteristics of a thin-client system.

1

2

[2]

- (b) One part of the school's LAN is a wired bus topology.

- (i) Explain how data is transmitted between two nodes in a bus topology using Ethernet.

[5]

- (ii) Give **two** drawbacks of using a bus topology.

1

2

[2]

- (c) Each of the school's computers has a static private IP address.

- (i) State what is meant by a static private IP address.

[1]

- (ii) The school's network uses subnetting.

Give **three** benefits of using subnetting in a network.

1

2

3

[3]

2. M/J 26/P12/Q6/d

- 6 A health monitor attaches to a person's arm and measures key health information, for example heart rate. The monitor transmits data to the user's smartphone. The results are analysed by the phone, which displays information to the user about their health.

- (d) The data from the health monitor is transmitted to the smartphone and stored in the cloud.

Discuss the benefits and drawbacks of storing the data from the health monitor in the cloud.

[4]

3. M/J 26/P13/Q3/c(ii,iii)

3 An animation is a sequence of bitmap images with a sound recording played at the same time.

(c) The sound and images are compressed independently before being transmitted over the internet to public cloud storage.

(ii) State what is meant by a public cloud.

[1]

(iii) Give **two** benefits to the animation creator of using public cloud storage for the animation.

1

2

[2]

4. O/N 25/P11/Q7/a,b,c

7 Web browsers use Internet Protocol (IP) addresses.

(a) Complete the table by describing the following types of IP address.

Type of IP address	Description
Static	
Public	

[2]

(b) Consider the following IP address:

256.0.0.A

Circle whether this IP address is IPv4, IPv6 or an invalid IP address.

IPv4

IPv6

Invalid

Justify your choice.

[2]

(c) A user types a Uniform Resource Locator (URL) into the address bar of a web browser to access a web page.

Explain how the web browser uses the URL to access the web page.

[4]

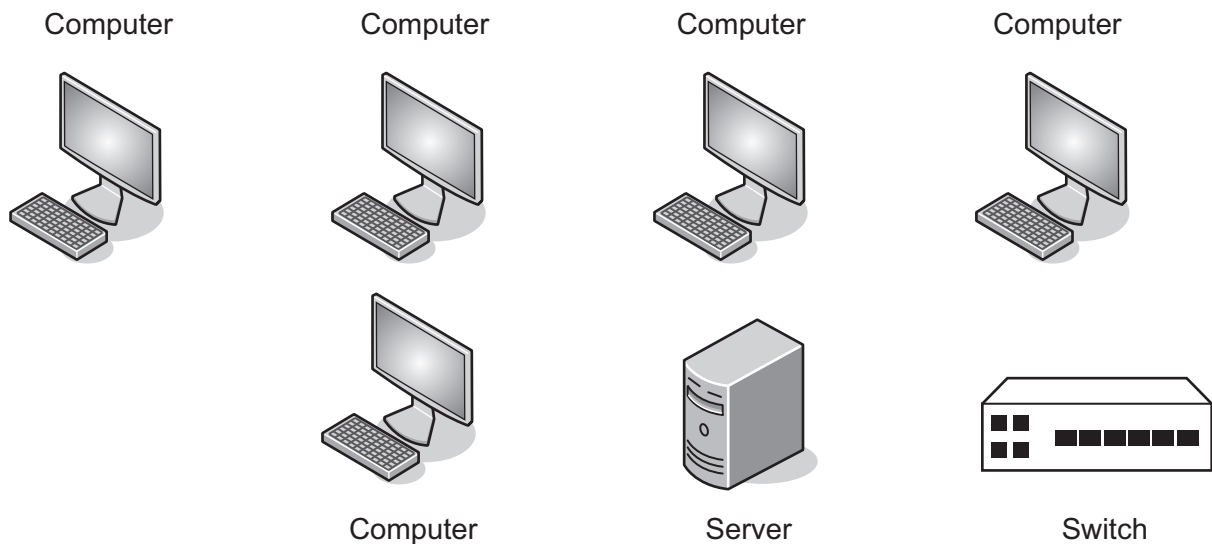
5. O/N 25/P12/Q5/a,b,c,d

5 A local area network (LAN) has five computers, one switch and one server.

(a) Describe the characteristics of a LAN.

[2]

(b) Complete the following diagram to show how these devices are connected in a star topology.



[2]

(c) Describe how packets are transmitted between two hosts using a star topology.

[3]

(d) Another type of network is a bus network.

Ethernet is used to transmit and receive data between the devices on the bus network.

Describe how collisions are detected and managed on this network.

[3]

6. O/N 25/P12/Q8/a

8 A restaurant accesses software and stores data using cloud computing.

(a) (i) Give **two** benefits of storing data using cloud computing.

1

2

[2]

(ii) Give **two** drawbacks of using cloud computing.

1

2

[2]

7. O/N 25/P13/Q4/a,b

- 4** A network engineer is setting up a Local Area Network (LAN) in an office. The network uses both wired and wireless connectivity.

(a) Give **three** advantages of a wireless network compared to a wired network.

- 1
- 2
- 3

[3]

(b) Give **one** drawback of using a wireless network.

[1]

8. M/J 25/P11/Q2/a,c

- 2** Programmers in a software development company take part in live video conferences to discuss their work.

The live video conferences take place using real-time bit streaming. The video is compressed before it is transmitted.

(a) Explain how data is transferred using real-time bit streaming.

[4]

(c) The video conference is accessed over the internet.

Complete the table by stating how modems **and** dedicated lines are used when data is transmitted over the internet.

Hardware	Use
modems	
dedicated lines	

[2]

9. M/J 25/P12/Q6/a,b,c,d,e

6 A company has multiple sites in different cities.

The company has drivers who deliver products to customers. Each driver can connect to the company's WAN (Wide Area Network) whilst out of the office using a smartphone.

(a) State **two** ways that a WAN is different to a LAN (Local Area Network).

1

2

[2]

(b) The smartphones use the cell phone network to connect to the WAN.

Explain how data is transmitted using the cell phone network.

[4]

(c) Devices that connect to the WAN have an IP address.

Complete the following description of IP addresses by writing the missing words.

IPv4 is displayed as four groups of 8-bit numbers separated by

Each IPv4 address is 32 bits.

IPv6 is made of eight groups of 4 numbers separated by colons.

Multiple consecutive groups of can be replaced with a double colon.

Each IPv6 address is bits.

A IP address can change each time the computer connects to a network.

A IP address can only be accessed by other devices in the same LAN and is assigned by the router within the LAN.

[6]

(d) The main office for the company has four wired computers, one server, two printers, one central switch and a device providing a single point of access for the internet. The network is set up as a star topology.

Draw the topology for the main office network. Label all devices.

[4]

(e) Describe the role of a switch in a network.

[3]

10. M/J 25/P13/Q3/c

3 A programmer is writing a program in a high-level language.

(c) The computer program allows users to play a game in a virtual world. Users can play the game from any computer with internet access using a web browser. The game allows users to interact with other users.

Explain the reasons why the statement 'This computer game uses a client-server model' is correct.

[4]

11. O/N 24/P13/Q9/a,b,c,d

- 9** A computer is connected to a Local Area Network (LAN) that connects to a Wide Area Network (WAN).

(a) Describe the characteristics of a WAN.

[2]

(b) Copper cable can be used to transmit data in a network.

Complete the table by identifying **and** describing **two other** transmission media that can be used to transfer data in the WAN.

Transmission medium	Description

[4]

(c) The computer on the LAN is used for real-time video conferences, where people connected to the internet communicate in real-time using video and audio.

Explain how bit streaming is used in a real-time video conference.

[4]

(d) The LAN has a router. The router has a public IP address and a private IP address.

(i) State the purpose of a public IP address and a private IP address.

Public IP address

Private IP address

[2]

(ii) The LAN uses subnetting.

Describe the purpose of subnetting in a network.

[2]

12. M/J 24/P11/Q2/e

- 2** A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:

- The digital camera in the doorbell starts recording a video.
- A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell's internal secondary storage device and overwritten when the secondary storage device is full.



- (e)** The video doorbell allows both real-time and on-demand bit streaming.

- (i)** State what is meant by bit streaming.

[1]

- (ii)** Give **two** differences between real-time and on-demand bit streaming.

1

2

[2]

13. M/J 24/P11/Q5/a

- 5** A bank allows customers to access their accounts using an application that they can download onto a device such as a smartphone.

- (a)** The system that allows customers to access their accounts using the application is a client-server model.

Describe the roles of the different devices in this model.

[4]

14. M/J 24/P11/Q8/a,b

8 A business is creating a local area network (LAN) in its office.

(a) The business is deciding which topology to use.

Tick (✓) **one or more** boxes in each row to identify the topology, or topologies, each statement describes.

Statement	Bus	Star	Mesh
all devices connect to one central device			
all devices connect to a central cable			
multiple paths for the packets to travel along			
robust against damage because if any line fails, the rest of the network retains full functionality			
most likely to lose data through collisions			

[5]

(b) The LAN will connect to the internet through a router. The router has a public IPv6 address.

(i) State why the router has a public IP address.

[1]

(ii) One difference between an IPv4 and IPv6 address is that the numbers in an IPv4 address are separated by full stops and in an IPv6 address they are separated by colons.

Identify **two other** differences between an IPv4 and IPv6 address.

1

2

[2]

15. M/J 24/P12/Q3/b,c

- 3 An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.
- (b) The exam markers use software that operates as a thin-client to mark the exam papers.

Complete the table by identifying **two** characteristics of a thin-client.

Describe how each characteristic will be used in this software.

	Thin-client characteristic	Description of use in this software
1		
2		

[4]

- (c) Data transmitted on the internet passes through multiple different systems.
- (i) Describe the role of routers in the transmission of data through the internet.
- [2]
- (ii) Describe the role of the PSTN (Public Switched Telephone Network) in the transmission of data through the internet.

[2]

16. M/J 24/P13/Q5/a,b,c,d

- 5** A multimedia design company has an office with a LAN (local area network). The LAN can have up to 20 devices connected with cables and other devices connected using wireless access.

(a) The company has private cloud storage for its employees to store their work.

(i) Define the term **private cloud**.

[1]

(ii) Describe the benefits to the company of using private cloud storage instead of public cloud storage.

[3]

(b) Part of the internal structure of the wired LAN is a star topology.

Explain how packets are transmitted between two devices in a star topology.

[2]



(c) A different part of the network uses the Ethernet protocol.

(i) A collision is detected.

Describe how the collision is managed using Carrier Sense Multiple Access/Collision Detection (CSMA/CD).

[2]

(ii) Identify **two** drawbacks of using CSMA/CD.

1

2

[2]

(d) The devices in the office have static private IP addresses.

State what is meant by a **static private IP address**.

[1]



17. O/N 23/P11/Q2/a,b,c,d

2 A school has a Local Area Network (LAN).

(a) The LAN connects to the internet using a router.

Describe the function of a router in a network.

[3]

(b) Complete the following table by writing the purpose of each of these other hardware devices used to support the LAN.

Hardware device	Purpose
switch	
Wireless Access Point (WAP)	
bridge	

[3]

(c) The students can save their school files on a public cloud.

Identify **two** drawbacks of the students storing their files on the public cloud.

1

2

[2]

(d) A new classroom is being set up with 20 computers and a switch.

Explain **one** advantage of implementing a star topology instead of a bus topology in the new classroom.

[2]

18. O/N 23/P12/Q7/a,b,c,d

7 A Local Area Network (LAN) contains four devices:

- a router
- two laptop computers
- a server.

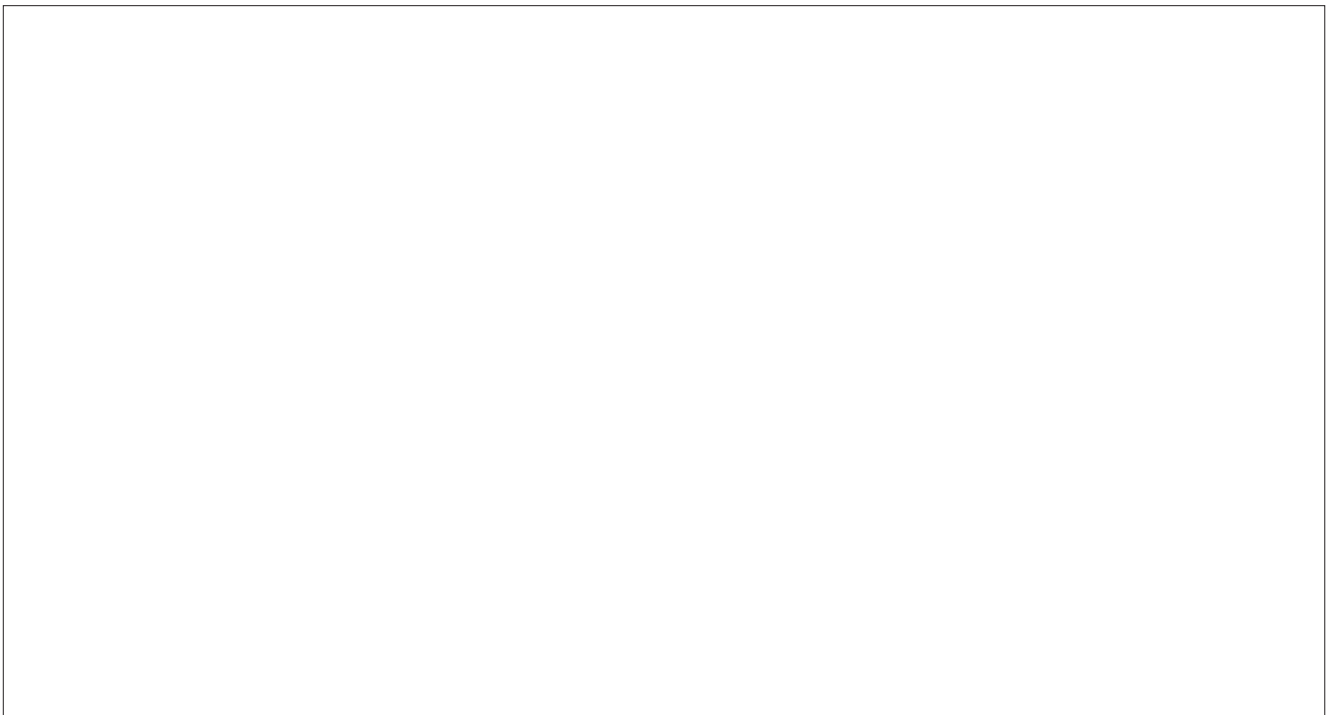
(a) The server has the IP address 192.168.3.2

Explain why this is **not** an IPv6 address.

[2]

(b) (i) The LAN is set up as a star topology.

Draw a diagram of the topology of the LAN.



[2]

(ii) Explain how data is transmitted between the two laptops in the LAN.

[2]

(iii) Subnetting can be used to separate a network into logical segments.

Describe **two other** reasons why subnetting is used in a network.

1

2

[4]

(c) State **three** tasks performed by devices to deal with collisions when using the Carrier Sense Multiple Access/Collision Detection (CSMA/CD) protocol in a network.

1

2

3

[3]

(d) The following incomplete table contains types of IP addresses and their descriptions.

Complete the table by writing the missing types of IP addresses and the missing descriptions.

Type of IP address	Description
.....	an IP address that is assigned to a device to allow direct access on the internet
static IP address	
.....	an IP address used for internal LAN communication only
dynamic IP address	

[4]

19. O/N 23/P13/Q3/a(i,ii)

3 A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

(a) (i) Explain why the company uses a public cloud to store these courses.

[2]

(ii) Describe **two** disadvantages of storing data on a public cloud compared to storing data on a server in a Local Area Network (LAN).

1

2

[4]

20. M/J 23/P11/Q4/d

- 4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

- (d) The CCTV cameras are connected to a network and transfer their data wirelessly to the central computer.

- (i) Each device on the network has an IP address.

Complete the description of IP addresses.

An IPv4 address contains groups of digits. Each group is represented in bits and the groups are separated by full stops.

An IPv6 address contains groups of digits. Each group is represented in bits. Multiple groups that only contain zeros can be replaced with a

[5]

- (ii) The network makes use of subnetting.

Describe **two** benefits of subnetting a network.

1

2

[4]

21. M/J 23/P12/Q1/a,b,c,d,e

1 A company has a LAN (local area network).

(a) Give **two** benefits of connecting computers to a LAN.

1

2

[2]

(b) Give **two** characteristics of a LAN.

1

2

[2]

(c) One of the company's offices has one server and four computers connected in a star topology.

Draw a diagram to show the layout of the office's star topology.

[2]

(d) Computers can be connected using Ethernet.

Describe what is meant by **Ethernet**.

[3]

(e) The network runs as a thick-client model.

Describe what is meant by a **thick-client** model.

[2]

22. M/J 23/P13/Q2/a,b,c,d,e

- 2** A university has two sites. Each site has several computer rooms. The computers are all connected as a WAN (wide area network).

(a) Identify **two** differences between a WAN and a LAN (local area network).

1

2

[2]

(b) The network uses different topologies in different areas of the sites. In one building there are five computers connected in a mesh topology.

(i) Describe what is meant by a mesh topology.

[2]

(ii) Give **two** advantages of using a mesh topology instead of a bus topology.

1

2

[2]

(c) The computers in one room are set up as thin-clients in a client-server model.

Describe the role of the different computers in this model.

[2]

(d) Students can connect their devices to the university network using cables or a wireless connection.

Explain the benefits to the students of allowing them to use **both** wired and wireless connections.

[4]

(e) One site has split the network into several subnetworks.

An IP address in a subnetwork is divided into two parts.

Identify **and** describe the **two** parts of an IP address in a subnetwork.

[3]

23. O/N 22/P11/Q8

- 8** A Local Area Network (LAN) uses a bus topology.

Describe how Carrier Sense Multiple Access/Collision Detection (CSMA/CD) is used in a bus network.

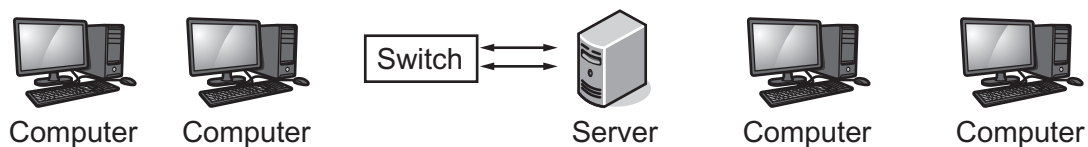
[4]

24. O/N 22/P12/Q10/a,b

10 A Local Area Network (LAN) consists of four computers, one server and a switch.

The LAN uses a star topology.

(a) Complete the following diagram to show how the hardware is connected.



[1]

(b) A router is attached to one of the devices on the LAN shown in **part (a)** to connect the LAN to the internet.

(i) Identify the device. Give a reason for your choice.

Device

Reason

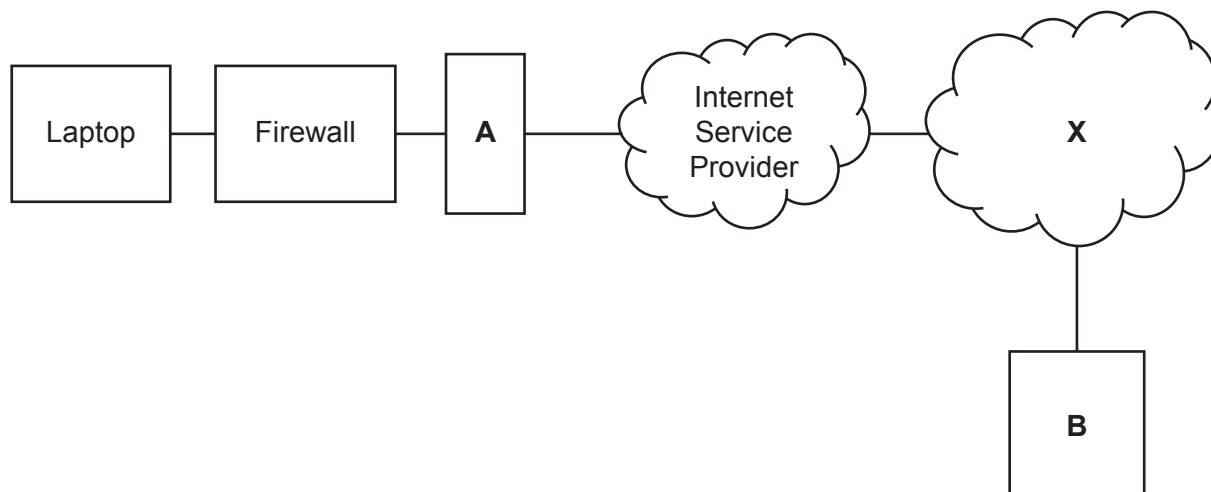
[2]

(ii) Describe the role **and** function of the router in the network.

[3]

25. O/N 22/P13/Q7/a,b,c

7 **(a)** The diagram shows the hardware and software used to retrieve files stored on the cloud.



Complete the following table by writing the answer for each row.

Answer

The name of device A that allows the laptop to connect to the internet	
A type of cloud, X	
An example of an application, B , that can run on the cloud	

[3]

- (b) Give **one** advantage and **two** disadvantages of transmitting data using satellites instead of copper cables.

Advantage 1

Disadvantage 1

Disadvantage 2

[3]

- (c) Local Area Networks (LANs) can be made up of several subnetworks.

- (i) Give **two** benefits of dividing a network into subnetworks by subnetting the LAN.

1

2

[2]

- (ii) A subnet mask is used when subnetting a LAN.

Two devices on the LAN are located in different subnetworks.

The IP addresses and corresponding subnet masks are shown:

Device IP address	Subnet mask
10.10.12.1	255.0.0.0
192.168.12.4	255.255.255.0

Identify the following network ID and host ID.

The network ID for the device with the IP address 10.10.12.1

The host ID for the device with the IP address 192.168.12.4

[2]

26. O/N 21/P11/Q8/a,b,c

8 A school is setting up a network within one of its buildings.

- (a) State whether the network will be a LAN (local area network) or a WAN (wide area network). Justify your choice.

[3]

- (b) One classroom in the building has 30 computers. The computers need to be connected to the network. Each computer has a network interface card (NIC).

Identify **two** possible devices that can be used to physically connect the 30 computers to the rest of the network.

1

2

[2]

- (c) The school has several laptops. Each laptop has a Wireless Network Interface Card (WNIC).

Describe the functions of a Wireless Network Interface Card.

[4]

27. O/N 21/P12/Q3/b

3 Andy likes to play computer games.

- (b) Andy plays some of the computer games over the internet. He has several devices that connect wirelessly to the router in his house.

- (i) Identify the topology of Andy's home network. Justify your choice.

Topology

Justification

[2]

- (ii) The router has a wireless access point (WAP) to allow the devices to connect wirelessly.

Identify **three** functions of the router in Andy's network.

1

2

3

[3]

28. O/N 21/P13/Q8/a,b,c

8 A school is setting up a network within one of its buildings.

- (a) State whether the network will be a LAN (local area network) or a WAN (wide area network). Justify your choice.

[3]

- (b) One classroom in the building has 30 computers. The computers need to be connected to the network. Each computer has a network interface card (NIC).

Identify **two** possible devices that can be used to physically connect the 30 computers to the rest of the network.

1

2

[2]

- (c) The school has several laptops. Each laptop has a Wireless Network Interface Card (WNIC).

Describe the functions of a Wireless Network Interface Card.

[4]

29. M/J 21/P11/Q4/a,b,c,d

4 Melinda and her friends set up a peer-to-peer network between their computers to share data.

(a) Describe the key features of a peer-to-peer network.

[2]

(b) Describe **two** drawbacks to Melinda and her friends of using a peer-to-peer network.

1

2

[4]

(c) Melinda connects her laptop to the internet through her router.

(i) Tick (✓) **one** box in each row to identify whether the task is performed by the router or not.

Task	Performed by router	Not performed by router
Receives packets from devices		
Finds the IP address of a Uniform Resource Locator (URL)		
Directs each packet to all devices attached to it		
Stores the IP and/or MAC address of all devices attached to it		

[2]

(ii) Melinda mainly uses the internet to watch films and play computer games.

Tick (✓) **one** box to identify whether Melinda should connect to the router using a wired or wireless network **and** justify your choice.

Wired	
Wireless	

Justification

[3]

(d) Melinda sends emails from her webmail account (email account accessed through a website).

Explain whether Melinda is using the internet, or the World Wide Web (WWW), or both.

[3]

30. M/J 21/P12/Q5/c,d

5 Seth uses a computer for work.

(c) Seth accesses both software and data using cloud computing.

(i) Give **two** benefits of storing data using cloud computing.

1

2

[2]

(ii) Give **two** drawbacks of Seth using cloud computing.

1

2

[2]

(d) Draw **one** line from each term to its **most appropriate** description.

Term

Description

Public IP address

It is only visible to devices within the Local Area Network (LAN)

Private IP address

It increments by 1 each time the device connects to the internet

Dynamic IP address

A new one is reallocated each time a device connects to the internet

Static IP address

It can only be allocated to a router

It is visible to any device on the internet

It does not change each time a device connects to the internet

[4]

31. M/J 21/P13/Q4/a,b,c,d

4 Melinda and her friends set up a peer-to-peer network between their computers to share data.

(a) Describe the key features of a peer-to-peer network.

[2]

(b) Describe **two** drawbacks to Melinda and her friends of using a peer-to-peer network.

1

2

[4]

(c) Melinda connects her laptop to the internet through her router.

(i) Tick (✓) **one** box in each row to identify whether the task is performed by the router or not.

Task	Performed by router	Not performed by router
Receives packets from devices		
Finds the IP address of a Uniform Resource Locator (URL)		
Directs each packet to all devices attached to it		
Stores the IP and/or MAC address of all devices attached to it		

[2]

(ii) Melinda mainly uses the internet to watch films and play computer games.

Tick (✓) **one** box to identify whether Melinda should connect to the router using a wired or wireless network **and** justify your choice.

Wired	
Wireless	

Justification

[3]

(d) Melinda sends emails from her webmail account (email account accessed through a website).

Explain whether Melinda is using the internet, or the World Wide Web (WWW), or both.

[3]

Answers Section

1. M/J 26/P11/Q7/a,b,c

7(a)	1 mark for each point to max 2 - Data is not stored on the client computer // all data is stored on the server - The client computer is reliant on access to server - The client computer is heavily reliant on the network connection - The client computer requires few local resources, for example, memory / processing power / storage - The client computer performs minimal functions / processes	2	
7(b)(i)	1 mark for each point to max 5 Bus: - Both nodes are connected to a bus/cable - Data is broadcast on the cable - but is only accepted by the intended recipient Ethernet: - Uses the IP/MAC addresses of the two nodes - Ethernet transmits data as frames - Ethernet uses Carrier Sense Multi-Access / Collision Detection / CSMA/CD - The transmitting node listens to see if the line is free and only submits data when the line is free - The node waits a random time if a collision is detected and then tries to send again	5	Award a maximum of 4 marks if only Ethernet referenced
7(b)(ii)	1 mark for each point to max 2 - Bus networks are inefficient with a large number of nodes - because only one transmission can be on the line at once - Bus networks are inefficient with large numbers of simultaneous data transmissions - because of the need for nodes to wait for the line to be free - Bus networks are not easily scalable - If the bus/cable goes down the whole/part of the network is unusable - Transmissions are broadcast to all nodes, so could be privacy/confidentiality/security issues	2	The answer spaces are numbered, so mark the first answer in each space
7(c)(i)	1 mark for a complete answer It does not change and it is only visible/accessible within the network/LAN	1	The answer needs both static and private
7(c)(ii)	1 mark for each point to max 3 For example: - Subnetting splits a larger network into smaller/more manageable parts - The network has increased efficiency – just faster is NE - Congestion is reduced // fewer collisions happen - Security is improved - It can be more straightforward to isolate faults - Easier to scale a network with subnets	3	Accept any sensible benefit.

2. M/J 26/P12/Q6/d

6(d)	1 mark for each correct point to max 4, max 3 if only either benefits or drawbacks discussed, not both For example: - The data can be accessed anywhere with internet access // without internet access the data cannot be accessed - The health monitor and smart phone can be mobile and still upload data - The data is shareable with medical professionals - Storing data in the cloud reduces the need for large amounts of internal storage on the phone - The user can upgrade/change their device and retain old data // the data can be accessed from different appropriate devices - The user has no control over security // There could be security issues with storing the data on the cloud - The data could be sold or misused by the cloud owner	4	Note: The mark points are examples only. Accept any sensible benefit or drawback Any answer about cost must be suitably qualified. Answers about no control need to be qualified Do not award two marks for a benefit and it's reverse as a drawback Just easier for hackers is NE
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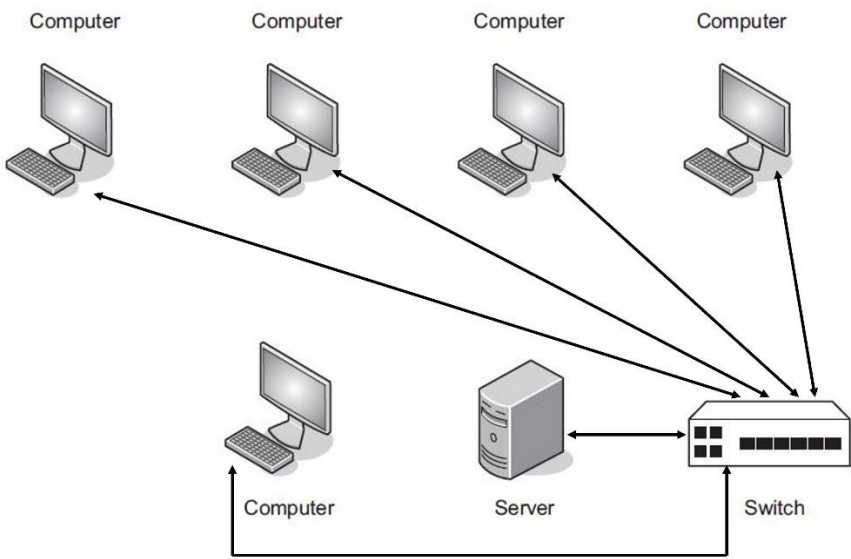
3. M/J 26/P13/Q3/c(ii,iii)

3(c)(ii)	1 mark for: Storage located on a remote server that anyone can sign up to and use	1	Not commercial sites
3(c)(iii)	1 mark each to max 2 - Animator can access anywhere that has internet access // animator can work on the animation in different locations as long as they have internet - Animator can access from any device // animator can work on different devices - Animator can give authentication/access to selected people without having to transfer the files/animation individually // Online collaboration with colleagues - Storing data in the cloud reduces the need for large amounts of internal storage on the local devices - The cloud company manages backups etc. - The Animator not responsible for security - More storage capacity readily available if needed	2	

4. O/N 25/P11/Q7/a,b,c

7(a)	1 mark for each correct description, max 2 marks	2						
	<table><tr><th>Type of IP address</th><th>Description</th></tr><tr><td>Static</td><td>IP address that does not change each time a device connects to the network</td></tr><tr><td>Public</td><td>IP address assigned to a device on a network to allow the device to be visible on the internet</td></tr></table>	Type of IP address	Description	Static	IP address that does not change each time a device connects to the network	Public	IP address assigned to a device on a network to allow the device to be visible on the internet	
Type of IP address	Description							
Static	IP address that does not change each time a device connects to the network							
Public	IP address assigned to a device on a network to allow the device to be visible on the internet							
7(b)	1 mark per bullet point for justification, max 2 marks Choice: Invalid (no mark for the choice) Justification: - It is not IPv4 because the maximum denary value should be 255, this uses 256 - It is not IPv6 because it uses a full stop to separate sections and not colon // It is not IPv6 because there are not enough groups	2						
7(c)	1 mark per bullet point, max 4 marks - The browser checks its cache for the URL - The browser parses the URL and splits it into its component parts - The browser finds the IP address for the domain name by querying a DNS server - The browser receives the matching IP address from the DNS server - The browser creates connection with the web server with the matching IP address - A request for the resource is sent to the web server with that IP address - The browser renders and displays the result - The IP address is stored in the browser cache for future use	4						

5. O/N 25/P12/Q5/a,b,c,d

5(a)	1 mark per bullet point, max 2 marks - Covers a small geographical area - Normally privately owned / dedicated infrastructure - Can use wired or wireless communication	2
5(b)	1 mark per bullet point, max 2 marks - All computers directly connected to switch and no other connections - Switch connected directly to server  <pre> graph TD C1[Computer] --> S[Switch] C2[Computer] --> S C3[Computer] --> S C4[Computer] --> S S <--> SR[Server] </pre> The diagram illustrates a star network topology. Four desktop computers are arranged in a horizontal row at the top. Below them, in the center, is a server tower. To the right of the server is a network switch. Arrows point from each of the four computers to the switch. A double-headed arrow connects the server and the switch, indicating bidirectional communication.	2
5(c)	1 mark per bullet point, max 3 marks - The sending device sends the packets to the switch - The switch checks the destination address on each packet ... and sends them only to the intended receiver	3
5(d)	1 mark per bullet point, max 3 marks - The use of <u>CSMA/CD</u> // Carrier Sense Multiple Access with Collision Detection - The workstations listen to the communication channel ... and send data only when there is no data being transmitted / the line is idle - If a collision is detected, transmission is aborted ...and a jamming signal is sent - The workstation calculates a random wait time before trying to re-transmit - The random time is increased if there are multiple collisions	3

6. O/N 25/P12/Q8/a

8(a)(i)	1 mark per bullet point, max 2 marks • Cloud storage can be free of charge for a limited amount • There is no need for personal storage devices • The data can be accessed from any computer with an internet connection • Cloud data services usually include data backup and recovery services • It is easier to share data / collaborate with colleagues • Cloud storage capacity can be quickly and easily increased if needed	2
8(a)(ii)	1 mark per bullet point, max 2 marks • The storage/cloud can only be accessed with an internet connection • There may be no backups // possibly no control over backups // reliance on third party for backups • Uploading / downloading data could take a long time • Potential issues with the cloud storage company impacting access e.g. downtime or technical problems • Potential compatibility issues if wanting to move the data • Limited storage availability without paying extra • ... meaning (possibly) more expensive in the long term • Possibly security issues	2

7. O/N 25/P13/Q4/a,b

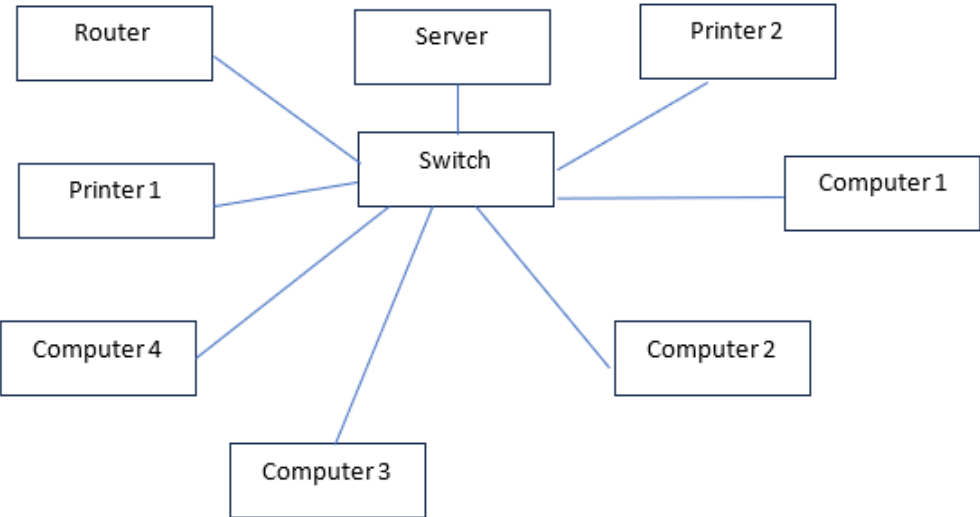
4(a)	1 mark for each benefit, max 3 • Devices can be more mobile as no physical connection is required • No cabling is needed so easier to set up • Easier to add additional devices • Multiple different device types can be connected	3
4(b)	1 mark for a drawback • Can be less secure • Transmission speed can be slow • Interference is possible • The signal degrades quickly without repeaters/boosters	1

8. M/J 25/P11/Q2/a,c

2(a)	1 mark each to max 4 • Video is transmitted continuously • as a series of bits • The video is uploaded to a media server • The users download from the media server • On download, the media server sends the data to a buffer on the user's device • buffer is used when there is a difference in speed between transmission and receipt • Buffer stores data from server until recipient can receive it • Recipient views bit stream from the buffer	4
2(c)	1 mark each modems: • Converts digital data into analogue for transmission down phone lines // Converts analogue data into digital after transmission down phone lines dedicated lines: • Used to provide a direct / private connection which therefore provides faster transmission	2

9. M/J 25/P12/Q6/a,b,c,d,e

6(a)	1 mark each to max 2 • In a WAN devices can connect over a larger geographical area in a LAN devices connect over a smaller geographical area • A WAN may use non-dedicated hardware the hardware in a LAN is usually dedicated • A WAN uses external / not company owned transmission media // the transmission media for a LAN is usually company owned	2
6(b)	1 mark each to max 4 e.g. • Land is split into cells designed for maximum line of sight • Each cell has a tower with an antenna which receives and transmits data • Data is transmitted between the tower and the phone • Data transmission is wireless using low power radio signals/frequencies • Multiple devices can communicate simultaneously with the same tower	4

6(c)	1 mark for each correct word • full stops • hexadecimal • zeros • 128 • dynamic • private IPv4 is made of four groups of 8-bit numbers separated by full stops. Each IPv4 address is 32 bits. IPv6 is made of eight groups of 4 hexadecimal numbers separated by colons. Multiple consecutive groups of zeros can be replaced with a double colon. Each IPv6 address is 128 bits. A dynamic IP address can change each time the computer connects to a network. A private IP address can only be accessed by other devices in the same LAN and is assigned by the router within the LAN.	6
6(d)	1 mark each • Server directly connected to the switch • Four labelled computers all connected directly to the switch • Two printers each directly connected to the switch / server / a computer • Router / modem directly connected to the switch / server for internet access e.g. Example only  <pre> graph TD Router[Router] --- Switch[Switch] Server[Server] --- Switch Printer1[Printer 1] --- Switch Printer2[Printer 2] --- Switch Computer1[Computer 1] --- Switch Computer2[Computer 2] --- Switch Computer3[Computer 3] --- Switch Computer4[Computer 4] --- Switch </pre>	4

6(e)	1 mark each to max 3 e.g. • To store the MAC addresses of devices connected to it • To receive packets from devices • To forward packets directly to the intended recipient • To provide a central point of connection for the network • to enable connected devices to communicate	3
------	---	----------

10. M/J 25/P13/Q3/c

3(c)	1 mark each to max 4 • A webserver stores all the data for each player • Each player is on a client computer // The player's web browser is the client • that sends requests over the internet to the web server • the server performs the required action in the game • the server updates the data in the game • the server sends the results to the player	4
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11. O/N 24/P13/Q9/a,b,c,d

9(a)	1 mark for each bullet point (max 2) • A WAN covers a large geographical area • External/public infrastructure is used • Non-dedicated hardware	2
9(b)	1 mark for correct media 1 mark for matching description e.g. • Fibre optic • Transmits data as pulses of light • Radio waves / microwaves • Transmits electromagnetic waves on different frequencies	4
9(c)	1 mark for each bullet point (max 4) • Data is compressed before transmitting • The video and audio are transmitted continuously as a series of bits • The video is uploaded to a media server • On download, the media server sends the data to a buffer • The buffer stores data from server until the receiving device can process it • The receiving device receives the bit stream from the buffer	4
9(d)(i)	1 mark for public IP address and 1 mark for private IP address: Public IP address: • So that the router is visible to the Internet/external network/WAN Private IP address: • So that the router is identified to computers within the LAN	2

9(d)(ii)	1 mark for each bullet point (max 2) • It allows the network to be divided into smaller networks • ... which reduces traffic in some parts of the network // reduces congestion • ... because traffic only travels through the parts necessary • ... which hides the complexity of network • ... and allows for easier maintenance of the network	2
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12. M/J 24/P11/Q2/e

2(e)(i)	1 mark for: Continuous ordered flow of bits over a communication path	1
2(e)(ii)	1 mark each to max 2 : • Real-time is direct from source whereas on-demand is pre-recorded/downloaded to view later • Real-time cannot be re-watched, on-demand can be paused, re-watched etc. • Real-time plays continually, on-demand downloads sections/blocks and cannot play until next section is downloaded	2

13. M/J 24/P11/Q5/a

5(a)	1 mark each: • Identification of server in the bank scenario • Description e.g. Receives requests, processes the requests • Identification of client in bank scenario • Description e.g. Sends request to the server, waits and outputs the response	4
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14. M/J 24/P11/Q8/a,b

8(a)	1 mark for each correct row:				5
		Statement	Bus	Star	Mesh
		all devices connect to one central device		✓	
		all devices connect to a central cable	✓		
		multiple paths for the packets to travel			✓
		robust against damage because if any line fails, the rest of the network retains full functionality		✓	✓
		most likely to lose data through collisions	✓		
8(b)(i)	1 mark for:				1
		to be visible to and accessible by other devices on the internet			

8(b)(ii)	1 mark each: - IPv4 has 4 groups of digits whilst IPv6 has 8 groups - IPv4 is usually represented in denary whilst IPv6 is usually represented in hexadecimal - IPv4 groups are between 0 and 255 whilst IPv6 is between 0 and FFFF - IPv4 is 32 bits whilst IPv6 is 128 bits	2
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15. M/J 24/P12/Q3/b,c

3(b)	1 mark for characteristic 1 mark for description of application to examination software: <table><tr><th>Thin-client characteristic</th><th>Description of use in this software</th></tr><tr><td>Data is not stored on the client computer</td><td>Exam papers are stored on the server and not on the examiner's computer // exam papers are not permanently stored on the examiners' computers</td></tr><tr><td>Client computer is reliant on access to server</td><td>Examiners cannot mark if their device cannot access the server / the server 'goes down'</td></tr><tr><td>Client computer heavily reliant on network/internet connection</td><td>The marking software will not operate without network/internet access</td></tr><tr><td>Client computer requires few local resources/memory</td><td>Examiners can use devices with low resources and the marking software will still function</td></tr><tr><td>Client computer performs minimal functions/processes</td><td>The marking software transmits requests, the server responds and sends the response to the user</td></tr></table>	Thin-client characteristic	Description of use in this software	Data is not stored on the client computer	Exam papers are stored on the server and not on the examiner's computer // exam papers are not permanently stored on the examiners' computers	Client computer is reliant on access to server	Examiners cannot mark if their device cannot access the server / the server 'goes down'	Client computer heavily reliant on network/internet connection	The marking software will not operate without network/internet access	Client computer requires few local resources/memory	Examiners can use devices with low resources and the marking software will still function	Client computer performs minimal functions/processes	The marking software transmits requests, the server responds and sends the response to the user	4
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Client computer performs minimal functions/processes	The marking software transmits requests, the server responds and sends the response to the user													
3(c)(i)	1 mark each to max 2: • Receives packets from internet • Analyses the destination IP address of each packet • Forwards the packet towards its destination • ... using the routing table • Maintains/updates the routing table • Finds the most efficient route to the destination	2												
3(c)(ii)	1 mark each to max 2: • The PSTN consists of many different types of communication lines • ... therefore the digital data may need to be converted into a different form/analogue signal • Data is transmitted in both directions at the same time // duplex data transmission • Using a PSTN the communication passes through different switching centres/ISPs	2												

16. M/J 24/P13/Q5/a,b,c,d

5(a)(i)	1 mark for: Dedicated/bespoke services/storage on a remote server only available to company	1
5(a)(ii)	1 mark each to max 3: e.g. • Not reliant on a third party • ... gives greater control over security/privacy • ... gives greater control over backup • Storage can be tailored/scalable to company requirements// an example e.g. the amount of storage accessible/ facilitating the sharing of files	3
5(b)	1 mark each: • Sending computer transmits packets directly to switch/router/central device • Switch/router/central device checks destination address of packet and forwards directly to that device	2
5(c)(i)	1 mark each to max 2: • Jamming signal is transmitted by the sending device • Transmission is aborted • The sending device waits a random time before trying to send data again ... • ... if further collisions occur the wait time is increased	2
5(c)(ii)	1 mark each to max 2: • Random time increased each time so can be infinite waiting • May be constant jamming signal so nothing ever sends • Certain nodes cannot be prioritised • High power consumption • Only suitable for short distance network // limited distance • Not scalable // more nodes means exponentially longer waiting times	2
5(d)	1 mark for: Static means the IP for that device does not change and Private means it can only be accessed/seen/used within the LAN	1

17. O/N 23/P11/Q2/a,b,c,d

2(a)	1 mark for each bullet point (max 3) • Receives packets from internet / external network • Implements a firewall • Analyses the destination IP address of each packet • Forwards the packet towards its destination // send packets onto local network or external network • ...using the routing table • Maintains / updates the routing • Allocates <u>private</u> IP addresses • Finds the most efficient route to the destination • Changes the packet format for transmission over the next network // • Network Address Translation (NAT): NAT is a technique used by routers to allow multiple devices in a private local area network (LAN) to share a single public IP address.	3
2(b)	Switch: 1 mark for each bullet point (max 1) • To allow two or more devices to communicate with one another • To connect individual devices to each other • To receive transmissions and forward them to their destination Wireless Access Point (WAP): 1 mark for each bullet point (max 1) • To allow connection of devices (to the central device) using radio signals / Wi-Fi • To allow the central device to send / receive radio signals / Wi-Fi signals • To allow wireless enabled devices to connect to a wired network Bridge: 1 mark for each bullet point (max 1) • To connect two LANs / segments with the same protocol • To transmit data between two networks with the same protocol	3
2(c)	1 mark for each bullet point (max 2) • The students cannot access their files without a reliable internet connection • The amount of space for no payment may be limited so students will have to purchase more space if needed • The students do not have control over the backup (or security) of their work // the students are dependent on a third party for the (security and) backing up of their work	2

2(d)	1 mark for each advantage and 1 mark for valid corresponding expansion - Star topology is more resilient to faults ...because there is no single cable and leads to less disruption to teaching - Higher performance as fewer collisions ...because each device in the classroom is only connected to the switch - Easier to add new nodes ...because each device in the classroom connects directly to the switch - Easier to fault find compared to bus topology	2
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18. O/N 23/P12/Q7/a,b,c,d

7(a)	1 mark for each bullet point (max 2) - Only has four groups of digits // IPv6 has eight groups - Uses dotted notation instead of colons - Because it is a 32 bit / 4 byte address // IPv6 is 128 bits / 16 bytes	2
7(b)(i)	1 mark for server only connected to router 1 mark for two laptop computers connected only to router <pre> graph TD Server[Server] --- Router[Router] Router --- Laptop1[Laptop 1] Router --- Laptop2[Laptop 2] </pre>	2
7(b)(ii)	1 mark for each bullet point (max 2) - The data from the sending laptop is transmitted to the router - The data has address of recipient - The router determines recipient's destination address ... by using a routing table - The router transmits data directly / only to recipient	2

7(b)(iii)	1 mark for each use (max 2) and 1 mark for corresponding expansion (max 2) • To improve the security of the LAN • ... so that devices do not receive unintended data • ... so that a compromised device does not expose the whole network • ... so not all devices can access all segments • To make the network management easier • ... because faults can be isolated more efficiently • ... by appropriate example • To make the network easier to expand // For better control of network growth • ... by allowing for greater range of IP addresses to be available • To improve network performance • To reduce network congestion • ... by localising network communications // by dividing data between segments • ... so that devices are not flooded with data • ... because data sent between devices on the same subnet stays within the subnet	4										
7(c)	1 mark for each bullet point (max 3) • To monitor the communications channel • To send data only when there is no data being transmitted / the line is quiet / idle • To detect a collision and then stop transmissions of further data // transmit a jamming signal • To calculate a random wait time / back-off time • ... then retransmit the data after that random wait time • Increase random time if multiple collisions	3										
7(d)	1 mark for each highlighted area <table><tr><th>Type of IP address</th><th>Description</th></tr><tr><td>public IP address</td><td>an IP address that is assigned to a device to allow direct access on the internet</td></tr><tr><td>static IP address</td><td>an IP address that is fixed / doesn't change each time a device re-joins a LAN / network</td></tr><tr><td>private IP address</td><td>an IP address used for internal LAN communication only</td></tr><tr><td>dynamic IP address</td><td>an IP address that may be refreshed / changed each time a device re-joins a LAN / network</td></tr></table>	Type of IP address	Description	public IP address	an IP address that is assigned to a device to allow direct access on the internet	static IP address	an IP address that is fixed / doesn't change each time a device re-joins a LAN / network	private IP address	an IP address used for internal LAN communication only	dynamic IP address	an IP address that may be refreshed / changed each time a device re-joins a LAN / network	4
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19. O/N 23/P13/Q3/a(i,ii)

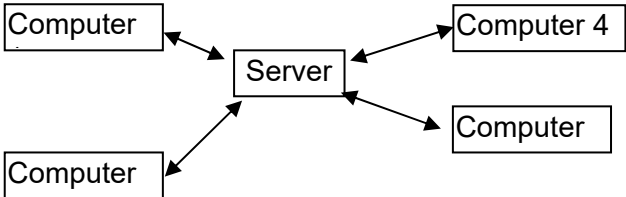
3(a)(i)	1 mark for each bullet point (max 2). • Courses must be available to anyone who wishes to follow them • Courses must be available on the internet • Company is willing to share infrastructure with other companies (public) • ...which is more economic for the company	2
3(a)(ii)	1 mark for each bullet point (max 2 for each disadvantage). • There could be a possible loss of control unlike the LAN • ...because the data is stored on a remote infrastructure / someone else's infrastructure • ...reliance on external agency to complete tasks, e.g. backups, security • Requires reliable internet connection • ...to ensure access to the remote data, more likely with LAN • Increased recurring costs • ...as cloud provider charges must be paid, costs for LAN once only.	4

20. M/J 23/P11/Q4/d

4(d)(i)	1 mark for each term An IPv4 address contains 4 groups of digits. Each group is represented in 8 bits and the groups are separated by full stops. An IPv6 address contains 8 groups of digits. Each group is represented in 16 bits. Multiple groups that only contain zeros can be replaced with a :: // double colon .	5
4(d)(ii)	1 mark for identification, 1 mark for expansion e.g. • Reduce amount of traffic in a network // improve network speed • Data stays in its subnet so it does not travel as far • Improves network security • .. so that not all devices can access all areas of the network • Allows for easier maintenance • ... because only one subnetwork may need taking down/changing while the rest of the network can continue	4

21. M/J 23/P12/Q1/a,b,c,d,e

1(a)	1 mark each to max 2 • Allows the sharing of files/data // Allows communication between the devices • Allows the sharing of resources e.g. hardware / software (applications) • Allows central management // by example, backup, security, etc.	2
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1(b)	1 mark each to max 2 - Covers a small geographical area - The infrastructure is privately owned // not controlled by external organisations	2
1(c)	1 mark each to max 2 - Each computer directly connected only to the server <u>all</u> components correctly labelled  <pre> graph TD S[Server] <--> C1[Computer] S <--> C2[Computer 4] S <--> C3[Computer] S <--> C4[Computer] </pre>	2
1(d)	1 mark each to max 3 - A protocol (suite) - For data transmission over standard / universal wired / cabled network connections - Uses Carrier Sense Multiple Access / Collision Detection (CSMA/CD) - Data is transmitted in frames ... each frame has a source and destination (IP/MAC) address ... and error checking data (so damaged frames can be resent)	3
1(e)	1 mark each - The server performs minimal / some processing for the client - The clients also do most of their own processing/work independently // most of the resources are installed locally	2

22. M/J 23/P13/Q2/a,b,c,d,e

2(a)	1 mark each to max 2 - WAN covers a large geographical area and LAN covers a small geographical area - LAN connections between devices are usually physical, whereas the WAN connections are often virtual - A LAN has a high data transfer rate, whereas a WAN has a low data transfer rate - The ownership of a LAN is private; the ownership of a WAN can be private or public - LAN is usually more secure than a WAN because protection is easier to implement	2
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2(b)(i)	1 mark each to max 2 - All computers are connected to at least one other device - There are multiple routes between devices - The computers can act as relays, passing packets on towards the final destination	2
2(b)(ii)	1 mark each to max 2 - If one line goes down there are more routes available - Improved security as not using one main line - No/fewer collisions - New nodes can be added without interruption or interfering with other nodes - More secure because data is sent over a dedicated connection	2
2(c)	1 mark each to max 2 - Server performs all processes required by the task and/or data storage - Clients only sends requests to the server and displays the returned results	2
2(d)	1 mark each to max 4 Examples: - Some students might only have one sort of connection on their device - Wired provides better performance for the student's device ... for example, enabling faster access to university databases - There will be less interference if students connect via a cable - Students can transmit private/confidential data/work securely ... for example, their final dissertation - Wireless connection means that the students can use their devices in different rooms/sites/outside/anywhere more freely // student devices can be portable - Wireless connection enables the students to bring multiple devices // bring their own devices // change devices	4
2(e)	1 mark for identification - IP address is made up of a <u>network ID</u> and a <u>host ID</u> 1 mark each to max 2 for description - Each device in a subnetwork has the same network ID // Each subnetwork has a different network ID - Every device in each subnetwork has a different host ID but the same network ID // the host ID uniquely identifies the device within the subnetwork	3

23. O/N 22/P11/Q8

8	1 mark for bullet point (max 4): - CSMA/CD is a protocol used to detect and prevent collisions in a bus topology - before transmitting, a device checks if the channel is busy - If the channel is busy the device waits // if the channel is free the data is sent - because there is more than one computer connected to the same transmission medium ... two workstations can start to transmit at the same time, causing a collision - If a collision is detected by the device, transmission is aborted / a jamming signal is transmitted - both devices wait a (different) random time and then try again	4
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24. O/N 22/P12/Q10/a,b

10(a)	all four computers directly connected to the switch and no other connections.	1
10(b)(i)	1 mark for the device. 1 mark for corresponding reason. - Device: Server - Reason: Server processes the requests and authorises traffic // firewall software on the server authorises traffic // server acts as the proxy - Device: Switch - Reason: Switch is connected to all the computers // to share access to the router on the network	2
10(b)(ii)	1 mark for each bullet point (max 3): - receive packets from devices / internet - find destination of packets using the IP address - forward packets to the destination - assign private IP addresses to devices on LAN - store/update/maintain a routing table - find most efficient path to destination - maintain table of MAC and IP addresses - provides the LAN with a public IP address - acts as a gateway - performs protocol conversion - acts as a firewall	3

25. O/N 22/P13/Q7/a,b,c

7(a)	1 mark for each correct answer: <table><tr><th colspan="2">Answer</th></tr><tr><td>The name of device A that allows the laptop to connect to the internet</td><td>Router</td></tr><tr><td>A type of cloud, X</td><td>Public (cloud)</td></tr><tr><td>An example of an application, B, that can run on the cloud,</td><td>Email / Graphics / Word processor / Spreadsheet / Game / Database, etc.</td></tr></table>	Answer		The name of device A that allows the laptop to connect to the internet	Router	A type of cloud, X	Public (cloud)	An example of an application, B, that can run on the cloud,	Email / Graphics / Word processor / Spreadsheet / Game / Database, etc.	3
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The name of device A that allows the laptop to connect to the internet	Router									
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An example of an application, B, that can run on the cloud,	Email / Graphics / Word processor / Spreadsheet / Game / Database, etc.									
7(b)	1 mark for a correct advantage: - not fixed to a single location - allows access in remote / rural areas 1 mark for each correct disadvantage (max 2): - high latency / lag / slow to connect - more expensive than wired methods, as need extra equipment - signal is affected by bad weather - the transmission speed is slower than fixed line broadband - direct line of sight needed	3								
7(c)(i)	1 mark for each bullet point (max 2) Examples: - improves security - reduces congestion - allows extension of the network / devices attached - aids day-to-day management - improves performance	2								
7(c)(ii)	1 mark for each correct answer: - network ID = 10 - host ID = 4	2								

26. O/N 21/P11/Q8/a,b,c

8(a)	1 mark per bullet point - LAN - Small geographical area - No leasing external infrastructure / transmission media // does not use internet to transmit within the building	3
8(b)	1 mark per item - router - switch - hub	2

8(c)	1 mark per bullet point to max 4 • Provide interface to wireless network • ... as an antenna • Receives analogue radio waves • ... convert them to digital / binary • Checks incoming transmissions for correct MAC / IP address • ... ignore transmissions not intended for it • Encrypts / encodes the data • Decrypts / decodes the data • Takes digital/binary input and converts to analogue waves • ... sends the radio waves via the antenna	4
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27. O/N 21/P12/Q3/b

3(b)(i)	1 mark for identification of star topology 1 mark for justification Devices are connected directly to the <u>router</u> independently // all devices are only connected to the <u>router</u>	2
3(b)(ii)	1 mark for each correct function to max 3 • To receive packets from devices or the Internet • To forward / route packets to the destination • To find the destination of the packet • To assign / allocate private IP addresses to devices on LAN • To store / update / maintain a routing table • To find the most efficient path to the destination • To maintain a table of MAC and IP addresses	3

28. O/N 21/P13/Q8/a,b,c

8(a)	1 mark per bullet point • LAN • Small geographical area • No leasing external infrastructure / transmission media // does not use internet to transmit within the building	3
8(b)	1 mark per item • router • switch • hub	2

8(c)	1 mark per bullet point to max 4 • Provide interface to wireless network • ... as an antenna • Receives analogue radio waves • ... convert them to digital / binary • Checks incoming transmissions for correct MAC / IP address • ... ignore transmissions not intended for it • Encrypts / encodes the data • Decrypts / decodes the data • Takes digital/binary input and converts to analogue waves • ... sends the radio waves via the antenna	4
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29. M/J 21/P11/Q4/a,b,c,d

4(a)	1 mark per bullet point to max 2 • All computers are of equal status • Each computer provides access to resources and data // data is distributed • Computers can communicate and share resources • Each computer is responsible for its own security	2
4(b)	1 mark per bullet point to max 2 per drawback • Reduced security // no central management of security • ... only as secure as the weakest computer on the network • ... each computer is at risk from viruses from other computers • No central management of backup • ... if the data from one computer is not backed up it is lost to all of them • No central management of files/software • ... consistency may be difficult to maintain • ... each computer may have different software from the others • Individual computers may respond slower • ... because they are being accessed by other computers • In order to share files etc. all the computers involved need to be switched on • ... so the files etc. may not be always available	4

4(c)(i)	1 mark for first 2 ticks, 1 mark for last 2 (shaded) <table border="1"> <thead> <tr> <th>Task</th><th>Performed by router</th><th>Not performed by router</th></tr> </thead> <tbody> <tr> <td>Receives packets from devices</td><td>✓</td><td></td></tr> <tr> <td>Finds the IP address of a Uniform Resource Locator (URL)</td><td></td><td>✓</td></tr> <tr> <td>Directs each packet to all devices attached to it</td><td></td><td>✓</td></tr> <tr> <td>Stores the IP and/or MAC address of all devices attached to it</td><td>✓</td><td></td></tr> </tbody> </table>	Task	Performed by router	Not performed by router	Receives packets from devices	✓		Finds the IP address of a Uniform Resource Locator (URL)		✓	Directs each packet to all devices attached to it		✓	Stores the IP and/or MAC address of all devices attached to it	✓		2
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4(c)(ii)	1 mark per bullet point for justification up to max 3 No mark for identification of wired/wireless Wired • Faster connection // higher bandwidth • ... needed as she is downloading/streaming large files • ... less time waiting / less latency / fewer delays • More reliable / stable connection • ... is less susceptible to issues with distance/walls/interference • More secure Wireless • Freedom of movement • ... can move between different rooms with a mobile device and still receive/transmit data • ... no need of a physical connection • Easily expanded if friends want to access the same network • Less cabling / expertise is needed • ... making the initial setup less expensive	3															
4(d)	1 mark for identifying that she is using both. 1 mark per bullet point for justification • using internet because sending data on the infrastructure • using WWW because accessing a website (that is stored on a web server operated by the webmail) that is part of the WWW	3															

30. M/J 21/P12/Q5/c,d

5(c)(i)	1 mark per bullet point to max 2 • Cloud storage can be free (for small quantities) • No need for separate (high capacity) storage devices // saves storage on existing devices • Can access data from any computer with internet access • Most cloud data services will have in-built backup/disaster recovery • Security could be better • Can easily increase capacity • Data can be easily shared	2														
5(c)(ii)	1 mark per bullet point to max 2: • Can only access (the cloud) with internet access • Security may not be strong // no control over security • There may not be any backups // no control over backups • It can take a long time to upload/download the data • It can be more expensive in the long term • There could be a limit to the amount of storage unless paid for • There could be compatibility/access issues • There could be issues with the company offering cloud services	2														
5(d)	1 mark for each correct line <table><thead><tr><th>Term</th><th>Description</th></tr></thead><tbody><tr><td>Public IP Address</td><td>It is only visible to devices within the Local Area Network (LAN)</td></tr><tr><td>Private IP address</td><td>It increments by 1 each time the device connects to the internet</td></tr><tr><td>Dynamic IP address</td><td>A new one is reallocated each time a device connects to the internet</td></tr><tr><td>Static IP address</td><td>It can only be allocated to a router</td></tr><tr><td></td><td>It is visible to any device on the internet</td></tr><tr><td></td><td>It does not change each time a device is connected to the internet</td></tr></tbody></table>	Term	Description	Public IP Address	It is only visible to devices within the Local Area Network (LAN)	Private IP address	It increments by 1 each time the device connects to the internet	Dynamic IP address	A new one is reallocated each time a device connects to the internet	Static IP address	It can only be allocated to a router		It is visible to any device on the internet		It does not change each time a device is connected to the internet	4
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31. M/J 21/P13/Q4/a,b,c,d

4(a)	1 mark per bullet point to max 2 • All computers are of equal status • Each computer provides access to resources and data // data is distributed • Computers can communicate and share resources • Each computer is responsible for its own security	2															
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Unit 3

HARDWARE

61 items · 307 marks

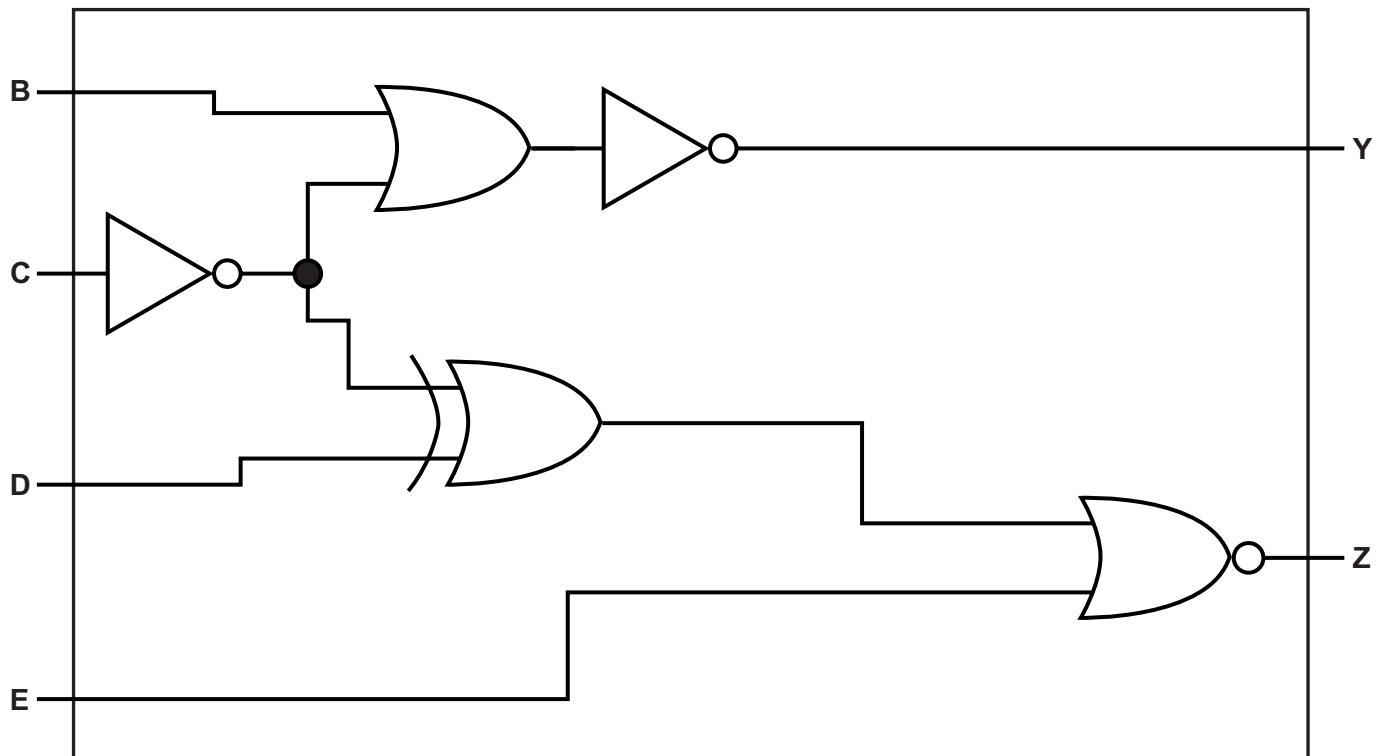
Syllabus sections in this unit

- 3.1 Computers and their components
- 3.2 Logic Gates and Logic Circuits

Unit 3: Hardware

1. M/J 26/P11/Q6/a,b

6 (a) Write the logic expressions for the following logic circuit.



Y =

Z =

[2]

(b) Create and complete the truth table for a two-input NAND gate.

[2]

2. M/J 26/P12/Q6/a,b,c

- 6** A health monitor attaches to a person's arm and measures key health information, for example heart rate. The monitor transmits data to the user's smartphone. The results are analysed by the phone, which displays information to the user about their health.

(a) Explain the reasons why the health monitor is an example of an embedded system.

[3]

(b) Tick (✓) **one** box to identify if the health monitor is an example of a monitoring system or a control system.

system	tick (✓)
monitoring	
control	

Justify your choice.

[2]

(c) The health monitor contains Random Access Memory (RAM) and Read Only Memory (ROM).

Describe **two** differences between RAM and ROM.

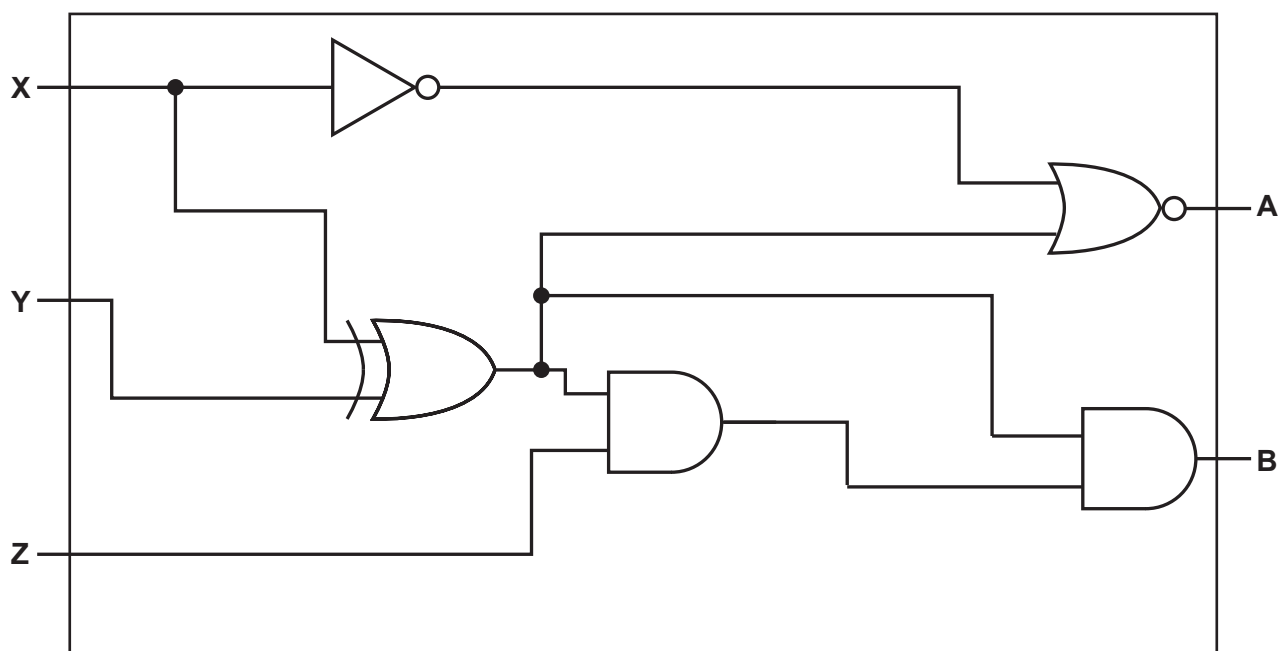
1

2

[2]

3. M/J 26/P12/Q8/a,b

- 8** (a) Complete the truth table for the given logic circuit.



X	Y	Z	working space	A	B
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

[2]

(b) Write the logic statement for the given truth table.

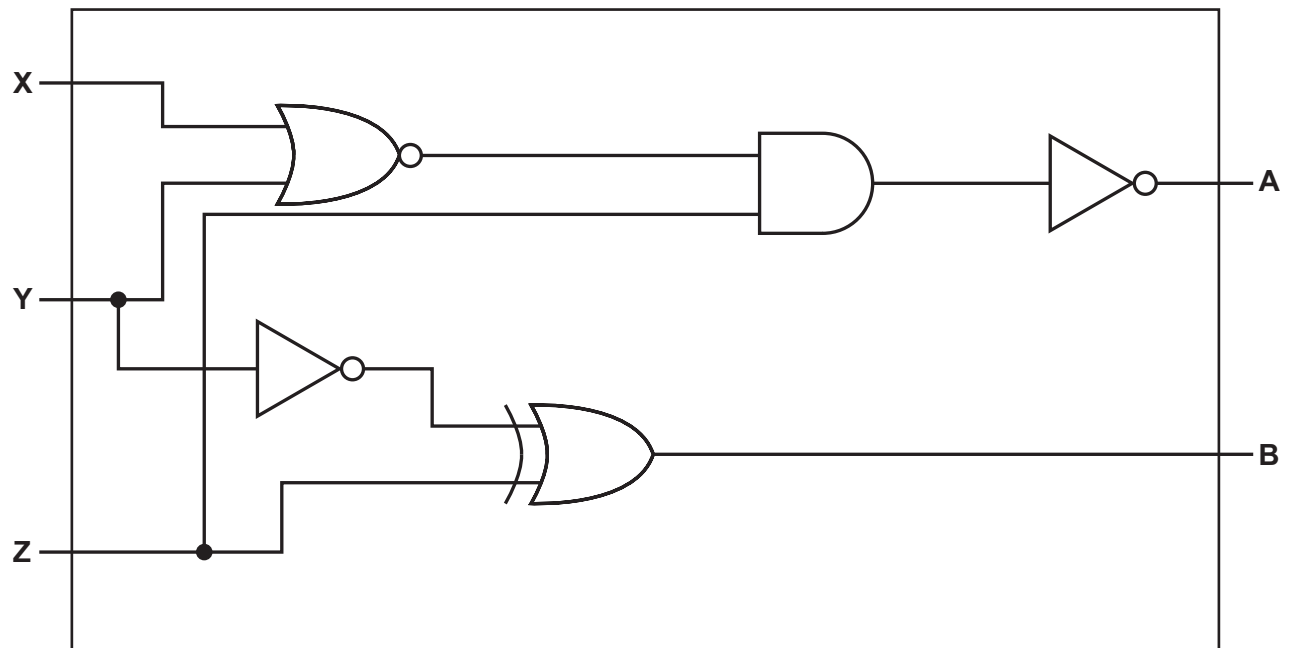
input			output
A	B	C	P
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

P =

[1]

4. M/J 26/P13/Q2/a,b

2 (a) Complete the truth table for the given logic circuit.



X	Y	Z	working space	A	B
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

[2]

(b) State **two** logic gates that are not in the circuit in **2(a)**.

[1]

5. M/J 26/P13/Q8/a,b,c

8 A computer is used to record sound.

(a) The computer has Programmable Read Only Memory (PROM).

Complete the table by naming and describing two other types of ROM.

type of ROM	description

[4]

(b) The sound is recorded using a microphone.

Describe the principal operation of a microphone.

[3]

(c) The computer outputs the sound through external speakers. The computer transfers the data to a buffer before the speakers output the sound.

Describe the purpose of the buffer.

[3]

6. O/N 25/P11/Q3/a,b

3 (a) Draw a logic circuit for the logic expression:

$$X = \text{NOT} ((A \text{ NAND } B) \text{ XOR } (\text{NOT } A \text{ OR } \text{NOT } C))$$



[2]

(b) Write the logic expression for the following truth table.

R	S	T	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Q =

[2]

7. O/N 25/P11/Q6/c

(c) State **three** differences between Dynamic RAM (DRAM) and Static RAM (SRAM).

- 1
- 2
- 3

[3]

8. O/N 25/P11/Q9

- 9 Describe the differences between a monitoring system and a control system.

[3]

9. O/N 25/P11/Q10/a,b

- 10 An architect needs a model of a building.

(a) Explain how the model will be printed using a 3D printer.

[3]

(b) State why a 3D printer needs a buffer.

[1]

10. O/N 25/P12/Q8/b(i)

- 8 A restaurant accesses software and stores data using cloud computing.

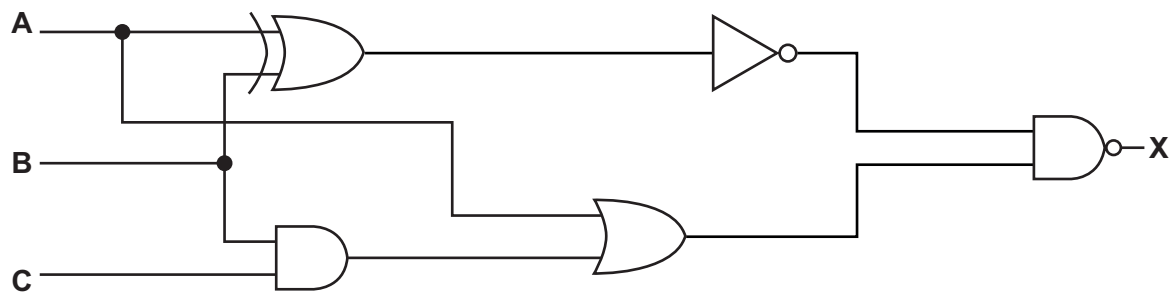
(b) The restaurant uses touchscreens as input devices to allow staff to select items from a menu.

(i) Explain how a touchscreen converts the point of touch to a selection from the menu.

[4]

11. O/N 25/P12/Q9/a,b

9 (a) Consider the following logic circuit:



Write the logic expression for the logic circuit. Do **not** simplify the expression.

X =

[2]

(b) Consider the following logic expression:

$$X = \text{NOT } ((A \text{ NAND } B) \text{ XOR } (\text{NOT } A \text{ OR } \text{NOT } C))$$

A truth table for the logic expression is given:

Row number	A	B	C	X
1	0	0	0	1
2	0	0	1	0
3	0	1	0	1
4	0	1	1	0
5	1	0	0	1
6	1	0	1	0
7	1	1	0	1
8	1	1	1	1

There are **three** errors in the truth table.

Identify the **three** errors in the truth table by writing the row numbers with an incorrect output.

Error 1 Row number

Error 2 Row number

Error 3 Row number

[2]

12. O/N 25/P12/Q11

- 11** An automated system opens doors when a person is detected within 2 metres. The system closes the doors when there is no longer a person within 2 metres.

Identify whether the automated system is an example of a monitoring system or a control system.

Justify your choice.

Monitoring or Control

Justification

[3]

13. O/N 25/P13/Q3/a,b

3 (a) Complete the truth table for the following logic expression:

$$X = (A \text{ NOR } B) \text{ NAND } (C \text{ XOR } B)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(b) Draw a logic circuit for the logic expression:

$$X = \text{NOT } (((A \text{ AND } B) \text{ OR } C) \text{ AND } (C \text{ NOR } D))$$



[2]

14. O/N 25/P13/Q4/c

- 4 A network engineer is setting up a Local Area Network (LAN) in an office. The network uses both wired and wireless connectivity.

(c) Complete the table by writing the name of the type of memory technology for each description.

Description	Memory technology
A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.	
A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.	
A read only memory where the contents can be written only once after manufacture.	

[3]

15. O/N 25/P13/Q7/d

- 7 A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

(d) Describe **two** reasons why the server that stores the videos uses magnetic hard disks instead of solid state (flash) memory.

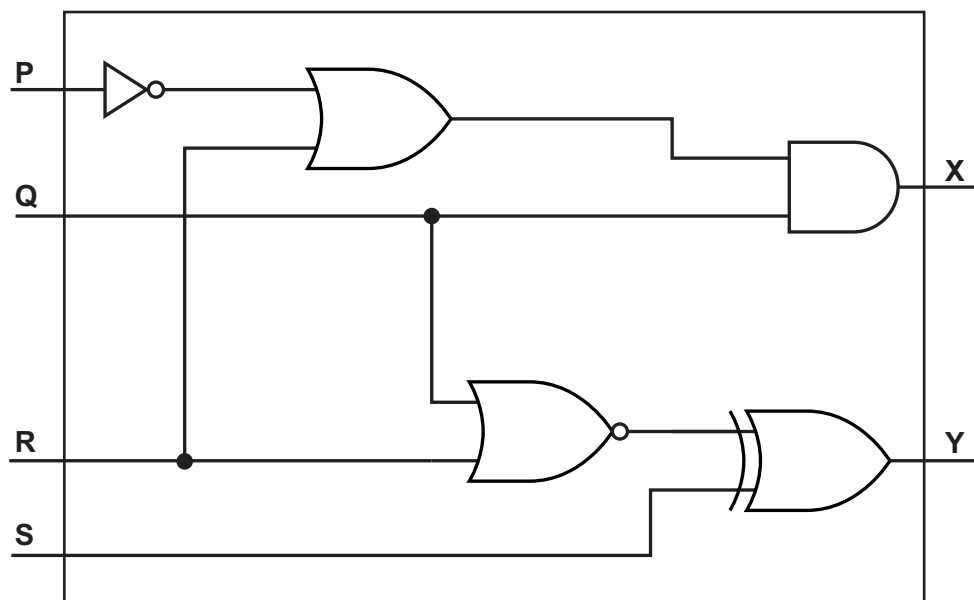
1

2

[4]

16. M/J 25/P11/Q1/a,b

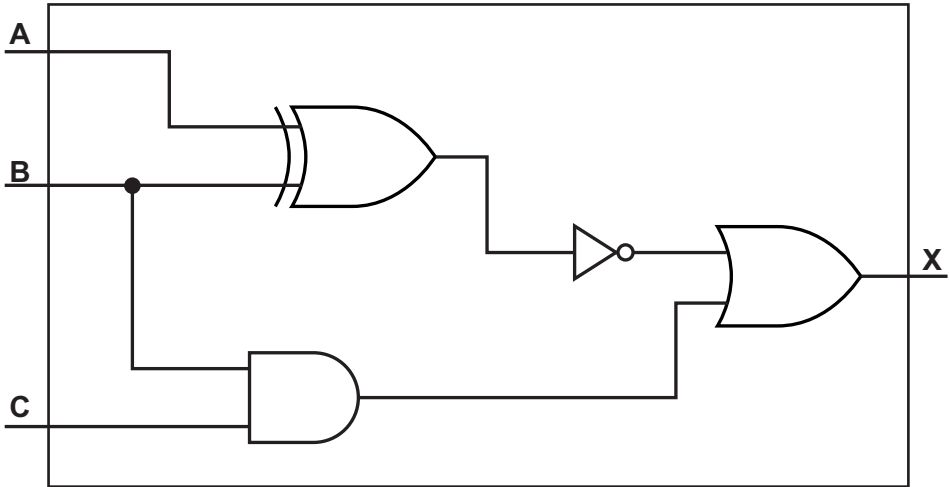
- 1 (a) Write the logic expressions for the following logic circuit.



X =
Y =

[2]

(b) Complete the truth table for the following logic circuit.



A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

17. M/J 25/P11/Q4/b

- 4 A shop installs a new system that allows users to purchase items without going through a manual checkout.

The new system:

- identifies customers when they enter the shop and matches them to their account
- prevents a customer from walking through the automatic barriers if they do not have an account
- automatically detects the items that a customer has taken from a shelf and charges these to the customer's account.

- (b) The new system uses sensors to identify the items taken from a shelf.

Identify **one** type of sensor that can be used in this new system.

State how the sensor can be used to identify the items taken from a shelf.

Sensor

Use

[2]

18. M/J 25/P11/Q6/a(i)

- 6 A programmer is buying a new computer.

- (a) The programmer is considering the following two computers:

Computer 1	Quad-core 2.2 GHz processor 16-bit architecture 1 GB Random Access Memory (RAM) 500 GB magnetic hard disk
Computer 2	Dual-core 3.8 GHz processor 32-bit architecture 2 GB RAM 500 GB solid state drive (SSD)

- (i) Computer 1 has a magnetic hard disk.

Complete the description of the principal operation of a magnetic hard disk by writing the missing words.

The magnetic hard disk has one or more that can be magnetised. These are mounted on a and rotate at high speed.

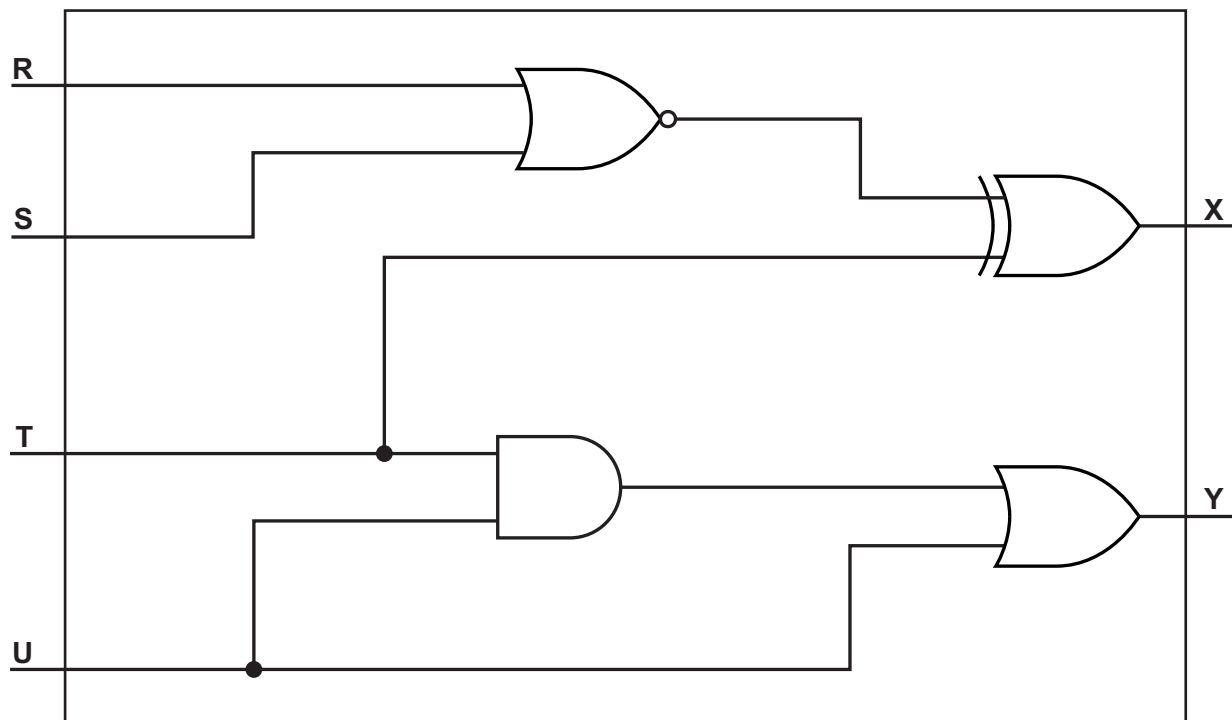
A is moved across the surface on an arm.

When data is read, the changes in the produce a change in the electric current.

[4]

19. M/J 25/P12/Q4/a,b

4 (a) Write the logic expressions for the following logic circuit.



X =

Y =

[2]

(b) Complete the truth table for the logic expression:

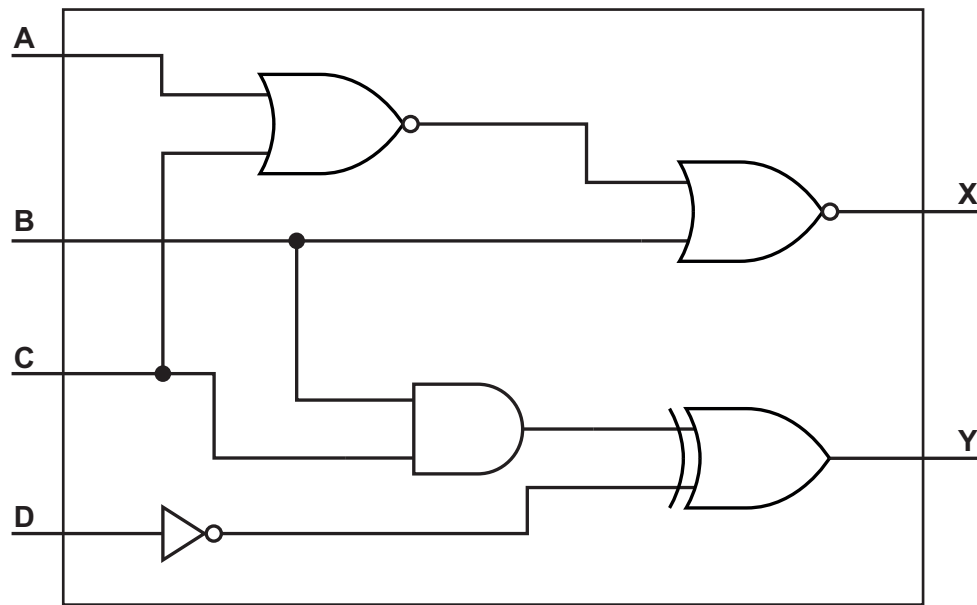
$$X = \text{NOT } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

20. M/J 25/P13/Q4/a,b

4 (a) Write the logic expressions for the following logic circuit.



X =

Y =

[2]

(b) Complete the truth table for the logic expression:

$$X = (A \text{ OR } B) \text{ XOR } (B \text{ OR } C) \text{ XOR } (\text{NOT } A \text{ NAND } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

21. O/N 24/P11/Q3/c,d

3 A computer system has a dual-core Central Processing Unit (CPU).

(c) Identify **two** disadvantages of using Dynamic RAM (DRAM) instead of Static RAM (SRAM) in a computer system.

1

2

[2]

(d) The computer system is used to store data received from a temperature sensor every five seconds. The data is stored on an optical disc using an optical disc reader/writer.

(i) Describe the principal operation of an optical disc reader/writer.

[4]

(ii) The computer uses a buffer when writing data to the optical disc.

Explain the use of a buffer when writing data to the optical disc.

[3]

22. O/N 24/P11/Q5/a,b,c

5 A security system has both a floodlight (very bright light) and an audio alarm.

The data from multiple sensors is analysed and used to:

- turn on the floodlight
- sound the audio alarm.

Sensors can be used to detect:

- if doors are open
- the external daylight level
- if people are detected within a set distance.

(a) Complete the table to identify the most appropriate type of sensor for each scenario.

Scenario	Type of sensor
A door is open.	
The external daylight level is below a set amount.	
A person is detected within 2 metres.	

[1]

(b) The floodlight (**X**) and audio alarm (**Y**) operate according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
A	external daylight level	1	Low
		0	High
B	front door	1	Open
		0	Closed
C	person is within 2 m	1	Detected
		0	Not detected
D	back door	1	Open
		0	Closed
E	security system	1	Switched on
		0	Switched off

The floodlight turns on (**X = 1**) if:

- the security system is switched on
and
- the external daylight level is low
and
- a person is detected within 2 m.

The audio alarm turns on (**Y = 1**) if:

- the security system is switched on
and
- one **or** more doors are open, **or** a person is detected within 2 m.

Write logic expressions for the security system.

X =

Y =

[2]

(c) Explain whether the security system is an example of a monitoring system or a control system.

[3]

23. O/N 24/P12/Q1/a,b

1 (a) Complete the truth table for the logic expression:

$$X = (A \text{ XOR } B) \text{ NAND } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(b) Draw the logic circuit for the logic expression:

$$W = P \text{ NAND } ((Q \text{ OR NOT } R) \text{ XOR } (P \text{ XOR } Q))$$



[2]

24. O/N 24/P12/Q2/a,b,c

2 Embedded systems are used in many electronic devices.

(a) Describe the drawbacks of embedded systems.

[3]

(b) An embedded system has Dynamic RAM (DRAM).

Identify **one** benefit of using DRAM instead of Static RAM (SRAM) in an embedded system.

[1]

(c) Give **two** differences between Erasable Programmable ROM (EPROM) and Electrically Erasable Programmable ROM (EEPROM).

Difference 1

Difference 2

[2]

25. O/N 24/P12/Q9/a,b

9 A road bridge has a weight limit and a height limit for vehicles. For example, a vehicle must weigh less than 10 000 kg and must have a height of less than 3 m.

The bridge has a warning system. If a vehicle is approaching the bridge and it exceeds one or both limits, a sign displays a warning telling the driver of the vehicle to stop.

(a) The bridge warning system uses sensors to detect if a vehicle exceeds the limits.

Complete the table by identifying **two** different sensors that could be used by the system **and** describe how each sensor is used by the system.

Sensor	Use in bridge warning system
.....	
.....	

[2]

(b) Explain whether the bridge warning system is an example of a monitoring system or of a control system.

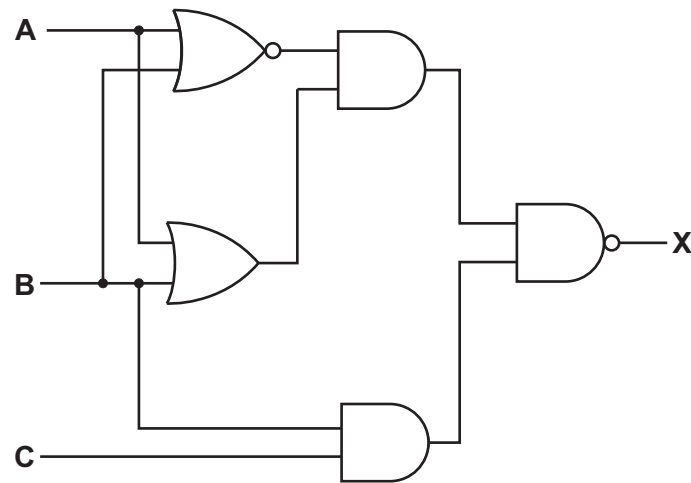
[2]

26. O/N 24/P13/Q1/a,b,c

- 1 (a) Describe the operation of a 2-input XOR gate.

[1]

- (b) Write the logic expression for the following logic circuit.



X =

[2]

- (c) Draw the logic circuit for the logic expression:

$$X = ((\text{NOT } A \text{ AND } (B \text{ AND } C)) \text{ OR } (B \text{ NAND } C)) \text{ AND NOT } A$$



[2]

27. O/N 24/P13/Q2/a,b

2 A memory buffer uses Dynamic RAM (DRAM).

(a) Identify **two** differences between DRAM and Static RAM (SRAM).

Difference 1

Difference 2

[2]

(b) Explain how a memory buffer is used when a computer is transferring data to its magnetic hard disk drive.

[4]

28. O/N 24/P13/Q3

3 A car has an automatic braking system.

A sensor is used to measure the distance to objects that are in front of the car.

The car automatically brakes if an object is too close to the front of the car or the distance between the car and the object is decreasing too quickly. The closer the object is to the front of the car, the harder the car brakes so that the car slows down more quickly.

Explain the reasons why the automatic braking system of the car is a control system.

[3]

29. M/J 24/P11/Q1/a,b

1 **(a)** Tick (✓) **one** box to identify the correct logic statement for this truth table.

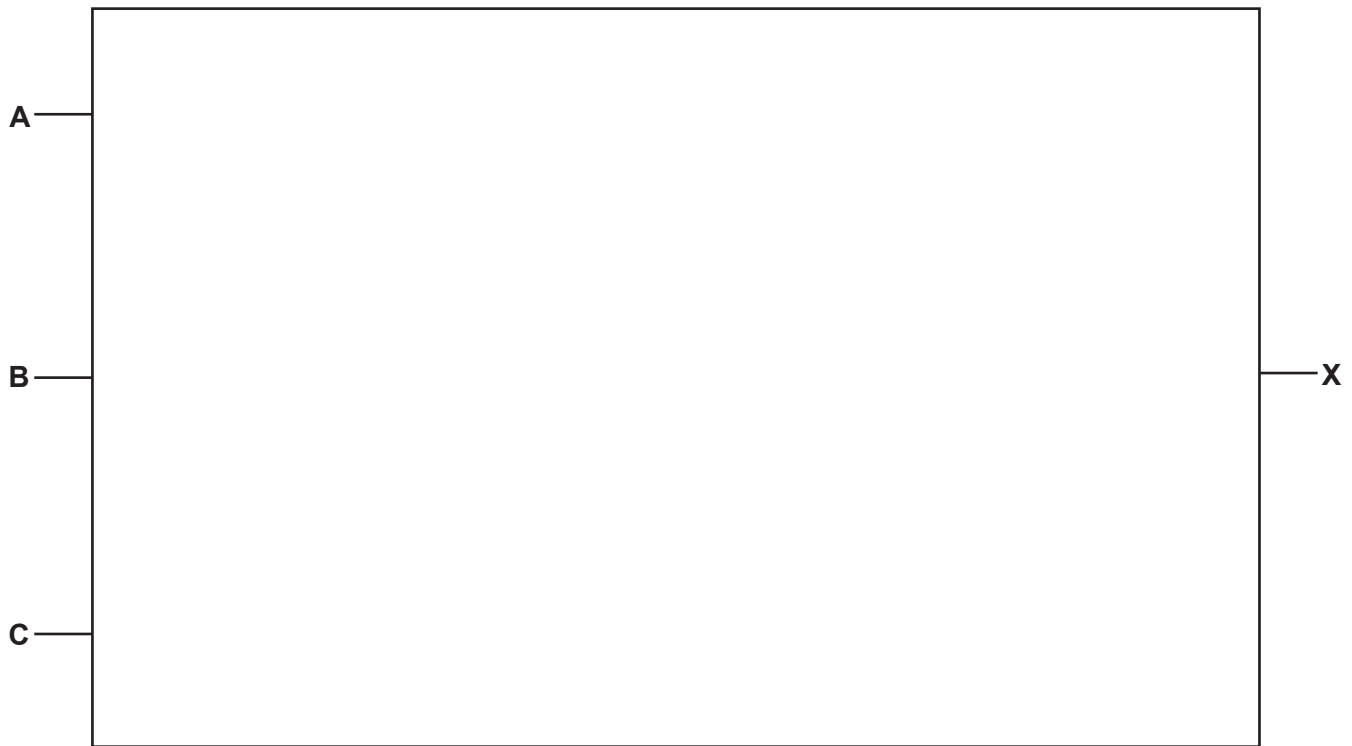
A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

- ☐ NOT (**A** AND **B** AND **C**)
- ☐ (**A** XOR **B**) NOR **C**
- ☐ (**A** OR **B** OR **C**) NOR **C**
- ☐ NOT **A** AND NOT **B** AND NOT **C**

[1]

(b) Draw a logic circuit for the logic expression:

$$X = \text{NOT} (\text{NOT } A \text{ AND } (\text{NOT } B \text{ XOR } C))$$



[2]

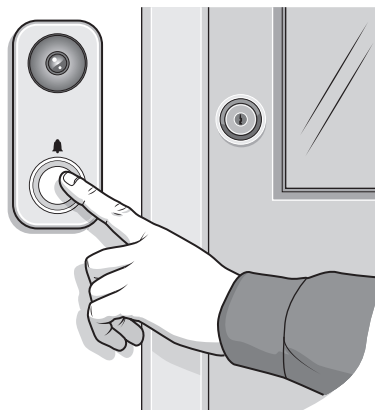
30. M/J 24/P11/Q2/a,b,c

- 2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:

- The digital camera in the doorbell starts recording a video.
- A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell's internal secondary storage device and overwritten when the secondary storage device is full.



- (a) The video doorbell can be considered an example of an embedded system.

Identify **two** characteristics of the doorbell that suggest it is an embedded system.

2

[2]

- (b) State whether the video doorbell is a monitoring system or a control system. Justify your choice.

Monitoring or control system

Justification

[2]

- (c) The video doorbell has both primary memory and secondary storage.

- (i) Identify **two** items of data that the video doorbell will store in primary memory.

1

2

[2]

- (ii) The video doorbell has a solid state (flash) secondary storage device.

Complete the table by writing the answer or answers to each statement about the principal operation of solid state (flash) memory.

Statement	Answer
the two types of logic gate that can be used to create solid state devices	1 2
the number of transistors contained in each cell	
the type of gate that can retain electrons without power	
the type of gate that allows or stops current from passing through	

[4]

- (iii) The video doorbell uses a buffer.

Describe how the video doorbell will use the buffer.

[2]

31. M/J 24/P12/Q1/a,b

- 1 (a) Describe the operation of each of the following logic gates:

NAND

NOR

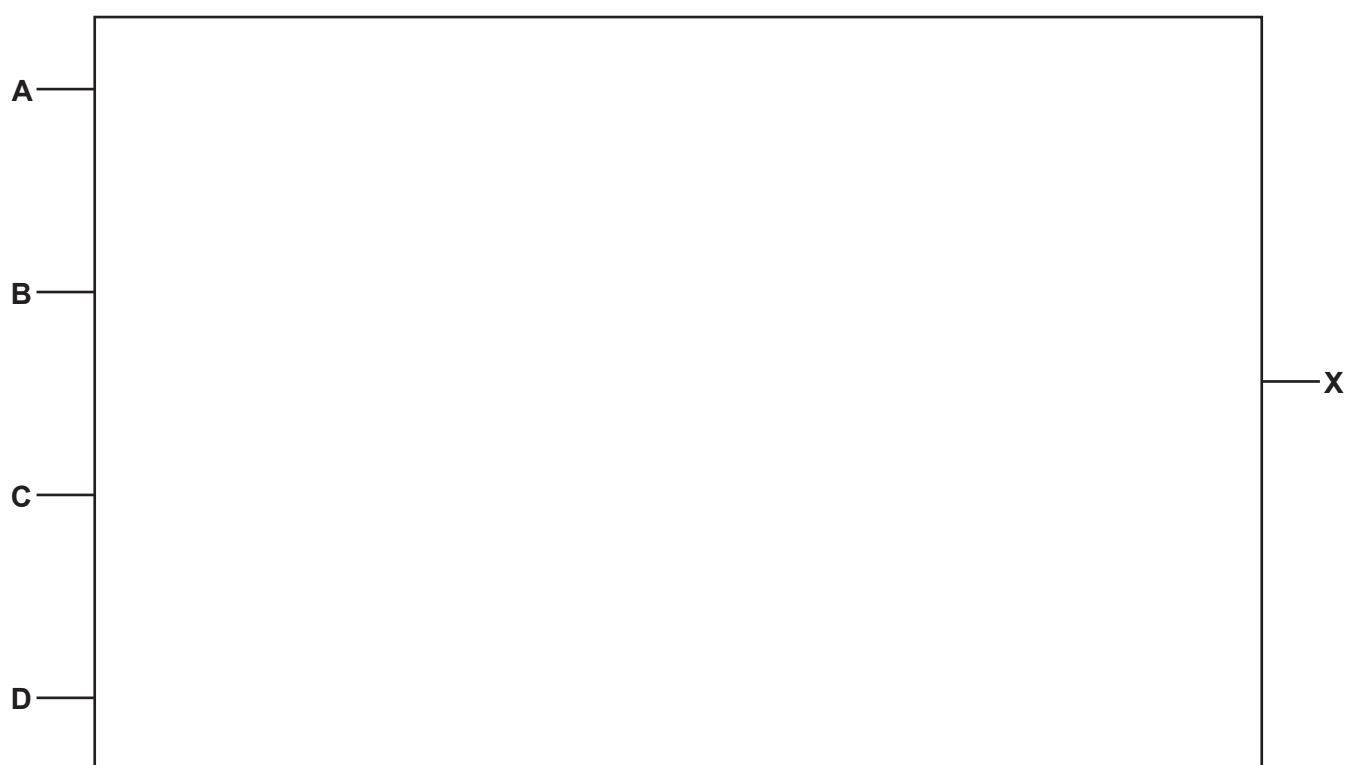
XOR

OR

[4]

- (b) Draw a logic circuit for this logic expression:

$$X = \text{NOT} ((A \text{ AND } B) \text{ OR } (C \text{ AND } D))$$



[2]

32. M/J 24/P12/Q2/a,b,c

- 2** A computer game is being designed that users will be able to play using a virtual reality (VR) headset.

(a) Complete the description of the principal operation of a VR headset.

A headset can have one or two that output the image to the user. The headset has speakers that output surround sound to give a realistic experience. The user's head movements are detected using a sensor.

This sensor is a The data is transmitted to a microprocessor that analyses the data to identify the of movement. Some headsets use that record the user's eye movements for analysis.

[4]

(b) The computer uses a buffer when transmitting data to the VR headset.

Explain how a buffer is used when data is transmitted between the computer and the VR headset.

[3]

(c) The VR headset has Electrically Erasable Programmable Read Only Memory (EEPROM).

Explain the benefits of using EEPROM instead of other types of Read Only Memory (ROM) in the VR headset.

[3]

33. M/J 24/P13/Q6

6 Each of the following truth tables has three inputs (**A**, **B** and **C**) and one output (**X**).

Draw **one** line to match each truth table with its logic expression.

Truth table

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

Logic expression

NOT (**A** XOR **B**) AND **C**

(**A** OR **C**) AND NOT **B**

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A NAND **B** NAND **C**

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

(**A** NAND **B**) OR **C**

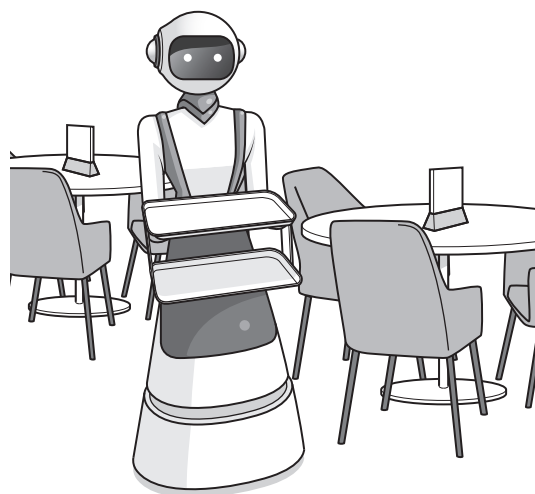
NOT (**A** AND **B** AND **C**)

[3]



34. M/J 24/P13/Q7/a,c,d

7 Robots are used to serve food and drink to customers at a restaurant.



(a) A robot navigates through the restaurant to the table it is serving.

Complete the table by identifying **two** sensors that can be included in the robot **and** the purpose of each sensor in the navigation system.

Sensor	Purpose of sensor in navigation system

[2]

(c) The navigation system can be considered an example of a control system.

Describe how feedback is used in a control system.

[2]

(d) The robot includes a touchscreen for the customer to make their payment.

Describe the principal operation of a touchscreen.

[4]



35. O/N 23/P11/Q4/a,b

4 (a) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ NAND } B) \text{ XOR } (\text{NOT } B \text{ AND } (B \text{ NOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(b) Draw a logic circuit for the logic expression:

$$X = (A \text{ AND NOT } (B \text{ OR } C)) \text{ AND } (B \text{ NOR } C)$$



[2]

36. O/N 23/P11/Q7/a,b,c

- 7 (a) Describe the principal operations of a 3D printer.

[3]

- (b) Describe the purpose of a temperature sensor within the 3D printer.

[2]

- (c) A 3D printer contains 1 GB of Dynamic RAM (DRAM) to store print data.

State **two** advantages of the printer having Dynamic RAM instead of Static RAM (SRAM).

1

2

[2]

37. O/N 23/P11/Q9/a,b

- 9 (a) Explain the importance of feedback in a control system.

[2]

- (b) Give **one** example of an embedded system **and** explain why it is an example of an embedded system.

Example

Explanation

[3]

38. O/N 23/P12/Q1/a,b,c

- 1** A factory makes chocolate bars.

The factory uses a conveyor belt that moves the products from one stage of production to the next stage.

- (a)** An automated system counts the number of chocolate bars made at the end of production.

The system includes a sensor positioned above the conveyor belt.

Identify **one** appropriate type of sensor that can be used.

[1]

- (b)** A second automated system removes chocolate bars with an incorrect weight from the production line.

Describe the role of an **actuator** in this second system.

[2]

- (c)** The factory has many different machines with embedded systems.

- (i)** Identify **two** features of embedded systems.

1

2

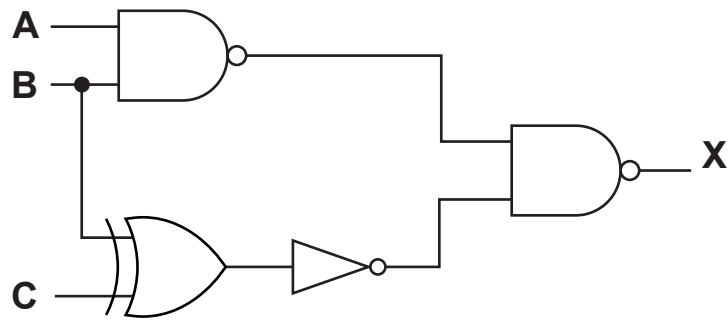
[2]

- (ii)** Identify **one** drawback of embedded systems.

[1]

39. O/N 23/P12/Q4/a,b

- 4 (a) Write the Boolean expression that corresponds to the following logic circuit.



[3]

- (b) Complete the truth table for the logic expression:

$$X = A \text{ XOR } (B \text{ AND } (A \text{ NAND } B) \text{ AND NOT } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

40. O/N 23/P13/Q4/a,b

- 4 (a) Complete the truth table for the logic expression:

$Y = ((P \text{ AND } Q) \text{ XOR } ((\text{NOT } Q) \text{ OR } R)) \text{ AND NOT } P$

P	Q	R	Working space	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(b) Draw a logic circuit for the logic expression:

$Y = ((P \text{ AND } Q) \text{ XOR } ((\text{NOT } Q) \text{ OR } R)) \text{ AND NOT } P$



[2]

41. O/N 23/P13/Q7/a,b,c

- 7 A laptop computer has Static RAM (SRAM).

A virtual reality headset and a laser printer are connected to the laptop.

- (a) Explain why Static RAM is used in the laptop instead of Dynamic RAM.

[2]

- (b) Identify **two** reasons for using Electrically Erasable Programmable ROM (EEPROM) in a virtual reality headset.

1

2

[2]

- (c) Describe how the laser printer makes use of a buffer.

[4]

42. M/J 23/P11/Q4/a,b

- 4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

- (a) This computer has both Read Only Memory (ROM) and Random Access Memory (RAM).

- (i) Describe the contents of the ROM in the central computer.

[2]

- (ii) The central computer has Dynamic RAM (DRAM).

Identify **two** advantages of using DRAM instead of Static RAM (SRAM).

1

2

[2]

- (b) The central computer stores the video files on secondary storage.

Describe **two** reasons why magnetic storage is more appropriate than solid state storage for this computer.

1

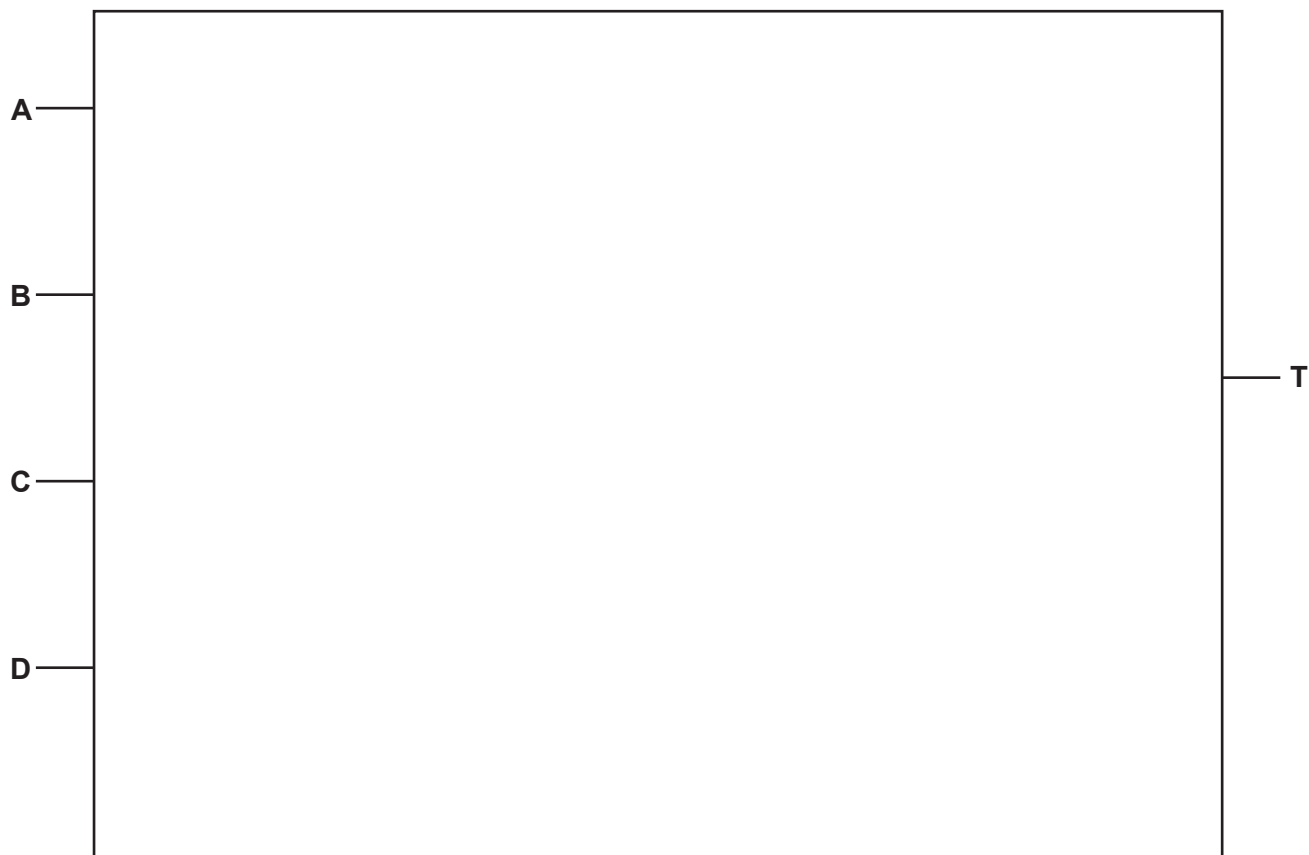
2

[4]

43. M/J 23/P11/Q5/a,b

5 (a) Draw the logic circuit for this logic expression:

$$T = (\text{NOT } A \text{ OR } B) \text{ XOR } (C \text{ NAND } D)$$



[2]

(b) Describe the function of the NAND and NOR logic gates.

NAND

NOR

[2]

44. M/J 23/P12/Q6/a,b

6 (a) Draw the logic circuit for this logic expression:

$Z = (R \text{ XOR } S) \text{ AND } (\text{NOT } T \text{ NOR } P)$



[2]

(b) Complete the truth table for this logic expression:

$Z = (\text{NOT } P \text{ OR } Q) \text{ XOR } (R \text{ NOR } Q)$

P	Q	R	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

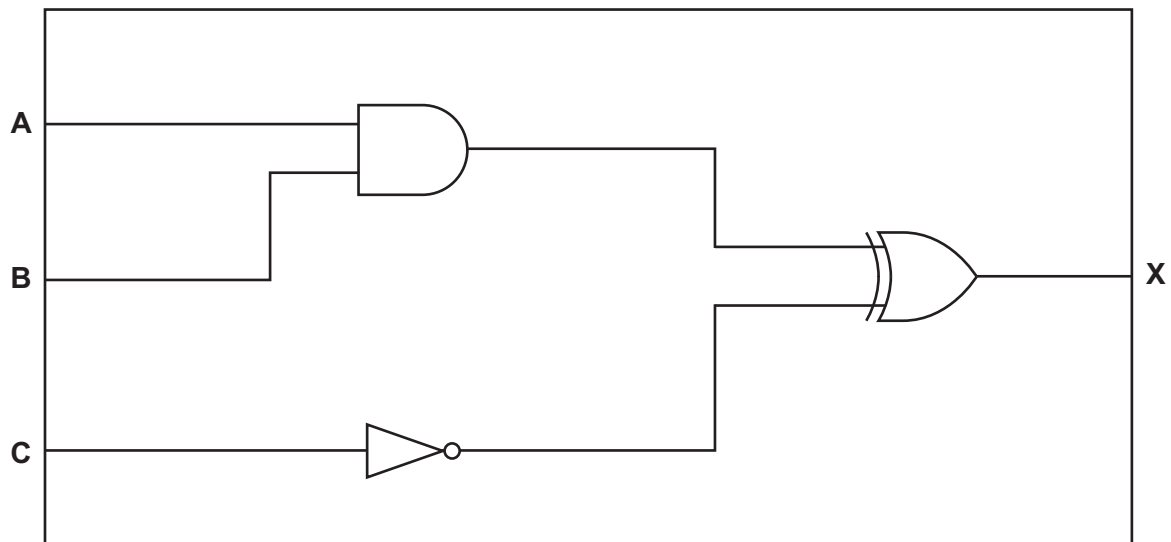
45. M/J 23/P13/Q1/a,b

1 (a) Write the logic expression for this truth table:

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

[1]

(b) Complete the truth table for this logic circuit:



A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

46. M/J 23/P13/Q3/a

3 A mobile telephone is used to record a video.

(a) The mobile telephone has a touchscreen. There are different types of touchscreen.

Complete the description of the principal operation of touchscreens.

A touchscreen has two layers. When the user touches the screen, the layers touch and a is completed.

A touchscreen has several layers. When the top layer is touched, there is a in the electric current.

A microprocessor identifies the of the touch.

[5]

47. O/N 22/P11/Q3/a,b

3 (a) Draw a logic circuit for the logic expression:

$$X = \text{NOT } ((\text{NOT } A \text{ AND NOT } B) \text{ OR } (\text{NOT } B \text{ AND NOT } C))$$



[2]

(b) Complete the truth table for the logic expression:

$$X = \text{NOT } ((\text{NOT } A \text{ AND NOT } B) \text{ OR } (\text{NOT } B \text{ AND NOT } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

48. O/N 22/P11/Q9/a,b

9 Many modern televisions are examples of embedded systems.

(a) Explain why these televisions are embedded systems.

[2]

(b) Embedded systems use Electrically Erasable Programmable ROM (EEPROM).

Describe **one** benefit of using EEPROMs in an embedded system.

[2]

49. O/N 22/P12/Q3/a,b,c

3 (a) A greenhouse has an automatic window.

The window (**X**) operates according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
T	Temperature	1	Too high
		0	Acceptable
W	Wind speed	1	Too high
		0	Acceptable
R	Rain	1	Detected
		0	Not detected
M	Manual override	1	On
		0	Off

The window opens (**X** = 1) if:

- the temperature is too high **and** the wind speed is acceptable
- **and**
- rain is not detected, **or** the manual override is off.

Draw a logic circuit to represent the operation of the window.



[3]

(b) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ OR } B \text{ OR } C) \text{ AND } (B \text{ NOR } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(c) Embedded systems contain Read Only Memory (ROM) and Random Access Memory (RAM).

Explain the reasons why ROM is used in an embedded system.

[2]

50. O/N 22/P13/Q1/c

1 A digital audio message needs to be recorded.

(c) A computer uses a buffer when playing the audio message.

Explain the purpose of a buffer in a computer system using **one other** example.

[3]

51. O/N 22/P13/Q5/a,b

5 (a) Draw a logic circuit for the logic expression:

$$X = \text{NOT} ((\text{NOT} (A \text{ AND } B)) \text{ OR } (\text{NOT} (B \text{ AND } C)))$$



[3]

(b) Complete the truth table for the logic expression:

$$Y = (\text{NOT } P \text{ AND } Q) \text{ OR } (Q \text{ AND } \text{NOT } R)$$

P	Q	R	Working space	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

52. O/N 22/P13/Q10/a,b

10 (a) Explain the importance of feedback in a control system.

[3]

(b) (i) Identify **one** sensor that could be used in a car alarm system.

Justify your choice.

Sensor

Justification

[2]

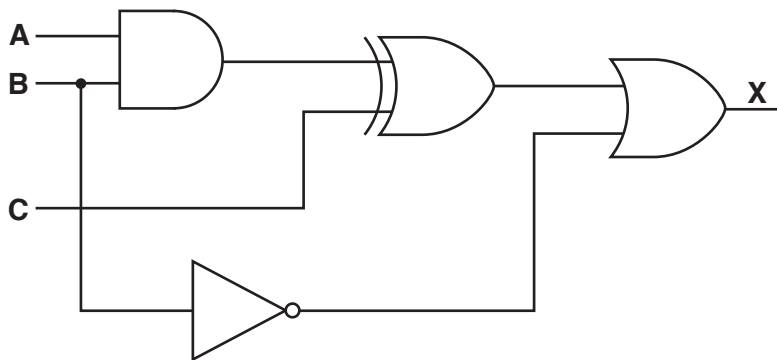
(ii) The car alarm is an example of an embedded system.

Describe the characteristics of an embedded system.

[3]

53. O/N 21/P11/Q3/a,b,c

3 A logic circuit is shown:



(a) Write the logic expression for the logic circuit.

[3]

(b) Complete the truth table for the given logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(c) Identify **one** logic gate **not** used in the given logic circuit. Draw the symbol for the logic gate **and** complete its truth table.

Logic gate:

Symbol:

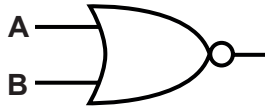
Truth table:

A	B	Output
0	0	
0	1	
1	0	
1	1	

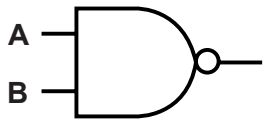
[3]

54. O/N 21/P12/Q2/a,b

- 2 (a) Complete the truth table for each of the following two logic gates.



A	B	Output
0	0	
0	1	
1	0	
1	1	



A	B	Output
0	0	
0	1	
1	0	
1	1	

[2]

- (b) Draw a logic circuit for the following logic expression.

$$X = \text{NOT}(\text{NOT}(A \text{ AND } B) \text{ AND } C)$$



[2]

55. O/N 21/P12/Q3/a

3 Andy likes to play computer games.

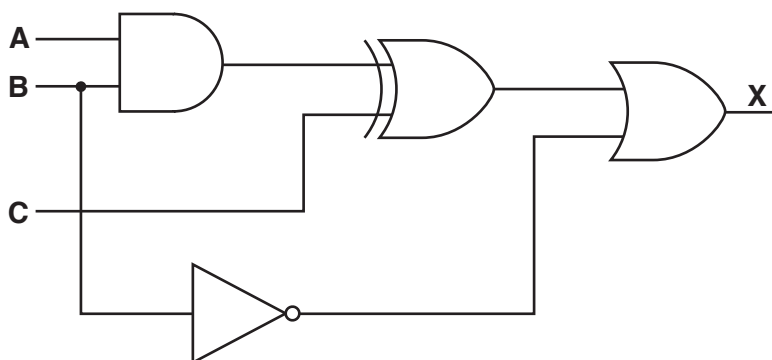
- (a) Andy uses several input devices to play the games. These include a keyboard and a microphone.

Describe the principal operation of a microphone.

[3]

56. O/N 21/P13/Q3/a,b,c

3 A logic circuit is shown:



- (a) Write the logic expression for the logic circuit.

[3]

- (b) Complete the truth table for the given logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

- (c) Identify **one** logic gate **not** used in the given logic circuit. Draw the symbol for the logic gate **and** complete its truth table.

Logic gate:

Symbol:

Truth table:

A	B	Output
0	0	
0	1	
1	0	
1	1	

[3]

57. M/J 21/P11/Q5/a,b,c

5 Kiara has a washing machine and a refrigerator.

(a) She has an embedded system in her washing machine.

Describe what is meant by an **embedded system**, using the washing machine as an example.

[2]

(b) The washing machine's embedded system makes use of both Random Access Memory (RAM) and Read Only Memory (ROM).

State the purpose of RAM and ROM within the washing machine's embedded system.

RAM

ROM

[2]

(c) The temperature in her refrigerator must be kept between 4 and 6 degrees Celsius.

The microprocessor in the refrigerator turns on the cooling if the temperature is too high, and turns off the cooling if the temperature is too low.

Explain why the system in the refrigerator is a control and not a monitoring system.

[2]

58. M/J 21/P11/Q8

8 Tick (✓) **one** box in each row to identify the logic gate that each statement describes.

Statement	AND	NAND	NOR	XOR	OR
The output is 1 only when both inputs are 1					
The output is 1 only when both inputs are different					
The output is 1 only when both inputs are 0					

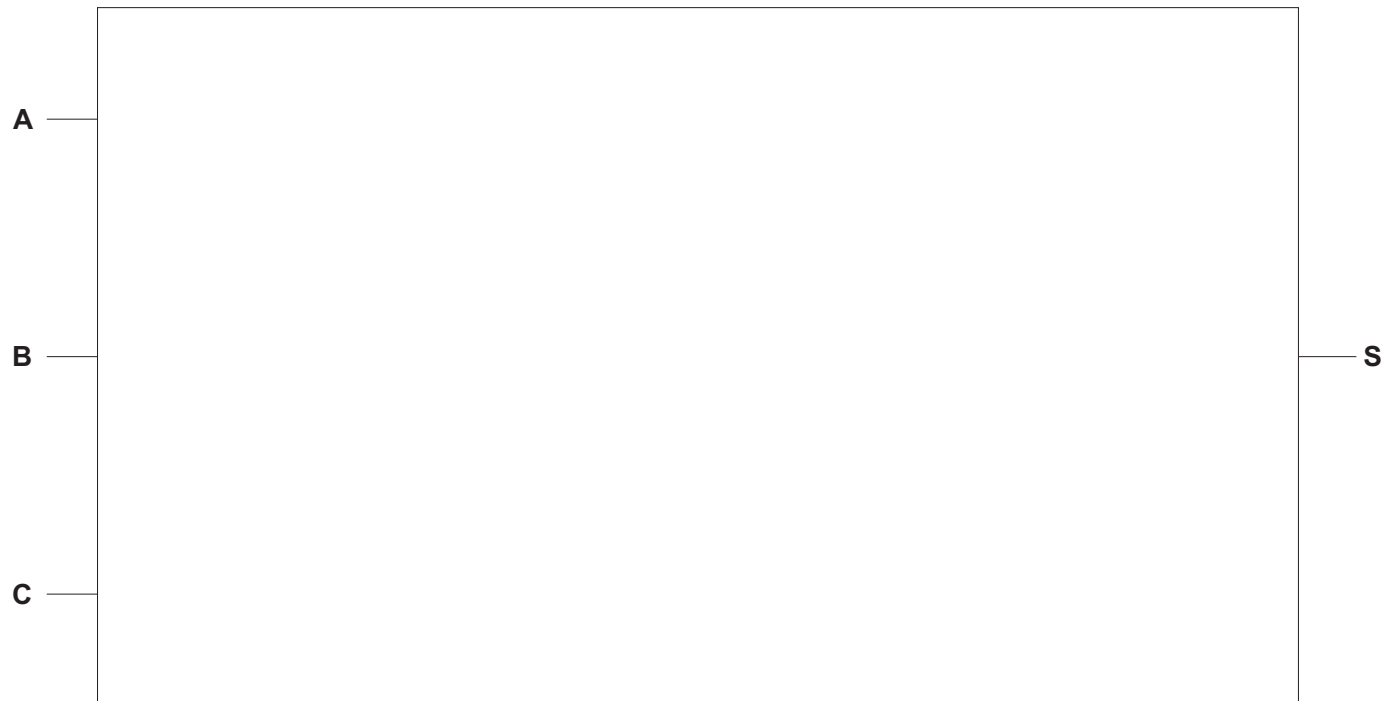
[3]

59. M/J 21/P12/Q3/a,b

3 A logic expression is given:

$$S = (A \text{ AND } B \text{ AND } C) \text{ OR } (B \text{ XOR } C)$$

(a) Draw the logic circuit for the given expression.



[4]

(b) Complete the truth table for the logic expression:

$$S = (A \text{ AND } B \text{ AND } C) \text{ OR } (B \text{ XOR } C)$$

A	B	C	Working space	S
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

60. M/J 21/P13/Q5/a,b,c

5 Kiara has a washing machine and a refrigerator.

(a) She has an embedded system in her washing machine.

Describe what is meant by an **embedded system**, using the washing machine as an example.

[2]

(b) The washing machine's embedded system makes use of both Random Access Memory (RAM) and Read Only Memory (ROM).

State the purpose of RAM and ROM within the washing machine's embedded system.

RAM

ROM

[2]

(c) The temperature in her refrigerator must be kept between 4 and 6 degrees Celsius.

The microprocessor in the refrigerator turns on the cooling if the temperature is too high, and turns off the cooling if the temperature is too low.

Explain why the system in the refrigerator is a control and not a monitoring system.

[2]

61. M/J 21/P13/Q8

8 Tick (✓) **one** box in each row to identify the logic gate that each statement describes.

Statement	AND	NAND	NOR	XOR	OR
The output is 1 only when both inputs are 1					
The output is 1 only when both inputs are different					
The output is 1 only when both inputs are 0					

[3]

Answers Section

1. M/J 26/P11/Q6/a,b

6(a)	1 mark for each correct answer Y = NOT (B OR NOT C) Z = E NOR (NOT C XOR D)	2 Accept any recognized notation Accept Y = B NOR NOT C															
6(b)	1 mark for drawing a truth table with correct inputs 1 mark for the correct outputs in the NAND column <table border="1"><thead><tr><th>Input 1</th><th>Input 2</th><th>NAND</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></tbody></table>	Input 1	Input 2	NAND	0	0	1	0	1	1	1	0	1	1	1	0	2 Ignore extra columns for working
Input 1	Input 2	NAND															
0	0	1															
0	1	1															
1	0	1															
1	1	0															

2. M/J 26/P12/Q6/a,b,c

6(a)	1 mark for each correct point to max 3 - It only has one function, to monitor health - All the hardware such as sensors and a microprocessor is dedicated to the task of monitoring health - The firmware cannot be easily amended or updated because its function is preset	3	
6(b)	No mark for the choice. 1 mark for each correct point to max 2 Monitoring: - The system only records/transmits the data - It does not make use of feedback // it does not change the environment - The output does not affect the next input - It does not contain an actuator	2	Not performing an action is NE for the no actuator mark
6(c)	1 mark for each correct point to max 2 - RAM can be changed; ROM cannot be changed // RAM can be read from and written to, ROM can only be read - RAM is volatile, ROM is non-volatile // RAM is temporary, ROM is permanent - RAM stores currently running data/instructions, ROM stores boot-up/firmware	2	The question asks for differences, so answers must refer to both RAM and ROM Do not accept the expansion of the acronyms, it is given in the question

3. M/J 26/P12/Q8/a,b

8(a)	1 mark for A column and 1 mark for B column <table><tr><th>X</th><th>Y</th><th>Z</th><th>Working space</th><th>A</th><th>B</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td><td>0</td></tr></table>	X	Y	Z	Working space	A	B	0	0	0		0	0	0	0	1		0	0	0	1	0		0	0	0	1	1		0	1	1	0	0		0	0	1	0	1		0	1	1	1	0		1	0	1	1	1		1	0	2
X	Y	Z	Working space	A	B																																																			
0	0	0		0	0																																																			
0	0	1		0	0																																																			
0	1	0		0	0																																																			
0	1	1		0	1																																																			
1	0	0		0	0																																																			
1	0	1		0	1																																																			
1	1	0		1	0																																																			
1	1	1		1	0																																																			
8(b)	1 mark for a correct answer • NOT (A XOR B) • NOT A XOR B // NOT B XOR A • NOT (A OR B) OR (A AND B) // (NOT A AND NOT B) OR (A AND B) • (NOT A AND NOT B AND NOT C) OR (NOT A AND NOT B AND C) OR (A AND B AND NOT C) OR (A AND B AND C) • A AND B OR (A NOR B)	1 Accept any correct Boolean notation For example: $\bar{A}.\bar{B}.\bar{C} + \bar{A}.\bar{B}.C + A.B.\bar{C} + A.B.C$ Or $A'B'C' + A'B'C + ABC' + ABC$																																																						

4. M/J 26/P13/Q2/a,b

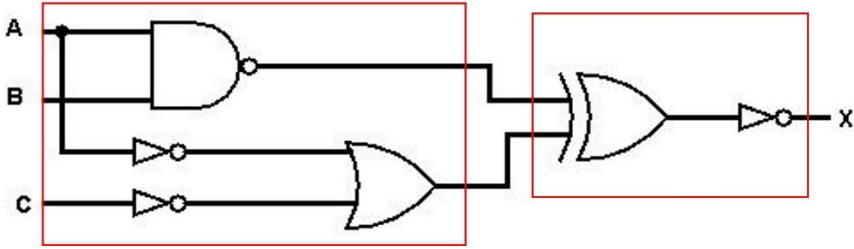
2(a)	1 mark for each correct column					2		
	X	Y	Z	Working space	A			B
	0	0	0		1			1
	0	0	1		0			0
	0	1	0		1			0
	0	1	1		1			1
	1	0	0		1			1
	1	0	1		1			0
	1	1	0		1			0
	1	1	1		1			1
2(b)	1 mark for two correct NAND and OR					1	Both are needed.	

5. M/J 26/P13/Q8/a,b,c

8(a)	1 mark for name, 1 mark for a matching description - EPROM - Erase using UV rays // needs removing to erase // have to erase whole device // supports many read/write cycles - EEPROM - Erase using electric signals // erase in situ // can update parts of memory // uses floating gate technology // supports many read/write cycles	4	No mark for expanding acronym for description.
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8(b)	1 mark each. - The microphone has a diaphragm / ribbon - The incoming sound waves cause vibrations of the diaphragm ... causing a coil to move past a magnet // causing a magnet to move past a coil (dynamic microphone). // changing the capacitance (condenser microphone) // deforms the crystal (crystal microphone) - An electrical signal is produced	3	For MP1 accept diaphragm equivalent e.g. cone.
8(c)	1 mark each - Temporary storage of data ...until the device is ready to receive it // until a sound player takes the data - Data may be sent faster than it can be received // Allows for different speed of transmission between devices - Allows processor to carry out other tasks	3	

6. O/N 25/P11/Q3/a,b

3(a)	One mark per bullet point, max 2 marks (A NAND B) and (NOT A OR NOT C) - XOR and NOT 	2
3(b)	1 mark per bullet point, max 2 marks - One correct term: (R AND S AND NOT T) // (NOT R AND NOT S AND T) - Second correct term plus the OR in correct place Answer: Q = (R AND S AND NOT T) OR (NOT R AND NOT S AND T)	2

7. O/N 25/P11/Q6/c

6(c)	1 mark per bullet point, max 3 marks - DRAM requires to be refreshed/charged SRAM does not request a refresh - DRAM stores each bit as charge SRAM uses flip-flop to store each bit - DRAM is less expensive to manufacture SRAM is more expensive to manufacture - DRAM has slower access speeds SRAM has faster access times - DRAM has higher <u>storage/bit/data density</u> SRAM has lower storage/bit/data density - DRAM is used in main memory SRAM is used in cache	3
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8. O/N 25/P11/Q9

9	1 mark per bullet point, max 3 marks - Monitoring systems do not take any action whereas control systems act autonomously to change the environment if the values are out of a prescribed range - Control systems use actuators monitoring systems do not have any actuators - Control systems make use of feedback monitoring systems do not make use of feedback - The output from a monitoring system does not affect the subsequent input whereas output from a control system affects the next input.	3
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9. O/N 25/P11/Q10/a,b

10(a)	1 mark per bullet point, max 3 marks Generic mark points: (max 3) - Additive manufacturing - Uses a digital file created from 3D modelling software or Computer Aided Design (CAD) software - The printer builds up the model one layer at a time - The process is repeated for each layer - Some materials may need time to cool and set or UV curing using LEDs for photopolymer resins Specific mark points: (max 1) e.g. - Fused Deposition Modelling (FDM) // Material is heated and pushed through nozzle / extruder - Stereolithography (SLA) // Photosensitive liquid resin is exposed to a UV-laser beam - Digital Light Processing (DLP) // Photosensitive liquid resin is exposed to a UV projected image of the layer to cure the resin - Selective Laser Sintering (SLS) // Uses a laser to form objects from powdered material, the material can be plastics or metals	3
10(b)	1 mark - To free up the processor to carry out other tasks as the rate data is received by the printer is different from the rate at which it can be processed - To manage the mismatch in speed between the processor and the peripheral / 3D printer	1

10. O/N 25/P12/Q8/b(i)

8(b)(i)	1 mark per bullet point, max 4 marks Methods to determine point of touch (Max 2) • Resistive // two layers of material make contact and complete a circuit • Capacitive // contact with the screen creates a change in charge • Infrared // the beams are broken by the pointing device • Optical imaging // a shadow is created when the surface is touched • Acoustic pulse // the acoustic wave is absorbed by the pointing device • Selection from the menu (Max 2) • The point of touch is used to determine the x and y coordinates • The menu item corresponding to the coordinate position is recognised and the menu item is added to the order	4
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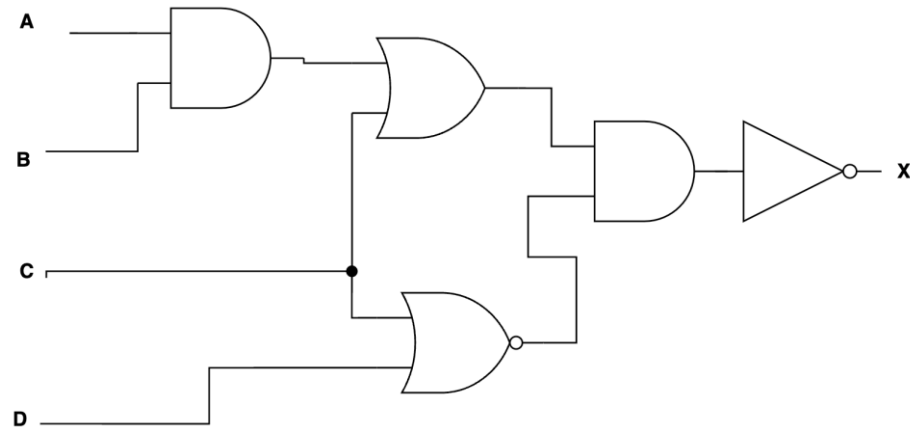
11. O/N 25/P12/Q9/a,b

9(a)	1 mark per bullet point, max 2 marks • NOT (A XOR B) • NAND ((B AND C) OR A) X = (NOT (A XOR B)) NAND ((B AND C) OR A) // X = ((B AND C) OR A) NAND (NOT (A XOR B))	2
9(b)	1 mark for 1 or 2 row numbers correct, 2 marks all three correct Row 2 Row 4 Row 7	2

12. O/N 25/P12/Q11

11	1 mark per bullet point, max 3 marks Identification: Control system (no mark) Justification: • The system uses an actuator to open or close the doors • The output / opening the doors allows people to move, which changes the input to the sensor • The system acts autonomously on the feedback from a sensor	3
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13. O/N 25/P13/Q3/a,b

3(a)	1 mark per shaded area, max 2 marks <table><tr><td>A</td><td>B</td><td>C</td><td></td><td>X</td></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C		X	0	0	0		1	0	0	1		0	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		1	2
A	B	C		X																																											
0	0	0		1																																											
0	0	1		0																																											
0	1	0		1																																											
0	1	1		1																																											
1	0	0		1																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		1																																											
3(b)	1 mark per bullet point, max 2 marks - A AND B OR C - C NOR D and final two gates 	2																																													

14. O/N 25/P13/Q4/c

4(c)	One mark for each correct row	3								
<table><tr><th>Description</th><th>Memory Technology</th></tr><tr><td>A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.</td><td>EEPROM</td></tr><tr><td>A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.</td><td>EPROM</td></tr><tr><td>A read only memory where the contents can be written only once after manufacture.</td><td>PROM</td></tr></table>			Description	Memory Technology	A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.	EEPROM	A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.	EPROM	A read only memory where the contents can be written only once after manufacture.	PROM
Description	Memory Technology									
A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.	EEPROM									
A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.	EPROM									
A read only memory where the contents can be written only once after manufacture.	PROM									

15. O/N 25/P13/Q7/d

7(d)	1 mark per reason, 1 mark for corresponding justification x2 max 4 marks • Lower cost per unit of storage • ... so the high capacity of storage required for large number of video files will be less costly • A large number of read/write operations are being performed continuously • ... and magnetic storage is likely to have a longer life span compared with solid state	4
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16. M/J 25/P11/Q1/a,b

1(a)	1 mark each: $X = Q \text{ AND } (\text{NOT } P \text{ OR } R)$ $Y = S \text{ XOR } (Q \text{ NOR } R)$	2
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1(b)	1 mark for first 4 rows, 1 mark for second 4 rows	2																																													
<table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>			A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		0	0	1	1		1	1	0	0		0	1	0	1		0	1	1	0		1	1	1	1		1
A	B	C	Working space	X																																											
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17. M/J 25/P11/Q4/b

4(b)	1 mark for sensor, 1 mark for corresponding use e.g. <table><tr><th>Sensor</th><th>Use</th></tr><tr><td>Pressure</td><td>Detects when the pressure of an item is removed from a shelf / put back on a shelf</td></tr><tr><td>Infrared</td><td>Detects when the beam is broken for an item removed / added</td></tr></table>	Sensor	Use	Pressure	Detects when the pressure of an item is removed from a shelf / put back on a shelf	Infrared	Detects when the beam is broken for an item removed / added	2
Sensor	Use							
Pressure	Detects when the pressure of an item is removed from a shelf / put back on a shelf							
Infrared	Detects when the beam is broken for an item removed / added							

18. M/J 25/P11/Q6/a(i)

6(a)(i)	1 mark for each correctly completed space • platters • spindle • read/write head • magnetic field The magnetic hard disk has one or more platters that can be magnetised. These are mounted on a spindle and rotate at high speed. A read/write head is moved across the surface on an arm. When data is read the changes in the magnetic field produce a change in the electric current.	4
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19. M/J 25/P12/Q4/a,b

4(a)	1 mark each: • $X = (R \text{ NOR } S) \text{ XOR } T$ • $Y = (T \text{ AND } U) \text{ OR } U$	2																																													
4(b)	1 mark for first four rows, 1 mark for second four rows <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		0	1	1	0		0	1	1	1		1	2
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20. M/J 25/P13/Q4/a,b

4(a)	1 mark for each correct statement X = B NOR (A NOR C) Y = (B AND C) XOR NOT D	2																																													
4(b)	1 mark for 1st 4 rows 1 mark for 2nd 4 rows <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		1	0	1	1		0	1	0	0		0	1	0	1		1	1	1	0		1	1	1	1		1	2
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21. O/N 24/P11/Q3/c,d

3(c)	1 mark for each bullet point (max 2) e.g. • DRAM requires constant refresh cycles unlike SRAM • DRAM has lower access speed than SRAM	2
3(d)(i)	1 mark for each bullet point (max 4) • The disc is spun at high speed • A laser is shone onto the disc to read / write • ... using optical head to move it into position • ... it follows the spiral track from the centre outwards • When writing the laser burns pits to represent the data • When reading the laser reflects from pits and lands • The reflection from a pit and a land is different • ... the differences are interpreted as 1 or 0	4
3(d)(ii)	1 mark for each bullet point (max 3) • The computer and optical disc reader/writer send and receive at different speeds • A buffer allows temporary storage of the data • ... so that the computer can transfer data to the buffer at the higher speed • ... and is not held up waiting for data to transfer // so the computer can carry on with other tasks • ... so the optical disc reader/writer is not overloaded • ... and so that data is transferred to optical disc reader/writer from the buffer at the slower rate	3

22. O/N 24/P11/Q5/a,b,c

5(a)	1 mark for all 3 appropriate sensors	1								
	<table><tr><th>Scenario</th><th>Type of sensor</th></tr><tr><td>A door is open.</td><td>Pressure sensor // Infra-red</td></tr><tr><td>The external daylight-level is below a set amount.</td><td>Light sensor</td></tr><tr><td>A person is detected within 2 metres.</td><td>Infra-red</td></tr></table>	Scenario	Type of sensor	A door is open.	Pressure sensor // Infra-red	The external daylight-level is below a set amount.	Light sensor	A person is detected within 2 metres.	Infra-red	
Scenario	Type of sensor									
A door is open.	Pressure sensor // Infra-red									
The external daylight-level is below a set amount.	Light sensor									
A person is detected within 2 metres.	Infra-red									
5(b)	1 mark for each correct expression X = E AND A AND C Y = E AND (B OR C OR D)	2								

5(c)	1 mark for each bullet point (max 3) for a correct justification No marks for the identification of the system Monitoring system - There is no use of feedback // The light and the alarm are just warnings - The output of the floodlight or the audio alarm does not affect the input of data from the sensors - The system does not have any actuators	3
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23. O/N 24/P12/Q1/a,b

1(a)	1 mark for each shaded part. <table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	A	B	C	X	0	0	0	1	0	0	1	1	0	1	0	1	0	1	1	1	1	0	0	1	1	0	1	0	1	1	0	1	1	1	1	1	2
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1(b)	1 mark for (P XOR Q) and (Q OR NOT R) 1 mark for second XOR gate and final NAND gate with appropriate inputs <pre>graph LR P --- J1(()) J1 --- XOR1[] J1 --- AND1[] Q --- J2(()) J2 --- XOR1 J2 --- XOR2[] R --- NOT1[] NOT1 --- XOR2 XOR1 --- XOR3[] XOR2 --- XOR3 XOR3 --- AND1 AND1 --- W</pre>	2																																				

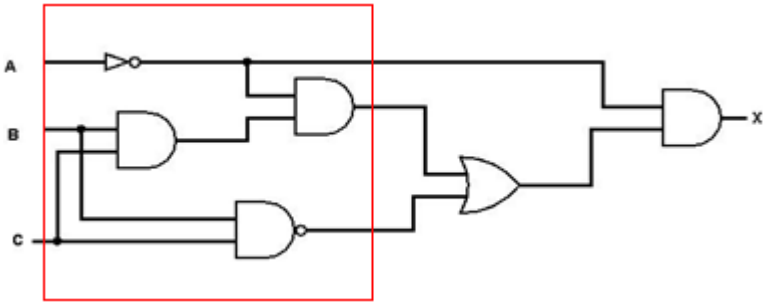
24. O/N 24/P12/Q2/a,b,c

2(a)	1 mark for each bullet point (max 3) e.g. • It is difficult to change / update the firmware by the user • Errors cannot be fixed easily // Troubleshooting / fault-finding / repairing is a specialist task / expensive • Functionality cannot be changed/extended easily // Cannot be easily adapted for another task • Faulty / outdated devices are often thrown away rather than repaired • ... leading to e-waste	3
2(b)	1 mark from: • DRAM is less expensive to manufacture/purchase than SRAM in the embedded system • DRAM has a higher bit density per chip // more data can be stored per chip	1
2(c)	1 mark for each bullet point (max 2) • EPROM uses ultraviolet light to erase data whilst EEPROM uses an electrical signal to do this • EPROM has to be removed from the circuit board when changing the data whilst EEPROM remains in the circuit when the data is changed • EPROM erases all the data, EEPROM can erase parts of the data	2

25. O/N 24/P12/Q9/a,b

9(a)	1 mark for sensor and its use in this system • Infra-red sensor • Measure / check the height of the vehicle • Pressure sensor • Measure / check the weight of the vehicle	2
9(b)	1 mark for each bullet point (max 2) for a correct justification No marks for the identification of the system Monitoring system • Because there is no use of feedback // the warning sign is only an indicator • ... the output of the turning on of the sign does not affect the input of data from the sensors • The system does not have any actuators	2

26. O/N 24/P13/Q1/a,b,c

1(a)	1 mark for: Output is only 1 if one input is 1 and the other is 0 // Output is only 1 if both inputs are different // Output is only 0 if both inputs are the same	1
1(b)	1 mark each: • $((A \text{ NOR } B) \text{ AND } (A \text{ OR } B))$ • $\text{NAND } (B \text{ AND } C)$ $X = ((A \text{ NOR } B) \text{ AND } (A \text{ OR } B)) \text{ NAND } (B \text{ AND } C)$	2
1(c)	1 mark for NOT gate, 2 correct AND gates and the NAND gate 1 mark for OR gate and the final AND gate 	2

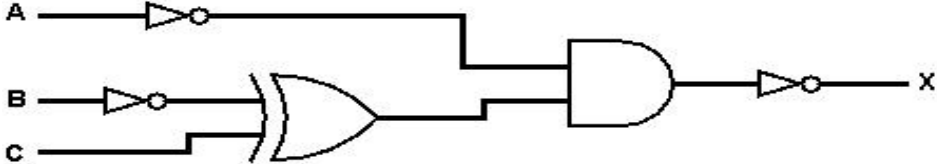
27. O/N 24/P13/Q2/a,b

2(a)	1 mark for each difference e.g. • DRAM requires constant refreshing, but SRAM does not need to be constantly refreshed • DRAM generally has greater capacities per chip, but SRAM generally has lower capacities per chip • DRAM has slower access times than SRAM	2
2(b)	1 mark for each bullet point (max 4) • The computer and the hard disk drive transmit and receive at different speeds // The computer transfers data faster than the HDD can receive • The buffer is used for temporary storage • ... so that the computer can transfer data to the buffer at the higher speed • ... and is not held up waiting for data to transfer • ... and so that data is transferred to hard disk drive from the buffer at the slower rate	4

28. O/N 24/P13/Q3

3	1 mark for each bullet point (max 3) - The system uses feedback - The system acts independently - Input data causes the car to brake ... braking decreases the/changes the distance to the object ... this new distance is used to determine whether/how hard the car needs to brake	3
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29. M/J 24/P11/Q1/a,b

1(a)	1 mark for: $(A \text{ XOR } B) \text{ NOR } C$	1
1(b)	1 mark for NOT B XOR C 1 mark for NOT A and final AND plus NOT 	2

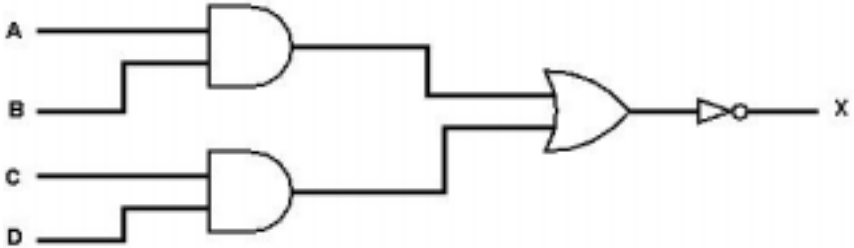
30. M/J 24/P11/Q2/a,b,c

2(a)	1 mark each to max 2 : - The doorbell only performs the specific tasks of motion detection/video recording/doorbell ringing - The motion sensor and digital camera are built into the doorbell - The CPU/memory/storage/software are all dedicated to this task only - Only a dedicated microprocessor is required due to the limited processing requirements	2
2(b)	No mark for identification of monitoring or control 1 mark each to max 2 for justification: Monitoring: - The turning on of the digital camera does not affect the input to the sensor/button - The transmission of the data/video does not affect the input to the sensor/button - The ringing of the doorbell does not affect the input to the button Control: - Video doorbell does not only store the values from the motion sensor - The data is processed, generating a signal to start the digital camera recording - Button pressed/motion detected causes a signal to be sent over a network to the smartphone	2

2(c)(i)	1 mark each to max 2: • Current reading/data from motion sensor • Current/recent video • Instructions being executed • Start-up/BIOS/boot-up instructions	2										
2(c)(ii)	1 mark for each row: <table><tr><th>Statement</th><th>Answer</th></tr><tr><td>The two types of logic gate that can be used to create solid state devices</td><td>NAND NOR</td></tr><tr><td>The number of transistors contained in each cell</td><td>2</td></tr><tr><td>The type of gate that can retain electrons without power</td><td>floating</td></tr><tr><td>The type of gate that allows or stops current from passing through</td><td>control</td></tr></table>	Statement	Answer	The two types of logic gate that can be used to create solid state devices	NAND NOR	The number of transistors contained in each cell	2	The type of gate that can retain electrons without power	floating	The type of gate that allows or stops current from passing through	control	4
Statement	Answer											
The two types of logic gate that can be used to create solid state devices	NAND NOR											
The number of transistors contained in each cell	2											
The type of gate that can retain electrons without power	floating											
The type of gate that allows or stops current from passing through	control											
2(c)(iii)	1 mark each to max 2: • Captured video is transmitted to buffer • ... video is transmitted from buffer to smartphone • Store recent data in a buffer for the user to rewind • ... instead of storing everything in secondary storage • Store readings from motion sensor • ... until the microprocessor can process them • Store video from digital camera • ... before moving it to secondary storage	2										

31. M/J 24/P12/Q1/a,b

1(a)	1 mark for each correct answer: <i>NAND</i> - The output is 0 when both inputs are 1, otherwise the output is 1 <i>NOR</i> - The output is 1 when both inputs are 0, otherwise the output is 0 <i>XOR</i> - The output is 1 when one of the inputs is 1 and the other input is 0, otherwise the output is 0 <i>OR</i> - The output is 0 when both inputs are 0, otherwise the output is 1	4
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1(b)	1 mark for both AND gates with correct inputs 1 mark for correct OR and NOT gates with correct inputs and no superfluous gates: 	2
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32. M/J 24/P12/Q2/a,b,c

2(a)	1 mark for each correctly completed statement: • (LCD) displays/screens/lenses • gyroscope/accelerometer • direction/speed • digital cameras A headset can have one or two (LCD) displays/screens/lenses that output the image to the user. The headset has speakers that output surround sound to give a realistic experience. The user's head movements are detected using a sensor. This sensor is a gyroscope/accelerometer. The data is transmitted to a microprocessor that analyses the data to identify the direction/speed of movement. Some headsets use digital cameras that record the user's eye movements for analysis.	4
2(b)	1 mark each to max 3: • The buffer is used as a temporary store for data going to the headset • Data is transferred into the buffer by the computer • Data is retrieved from the buffer by the headset • When the buffer is empty/full an interrupt is sent to the computer requesting more data/stopping further data being sent • When the headset has enough data/needs more data, an interrupt is sent by the headset to the computer to stop sending data from buffer	3
2(c)	1 mark each to max 3: • EEPROM allows frequent/multiple read/write/erase operations • ... so the headset can take advantage of new features • ... without fully erasing the contents of the firmware in the headset first // can erase a particular byte or the whole EEPROM • ... without removing the chip(s)/firmware from the headset • ... the contents of the firmware in the headset can be changed by the user without technical expertise • Cheaper to manufacture so headset will be cheaper to purchase	3

33. M/J 24/P13/Q6

6

1 mark for each correct line:

3

Truth table

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Logic expression

NOT (A XOR B) AND C

(A OR C) AND NOT B

A NAND B NAND C

(A NAND B) OR C

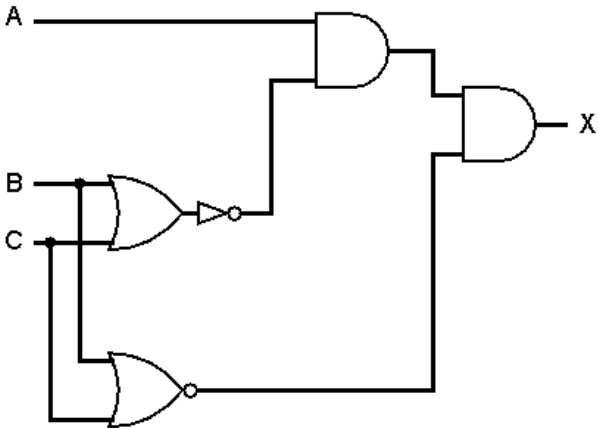
NOT (A AND B AND C)

34. M/J 24/P13/Q7/a,c,d

7(a)	1 mark for sensor and matching purpose to max 2: <table><tr><th>Sensor</th><th>Purpose of sensor in navigation system</th></tr><tr><td>Pressure</td><td>To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on</td></tr><tr><td>Infra-red</td><td>To detect if there is an obstacle in the way // to indicate that it has reached the desired table</td></tr><tr><td>Sound</td><td>To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop</td></tr></table>	Sensor	Purpose of sensor in navigation system	Pressure	To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on	Infra-red	To detect if there is an obstacle in the way // to indicate that it has reached the desired table	Sound	To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop	2
Sensor	Purpose of sensor in navigation system									
Pressure	To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on									
Infra-red	To detect if there is an obstacle in the way // to indicate that it has reached the desired table									
Sound	To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop									
7(c)	1 mark each to max 2: • Feedback ensures that a system operates within set criteria / constraints • ... by enabling system output to affect subsequent system input • ... thus allowing conditions to be automatically adjusted	2								

7(d)	1 mark each to max 4 : - Resistive: The space between the conductive layers is removed/the layers touch and a circuit is completed - Capacitive: The electrical charge changes where the user pressed - The point of contact is identified ... from the change in electrical field - The software/microprocessor calculates the coordinates	4
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35. O/N 23/P11/Q4/a,b

4(a)	1 mark for each set of 4 rows (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	A	B	C	X	0	0	0	1	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	1	1	0	1	0	1	1	0	1	1	1	1	1	2
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1	1	0	1																																			
1	1	1	1																																			
4(b)	1 mark for both AND gates 1 mark for NOT gate <u>and</u> OR gate <u>and</u> NOR gate 	2																																				

36. O/N 23/P11/Q7/a,b,c

7(a)	1 mark for each bullet point (max 3) Generic mark points: - Additive manufacturing - Uses a digital 3D model or a Computer Aided Design (CAD) (file) - Builds up the model one layer at a time ...starting from the bottom ...using x, y and z co-ordinates - The material is fused / cured together layer by layer Specific mark points: Fused Deposition Modelling (FDM) - Material is heated and pushed through nozzle / extruder Stereolithography (SLA) - Photosensitive liquid resin is exposed to a UV-laser beam Digital Light Processing (DLP) - Uses liquid plastic resin melted with arc lamps Selective Laser Sintering (SLS) - Uses a laser to form objects from powdered material	3
7(b)	1 mark for each bullet point (max 2) - To prevent overheating // ensure material is hot enough ...by identifying the temperature of the object (being printed) ...by identifying the temperature of the material being used	2
7(c)	1 mark for each bullet point (max 2) - Dynamic RAM has lower cost per unit - A fast access speed is not needed - Higher bit density // more data can be stored per chip	2

37. O/N 23/P11/Q9/a,b

9(a)	One mark for each bullet point (max 2) - Feedback ensures that a system operates within set criteria / constraints ...by enabling system output to affect (subsequent) system input ...thus allowing conditions to be <u>automatically</u> adjusted	2
9(b)	One mark for each reason to max 3 to match the example given - Dedicated to one task applied to example - Does not require much processing power applied to example - Built into a larger system applied to example - Contains firmware that cannot be easily updated applied to example - The system does not have its own operating system - An embedded system must contain a processor, memory and an I/O capability // Dedicated hardware	3

38. O/N 23/P12/Q1/a,b,c

1(a)	1 mark for: infra-red / proximity (sensor)	1
1(b)	1 mark for each bullet point (max 2) - Actuator generates a signal / causes an action / converts electrical energy into a mechanical force ... to push an arm // to open a trap door // to pick up the chocolate bar with the incorrect weight	2
1(c)(i)	1 mark for each feature (max 2) - Dedicated to a single task // limited number of functions - Built into a larger system // integrated into a larger system - An embedded system must contain a processor, memory and an I/O capability // Dedicated hardware	2
1(c)(ii)	1 mark for each bullet point (max 1) - Difficult to change / update the firmware by the user // Difficult to upgrade devices to take advantage of new technology - Cannot be easily adapted for another task - Troubleshooting faults/upgrading is a specialist task - Devices are often thrown away rather than repaired as difficult to upgrade or repair	1

39. O/N 23/P12/Q4/a,b

4(a)	1 mark for each bullet point - A NAND B - NOT(B XOR C) - Final NAND $X = A \text{ NAND } B \text{ NAND } (\text{NOT}(B \text{ XOR } C))$	3
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4(b)	1 mark for each set of rows (shaded)	2																																				
<table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>			A	B	C	X	0	0	0	0	0	0	1	0	0	1	0	1	0	1	1	0	1	0	0	1	1	0	1	1	1	1	0	1	1	1	1	1
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40. O/N 23/P13/Q4/a,b

4(a)	1 mark for each set of highlighted rows	2																																				
<table><tr><th>P</th><th>Q</th><th>R</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>			P	Q	R	Y	0	0	0	1	0	0	1	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	1	0	1	1	0	0	1	1	1	0
P	Q	R	Y																																			
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4(b)	1 mark for P AND Q and NOT Q OR R 1 mark for NOT gate and AND gate and XOR gate	2

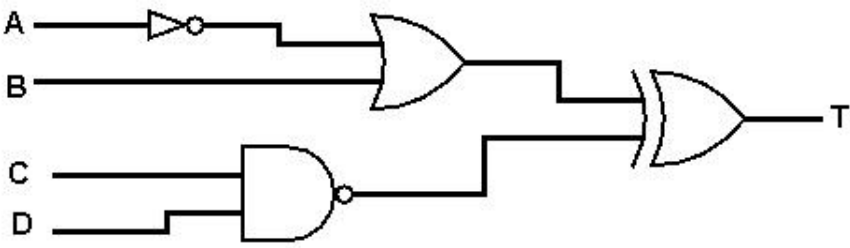
41. O/N 23/P13/Q7/a,b,c

7(a)	1 mark for each bullet point (max 2). - Static RAM has faster access time ...because it does not need to be refreshed ...used on the CPU for improvement of CPU cache speed - Static RAM has lower	2
7(b)	1 mark for each bullet point (max 2) - EEPROM allows frequent / multiple read / write / erase operations ...which means that the contents of the firmware in the headset can be changed easily ...without fully erasing the contents of the firmware in the headset first	2
7(c)	1 mark for each bullet point (max 4) - The print instructions and data are sent by the laptop to a buffer (at laptop speed) - The data is transferred from the buffer to the printer (at printer speed) ...allowing user to continue using the laptop // and allowing processor to continue processing ...instead of waiting for relatively slower printer - When the buffer is empty an interrupt is sent to the laptop ...requesting more data	4

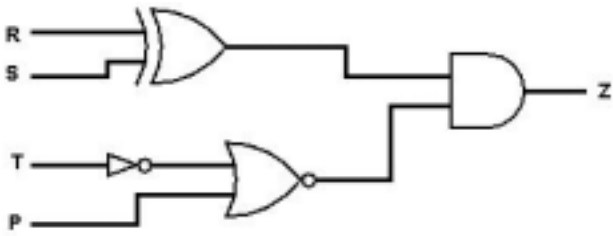
42. M/J 23/P11/Q4/a,b

4(a)(i)	1 mark each to max 2 - Stores the bootstrap program // start-up instructions for the central computer // BIOS - Stores the start-up instructions for the CCTV system/cameras // firmware for CCTV - Stores the kernel of the Operating System // stores parts of the Operating System	2
4(a)(ii)	1 mark each to max 2 - Costs less per unit - Higher storage density - Simple design – uses fewer transistors	2
4(b)	1 mark for reason, 1mark for application/justification - The computer will have a large number of read/write operations because it is working all the time .. magnetic storage has more longevity - Magnetic storage costs less per storage unit ... videos are large files and therefore very large storage capacity is required	4

43. M/J 23/P11/Q5/a,b

5(a)	1 mark for 2 gates 2 marks for all 4 gates 	2
5(b)	1 mark each NAND 0 is only output when both inputs are 1 // 1 is only output when none, or (either) one of the inputs is 1 NOR 1 is only output when both inputs are 0 // 0 is only output when (either) one or both inputs are 1	2

44. M/J 23/P12/Q6/a,b

6(a)	1 mark for correct XOR and AND gates, with correct inputs 1 mark for correct NOT and NOR gates with correct inputs 	2
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6(b)

1 mark for first 4 rows
1 mark for last 4 rows

P	Q	R	Working space	Z
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		1
1	0	0		1
1	0	1		0
1	1	0		1
1	1	1		1

2

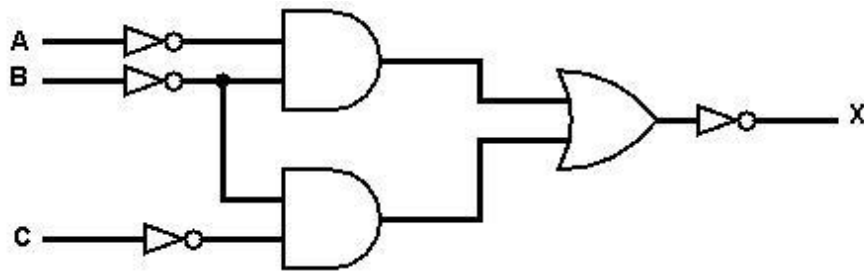
45. M/J 23/P13/Q1/a,b

1(a)	NOT B	1																																				
1(b)	1 mark for first 4 rows correct; 1 mark for second 4 rows correct <table><tr><td>A</td><td>B</td><td>C</td><td>X</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	A	B	C	X	0	0	0	1	0	0	1	0	0	1	0	1	0	1	1	0	1	0	0	1	1	0	1	0	1	1	0	0	1	1	1	1	2
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46. M/J 23/P13/Q3/a

3(a)	1 mark for each term • Resistive • Circuit • Capacitive • Change • Coordinates A resistive touchscreen has two layers. When the user touches the screen, the layers touch and a circuit is completed. A capacitive touchscreen has several layers. When the top layer is touched there is a change/increase/decrease in the electric current. A microprocessor identifies the coordinates of the touch.	5
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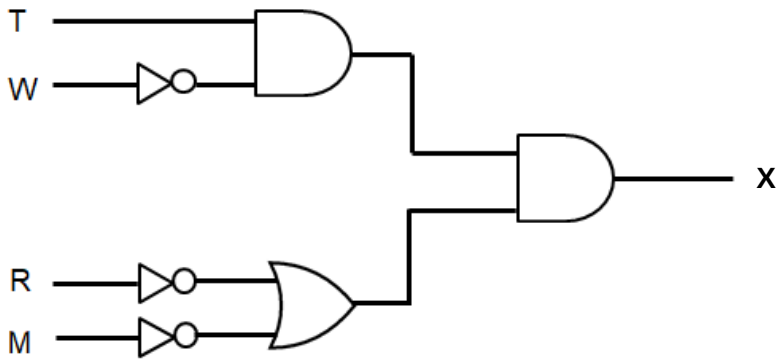
47. O/N 22/P11/Q3/a,b

3(a)	1 mark for each bullet point: • NOT A AND NOT B and NOT B AND NOT C // A NOR B and B NOR C • final OR and NOT gates (with correct inputs) // NOR gate (with correct inputs) 	2																																				
3(b)	1 mark for each set of rows as highlighted: <table><thead><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></tbody></table>	A	B	C	X	0	0	0	0	0	0	1	0	0	1	0	1	0	1	1	1	1	0	0	0	1	0	1	1	1	1	0	1	1	1	1	1	2
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48. O/N 22/P11/Q9/a,b

9(a)	1 mark for each bullet point (max 2): - the embedded system is built into / integrated into the TV - combination of hardware and software designed for a specific function - The system is not easily changed/updated by the TV owner	2
9(b)	1 mark for each benefit and 1 mark for corresponding expansion (max 2). For example: - no additional equipment is needed to change ... enables firmware updates by non-technical users - can be erased and reprogrammed several times ... so firmware can be updated ... can erase a particular byte or the whole EEPROM - possible to reprogram / update ... without removing it from the device	2

49. O/N 22/P12/Q3/a,b,c

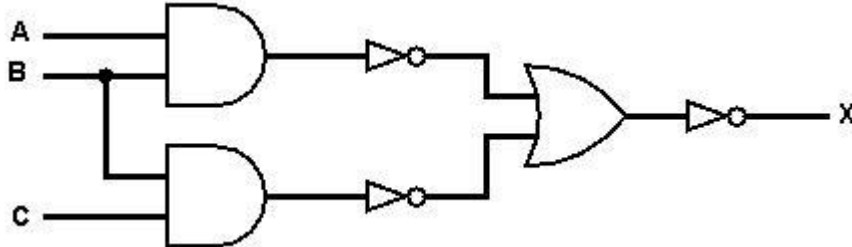
3(a)	1 mark for T AND NOT W 1 mark for NOT R OR NOT M 1 mark for final AND 	3
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3(b)	1 mark for each set of rows as highlighted:	2																																				
	<table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>	A	B	C	X	0	0	0	1	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0	1	0	1	0	1	1	0	0	1	1	1	0	
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3(c)	1 mark for each bullet point: - to store data that does not change - data must be stored even when device is without power - to store boot up instructions / system software / firmware / BIOS	2																																				

50. O/N 22/P13/Q1/c

1(c)	1 mark for purpose (max 2): Purpose: - to act as temporary storage // to store (downloaded) data ... before it is used by the receiving device ... to allow processes / devices to operate at different speeds // independently of each other 1 mark for each example (max 1): Examples: - printer buffer used when data is transferred from a computer to a printer - video buffer when streaming videos - keyboard buffer when performing data entry	3
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51. O/N 22/P13/Q5/a,b

5(a)	1 mark for each bullet point: • NOT (A AND B) • NOT (B AND C) • NOT(NOT(A AND B) OR NOT(B AND C)) 	3																																				
5(b)	1 mark for each set of highlighted rows. <table data-bbox="280 761 839 1348"><tr><th>P</th><th>Q</th><th>R</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>	P	Q	R	Y	0	0	0	0	0	0	1	0	0	1	0	1	0	1	1	1	1	0	0	0	1	0	1	0	1	1	0	1	1	1	1	0	2
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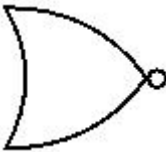
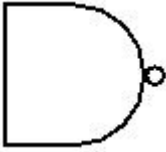
52. O/N 22/P13/Q10/a,b

10(a)	1 mark for each bullet point: • to ensure the system operates with the given criteria • ... by enabling system output to affect subsequent system input • ... thus allowing conditions to be <u>automatically</u> adjusted	3
10(b)(i)	1 mark for identification of a suitable sensor 1 mark for corresponding justification Example: • sound sensor • if a sound occurs inside the car the alarm is activated • infra-red sensor • senses the heat of person in the car / infra-red beams are broken • pressure sensor • an intruder sits in the driver's seat	2

10(b)(ii)	1 mark for each bullet point (max 3): - the embedded system is built into / integrated (into the car alarm) - combination of hardware and software designed for a specific function - must have a processor, memory and input / output - The system is not easily changed/updated by the car owner	3
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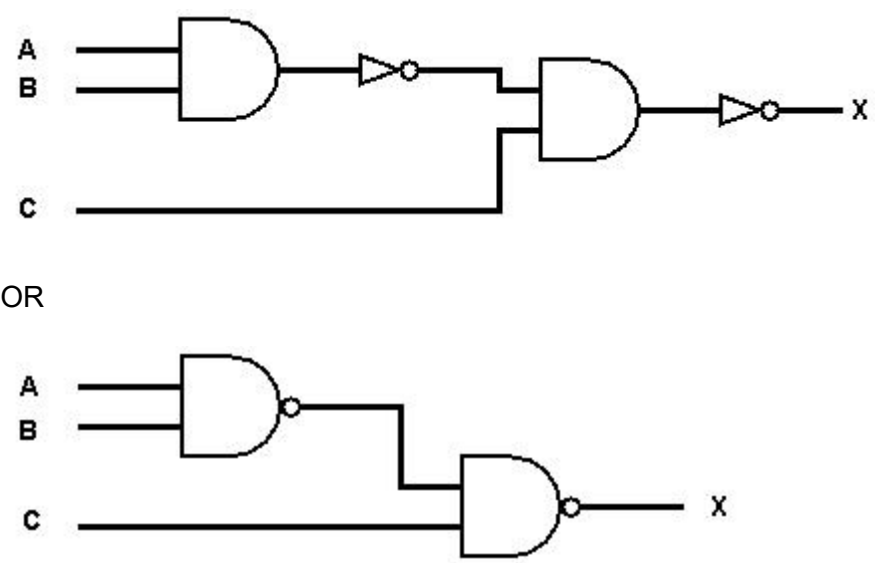
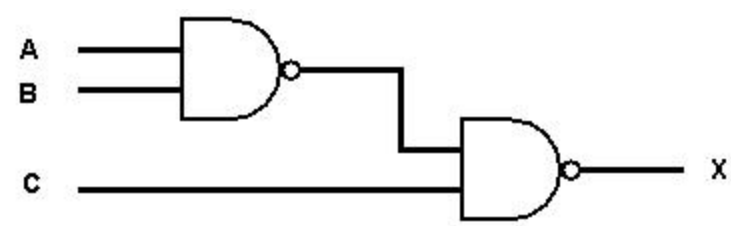
53. O/N 21/P11/Q3/a,b,c

3(a)	1 mark per bullet point - A AND B XOR C OR NOT B ((A AND B) XOR C) OR NOT B	3																																													
3(b)	1 mark for each set of 4 rows (shaded) <table><thead><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></tbody></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		0	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		0	2
A	B	C	Working space	X																																											
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3(c)	1 mark for gate, 1 mark for matching symbol, 1 mark for matching truth table NOR  <table border="1" data-bbox="683 309 1149 631"> <thead> <tr> <th>A</th><th>B</th><th>OUTPUT</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table> NAND  <table border="1" data-bbox="678 701 1144 1023"> <thead> <tr> <th>A</th><th>B</th><th>OUTPUT</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	A	B	OUTPUT	0	0	1	0	1	0	1	0	0	1	1	0	A	B	OUTPUT	0	0	1	0	1	1	1	0	1	1	1	0	3
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54. O/N 21/P12/Q2/a,b

2(a)	1 mark for each completely correct truth table NOR <table border="1" data-bbox="287 1243 617 1568"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table> NAND <table border="1" data-bbox="287 1635 617 1960"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	A	B	Output	0	0	1	0	1	0	1	0	0	1	1	0	A	B	Output	0	0	1	0	1	1	1	0	1	1	1	0	2
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A	B	Output																														
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0	1	1																														
1	0	1																														
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2(b)	1 mark for each correct bullet point • NOT (A AND B) // A NAND B • NOT the result AND C // the result NAND C  OR 	2
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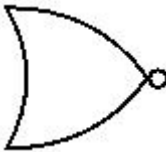
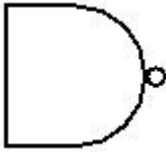
55. O/N 21/P12/Q3/a

3(a)	1 mark for each bullet point to max 3 • The microphone has a diaphragm / ribbon • The incoming sound waves cause vibrations of the diaphragm • ... causing a coil to move past a magnet // causing a magnet to move past a coil (dynamic microphone) // changing the capacitance (condenser microphone) // deforms the crystal (crystal microphone) • An electrical signal is produced	3
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56. O/N 21/P13/Q3/a,b,c

3(a)	1 mark per bullet point • A AND B ... • ... XOR C ... • ... OR NOT B ((A AND B) XOR C) OR NOT B	3
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3(b)	1 mark for each set of 4 rows (shaded) <table><tr><td>A</td><td>B</td><td>C</td><td>Working space</td><td>X</td></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		0	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		0	2
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3(c)	1 mark for gate, 1 mark for matching symbol, 1 mark for matching truth table NOR  <table><tr><td>A</td><td>B</td><td>OUTPUT</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table> NAND  <table><tr><td>A</td><td>B</td><td>OUTPUT</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	OUTPUT	0	0	1	0	1	0	1	0	0	1	1	0	A	B	OUTPUT	0	0	1	0	1	1	1	0	1	1	1	0	3
A	B	OUTPUT																														
0	0	1																														
0	1	0																														
1	0	0																														
1	1	0																														
A	B	OUTPUT																														
0	0	1																														
0	1	1																														
1	0	1																														
1	1	0																														

57. M/J 21/P11/Q5/a,b,c

5(a)	1 mark per bullet point to max 2 - Definition: Microprocessor/microcontroller within a larger system // microprocessor/microcontroller that performs one specific task - Example: e.g. Embedded system in washing machine only controls the programs for the washing cycle // it is part of the washing machine but does not perform any other function within it	2
5(b)	1 mark for RAM, 1 mark for ROM RAM: Store the choices/wash program the user has entered // stores the data read from the sensors // stores the time left in the program // by example ROM: Store the start-up instructions (for the washing cycles)	2
5(c)	1 mark per bullet point - The system uses feedback - The system causes the temperature to change // produces an action	2

58. M/J 21/P11/Q8

8	1 mark per correct row	3																								
<table><tr><th>Statement</th><th>AND</th><th>NAND</th><th>NOR</th><th>XOR</th><th>OR</th></tr><tr><td>The output is 1 only when both inputs are 1</td><td>✓</td><td></td><td></td><td></td><td></td></tr><tr><td>The output is 1 only when both inputs are different</td><td></td><td></td><td></td><td>✓</td><td></td></tr><tr><td>The output is 1 only when both inputs are 0</td><td></td><td></td><td>✓</td><td></td><td></td></tr></table>			Statement	AND	NAND	NOR	XOR	OR	The output is 1 only when both inputs are 1	✓					The output is 1 only when both inputs are different				✓		The output is 1 only when both inputs are 0			✓		
Statement	AND	NAND	NOR	XOR	OR																					
The output is 1 only when both inputs are 1	✓																									
The output is 1 only when both inputs are different				✓																						
The output is 1 only when both inputs are 0			✓																							

59. M/J 21/P12/Q3/a,b

3(a)	1 mark for each correct gate, with correct inputs <pre>graph LR; A --- AND1[AND]; B --- AND1; B --- AND2[AND]; C --- AND2; C --- OR1[OR]; AND1 --- AND2; AND2 --- OR2[OR]; OR1 --- OR2; OR2 --- S</pre>	4																																													
3(b)	1 mark for each half (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>S</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	S	0	0	0		0	0	0	1		1	0	1	0		1	0	1	1		0	1	0	0		0	1	0	1		1	1	1	0		1	1	1	1		1	2
A	B	C	Working space	S																																											
0	0	0		0																																											
0	0	1		1																																											
0	1	0		1																																											
0	1	1		0																																											
1	0	0		0																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		1																																											

60. M/J 21/P13/Q5/a,b,c

5(a)	1 mark per bullet point to max 2 - Definition: Microprocessor/microcontroller within a larger system // microprocessor/microcontroller that performs one specific task - Example: e.g. Embedded system in washing machine only controls the programs for the washing cycle // it is part of the washing machine but does not perform any other function within it	2
5(b)	1 mark for RAM, 1 mark for ROM RAM: Store the choices/wash program the user has entered // stores the data read from the sensors // stores the time left in the program // by example ROM: Store the start-up instructions (for the washing cycles)	2

5(c)	1 mark per bullet point - The system uses feedback - The system causes the temperature to change // produces an action	2
------	--	----------

61. M/J 21/P13/Q8

8	1 mark per correct row	3																								
<table><tr><th>Statement</th><th>AND</th><th>NAND</th><th>NOR</th><th>XOR</th><th>OR</th></tr><tr><td>The output is 1 only when both inputs are 1</td><td>✓</td><td></td><td></td><td></td><td></td></tr><tr><td>The output is 1 only when both inputs are different</td><td></td><td></td><td></td><td>✓</td><td></td></tr><tr><td>The output is 1 only when both inputs are 0</td><td></td><td></td><td>✓</td><td></td><td></td></tr></table>			Statement	AND	NAND	NOR	XOR	OR	The output is 1 only when both inputs are 1	✓					The output is 1 only when both inputs are different				✓		The output is 1 only when both inputs are 0			✓		
Statement	AND	NAND	NOR	XOR	OR																					
The output is 1 only when both inputs are 1	✓																									
The output is 1 only when both inputs are different				✓																						
The output is 1 only when both inputs are 0			✓																							

Unit 4

PROCESSOR FUNDAMENTALS

48 items · 357 marks

Syllabus sections in this unit

- 4.1 Central Processing Unit (CPU) Architecture
- 4.2 Assembly Language
- 4.3 Bit manipulation

Unit 4: Processor Fundamentals

1. M/J 26/P11/Q3/a,b,c

- 3 The following table shows part of the instruction set for a processor. The processor has two registers: the Accumulator (ACC) and an Index Register (IX).

instruction		explanation
opcode	operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
STO	<address>	Store the contents of ACC at the given address
ADD	#n / Bn / &n	Add the number n to the ACC
ADD	<address>	Add the contents of the given address to the ACC
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
END		Return control to the operating system
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

The current contents of memory are:

address	data
100	200
101	205
102	100
103	50
...	
200	101
201	100
202	20
203	13
204	220
205	30

- (a) Complete the table to identify the data that will be stored in the accumulator after each set of instructions has been executed.

accumulator current data	set of instructions	new accumulator data
B11001010	ADD B00000010 INC ACC	
#100	LDM #200 ADD 200	
#103	LDD 103 DEC ACC DEC ACC	

[3]

(b) The instruction set also contains the following instructions:

instruction		explanation
opcode	operand	
AND	#n / Bn / &n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n / Bn / &n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n / Bn / &n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left hand end
CSL	#n	Bits in ACC are shifted n places to the left using a cyclic shift
CSR	#n	Bits in ACC are shifted n places to the right using a cyclic shift
ASL	#n	Bits in ACC are shifted n places to the left using an arithmetic shift
ASR	#n	Bits in ACC are shifted n places to the right using an arithmetic shift
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

- (i) Write the bit manipulation instruction that will set the ACC to $\&0$

[2]

- (ii) The ACC currently contains the following binary value.

0	0	1	1	0	1	0	1
---	---	---	---	---	---	---	---

Write the result after the instruction `LSR #3` is run.

--	--	--	--	--	--	--	--

[1]

- (c) Immediate, relative and direct are three modes of addressing used in assembly language programming.

Identify and describe **one** additional mode of addressing used in assembly language programming.

Include an example of an instruction that uses your selected mode of addressing in your answer.

mode of addressing

description with example

[3]

2. M/J 26/P13/Q7/a,b,c

7 A program is written in assembly language.

(a) Complete the table by writing the name of each mode of addressing.

mode of addressing	description
	The operand is an address where the data is stored.
	The operand is an address that stores the address where the data is stored.
	The operand is an address. The address where the data is stored is calculated by adding the operand to the content of the index register.
	The operand is the data to be used.

[4]

(b) Explain how bit manipulation can be used to monitor or control a device.

[3]

(c) Identify **one** similarity and **one** difference between a right logical shift and a right arithmetic shift.

similarity

difference

[2]

3. O/N 25/P11/Q4/a,b,c,d

4 The table shows part of the instruction set for a processor. The processor has one register, the Accumulator (ACC).

Instruction		Explanation
Opcode	Operand	
AND	#n / Bn / &n	Bitwise AND operation of the contents of the ACC with the operand
AND	<address>	Bitwise AND operation of the contents of the ACC with the contents of <address>
XOR	#n / Bn / &n	Bitwise XOR operation of the contents of the ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of the ACC with the contents of <address>
OR	#n / Bn / &n	Bitwise OR operation of the contents of the ACC with the operand
OR	<address>	Bitwise OR operation of the contents of the ACC with the contents of <address>
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right-hand end.
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left-hand end.
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #127 B denotes a binary number, e.g. B10010001 & denotes a hexadecimal number, e.g. &4A		

(a) The ACC currently contains the following positive binary integer:

0	0	0	1	1	1	1	0
---	---	---	---	---	---	---	---

Write a bit manipulation instruction that uses a binary shift to change the contents of the ACC to:

0	1	1	1	1	0	0	0
---	---	---	---	---	---	---	---

Instruction

[1]

(b) The ACC currently contains the following positive binary integer:

1	1	1	0	0	0	1	1
---	---	---	---	---	---	---	---

Write the contents of the ACC after the instruction `XOR &12` is carried out.

--	--	--	--	--	--	--	--

[1]

(c) The ACC currently contains the following positive binary integer:

1	1	1	0	0	0	1	1
---	---	---	---	---	---	---	---

Write the contents of the ACC after the instruction `AND #63` is carried out.

--	--	--	--	--	--	--	--

[1]

(d) The ACC currently contains the following positive binary integer:

1	1	1	0	0	0	1	1
---	---	---	---	---	---	---	---

The current contents of memory are:

Address	Data
98	00100100
99	00110001
100	00110011
101	10100011
102	10101100

Write the contents of the ACC after the instruction `OR 100` is carried out.

--	--	--	--	--	--	--	--

[1]

4. O/N 25/P11/Q6/a,b

- 6 (a) The processor uses several registers, including the Accumulator (ACC) and the Current Instruction Register (CIR).

Complete the table by describing the role of each register.

Register	Role
ACC	
CIR	

[2]

- (b) Increasing the number of cores in a processor can affect the performance of a computer.

Describe the drawbacks of increasing the number of cores in a processor.

[2]

5. O/N 25/P12/Q3/a,b,c

- 3 (a) (i) State what is meant by relative addressing.

[1]

- (ii) Registers such as the Accumulator (ACC) and the Index Register (IX) are used in the CPU.

Identify **two** special purpose registers used in the CPU. Do **not** include the ACC or IX in your answers.

1

2

[2]

- (b) The following table shows part of the instruction set for a processor. The processor has two registers: the ACC and an IX.

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #127		

The current contents of the main memory and the index register are shown.

Address	Instruction
98	8
99	16
100	3
101	98
102	32
IX	2

Write the contents of the ACC after each instruction is executed.

Instruction	Value in ACC
LDM #98	
LDI 101	
LDX 100	

[3]

- (c) A student buys a new computer. The table shows the specifications of the old computer and the new computer.

Old computer	New computer
1.8 GHz dual core processor	2.3 GHz dual core processor
16 MB cache	32 MB cache

Explain why increasing the clock speed **and** increasing the cache memory will improve the performance of the computer.

Clock speed

Cache memory

[4]

6. O/N 25/P13/Q2/c

- (c) A three-place arithmetic shift to the right is performed on the following two's complement negative integer.

Show the result of this arithmetic shift.

10010011

[1]

7. O/N 25/P13/Q6/a,b,c

- 6 The following table shows part of the instruction set for a processor. The processor has two registers: the Accumulator (ACC) and the Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to the ACC
AND	#n / Bn / &n	Bitwise AND operation of the contents of the ACC with the operand
AND	<address>	Bitwise AND operation of the contents of the ACC with the contents of <address>
XOR	#n / Bn / &n	Bitwise XOR operation of the contents of the ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of the ACC with the contents of <address>
OR	#n / Bn / &n	Bitwise OR operation of the contents of the ACC with the operand
OR	<address>	Bitwise OR operation of the contents of the ACC with the contents of <address>
CMP	#n	Compare the contents of the ACC with number n
CMP	<address>	Compare the contents of the ACC with the contents of <address>
LSL	#n	Bits in the ACC are shifted logically n places to the left. Zeros are introduced on the right-hand end
LSR	#n	Bits in the ACC are shifted logically n places to the right. Zeros are introduced on the left-hand end
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #127 B denotes a binary number, e.g. B10010001 & denotes a hexadecimal number, e.g. &4A		

The current contents of main memory are shown:

Address	Data
100	0000 0011
101	1010 1110
102	1100 1100
103	1111 1111
104	1100 1100

- (a) Complete the table by writing the contents of the ACC after the execution of each instruction.

Current contents of the ACC	Instruction	Contents of the ACC after the execution of the instruction
0000 1111	AND 101	
0000 0000	LDM #100	
0000 0001	XOR &F1	
0001 0001	CMP 101	

[4]

- (b) The Von Neumann model for a computer system uses registers.

Describe the role of the Memory Address Register (MAR) and Memory Data Register (MDR) in the fetch-execute (F-E) cycle.

[4]

- (c) Assembly language instructions are grouped.

Complete each statement by writing the name of the appropriate instruction group.

Loading data into the accumulator is an example of an instruction in the

..... group. Incrementing the index register is an example of an instruction in the group. Branching to another address is an example of an instruction in the group.

[3]

8. M/J 25/P11/Q6/a(ii,iii)

6 A programmer is buying a new computer.

(a) The programmer is considering the following two computers:

Computer 1	Quad-core 2.2 GHz processor 16-bit architecture 1 GB Random Access Memory (RAM) 500 GB magnetic hard disk
Computer 2	Dual-core 3.8 GHz processor 32-bit architecture 2 GB RAM 500 GB solid state drive (SSD)

(ii) The two computers have different amounts of RAM.

Explain how different amounts of RAM affect the performance of a computer.

[3]

(iii) The two computers have different bus widths.

Explain how different bus widths affect the performance of a computer.

[2]

9. M/J 25/P11/Q8/a,b

8 (a) The following table shows part of the instruction set for a processor. The processor has two registers: the Accumulator (ACC) and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
STO	<address>	Store the contents of ACC at the given address
ADD	#n/Bn/&n	Add the number n to the ACC
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
JMP	<address>	Jump to the given address
CMP	<address>	Compare the contents of ACC with the contents of <address>
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
END		Return control to the operating system
ACC denotes Accumulator <address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

The current contents of memory are:

address Instruction

80	10
81	8
82	80
83	81
...	
200	LDD 81
201	INC ACC
202	STO 83
203	LDI 82
204	CMP 83
205	JPE 209
206	LDD 83
207	ADD #10
208	JMP 210
209	DEC ACC
210	STO 81
211	END

Trace the program currently in memory using the following trace table.

Instruction address	ACC	Memory address			
		80	81	82	83
		10	8	80	81

[4]

- (b) The table shows part of the instruction set for a processor. The processor has one register: the Accumulator (ACC).

Instruction		Explanation
Opcode	Operand	
AND	#n/Bn/&n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n/Bn/&n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n/Bn/&n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right-hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left-hand end
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

- (i) Write the bit manipulation instruction that can be used to set the least significant bit to 1 in an 8-bit register. All other bits must remain unchanged.

The instruction needs to work on a register that contains any 8-bit binary number.

[1]

- (ii) The ACC currently contains the following binary value.

0	1	0	1	0	1	0	1
---	---	---	---	---	---	---	---

Write the result after the instruction `XOR &FE` is run.

--	--	--	--	--	--	--	--

[1]

- (iii) The ACC currently contains the following binary value.

0	1	1	0	1	0	1	1
---	---	---	---	---	---	---	---

Write the result after the instruction `LSR #5` is run.

--	--	--	--	--	--	--	--

[1]

10. M/J 25/P12/Q1/a,b,c

- 1 (a) The table has six statements about the Von Neumann model for a computer system. Three of the statements are incorrect.

Statement number	Statement
1	The Program Counter (PC) stores the next instruction to be fetched from memory.
2	The Arithmetic and Logic Unit (ALU) performs mathematical and logical operations.
3	The Control Unit (CU) sends signals to other components on the data bus.
4	The Memory Data Register (MDR) transfers data to the memory address stored in the Memory Address Register (MAR).
5	The MAR stores an address from memory.
6	The Accumulator (ACC) stores the result of calculations.

Complete the table by writing the **three** incorrect statement numbers and the corrected statements.

Incorrect statement number	Corrected statement

[3]

(b) Registers that are used in the Fetch-Execute (F-E) cycle include the PC, MAR, MDR and the ACC.

Identify **one** other register and describe its role in the Fetch-Execute (F-E) cycle.

Register

Role

[2]

(c) Explain how an interrupt from an input device will be detected and handled in the F-E cycle.

[4]

11. M/J 25/P13/Q2/a,b,c

2 A computer has a processor.

(a) The processor has a Control Unit (CU) and system clock.

(i) Explain how the CU and the system clock work together.

[2]

(ii) The processor runs the Fetch-Execute (F-E) cycle.

Write the stages of the F-E cycle using register transfer notation.

[4]

(b) The computer has cache memory.

Describe **one** benefit of the computer using cache memory.

[2]

(c) The computer connects to a monitor using a High Definition Multimedia Interface (HDMI) cable that connects into an HDMI port.

Explain how HDMI provides connection to peripheral devices.

[2]

12. M/J 25/P13/Q5/a,b

5 The following table shows part of the instruction set for a processor. The processor has two registers: the Accumulator (ACC) and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
STO	<address>	Store the contents of ACC at the given address
ADD	#n/Bn/&n	Add the number n to the ACC
ADD	<address>	Add the contents of the given address to the ACC
SUB	<address>	Subtract the contents of the given address from the ACC
SUB	#n/Bn/&n	Subtract the number n from the ACC
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
JMP	<address>	Jump to the given address
CMP	<address>	Compare the contents of ACC with the contents of <address>
CMP	#n	Compare the contents of ACC with number n
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
END		Return control to the operating system

ACC denotes Accumulator
 <address> can be an absolute or a symbolic address
 # denotes a denary number, e.g. #123
 B denotes a binary number, e.g. B01001010
 & denotes a hexadecimal number, e.g. &4A

(a) The current contents of memory are:

Address	Instruction
50	47
51	48
52	49
53	50
...	
500	LDM #50
501	DEC ACC
502	CMP 51
503	JPE 505
504	JMP 501
505	STO 50
506	SUB #10
507	STO 51
508	END

Mode of addressing	Description
.....	
.....	

[4]

- (b) The table shows part of the instruction set for a processor. The processor has one register: the Accumulator (ACC).

Instruction		Explanation
Opcode	Operand	
AND	#n/Bn/&n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n/Bn/&n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n/Bn/&n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

Write the bit manipulation instructions that can be used to set only the most significant bit to 1 in an 8-bit register.

The instructions need to work on a register that contains any 8-bit binary number.

[2]

13. M/J 25/P13/Q7/a

7 (a) A computer can perform logical and arithmetic shifts.

- (i) Show the result of a logical left shift of 2 places on the two's complement binary integer 11001010

[1]

- (ii) Show the result of an arithmetic right shift of 3 places on the two's complement binary integer 10011110

[1]

14. O/N 24/P11/Q3/a,b

3 A computer system has a dual-core Central Processing Unit (CPU).

- (a) State the purpose of the system clock and the Control Unit (CU) in a CPU.

System clock

CU

[2]

- (b) (i) The number of cores in the processor affects the performance of the computer system.

Identify **one other** feature of a processor that can affect the performance of a computer system **and** state why it affects the performance.

Feature

Reason

[2]

- (ii) A solid state (flash) memory drive is automatically recognised by the computer when it is plugged into a port in the computer.

Identify an appropriate type of port to connect the solid state memory drive to the computer.

Explain how this port provides an automatic connection.

Port

Explanation

[3]

15. O/N 24/P11/Q8/a,b

- 8** A computer designed using the Von Neumann model for a computer system contains general purpose registers and special purpose registers.

(a) Describe the purpose of the Status Register (SR).

[2]

(b) Identify **two** differences between general purpose registers and special purpose registers.

1

2

[2]

16. O/N 24/P12/Q3/a,b

3 A student has a computer.

(a) The computer is designed using the Von Neumann model for a computer system.

Complete the table by describing the purpose of each of the given registers.

Register	Purpose
Program Counter (PC)	
Memory Address Register (MAR)	
Memory Data Register (MDR)	
Index Register (IX)	

[4]

(b) The student needs to connect the computer to a monitor that has a screen resolution of 2560×1600 pixels. The monitor also has built-in speakers.

The computer has a Video Graphics Array (VGA) port and a High Definition Multimedia Interface (HDMI) port.

Explain the benefits of connecting the monitor to the computer using the HDMI port instead of the VGA port.

[4]

17. O/N 24/P12/Q8/a,b

8 The following table shows part of the instruction set for a processor. The processor has two registers, the Accumulator (ACC) and the Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
ADD	#n/Bn/&n	Add the number n to the ACC
ADD	<address>	Add the contents of the given address to the ACC
SUB	#n/Bn/&n	Subtract the number n from the ACC
SUB	<address>	Subtract the contents of the given address from the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

(a) The current contents of memory are shown:

Address	Data
19	25
20	23
21	2
22	4
23	15
24	50
25	22

The current contents of the ACC and IX are shown:

ACC	50
IX	20

Complete the table by writing the content of the ACC and the IX after each set of instructions has run.

	Instructions	ACC content	IX content
1	LDM #19 DEC ACC		
2	LDD 23 ADD 19		
3	LDI 25 INC ACC		
4	LDR #21 LDX 2		

[5]

(b) The instruction set also includes these bit manipulation instructions:

Instruction		Explanation
Opcode	Operand	
AND	#n/Bn/&n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n/Bn/&n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n/Bn/&n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right-hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left-hand end.
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

The current content of the ACC is shown:

ACC	1	0	0	1	1	0	1	0
------------	---	---	---	---	---	---	---	---

(i) The table has three sets of instructions. The binary number 10011010 is reloaded into the ACC before each set of instructions is run.

Complete the table by writing the content of the ACC after each set of instructions has run.

	Instructions	ACC content
1	LSL #2	
2	ADD #5 AND #30	
3	OR B11110010 INC ACC	

[3]

- (ii) Explain how bit manipulation can be used to test whether the binary number stored in the ACC represents an odd denary number.

Write the bit manipulation instruction that will be used.

Explanation

Instruction

[3]

18. O/N 24/P13/Q7/a,b

- 7 The following table shows part of the instruction set for a processor. The processor has two registers, the Accumulator (ACC) and the Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
ADD	#n/Bn/&n	Add the number n to the ACC
ADD	<address>	Add the contents of the given address to the ACC
SUB	#n/Bn/&n	Subtract the number n from the ACC
SUB	<address>	Subtract the contents of the given address from the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

(a) The current contents of memory are shown:

Address	Data
50	54
51	55
52	50
53	52
54	100
55	25
56	50

The current contents of the ACC and IX are shown:

ACC	50
IX	45

Complete the table by writing the content of the ACC after each program has run.

	Instructions	ACC content
1	LDD 50 ADD #4 ADD 54	
2	LDI 53 DEC ACC ADD 56	
3	LDM #55 SUB #5	

(b) The instruction set also includes these bit manipulation instructions:

Instruction		Explanation
Opcode	Operand	
AND	#n/Bn/&n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n/Bn/&n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n/Bn/&n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

Explain how bit manipulation can be used to clear the data in an 8-bit register.

Write the bit manipulation instruction that will be used.

Explanation

Instruction

[3]

19. O/N 24/P13/Q8/c

(c) Describe the difference between a right logical binary shift and a right arithmetic binary shift.

[2]

20. M/J 24/P11/Q4/a,b

- 4 The following table shows part of the instruction set for a processor. The processor has two registers: the Accumulator (ACC) and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the Index Register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
ADD	#n/Bn/&n	Add the number n to the ACC
ADD	<address>	Add the contents of the given address to the ACC
SUB	#n/Bn/&n	Subtract the number n from the ACC
SUB	<address>	Subtract the contents of the given address from the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

(a) The current contents of memory are shown:

Address	Data
19	24
20	2
21	1
22	3
23	5
24	4
25	22

The current contents of the ACC and IX are shown:

ACC	12
IX	1

Complete the table by writing the content of the ACC after each program has run.

Program number	Code	ACC content
1	LDD 20 ADD #2	
2	LDX 22	
3	LDI 25 INC ACC SUB 22	
4	LDD 19 LDM #5 LDM #25	

[4]

(b) The processor includes these bit manipulation instructions:

Instruction		Explanation
Opcode	Operand	
AND	#n/Bn/&n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n/Bn/&n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n/Bn/&n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

The current contents of memory are shown:

Address	Data
30	01110101
31	11111111
32	00000000
33	11001100
34	10101010

The current content of the ACC is shown:

1	0	0	1	1	0	1	0
---	---	---	---	---	---	---	---

Complete the table by writing the content of the ACC after each program has run.

The binary number 10011010 is reloaded into the ACC before each program is run.

Program number	Code	ACC content
1	AND 31	
2	XOR B01001111	
3	OR #30	

21. M/J 24/P12/Q5/a,b

- 5 The following table shows part of the instruction set for a processor. The processor has two registers: the Accumulator (ACC) and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
ADD	#n/Bn/&n	Add the number n to the ACC
ADD	<address>	Add the contents of the given address to the ACC
SUB	#n/Bn/&n	Subtract the number n from the ACC
SUB	<address>	Subtract the contents of the given address from the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

- (a) The current contents of memory are shown:

Address	Data
10	1
11	3
12	5
13	11
14	10
15	16
16	12

The current contents of the ACC and IX are shown:

ACC	10
IX	0

Complete the table by writing the content of the ACC after each program has run.

Program number	Code	ACC content
1	LDI 15 SUB #1	
2	LDD 14 ADD 11	
3	LDM #11 ADD #3 SUB 16	
4	LDR #2 LDX 14 ADD #2	

(b) The processor includes these bit manipulation instructions:

Instruction		Explanation
Opcode	Operand	
AND	#n/Bn/&n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n/Bn/&n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n/Bn/&n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

The current contents of memory are shown:

Address	Data
25	11000110
26	11100001
27	10000001
28	11001101
29	00001111

The current content of the ACC is shown:

0	1	0	0	0	1	1	0
---	---	---	---	---	---	---	---

Complete the table by writing the content of the ACC after each program has run.

The binary number 01000110 is reloaded into the ACC before each program is run.

Program number	Code	ACC content
1	XOR 29	
2	AND #29	
3	OR B11111111	

22. M/J 24/P13/Q3/a,b

- 3 The following table shows part of the instruction set for a processor. The processor has two registers: the Accumulator (ACC) and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
ADD	#n/Bn/&n	Add the number n to the ACC
ADD	<address>	Add the contents of the given address to the ACC
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
SUB	#n/Bn/&n	Subtract the number n from the ACC
SUB	<address>	Subtract the contents of the given address from the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		



- (a) The current contents of memory are shown:

Address	Data
48	51
49	6
50	48
51	50
52	49
53	50
54	6

The current contents of the ACC and IX are shown:

ACC	2
IX	50

Complete the table by writing the content of the ACC after each program has run.

Program number	Code	ACC content
1	LDM #50 INC ACC SUB #1	
2	LDI 51 ADD 52	
3	LDR #2 LDX 50 DEC ACC	
4	LDD 52 SUB 54 INC ACC	

[4]



(b) The processor includes these bit manipulation instructions:

Instruction		Explanation
Opcode	Operand	
AND	#n/Bn/&n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n/Bn/&n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n/Bn/&n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right-hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left-hand end
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

The current contents of memory are shown:

Address	Data
100	00001101
101	10111110
102	11110011
103	00110111
104	00000000

The current content of the ACC is shown:

1	1	1	1	1	1	1	1
---	---	---	---	---	---	---	---



Complete the table by writing the content of the ACC after each instruction has run.

The binary number 11111111 is reloaded into the ACC before each instruction is run.

Instruction number	Instruction	ACC content
1	LSL #2	
2	XOR 100	
3	AND 103	

[3]



23. O/N 23/P11/Q5/a,b,c

- 5 The Central Processing Unit (CPU) of the basic Von Neumann model for a computer system contains several special purpose registers.

- (a) The Memory Data Register (MDR), Index Register (IX) and the Accumulator (ACC) are examples of special purpose registers.

Identify **two other** special purpose registers **and** state their role in the CPU.

Special purpose register 1

Role

Special purpose register 2

Role

[4]

- (b) Describe what is meant by the **Immediate Access Store (IAS)** in a computer system.

[2]

- (c) A computer has a single 2.1 GHz CPU.

- (i) Describe how increasing the clock speed to 4 GHz can increase the performance of the computer.

[1]

- (ii) A second computer has a CPU with two 2.1 GHz cores.

Explain why the second computer does not always run twice as fast as the computer with one 2.1 GHz CPU.

[5]

24. O/N 23/P11/Q8/a,b,c

- 8 (a) Identify the purpose of the first pass of a two-pass assembler.

[1]

- (b) The following table shows part of the instruction set for a processor. The processor has two registers, the Accumulator (ACC) and the Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDR	#n	Immediate addressing. Load the number n to IX
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
CMP	#n	Compare the contents of ACC with number n
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
OUT		Output to the screen the character whose ASCII value is stored in ACC
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123		

- (i) Give **one** example of an instruction that belongs to **each** of the following instruction groups.

Only use the instructions given in the table. Each instruction must have a suitable operand.

Data movement

Arithmetic operation

Conditional instruction

[3]

- (ii) The instruction `LDR #2` uses immediate addressing.

Give **one** similarity and **one** difference between direct addressing and indexed addressing.

Similarity

Difference

[2]

- (iii) Identify **one other** mode of addressing.

[1]

(c) The following table shows another part of the instruction set for the same processor.

Instruction		Explanation
Opcode	Operand	
AND	Bn	Bitwise AND operation of the contents of ACC with the operand
XOR	Bn	Bitwise XOR operation of the contents of ACC with the operand
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left hand end
# denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001101		

(i) The current contents of the ACC are:

0	1	0	0	1	1	1	1
---	---	---	---	---	---	---	---

Show the contents of the ACC after the execution of the following instruction.

AND B10100101

--	--	--	--	--	--	--	--

[1]

(ii) The current contents of the ACC are:

0	0	0	1	0	1	1	1
---	---	---	---	---	---	---	---

Show the contents of the ACC after the execution of the following instruction.

LSR #3

--	--	--	--	--	--	--	--

[1]

(iii) The current contents of the ACC are:

1	1	1	1	0	1	1	1
---	---	---	---	---	---	---	---

Show the contents of the ACC after the execution of the following instruction.

XOR B00100101

--	--	--	--	--	--	--	--

[1]

25. O/N 23/P12/Q8/c

8 A computer has an Operating System (OS).

(c) An optical disc reader/writer is connected to the computer.

(i) Give the name of **one** port that can provide a connection for the optical disc reader/writer.

[1]

(ii) Describe the roles of the address bus, the data bus **and** buffers in the process of writing data to the optical disc reader/writer.

[3]

26. O/N 23/P12/Q9/a,b

9 The following table shows part of the instruction set for a processor. The processor has two registers, the Accumulator (ACC) and the Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
STO	<address>	Store the contents of ACC at the given address
ADD	#n	Add the denary number n to the ACC
JMP	<address>	Jump to the given address
INC	<register>	Add 1 to the contents of the register (ACC or IX)
CMP	<address>	Compare the contents of ACC with the contents of <address>
CMI	<address>	Indirect addressing. The address to be used is at the given address. Compare the contents of ACC with the contents of this second address
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
IN		Key in a character and store its ASCII value in ACC
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123		

(a) The instructions in the processor's instruction set can be grouped according to their function.

Identify the instruction group for each of the following opcodes.

IN

ADD

JPE

CMI

[4]

- (b)** The current contents of main memory and selected values from the ASCII character set are given on page 15.

Trace the program currently in memory using the trace table when the input is '1'.

Address	Instruction	Instruction address	ACC	IX	Memory address				Output
					100	101	110	111	
10	LDR #0				0	0	51	65	
11	IN								
12	STO 101								
13	LDX 110								
14	CMP 100								
15	JPE 21								
16	LDD 101								
17	ADD #16								
18	INC IX								
19	STO 100								
20	JMP 13								
21	OUT								
22	END								
...	↗								
100	0								
101	0								
...	↗								
110	51								
111	65								

ASCII value	Character
49	1
50	2
51	3
52	4
...	↗
65	A
66	B
67	C
68	D

27. O/N 23/P13/Q7/d

- 7** A laptop computer has Static RAM (SRAM).

A virtual reality headset and a laser printer are connected to the laptop.

- (d)** Identify **one** port that could be used to connect the virtual reality headset to the laptop.

Justify your choice.

Port

Justification

[3]

28. O/N 23/P13/Q9/a,b,c,d

- 9** A computer system is designed using the basic Von Neumann model.

Registers and buses are components in the Von Neumann model.

- (a) (i)** Identify **three other** components in the Von Neumann model of a computer system.

Do not include registers or buses in your answers.

1

2

3

[3]

- (ii)** Identify **two** differences between special purpose registers and general purpose registers.

1

2

[2]

- (b)** The following incomplete table contains steps of the Fetch-Execute (F-E) cycle and their descriptions.

Complete the table by writing the missing steps using register transfer notation **and** the missing descriptions.

Step	Description
.....	The address in PC is incremented.
$\text{MDR} \leftarrow [[\text{MAR}]]$	
$\text{MAR} \leftarrow [\text{PC}]$	
.....	The contents of MDR are copied into CIR.

[4]

- (c) Interrupts can be caused by software programs or hardware devices.

State **one** cause of a software interrupt.

[1]

- (d) The following statements describe the stages that the CPU performs when an interrupt is detected.

There are **three** missing statements.

Write the letter of the missing statements from the table in the correct place to complete the description.

- 1 At the end of each Fetch-Execute (F-E) cycle, the processor checks if an interrupt flag is set.
- 2
- 3 If the interrupt priority is high enough, the processor saves the current contents of the registers.
- 4
- 5 When servicing of the interrupt is complete, the processor restores the registers.

6

Letter	Stage
A	The address of the Interrupt Service (ISR) handling routine is loaded into the Program Counter (PC).
B	Lower priority interrupts are re-enabled.
C	The device causing the interrupt transfers data to the CPU.
D	The processor identifies the source of the interrupt and checks the priority of the interrupt.
E	The ISR is incremented.

[3]

29. M/J 23/P11/Q3/d(vi)

3 A computer has an Operating System (OS).

(d) The computer stores data in binary form.

(vi) Show the result of a 3-place right logical shift on the binary number:

11001100

[1]

30. M/J 23/P11/Q6

6 An interrupt is generated when a key is pressed on a computer keyboard.

Explain how the computer handles this interrupt.

[5]

31. M/J 23/P12/Q3/a,b

3 A program is written in assembly language.

(a) The program is converted into machine code by a two-pass assembler.

Draw **one or more** lines to identify the pass or passes in which each action takes place.

Action	Pass
generates object code	
reads the source code one line at a time	first
removes white space	second
adds labels to the symbol table	

[3]

(b) Assembly language statements can use different modes of addressing.

Complete the following description of addressing modes.

..... addressing is when the operand holds the memory address of the data.

..... addressing is when the operand holds a memory address that stores the memory address of the data.

..... addressing is when the operand is the data.

[3]

32. M/J 23/P12/Q4/d

4 Data in a computer is stored in binary form.

(d) Show the result of a 2-place left logical shift on the binary number:

01001111

[1]

33. M/J 23/P12/Q5/a,b,c

5 A student has purchased a new laptop.

(a) The laptop is designed using the Von Neumann model for a computer system.

Identify **two** types of signal that a control bus can transfer.

1

2

[2]

(b) Describe **two** ways the hardware of a laptop can be upgraded to improve the performance **and** explain how each upgrade improves the performance.

1

2

[4]

(c) Peripherals are connected to the laptop using ports.

(i) A printer is connected to a Universal Serial Bus (USB) port.

Describe how data is transmitted through a USB port.

[1]

(ii) A monitor is connected to the laptop using a different type of port.

Identify **one** other type of port that can be used to connect the monitor.

[1]

34. M/J 23/P13/Q7/b,c

7 A computer stores data in binary form.

(b) The computer has a Control Unit (CU), system clock and control bus.

Explain how the CU, system clock and control bus operate to transfer data between the components of the computer system.

[4]

(c) Complete the table by writing the register transfer notation for each stage of the Fetch-Execute (F-E) cycle given in the table.

Stage description	Register transfer notation
The Program Counter (PC) is incremented	
The data in the address stored in the Memory Address Register (MAR) is copied to the Memory Data Register (MDR)	

[2]

35. O/N 22/P11/Q5/a,b,c,d

- 5 (a) State what is meant by the **stored program concept** in the Von Neumann model of a computer system.

[1]

- (b) A Central Processing Unit (CPU) contains several special purpose registers and other components.

- (i) State the role of the following registers.

Program Counter (PC)

Index Register (IX)

Status Register (SR)

[3]

- (ii) Tick (✓) **one** box in each row to identify the system bus used by each CPU component.

CPU component	Data bus	Address bus	Control bus
System clock			
Memory Address Register (MAR)			

[1]

- (iii) Describe the purpose of the Control Unit (CU) in a CPU.

[2]

- (c) Describe the purpose of an interrupt in a computer system.

[2]

- (d) Identify **two** causes of a software interrupt.

1

2

[2]

36. O/N 22/P11/Q6/a,b,c

- 6 The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
ADD	#n	Add the denary number n to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
CMP	#n	Compare the contents of ACC with number n
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001101		

- (a) The current contents of main memory and selected values from the ASCII character set are given.

(ii) The following instructions are repeated for your reference.

Instruction		Explanation
Opcode	Operand	
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
STO	<address>	Store contents of ACC at the given address

State the purpose of this part of an assembly language program.

```
LDD 100
STO 165
LDD 101
STO 100
LDD 165
STO 101
```

[1]

(b) The following table shows another part of the instruction set for the processor.

Instruction		Explanation
Opcode	Operand	
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	Bn	Bitwise AND operation of the contents of ACC with the binary number n
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	Bn	Bitwise XOR operation of the contents of ACC with the binary number n
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	Bn	Bitwise OR operation of the contents of ACC with the binary number n
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left-hand end
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001101		

(i) The current contents of the ACC are:

1	0	0	1	0	0	1	1
---	---	---	---	---	---	---	---

Show the result after the execution of the following instruction.

```
XOR B00011111
```

--	--	--	--	--	--	--	--

[1]

(ii) The current contents of the ACC are:

1	0	0	1	0	0	1	1
---	---	---	---	---	---	---	---

Show the result after the execution of the following instruction.

AND B11110000

--	--	--	--	--	--	--	--

[1]

(iii) The current contents of the ACC are:

1	0	0	1	0	0	1	1
---	---	---	---	---	---	---	---

Show the result after the execution of the following instruction.

OR B11001100

--	--	--	--	--	--	--	--

[1]

(iv) The current contents of the ACC are:

1	0	0	1	0	0	1	1
---	---	---	---	---	---	---	---

Show the result after the execution of the following instruction.

LSR #2

--	--	--	--	--	--	--	--

[1]

(c) Tick (✓) **one or more** boxes in each row to indicate whether the task is performed in the first pass or the second pass of a two-pass assembler.

Task	First pass	Second pass
Remove comments.		
Read the assembly language program one line at a time.		
Generate the object code.		
Check the opcode is in the instruction set.		

[2]

37. O/N 22/P11/Q7/d,e

- (d) Explain the reasons why increasing the amount of cache memory can improve the performance of a CPU.

[2]

- (e) State the name of a peripheral device port that provides a physical connection in the computer for each of these peripherals.

3D printer

Monitor

[2]

38. O/N 22/P12/Q7/a,b,c

- 7 The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
STO	<address>	Store the contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
ADD	#n	Add the denary number n to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
JMP	<address>	Jump to the given address
CMP	<address>	Compare the contents of ACC with the contents of <address>
CMI	<address>	Indirect addressing. The address to be used is at the given address. Compare the contents of ACC with the contents of this second address
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
END		Return control to the operating system
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001101		

- (a) Trace the program currently in memory using the trace table, stopping when line 90 is executed for a second time.

[5]

(b) The following table shows another part of the instruction set for the processor.

Instruction		Explanation
Opcode	Operand	
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	Bn	Bitwise XOR operation of the contents of ACC with the binary number n
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right-hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left-hand end
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001101		

The contents of memory addresses 50 and 51 are shown:

Memory address	Data value
50	01001101
51	10001111

(i) The current contents of the ACC are:

0	1	0	1	0	0	1	1
---	---	---	---	---	---	---	---

Show the contents of the ACC after the execution of the following instruction.

XOR B00011111

--	--	--	--	--	--	--	--

[1]

(ii) The current contents of the ACC are:

0	1	0	1	0	0	1	1
---	---	---	---	---	---	---	---

Show the contents of the ACC after the execution of the following instruction.

AND 50

--	--	--	--	--	--	--	--

[1]

(iii) The current contents of the ACC are:

0	1	0	1	0	0	1	1
---	---	---	---	---	---	---	---

Show the contents of the ACC after the execution of the following instruction.

LSL #3

--	--	--	--	--	--	--	--

[1]

(iv) The current contents of the ACC are:

0	1	0	1	0	0	1	1
---	---	---	---	---	---	---	---

Show the contents of the ACC after the execution of the following instruction.

OR 51

--	--	--	--	--	--	--	--

[1]

(c) Write the register transfer notation for each of the stages in the fetch-execute cycle described in the table.

Description	Register transfer notation
Copy the address of the next instruction into the Memory Address Register.	
Increment the Program Counter.	
Copy the contents of the Memory Data Register into the Current Instruction Register.	

[3]

39. O/N 22/P13/Q4/a,b,c

- 4 (a)** A Central Processing Unit (CPU) contains several special purpose registers and other components.

- (i)** State the roles of the following registers.

Memory Address Register (MAR)

Memory Data Register (MDR)

[2]

- (ii)** State when interrupts are detected during the Fetch-Execute (F-E) cycle.

[1]

- (b)** A computer system contains a system clock.

Describe the purpose of the system clock.

[2]

- (c)** Upgrading secondary storage to solid state typically improves the performance of computer systems.

Identify **one other** upgrade to the hardware **and** explain why it improves the performance of a computer system.

Upgrade

Explanation

[2]

40. O/N 22/P13/Q6/a,b,c

- 6** The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
MOV	<register>	Moves the contents of the accumulator to the given register (IX)
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
ADD	#n	Add the denary number n to the ACC
SUB	#n	Subtract the denary number n from the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
JMP	<address>	Jump to the given address
CMP	#n	Compare the contents of ACC with number n
CMI	<address>	Indirect addressing. The address to be used is at the given address. Compare the contents of ACC with the contents of this second address
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001101		

- (a) The current contents of main memory and selected values from the ASCII character set are given.

(b) The following table shows another part of the instruction set for the processor.

Instruction		Explanation
Opcode	Operand	
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	Bn	Bitwise AND operation of the contents of ACC with the binary number n
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	Bn	Bitwise OR operation of the contents of ACC with the binary number n
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right-hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left-hand end
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001101		

(i) The current contents of the ACC are:

0	0	1	1	0	1	1	0
---	---	---	---	---	---	---	---

Show the contents of the ACC after the execution of the following instruction.

AND B01001100

--	--	--	--	--	--	--	--

[1]

(ii) The current contents of the ACC are:

1	0	0	1	0	1	0	1
---	---	---	---	---	---	---	---

Show the contents of the ACC after the execution of the following instruction.

OR B01001111

--	--	--	--	--	--	--	--

[1]

(iii) The current contents of the ACC are:

1	0	0	1	1	1	0	1
---	---	---	---	---	---	---	---

Show the contents of the ACC after the execution of the following instruction.

LSR #2

--	--	--	--	--	--	--	--

[1]

(c) One instruction group is data movement.

Give the name of **one other** instruction group.

[1]

41. O/N 21/P11/Q6/a,b

6 (a) There are **two** errors in the following register transfer notation for the fetch-execute cycle.

- 1 MAR $\leftarrow$ [PC]
- 2 PC $\leftarrow$ [PC] - 1
- 3 MDR $\leftarrow$ [MAR]
- 4 CIR $\leftarrow$ [MDR]

Complete the following table by:

- identifying the line number of each error
- describing the error
- writing the correct statement.

Line number	Description of the error	Correct statement

[4]

(b) The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
STO	<address>	Store the contents of ACC at the given address
INC	<register>	Add 1 to the contents of the register (ACC or IX)
CMP	<address>	Compare the contents of ACC with the contents of <address>
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
JMP	<address>	Jump to the given address
IN		Key in a character and store its ASCII value in ACC
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left hand end
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001101		

The current contents of main memory are shown:

Address	Data
100	00001111
101	11110000
102	01010101
103	11111111
104	00000000

Each row of the following table shows the current contents of ACC in binary and the instruction that will be performed on those contents.

Complete the table by writing the new contents of the ACC after the execution of each instruction.

Current contents of the ACC	Instruction	New contents of the ACC
11111111	OR 101	
00000000	XOR #15	
10101010	LSR #2	
01010101	AND 104	

[4]

42. O/N 21/P12/Q7/a

7 A computer has system software.

(a) The Operating System handles interrupts.

Tick (✓) **one** box in each row to identify whether each event is an example of a hardware interrupt or a software interrupt.

Event	Hardware interrupt	Software interrupt
Buffer full		
Printer is out of paper		
User has pressed a key on the keyboard		
Division by zero		
Power failure		
Stack overflow		

[3]

43. O/N 21/P12/Q8/a,b

8 The Von Neumann model for a computer system uses registers.

(a) Describe the role of the following special purpose registers in the fetch-execute (F-E) cycle.

(i) Memory Address Register (MAR)

Memory Data Register (MDR)

[4]

(ii) Another special purpose register is the Index Register.

Identify **one other** special purpose register used in the Von Neumann model for a computer system.

[1]

- (b) The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
STO	<address>	Store the contents of ACC at the given address
INC	<register>	Add 1 to the contents of the register (ACC or IX)
CMP	<address>	Compare the contents of ACC with the contents of <address>
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
JMP	<address>	Jump to the given address
IN		Key in a character and store its ASCII value in ACC
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left hand end
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123		

The current contents of main memory are shown:

Address	Data
100	01010101
101	11110000
102	00001111
103	00000000
104	11111111

- (i) In the following table, each row shows the current contents of the ACC in binary and the instruction that will be performed on those contents.

Complete the table by writing the new contents of the ACC after the execution of each instruction.

Current contents of the ACC	Instruction	New contents of the ACC
01010101	XOR 101	
11110000	AND 104	
00001111	LSL #4	
11111111	OR 102	

[4]

- (ii) The following table contains five assembly language instruction groups.

Write an appropriate assembly language instruction for each instruction group, using the given instruction set. The first one has been completed for you.

Instruction Group	Instruction
Data movement	LDM #2
Input and output of data	
Arithmetic operations	
Unconditional and conditional instructions	
Compare instructions	

[4]

- (iii) The opcode **LDM** uses immediate addressing. The opcode **LDD** uses direct addressing.

Identify **and** describe **one additional** mode of addressing.

Mode of addressing

Description

[2]

44. O/N 21/P13/Q6/a,b

- 6 (a) There are **two** errors in the following register transfer notation for the fetch-execute cycle.


```
1  MAR ← [PC]

2  PC ← [PC] - 1

3  MDR ← [MAR]

4  CIR ← [MDR]
```

- Complete the following table by:
- identifying the line number of each error
 - describing the error
 - writing the correct statement.

Line number	Description of the error	Correct statement

[4]

(b) The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
STO	<address>	Store the contents of ACC at the given address
INC	<register>	Add 1 to the contents of the register (ACC or IX)
CMP	<address>	Compare the contents of ACC with the contents of <address>
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
JMP	<address>	Jump to the given address
IN		Key in a character and store its ASCII value in ACC
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left hand end
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001101		

The current contents of main memory are shown:

Address	Data
100	00001111
101	11110000
102	01010101
103	11111111
104	00000000

Each row of the following table shows the current contents of ACC in binary and the instruction that will be performed on those contents.

Complete the table by writing the new contents of the ACC after the execution of each instruction.

Current contents of the ACC	Instruction	New contents of the ACC
11111111	OR 101	
00000000	XOR #15	
10101010	LSR #2	
01010101	AND 104	

[4]

45. M/J 21/P11/Q3/a,b,c

- 3** A processor has one general purpose register, the Accumulator (ACC), and several special purpose registers.

- (a)** Complete the following description of the role of the registers in the fetch-execute cycle by writing the missing registers.

The holds the address of the next instruction to be loaded. This address is sent to the

The holds the data fetched from this address.

This data is sent to the and the Control Unit decodes the instruction's opcode.

The is incremented.

[5]

- (b)** The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDI	<address>	Indirect addressing: The address to be used is at the given address. Load the contents of this second address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the Index Register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
MOV	<register>	Move the contents of the accumulator to the given register (IX)
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
CMP	<address>	Compare the contents of ACC with the contents of <address>
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
JMP	<address>	Jump to the given address
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left hand end
<address> can be an absolute address or a symbolic address # denotes a denary number, e.g. #123		

The current contents of the main memory and selected values from the ASCII character set are shown.

Address	Instruction
200	LDD 365
201	CMP 366
202	JPE 209
203	INC ACC
204	STO 365
205	MOV IX
206	LDX 365
207	OUT
208	JMP 200
209	END
...	
365	1
366	3
367	65
368	66
IX	0

ASCII code table (selected codes only)	
ASCII code	Character
65	A
66	B
67	C
68	D

Write the contents of the Accumulator after the processor has executed the following instruction:

LSL #2

--	--	--	--	--	--	--	--

[1]

(ii) The Accumulator currently contains the binary number:

0	0	1	1	0	1	0	1
---	---	---	---	---	---	---	---

Identify the mathematical operation that the following instruction will perform on the contents of the accumulator.

LSR #3

[1]

46. M/J 21/P12/Q4/a,b(ii,iii)

- 4 The table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
CMP	<address>	Compare the contents of ACC with the contents of <address>
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
JMP	<address>	Jump to the given address
IN		Key in a character and store its ASCII value in ACC
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
# denotes a denary number, e.g. #123		

The current contents of the main memory and selected values from the ASCII character set are:

Address Instruction

70	IN
71	CMP 100
72	JPE 80
73	CMP 101
74	JPE 76
75	JMP 80
76	LDD 102
77	INC ACC
78	STO 102
79	JMP 70
80	LDD 102
81	DEC ACC
82	STO 102
83	JMP 70
...	
100	68
101	65
102	100

ASCII code table (selected codes only)

ASCII code	Character
65	A
66	B
67	C
68	D

- (a) Complete the trace table for the program currently in main memory when the following characters are input:

A D

(b) Some bit manipulation instructions are shown in the table:

Instruction		Explanation
Opcode	Operand	
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
<address> can be an absolute address or a symbolic address # denotes a denary number, e.g. #123		

The contents of the memory address 300 are shown:

Bit Number	7	6	5	4	3	2	1	0
300	0	1	1	0	0	1	1	0

- (ii) An assembly language program needs to test if bit number 2 in memory address 300 is a 1.

Complete the assembly language instruction to perform this test.

#4

[1]

- (iii) An assembly language program needs to set bit numbers 4, 5, 6 and 7 to 0, but keep bits 0 to 3 with their existing values.

Write the assembly language instruction to perform this action.

[2]

47. M/J 21/P12/Q5/a,b

5 Seth uses a computer for work.

- (a) Complete the following descriptions of internal components of a computer by writing the missing terms.

The transmits the signals to coordinate events based on the electronic pulses of the

The carries data to the components, while the carries the address where data needs to be written to or read from.

The performs mathematical operations and logical comparisons.

[5]

- (b) Describe the ways in which the following factors can affect the performance of his laptop computer.

Number of cores

Clock speed

[4]

48. M/J 21/P13/Q3/a,b,c

3 A processor has one general purpose register, the Accumulator (ACC), and several special purpose registers.

- (a) Complete the following description of the role of the registers in the fetch-execute cycle by writing the missing registers.

The holds the address of the next instruction to be loaded. This address is sent to the

The holds the data fetched from this address.

This data is sent to the and the Control Unit decodes the instruction's opcode.

The is incremented.

[5]

- (b) The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
LDI	<address>	Indirect addressing: The address to be used is at the given address. Load the contents of this second address to ACC
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the Index Register. Copy the contents of this calculated address to ACC
LDR	#n	Immediate addressing. Load the number n to IX
MOV	<register>	Move the contents of the accumulator to the given register (IX)
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
CMP	<address>	Compare the contents of ACC with the contents of <address>
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
JMP	<address>	Jump to the given address
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left hand end
<address> can be an absolute address or a symbolic address # denotes a denary number, e.g. #123		

The current contents of the main memory and selected values from the ASCII character set are shown.

Address	Instruction
200	LDD 365
201	CMP 366
202	JPE 209
203	INC ACC
204	STO 365
205	MOV IX
206	LDX 365
207	OUT
208	JMP 200
209	END
...	
365	1
366	3
367	65
368	66
IX	0

ASCII code table (selected codes only)	
ASCII code	Character
65	A
66	B
67	C
68	D

Write the contents of the Accumulator after the processor has executed the following instruction:

LSL #2

--	--	--	--	--	--	--	--

[1]

(ii) The Accumulator currently contains the binary number:

0	0	1	1	0	1	0	1
---	---	---	---	---	---	---	---

Identify the mathematical operation that the following instruction will perform on the contents of the accumulator.

LSR #3

[1]

Answers Section

1. M/J 26/P11/Q3/a,b,c

3(a)	1 mark for each correct row <table border="1" data-bbox="236 349 1018 674"> <thead> <tr> <th>accumulator current data</th><th>set of instructions</th><th>new accumulator data</th></tr> </thead> <tbody> <tr> <td>B11001010</td><td>ADD B00000010 INC ACC</td><td>B11001101</td></tr> <tr> <td>#100</td><td>LDM #200 ADD 200</td><td>#301</td></tr> <tr> <td>#103</td><td>LDD 103 DEC ACC DEC ACC</td><td>#48</td></tr> </tbody> </table>	accumulator current data	set of instructions	new accumulator data	B11001010	ADD B00000010 INC ACC	B11001101	#100	LDM #200 ADD 200	#301	#103	LDD 103 DEC ACC DEC ACC	#48	3 Correct answers only
accumulator current data	set of instructions	new accumulator data												
B11001010	ADD B00000010 INC ACC	B11001101												
#100	LDM #200 ADD 200	#301												
#103	LDD 103 DEC ACC DEC ACC	#48												
3(b)(i)	1 mark for AND 1 mark for #0 // &0 // B00000000 Answer: AND B00000000 // AND &0 // AND #0	2 Accept correct shift instructions Bit manipulation so no load commands												
3(b)(ii)	1 mark for the correct answer 0000 0110	1 Correct answer only												
3(c)	1 mark for a correct name 1 mark for a matching description 1 mark for a reasonable example - Indirect - the operand stores the address where the data is found // the data is held in the address stored in the given address - example: LDI 23 loads the data in the address stored in address 23 - Indexed - the address is calculated by adding the operand to the contents of the index register - example: LDX 20 adds 20 to the contents of the index register to calculate the address	3 Do not accept Immediate, relative, direct, absolute, symbolic.												

2. M/J 26/P13/Q7/a,b,c

7(a)	1 mark for each correct answer	4	No mnemonics / opcodes										
	<table><tr><td>Mode of addressing</td><td>Description</td></tr><tr><td>Direct</td><td>The operand is an address where the data is stored.</td></tr><tr><td>Indirect</td><td>The operand is an address that stores the address where the data is stored.</td></tr><tr><td>Indexed</td><td>The operand is an address. The address where the data is stored is calculated by adding the operand to the content of the index register.</td></tr><tr><td>Immediate</td><td>The operand is the data to be used.</td></tr></table>			Mode of addressing	Description	Direct	The operand is an address where the data is stored.	Indirect	The operand is an address that stores the address where the data is stored.	Indexed	The operand is an address. The address where the data is stored is calculated by adding the operand to the content of the index register.	Immediate	The operand is the data to be used.
	Mode of addressing			Description									
	Direct			The operand is an address where the data is stored.									
	Indirect			The operand is an address that stores the address where the data is stored.									
	Indexed			The operand is an address. The address where the data is stored is calculated by adding the operand to the content of the index register.									
Immediate	The operand is the data to be used.												

7(b)	1 mark each to max 3 - Bit manipulation allows a change/evaluation of the state/value of a bit // masking Monitoring: - When monitoring, bit manipulation can be used to establish the state of a device (read inputs) ... by testing/observing the state of an individual bit Controlling: - When controlling bit manipulation can be used to change the state of a device (control output) ... by setting/clearing/toggling a single bit ... or by clearing a byte	3	
7(c)	1 mark for similarity: - Both lose the Least Significant Bit(s) (LSB) // both lose bit(s) from the right of the number 1 mark for difference: - Arithmetic retains sign/MSB while logical does not retain sign/MSB - MSBs are filled with 0 for logical and copies of the sign bit for arithmetic.	2	For difference, must have both sides

3. O/N 25/P11/Q4/a,b,c,d

4(a)	1 mark for: LSL #2	1								
4(b)	<table><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr></table>	1	1	1	1	0	0	0	1	1
1	1	1	1	0	0	0	1			
4(c)	<table><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table>	0	0	1	0	0	0	1	1	1
0	0	1	0	0	0	1	1			
4(d)	<table><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table>	1	1	1	1	0	0	1	1	1
1	1	1	1	0	0	1	1			

4. O/N 25/P11/Q6/a,b

6(a)	1 mark for each correct description, max 2 marks		2
	Register	Role	
	ACC	stores the intermediate results of arithmetic and logical operations // holds the result of a calculation	
	CIR	holds the instruction currently being decoded and/or executed	

6(b)	1 mark per bullet point, max 2 marks e.g. • Latency may be increased • ... because the cores must communicate with one another • There is a potential for dead-lock situations • ... where one core may wait for information from other cores which in turn are waiting for the first one • Not all software is designed to use multi-cores • ... so some of the additional cores would be idle • Increased heat generation • ... which could cause damage to other components	2
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5. O/N 25/P12/Q3/a,b,c

3(a)(i)	1 mark The value of the operand is an offset value which is added to another base value to give the address from which the contents are loaded to the accumulator	1								
3(a)(ii)	1 mark per bullet point, max 2 marks - Program counter (PC) - Memory Data Register (MDR) - Memory Address Register (MAR) - Current Instruction Register (CIR) - Status Register	2								
3(b)	1 mark for each correct value, max 3 marks <table><tr><th>Instruction</th><th>Value in ACC</th></tr><tr><td>LDM #98</td><td>98</td></tr><tr><td>LDI 101</td><td>8</td></tr><tr><td>LDX 100</td><td>32</td></tr></table>	Instruction	Value in ACC	LDM #98	98	LDI 101	8	LDX 100	32	3
Instruction	Value in ACC									
LDM #98	98									
LDI 101	8									
LDX 100	32									
3(c)	1 mark per bullet point, max 2 marks for each Clock Speed: - Processor can perform more F-E cycles per second ... so more instructions / data can be processed each second Cache Memory: - Can store more of the most frequently used instructions ... which reduces the need to access slower RAM	4								

6. O/N 25/P13/Q2/c

2(c)	1 mark 1111 0010	1
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7. O/N 25/P13/Q6/a,b,c

6(a)	1 mark for each correct row <table border="1"> <thead> <tr> <th>Current contents of the ACC</th><th>Instruction</th><th>Contents of the ACC after the execution of the instruction</th></tr> </thead> <tbody> <tr> <td>0000 1111</td><td>AND 101</td><td>0000 1110</td></tr> <tr> <td>0000 0000</td><td>LDM #100</td><td>0110 0100</td></tr> <tr> <td>0000 0001</td><td>XOR &F1</td><td>1111 0000</td></tr> <tr> <td>0001 0001</td><td>CMP 101</td><td>0001 0001</td></tr> </tbody> </table>	Current contents of the ACC	Instruction	Contents of the ACC after the execution of the instruction	0000 1111	AND 101	0000 1110	0000 0000	LDM #100	0110 0100	0000 0001	XOR &F1	1111 0000	0001 0001	CMP 101	0001 0001	4
Current contents of the ACC	Instruction	Contents of the ACC after the execution of the instruction															
0000 1111	AND 101	0000 1110															
0000 0000	LDM #100	0110 0100															
0000 0001	XOR &F1	1111 0000															
0001 0001	CMP 101	0001 0001															
6(b)	1 mark for each bullet point, max 4 marks - MAR stores the <u>address</u> of the next instruction / data to be read from / or written to memory - The address is received from the Program Counter (PC) - The MDR stores the data / instruction in the address stored in the MAR which has been read / written - The instruction passes to the CIR for decoding and executing	4															
6(c)	1 mark for each bullet point, max 3 marks - Data movement - Arithmetic operations - Conditional and unconditional (jump) instructions	3															

8. M/J 25/P11/Q6/a(ii,iii)

6(a)(ii)	1 mark each to max 3 - More RAM means more currently running data and instructions can be stored - without needing to use virtual memory - without having to fetch the data from secondary storage first - which has a slower access time - Less latency / delay waiting for instructions / data	3
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6(a)(iii)	1 mark each to max 2 e.g. • Wider data bus means more data can be transferred between components at a time • there is less delay / latency when fetching data for a running process • Wider address bus means larger memory addresses can be used • allowing more memory locations to be accessed directly • so less likely to run out of memory	2
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9. M/J 25/P11/Q8/a,b

8(a)	1 mark for each shaded section					4	
	Instruction address	ACC	Memory address				
			80	81	82		83
			10	8	80		81
	200	8					
	201	9					
	202						9
	203	10					
	204						
	206	9					
	207	19					
	208						
	210			19			
8(b)(i)	1 mark for: OR B0000 0001 // OR #1 // OR &1					1	
8(b)(ii)	1 mark for: 1010 1011					1	
8(b)(iii)	1 mark for: 0000 0011					1	

10. M/J 25/P12/Q1/a,b,c

1(a)	1 mark for each correct row <table><tr><th>Incorrect statement number</th><th>Corrected statement</th></tr><tr><td>1</td><td>The Program Counter (PC) stores the address of the next instruction to be fetched from memory.</td></tr><tr><td>3</td><td>The Control Unit (CU) sends signals to other components on the control bus.</td></tr><tr><td>4</td><td>The Memory Data Register (MDR) holds data to be stored in the memory address in the MAR. // The Memory Data Register (MDR) holds data read from the memory address in the MAR</td></tr></table>	Incorrect statement number	Corrected statement	1	The Program Counter (PC) stores the address of the next instruction to be fetched from memory.	3	The Control Unit (CU) sends signals to other components on the control bus.	4	The Memory Data Register (MDR) holds data to be stored in the memory address in the MAR. // The Memory Data Register (MDR) holds data read from the memory address in the MAR	3
Incorrect statement number	Corrected statement									
1	The Program Counter (PC) stores the address of the next instruction to be fetched from memory.									
3	The Control Unit (CU) sends signals to other components on the control bus.									
4	The Memory Data Register (MDR) holds data to be stored in the memory address in the MAR. // The Memory Data Register (MDR) holds data read from the memory address in the MAR									
1(b)	1 mark for name, 1 mark for corresponding role e.g. • Current Instruction Register // CIR • To store the instruction to be decoded / executed next • Status register // SR • To contain bits that can be referenced individually to indicate a state or event • Interrupt register • To store details of any interrupts that have occurred	2								
1(c)	1 mark each to max 4 • At the start / end of FE cycle the interrupt register is checked • The priority of any interrupts waiting is checked • if the priority of the interrupt is higher than the current process • the contents of the registers are stored on the stack • The relevant Interrupt Service Routine (ISR) / interrupt handler is called to process the interrupt • When the ISR has finished, a further check is made for higher priority interrupts • if no more interrupts of higher priority, the register contents are restored and the next FE cycle continues	4								

11. M/J 25/P13/Q2/a,b,c

2(a)(i)	1 mark each to max 2 - max 1 for each set of marks • The control unit synchronises the actions of the processor • by sending a command / signal on each timing signal produced by the system clock • using / along the control bus	2
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2(a)(ii)	1 mark each • $MAR \leftarrow [PC]$ and $PC \leftarrow [PC] + 1$ • $MDR \leftarrow [[MAR]]$ • $CIR \leftarrow [MDR]$ • Correct order • For example: • $MAR \leftarrow [PC]$ • $PC \leftarrow [PC] + 1$ • $MDR \leftarrow [[MAR]]$ • $CIR \leftarrow [MDR]$	4
2(b)	1 mark each • Using cache memory improves system performance • because cache is fast access memory close to the CPU • which stores frequently used instructions / data • so that they can be accessed faster than from RAM	2
2(c)	1 mark each to max 2 e.g. • HDMI transfers both audio and video using a single cable • HDMI has a high bandwidth • Data is transmitted in a stream • of uncompressed digital signals • HDMI uses a technology called Transition-Minimized Differential Signalling (TMDS)	2

12. M/J 25/P13/Q5/a,b

5(a)(i)	1 mark for each shaded section <table><tr><th rowspan="2">Instruction address</th><th rowspan="2">ACC</th><th colspan="4">Memory address</th></tr><tr><th>50</th><th>51</th><th>52</th><th>53</th></tr><tr><td></td><td></td><td>47</td><td>48</td><td>49</td><td>50</td></tr><tr><td>500</td><td>50</td><td></td><td></td><td></td><td></td></tr><tr><td>501</td><td>49</td><td></td><td></td><td></td><td></td></tr><tr><td>502</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>504</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>501</td><td>48</td><td></td><td></td><td></td><td></td></tr><tr><td>502</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>503</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>505</td><td></td><td>48</td><td></td><td></td><td></td></tr><tr><td>506</td><td>38</td><td></td><td></td><td></td><td></td></tr><tr><td>507</td><td></td><td></td><td>38</td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Instruction address	ACC	Memory address				50	51	52	53			47	48	49	50	500	50					501	49					502						504						501	48					502						503						505		48				506	38					507			38															3
Instruction address	ACC			Memory address																																																																																						
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507			38																																																																																							
5(a)(ii)	1 mark for addressing mode, 1 mark for matching description • Direct • The operand is the address of the data • Indirect • The operand points to the memory location which is the address of the data • Indexed • The address of the data is formed by adding the contents of the Index Register (IX) to the operand	4																																																																																								
5(b)	1 mark for each correct instruction e.g. AND B00000000 / #0 / &0 OR B10000000 / #128 / &80	2																																																																																								

13. M/J 25/P13/Q7/a

7(a)(i)	1 mark for 0010 1000	1
7(a)(ii)	1 mark for 1111 0011	1

14. O/N 24/P11/Q3/a,b

3(a)	1 mark for system clock and 1 mark for Control Unit System clock • To synchronise operations • ... by creating and transmitting timing signals on the control bus Control Unit • Sends/receives control signals along control bus • Reads an instruction from the contents of the memory location whose address is stored in PC • Coordinates/synchronises the activity of other components in the CPU • Manages the execution of instructions • Controls communication between the components in the CPU	2
3(b)(i)	1 mark for the feature, 1 mark max for the matching reason e.g. Feature: clock speed • Higher clock speed means that more F-E cycles are executed per second // Higher clock speed results in more throughput Feature: bus width • Larger bus width means that more data transferred at the same time	2
3(b)(ii)	1 mark for correct port. 2 marks for explanation Port: USB / Universal Serial Bus Explanation: • A voltage change occurs when the drive is plugged in • The computer detects this voltage change • The code of the device is transferred to computer • ... the OS finds the code of the device in the list of devices • ... and loads the appropriate device driver	3

15. O/N 24/P11/Q8/a,b

8(a)	1 mark for each bullet point (max 2) - To store the value of flags/bits ... that can be changed/set/cleared after arithmetic / logical operations - To allow flags to be checked ... to change instruction sequence	2
8(b)	1 mark for each bullet point (max 2) - Special purpose registers have a specified role in the machine whereas general purpose registers can be used for all purposes defined by the programmer - Special purpose registers hold the state of the program's execution while general purpose registers hold the program's data during operations	2

16. O/N 24/P12/Q3/a,b

3(a)	1 mark for each correct purpose	4										
<table><tr><th>Register</th><th>Purpose</th></tr><tr><td>Program Counter (PC)</td><td>Stores the address of the next instruction to be fetched/executed</td></tr><tr><td>Memory Address Register (MAR)</td><td>Stores the address of the memory location where data will be read from/written to</td></tr><tr><td>Memory Data Register (MDR)</td><td>Stores the data read from the address in the MAR // stores the data to be written to the address in the MAR</td></tr><tr><td>Index Register (IX)</td><td>Stores a number that will be added to the operand, to form the address of the data</td></tr></table>			Register	Purpose	Program Counter (PC)	Stores the address of the next instruction to be fetched/executed	Memory Address Register (MAR)	Stores the address of the memory location where data will be read from/written to	Memory Data Register (MDR)	Stores the data read from the address in the MAR // stores the data to be written to the address in the MAR	Index Register (IX)	Stores a number that will be added to the operand, to form the address of the data
Register	Purpose											
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Memory Address Register (MAR)	Stores the address of the memory location where data will be read from/written to											
Memory Data Register (MDR)	Stores the data read from the address in the MAR // stores the data to be written to the address in the MAR											
Index Register (IX)	Stores a number that will be added to the operand, to form the address of the data											
3(b)	1 mark for each bullet point (max 4) e.g. • HDMI has faster transfer rates than VGA • ... needed due to high resolution / large number of pixels of monitor // HDMI supports the high resolution of the monitor • HDMI supports video and audio transfer between computer and monitor speakers • ... so no separate sound cable is needed unlike VGA • HDMI is digital interface therefore no data is lost in transfer to analogue and back • HDMI is less prone to error/crosstalk/external interference	4										

17. O/N 24/P12/Q8/a,b

8(a)	1 mark for each correct content of the ACC (4) 1 mark for correct IX column <table><thead><tr><th></th><th>Instructions</th><th>ACC content</th><th>IX content</th></tr></thead><tbody><tr><td>1</td><td>LDM #19 DEC ACC</td><td>18</td><td>20</td></tr><tr><td>2</td><td>LDD 23 ADD 19</td><td>40</td><td>20</td></tr><tr><td>3</td><td>LDI 25 INC ACC</td><td>5</td><td>20</td></tr><tr><td>4</td><td>LDR #21 LDX 2</td><td>15</td><td>21</td></tr></tbody></table>		Instructions	ACC content	IX content	1	LDM #19 DEC ACC	18	20	2	LDD 23 ADD 19	40	20	3	LDI 25 INC ACC	5	20	4	LDR #21 LDX 2	15	21	5
	Instructions	ACC content	IX content																			
1	LDM #19 DEC ACC	18	20																			
2	LDD 23 ADD 19	40	20																			
3	LDI 25 INC ACC	5	20																			
4	LDR #21 LDX 2	15	21																			
8(b)(i)	1 mark each correct content of ACC <table><thead><tr><th></th><th>Instructions</th><th>ACC content</th></tr></thead><tbody><tr><td>1</td><td>LSL #2</td><td>0110 1000</td></tr><tr><td>2</td><td>ADD #5 AND #30</td><td>0001 1110</td></tr><tr><td>3</td><td>OR B11110010 INC ACC</td><td>1111 1011</td></tr></tbody></table>		Instructions	ACC content	1	LSL #2	0110 1000	2	ADD #5 AND #30	0001 1110	3	OR B11110010 INC ACC	1111 1011	3								
	Instructions	ACC content																				
1	LSL #2	0110 1000																				
2	ADD #5 AND #30	0001 1110																				
3	OR B11110010 INC ACC	1111 1011																				
8(b)(ii)	1 mark each to max 2 for explanation • An odd binary number will have a 1 in the Least Significant Bit (LSB) • A bit manipulation operation is required to access/mask only the LSB and clear all the others • Compare the result of the masking with denary 1 • ... the result of the comparison will be true if the number is odd 1 mark for correct instruction • AND B00000001 // AND #1 // AND &01	3																				

18. O/N 24/P13/Q7/a,b

7(a)	1 mark each: <table border="1" data-bbox="284 282 1046 730"> <thead> <tr> <th></th><th>Instructions</th><th>ACC content</th></tr> </thead> <tbody> <tr> <td>1</td><td>LDD 50 ADD #4 ADD 54</td><td>158</td></tr> <tr> <td>2</td><td>LDI 53 DEC ACC ADD 56</td><td>99</td></tr> <tr> <td>3</td><td>LDM #55 SUB #5</td><td>50</td></tr> </tbody> </table>		Instructions	ACC content	1	LDD 50 ADD #4 ADD 54	158	2	LDI 53 DEC ACC ADD 56	99	3	LDM #55 SUB #5	50	3
	Instructions	ACC content												
1	LDD 50 ADD #4 ADD 54	158												
2	LDI 53 DEC ACC ADD 56	99												
3	LDM #55 SUB #5	50												
7(b)	1 mark for each bullet point for the explanation 1 mark for correct instruction • A bit manipulation operation is required to set all the bits to zero • Compare the result of the masking with 0 • ... the result of comparison will be true if the register is cleared • AND B00000000 / #00 / &00	3												

19. O/N 24/P13/Q8/c

8(c)	1 mark each: • A logical shift moves all bits to the right and inserts zeros in the appropriate leftmost bits • An arithmetic shift moves all bits to the right but copies the sign bit into the Most Significant Bit (MSB)	2
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20. M/J 24/P11/Q4/a,b

4(a)	1 mark for each correct answer: <table border="1" data-bbox="284 1552 861 2074"> <thead> <tr> <th>Program Number</th><th>Code</th><th>ACC Content</th></tr> </thead> <tbody> <tr> <td>1</td><td>LDD 20 ADD #2</td><td>4</td></tr> <tr> <td>2</td><td>LDX 22</td><td>5</td></tr> <tr> <td>3</td><td>LDI 25 INC ACC SUB 22</td><td>1</td></tr> <tr> <td>4</td><td>LDD 19 LDM #5 LDM #25</td><td>25</td></tr> </tbody> </table>	Program Number	Code	ACC Content	1	LDD 20 ADD #2	4	2	LDX 22	5	3	LDI 25 INC ACC SUB 22	1	4	LDD 19 LDM #5 LDM #25	25	4
Program Number	Code	ACC Content															
1	LDD 20 ADD #2	4															
2	LDX 22	5															
3	LDI 25 INC ACC SUB 22	1															
4	LDD 19 LDM #5 LDM #25	25															

4(b)	1 mark for each correct answer: <table border="1" data-bbox="284 237 951 533"> <thead> <tr> <th>Program Number</th><th>Code</th><th>ACC Content</th></tr> </thead> <tbody> <tr> <td>1</td><td>AND 31</td><td>1001 1010</td></tr> <tr> <td>2</td><td>XOR B01001111</td><td>1101 0101</td></tr> <tr> <td>3</td><td>OR #30</td><td>1001 1110</td></tr> </tbody> </table>	Program Number	Code	ACC Content	1	AND 31	1001 1010	2	XOR B01001111	1101 0101	3	OR #30	1001 1110	3
Program Number	Code	ACC Content												
1	AND 31	1001 1010												
2	XOR B01001111	1101 0101												
3	OR #30	1001 1110												

21. M/J 24/P12/Q5/a,b

5(a)	1 mark for each correct answer: <table border="1" data-bbox="284 714 1123 1279"> <thead> <tr> <th>Program Number</th><th>Code</th><th>ACC Content</th></tr> </thead> <tbody> <tr> <td>1</td><td>LDI 15 SUB #1</td><td>11</td></tr> <tr> <td>2</td><td>LDD 14 ADD 11</td><td>13</td></tr> <tr> <td>3</td><td>LDM #11 ADD #3 SUB 16</td><td>2</td></tr> <tr> <td>4</td><td>LDR #2 LDX 14 ADD #2</td><td>14</td></tr> </tbody> </table>	Program Number	Code	ACC Content	1	LDI 15 SUB #1	11	2	LDD 14 ADD 11	13	3	LDM #11 ADD #3 SUB 16	2	4	LDR #2 LDX 14 ADD #2	14	4
Program Number	Code	ACC Content															
1	LDI 15 SUB #1	11															
2	LDD 14 ADD 11	13															
3	LDM #11 ADD #3 SUB 16	2															
4	LDR #2 LDX 14 ADD #2	14															
5(b)	1 mark for each correct answer: <table border="1" data-bbox="284 1386 1123 1686"> <thead> <tr> <th>Program Number</th><th>Code</th><th>ACC Content</th></tr> </thead> <tbody> <tr> <td>1</td><td>XOR 29</td><td>0100 1001</td></tr> <tr> <td>2</td><td>AND #29</td><td>0000 0100</td></tr> <tr> <td>3</td><td>OR B11111111</td><td>1111 1111</td></tr> </tbody> </table>	Program Number	Code	ACC Content	1	XOR 29	0100 1001	2	AND #29	0000 0100	3	OR B11111111	1111 1111	3			
Program Number	Code	ACC Content															
1	XOR 29	0100 1001															
2	AND #29	0000 0100															
3	OR B11111111	1111 1111															

22. M/J 24/P13/Q3/a,b

3(a)	1 mark for each correct answer: <table border="1" data-bbox="284 280 842 869"> <thead> <tr> <th>Program Number</th><th>Code</th><th>ACC Content</th></tr> </thead> <tbody> <tr> <td>1</td><td>LDM #50 INC ACC SUB #1</td><td>50</td></tr> <tr> <td>2</td><td>LDI 51 ADD 52</td><td>97</td></tr> <tr> <td>3</td><td>LDR #2 LDX 50 DEC ACC</td><td>48</td></tr> <tr> <td>4</td><td>LDD 52 SUB 54 INC ACC</td><td>44</td></tr> </tbody> </table>	Program Number	Code	ACC Content	1	LDM #50 INC ACC SUB #1	50	2	LDI 51 ADD 52	97	3	LDR #2 LDX 50 DEC ACC	48	4	LDD 52 SUB 54 INC ACC	44	4
Program Number	Code	ACC Content															
1	LDM #50 INC ACC SUB #1	50															
2	LDI 51 ADD 52	97															
3	LDR #2 LDX 50 DEC ACC	48															
4	LDD 52 SUB 54 INC ACC	44															
3(b)	1 mark for each correct answer: <table border="1" data-bbox="284 981 1008 1276"> <thead> <tr> <th>Instruction Number</th><th>Instruction</th><th>ACC Content</th></tr> </thead> <tbody> <tr> <td>1</td><td>LSL #2</td><td>1111 1100</td></tr> <tr> <td>2</td><td>XOR 100</td><td>1111 0010</td></tr> <tr> <td>3</td><td>AND 103</td><td>0011 0111</td></tr> </tbody> </table>	Instruction Number	Instruction	ACC Content	1	LSL #2	1111 1100	2	XOR 100	1111 0010	3	AND 103	0011 0111	3			
Instruction Number	Instruction	ACC Content															
1	LSL #2	1111 1100															
2	XOR 100	1111 0010															
3	AND 103	0011 0111															

23. O/N 23/P11/Q5/a,b,c

5(a)	1 mark for identification of the register and 1 mark for role (max 2 for each register) • Program Counter (PC) • stores the <u>address</u> where the <u>next</u> instruction is to be read from • Memory Address Register (MAR) • stores the address of the memory location (or an I/O component) currently being read from or written to • Current Instruction Register (CIR) • holds the instruction currently being decoded and/or executed • Status Register • Contains bits which can be referenced individually and set or cleared depending on the operation e.g. overflow, underflow	4
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5(b)	1 mark for each bullet point (max 2) - Immediate Access Store holds all the data / instructions / programs currently in use - Immediate Access Store is volatile memory - Immediate Access Store has fast access times	2
5(c)(i)	1 mark for each bullet point (max 1) - The CPU can now perform nearly twice as many F-E cycles per second - Instead of 2.1 billion F-E cycles per second, the CPU can now perform 4 billion FE cycles per second	1
5(c)(ii)	1 mark for each bullet point (max 5) - Multiple cores introduce additional overheads ...because of the need for communication between cores - Software may not be designed for multiple cores... ...so one of the cores will be left idle - Memory access speed may not match speed of cores... ...so causing delay - The two computers may have more differences than just the cores ...one may have more RAM which allows faster multitasking ...one may have a GPU ...etc.	5

24. O/N 23/P11/Q8/a,b,c

8(a)	1 mark for: To create a symbol table	1
8(b)(i)	1 mark for each bullet point - Data movement: e.g. LDR #50 // STO 201 - Arithmetic operation: e.g. ADD 100 // INC IX - Conditional instruction: e.g. JPE 96	3
8(b)(ii)	1 mark for each bullet point (max 2) Similarity: - both load the contents of an <u>address</u> into the <u>Accumulator</u> Difference: - direct accesses the address given by the operand whereas indexed adds the contents of IX to the operand and accesses the data at that calculated address	2
8(b)(iii)	1 mark for - Indirect (addressing) - Relative (addressing)	1
8(c)(i)	0000 0101	1

8(c)(ii)	0000 0010	1
8(c)(iii)	1101 0010	1

25. O/N 23/P12/Q8/c

8(c)(i)	1 mark for each bullet point (max 1) • USB / Universal Serial Bus • HDMI	1
8(c)(ii)	1 mark for each component (max 3) Buffers • A buffer temporarily holds data until it is ready to be transmitted to the device Address Bus • The address of the data to be written to the device (in RAM) is carried on the address bus Data Bus • All data to be written to the device / buffer is carried on the data bus	3

26. O/N 23/P12/Q9/a,b

9(a)	1 mark for each bullet point • IN - Input and output of data • ADD - Arithmetic operations • JPE - Unconditional and conditional instructions • CMI - Compare instructions	4
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9(b)

1 mark for each set of shaded rows

4

Instruction address	ACC	IX	Memory address				Output
			100	101	110	111	
			0	0	51	65	
10		0					
11	49						
12				49			
13	51						
14							
15							
16	49						
17	65						
18		1					
19			65				
20							
13	65						
14							
15							
21							A
22							

27. O/N 23/P13/Q7/d

7(d)	1 mark for naming a correct port 2 marks for matching justification • USB • ...has fast data transfer speeds for data (to the headset) • ...is a universal/popular cable // universal standard • HDMI • ...allows video and audio to be transferred (on the same cable) • ...convenience of HDMI as no need for two cables	3
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28. O/N 23/P13/Q9/a,b,c,d

9(a)(i)	1 mark for each bullet point (max 3). • Control unit (CU) • Arithmetic and Logic Unit (ALU) • Immediate Access Store (IAS) • (System clock)	3
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9(a)(ii)	1 mark for each bullet point (max 2). - Special purpose registers hold the status of a program whereas general purpose registers hold the temporary data while performing operations. - Special purpose registers are specialised for a specific use, whereas general purpose registers are used for any purpose. - General purpose registers can be used by most instructions, whereas special purpose can only be used by certain instruction	2										
9(b)	1 mark for each correct answer (shaded cells) <table><tr><th>Step</th><th>Description</th></tr><tr><td>PC ← [PC] + 1</td><td>The address in PC is incremented.</td></tr><tr><td>MDR ← [[MAR]]</td><td>The data in the location pointed to by the MAR is copied to the MDR.</td></tr><tr><td>MAR ← [PC]</td><td>The contents of PC are copied to the MAR.</td></tr><tr><td>CIR ← [MDR]</td><td>The contents of MDR are copied into CIR.</td></tr></table>	Step	Description	PC ← [PC] + 1	The address in PC is incremented.	MDR ← [[MAR]]	The data in the location pointed to by the MAR is copied to the MDR.	MAR ← [PC]	The contents of PC are copied to the MAR.	CIR ← [MDR]	The contents of MDR are copied into CIR.	4
Step	Description											
PC ← [PC] + 1	The address in PC is incremented.											
MDR ← [[MAR]]	The data in the location pointed to by the MAR is copied to the MDR.											
MAR ← [PC]	The contents of PC are copied to the MAR.											
CIR ← [MDR]	The contents of MDR are copied into CIR.											
9(c)	1 mark for an appropriate example. Examples: - Division by zero - Runtime error - Out of memory bounds - Program requesting an external device / input - Buffer overflow	1										
9(d)	1 mark for each letter in the correct position (2, 4 and 6) D A B	3										

29. M/J 23/P11/Q3/d(vi)

3(d)(vi)	00011001	1
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30. M/J 23/P11/Q6

6	1 mark each to max 5 • An interrupt flag is raised in the (interrupt) register • At the end of the current FE cycle // at the start of the next FE cycle • The system checks the interrupt register for higher priority interrupts than current process • If true, it stores the current contents of the registers on the stack • The appropriate interrupt service routine (ISR) for the key press is called • The input data from the keyboard is processed • The contents of the registers are restored from the stack • ... and control is passed back to previous process	5
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31. M/J 23/P12/Q3/a,b

3(a)	1 mark for generates object code to second pass 1 mark for reads source code one line at a time to both boxes 1 mark for removes white space and adds labels to first pass Action generates object code reads the source code one line at a time removes white space adds labels to the symbol table Pass first second <pre> graph LR A1[generates object code] --> P2[second] A2[reads the source code one line at a time] --> P1[first] A2 --> P2[second] A3[removes white space] --> P1[first] A4[adds labels to the symbol table] --> P1[first] </pre>	3
3(b)	1 mark for each correct term Direct addressing is when the operand holds the memory address of the data. Indirect addressing is when the operand holds a memory address that stores the memory address of the data. Immediate addressing is when the operand is the data.	3

32. M/J 23/P12/Q4/d

4(d)	0011 1100	1
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33. M/J 23/P12/Q5/a,b,c

5(a)	1 mark each to max 2 Examples: • Interrupt • Timing • Read • Write	2
5(b)	1 mark for description; 1 mark for corresponding explanation Examples • Increase number of cores • Each core can independently carry out a process at the same time // so that more instructions are performed in parallel • Increase RAM capacity • ... allowing more applications to reside in memory at the same time, saving disk access times • Increase cache memory • More data can be stored in fast access so less time is spent accessing from RAM • Increase clock speed • More Fetch-Decode-Execute (FDE) cycles can run each second / per unit time	4
5(c)(i)	1 mark for a correct answer • 1 bit is transferred at a time • Can be synchronous or asynchronous • USB-3 is full duplex and earlier versions are half-duplex	1
5(c)(ii)	1 mark for identification of a suitable port Examples • HDMI • DisplayPort	1

34. M/J 23/P13/Q7/b,c

7(b)	1 mark each to max 4 • The system clock gives out timing signals • ... which are sent on the control bus • ...to synchronise the other system components • The Control Unit initiates data transfer • ...by generating signals that are sent on the control bus to other components	4						
7(c)	1 mark for each Register transfer notation <table><tr><th>Stage description</th><th>Register transfer notation</th></tr><tr><td>The Program Counter (PC) is incremented</td><td>$PC \leftarrow [PC] + 1$</td></tr><tr><td>The data in the address stored in the Memory Address Register (MAR) is copied to the Memory Data Register (MDR)</td><td>$MDR \leftarrow [[MAR]]$</td></tr></table>	Stage description	Register transfer notation	The Program Counter (PC) is incremented	$PC \leftarrow [PC] + 1$	The data in the address stored in the Memory Address Register (MAR) is copied to the Memory Data Register (MDR)	$MDR \leftarrow [[MAR]]$	2
Stage description	Register transfer notation							
The Program Counter (PC) is incremented	$PC \leftarrow [PC] + 1$							
The data in the address stored in the Memory Address Register (MAR) is copied to the Memory Data Register (MDR)	$MDR \leftarrow [[MAR]]$							

35. O/N 22/P11/Q5/a,b,c,d

5(a)	Instructions and data are stored in the same memory space / in main memory.	1												
5(b)(i)	1 mark for each special purpose register: Program Counter (PC): - to store the address / location / memory location of the <u>next</u> instruction to be fetched Index Register (IX): - to store a value that is added to an address to give another address Status Register (SR): - to store flags which are set by events // from the results of arithmetic and logic operations and interrupt flags	3												
5(b)(ii)	1 mark for both rows: <table><tr><th>CPU component</th><th>Data bus</th><th>Address bus</th><th>Control bus</th></tr><tr><td>System clock</td><td></td><td></td><td>✓</td></tr><tr><td>Memory Address Register (MAR)</td><td></td><td>✓</td><td></td></tr></table>	CPU component	Data bus	Address bus	Control bus	System clock			✓	Memory Address Register (MAR)		✓		1
CPU component	Data bus	Address bus	Control bus											
System clock			✓											
Memory Address Register (MAR)		✓												

5(b)(iii)	1 mark for each bullet point (max 2): • to coordinate / synchronise the actions of other components in the CPU • to send / receive control signals along the control bus • to manage the execution of instructions (in sequence) • to control the communication between the components of the CPU	2
5(c)	1 mark for each bullet point: • to send a signal from a device or process • ... seeking the attention of the processor	2
5(d)	1 mark for each bullet point (max 2). For example: • division by zero // runtime error in a program • attempt to access an invalid memory location • array index out of bounds • stack overflow	2

36. O/N 22/P11/Q6/a,b,c

6(a)(i)	1 mark for each set of highlighted rows.	4																																																																																																																																																																																																																																																																	
	<table><tr><th rowspan="2">Instruction address</th><th rowspan="2">ACC</th><th rowspan="2">IX</th><th colspan="5">Memory address</th><th rowspan="2">Output</th></tr><tr><th>100</th><th>101</th><th>110</th><th>111</th><th>112</th></tr><tr><td></td><td></td><td></td><td>0</td><td>0</td><td>66</td><td>65</td><td>35</td><td></td></tr><tr><td>77</td><td></td><td>0</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>78</td><td>66</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>79</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>80</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>81</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>82</td><td></td><td></td><td></td><td>66</td><td></td><td></td><td></td><td></td></tr><tr><td>83</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>84</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>85</td><td></td><td></td><td>1</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>86</td><td></td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>87</td><td>65</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>88</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>89</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>81</td><td>66</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>82</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>83</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>84</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>85</td><td></td><td></td><td>2</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>86</td><td></td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>87</td><td>35</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>88</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>89</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>90</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>91</td><td>50</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>92</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>2</td></tr><tr><td>93</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Instruction address	ACC	IX	Memory address					Output	100	101	110	111	112				0	0	66	65	35		77		0							78	66								79									80									81									82				66					83	1								84									85			1						86		1							87	65								88									89									81	66								82									83	1								84	2								85			2						86		2							87	35								88									89									90	2								91	50								92								2	93									
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6(b)(ii)	1001 0000	1															
6(b)(iii)	1101 1111	1															
6(b)(iv)	0010 0100	1															
6(c)	1 mark for each pair of highlighted rows <table> <tr> <th>Task</th><th>First pass</th><th>Second pass</th></tr> <tr> <td>Remove comments.</td><td>✓</td><td></td></tr> <tr> <td>Read the assembly language program one line at a time.</td><td>✓</td><td>✓</td></tr> <tr> <td>Generate the object code.</td><td></td><td>✓</td></tr> <tr> <td>Check the opcode is in the instruction set.</td><td>✓</td><td></td></tr> </table>	Task	First pass	Second pass	Remove comments.	✓		Read the assembly language program one line at a time.	✓	✓	Generate the object code.		✓	Check the opcode is in the instruction set.	✓		2
Task	First pass	Second pass															
Remove comments.	✓																
Read the assembly language program one line at a time.	✓	✓															
Generate the object code.		✓															
Check the opcode is in the instruction set.	✓																

37. O/N 22/P11/Q7/d,e

7(d)	1 mark for each bullet point (max 2): • cache is fast access memory (close to the CPU) • cache stores frequently used instructions / data • ... more cache means more instructions / data can be transferred faster • ... less swapping between RAM and cache • prevents the CPU idling while waiting for data	2
7(e)	1 mark for each device. • 3D printer: USB port / COM port • Monitor: HDMI / VGA / USB / DisplayPort	2

38. O/N 22/P12/Q7/a,b,c

7(a)	1 mark for each set of highlighted rows.										5																																																																																																																																																																																																																																																																																																																											
<table><tr><th rowspan="2">Instruction address</th><th rowspan="2">ACC</th><th rowspan="2">IX</th><th colspan="7">Memory address</th></tr><tr><th>100</th><th>101</th><th>102</th><th>103</th><th>110</th><th>111</th><th>112</th></tr><tr><td></td><td></td><td></td><td>0</td><td>0</td><td>112</td><td>4</td><td>1</td><td>4</td><td>0</td><td></td></tr><tr><td>75</td><td></td><td>0</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>76</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>77</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>78</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>79</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>80</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>84</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>85</td><td></td><td></td><td></td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>86</td><td>0</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>87</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>88</td><td></td><td></td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>89</td><td></td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>90</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>76</td><td>4</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>77</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>78</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>79</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>80</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>81</td><td>6</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>82</td><td></td><td></td><td></td><td>6</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>83</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>86</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>87</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>88</td><td></td><td></td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>89</td><td></td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>90</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>												Instruction address	ACC	IX	Memory address							100	101	102	103	110	111	112				0	0	112	4	1	4	0		75		0									76	1										77											78											79											80											84	2										85				2							86	0										87	1										88			1								89		1									90											76	4										77											78											79											80											81	6										82				6							83											86	1										87	2										88			2								89		2									90											
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7(b)(iii)	1001 1000	1								
7(b)(iv)	1101 1111	1								
7(c)	1 mark for each correct row: <table><thead><tr><th>Description</th><th>Register transfer notation</th></tr></thead><tbody><tr><td>Copy the address of the next instruction into the Memory Address Register.</td><td>MAR ← [PC]</td></tr><tr><td>Increment the Program Counter.</td><td>PC ← [PC] + 1</td></tr><tr><td>Copy the contents of the Memory Data Register into the Current Instruction Register.</td><td>CIR ← [MDR]</td></tr></tbody></table>	Description	Register transfer notation	Copy the address of the next instruction into the Memory Address Register.	MAR ← [PC]	Increment the Program Counter.	PC ← [PC] + 1	Copy the contents of the Memory Data Register into the Current Instruction Register.	CIR ← [MDR]	3
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Copy the contents of the Memory Data Register into the Current Instruction Register.	CIR ← [MDR]									

39. O/N 22/P13/Q4/a,b,c

4(a)(i)	1 mark for each register: MAR: holds address in memory from which data will be read / to which data will be written MDR: holds the data/instructions which has been read from or is to be written to the address in the MAR	2
4(a)(ii)	after completion of the execute stage // before the cycle begins	1
4(b)	1 mark for each bullet point (max 2): - synchronise operations ... by creating timing signals - to keep track of the date and time / timestamp files - to process operations in the correct order / sequence	2
4(c)	1 mark for identification of a correct upgrade: 1 mark for a corresponding explanation: Examples: - increase quantity of RAM ... so allowing more applications to reside in memory at the same time, saving disk access times - increase the size of cache memory ... so that the CPU can continue working without waiting for data - increase clock speed ... so that more instructions are performed in a time period - increase the number of processors / cores ... so that more instructions are performed in parallel	2

40. O/N 22/P13/Q6/a,b,c

6(a)(i)	1 mark for each set of highlighted rows: <table><tr><th rowspan="2">Instruction address</th><th rowspan="2">ACC</th><th rowspan="2">IX</th><th colspan="5">Memory address</th><th rowspan="2">Output</th></tr><tr><th>100</th><th>101</th><th>110</th><th>111</th><th>112</th></tr><tr><td></td><td></td><td></td><td>1</td><td>0</td><td>97</td><td>98</td><td>97</td><td></td></tr><tr><td>75</td><td></td><td>0</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>76</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>77</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>78</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>79</td><td>97</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>80</td><td>65</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>81</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>82</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>83</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>84</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>85</td><td></td><td></td><td></td><td>1</td><td></td><td></td><td></td><td></td></tr><tr><td>86</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>87</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>88</td><td></td><td></td><td>2</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>89</td><td></td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>90</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>76</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>77</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>78</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>91</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>92</td><td>49</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>93</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>1</td></tr><tr><td>94</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Instruction address	ACC	IX	Memory address					Output	100	101	110	111	112				1	0	97	98	97		75		0							76	1								77									78									79	97								80	65								81									82									83	1								84									85				1					86									87	2								88			2						89		1							90									76									77									78									91	1								92	49								93								1	94									4
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6(a)(ii)	1 mark for each bullet point: To allow for re-locatable code ... because all (target) addresses can be specified by the base address + offset	2																																																																																																																																																																																																																																						
6(b)(i)	0000 0100	1																																																																																																																																																																																																																																						

6(b)(ii)	1101 1111	1
6(b)(iii)	0010 0111	1
6(c)	1 mark for a correct name: - input and output of data - arithmetic operations - unconditional and conditional instructions - compare instructions	1

41. O/N 21/P11/Q6/a,b

6(a)	1 mark for identification of line and description of error 1 mark for the correct statement			4
	Line number	Description of the error	Correct statement	
	2	Program Counter should be incremented, not decremented	$PC \leftarrow [PC] + 1$	
	3	It should be the contents of the address in the MAR	$MDR \leftarrow [[MAR]]$	
6(b)	1 mark for each correct row			4
	Current contents of the ACC	Instruction	New contents of the ACC	
	11111111	OR 101	11111111	
	00000000	XOR #15	00001111	
	10101010	LSR #2	00101010	
	01010101	AND 104	00000000	

42. O/N 21/P12/Q7/a

7(a)	1 mark per pair of rows (shaded & unshaded)			3
Event		Hardware Interrupt	Software Interrupt	
Buffer full			✓	} 1
Printer is out of paper		✓		
User has pressed a key on the keyboard		✓		} 1
Division by zero			✓	
Power failure		✓		} 1
Stack overflow			✓	

43. O/N 21/P12/Q8/a,b

8(a)(i)	1 mark for each bullet point to max 2 for each register MAR • Stores the next <u>address</u> to be fetched • ... held in the Program Counter (PC) • The data at this address is then fetched MDR • Stores the data from the address pointed to by the MAR • The data in it is copied to the Current Instruction Register (CIR)	4															
8(a)(ii)	1 mark for a correct register e.g. Program Counter (PC) Current Instruction Register (CIR) Status register Interrupt register	1															
8(b)(i)	1 mark for each correct answer <table border="1"><thead><tr><th>Current contents of the ACC</th><th>Instruction</th><th>New contents of the ACC</th></tr></thead><tbody><tr><td>01010101</td><td>XOR 101</td><td>1010 0101</td></tr><tr><td>11110000</td><td>AND 104</td><td>1111 0000</td></tr><tr><td>00001111</td><td>LSL #4</td><td>1111 0000</td></tr><tr><td>11111111</td><td>OR 102</td><td>1111 1111</td></tr></tbody></table>	Current contents of the ACC	Instruction	New contents of the ACC	01010101	XOR 101	1010 0101	11110000	AND 104	1111 0000	00001111	LSL #4	1111 0000	11111111	OR 102	1111 1111	4
Current contents of the ACC	Instruction	New contents of the ACC															
01010101	XOR 101	1010 0101															
11110000	AND 104	1111 0000															
00001111	LSL #4	1111 0000															
11111111	OR 102	1111 1111															

8(b)(ii)	1 mark for each correct instruction <table> <tr> <th>Instruction Group</th> <th>Instruction</th> </tr> <tr> <td>Data movement</td> <td>LDM #2</td> </tr> <tr> <td>Input and output of data</td> <td>IN / OUT</td> </tr> <tr> <td>Arithmetic Operations</td> <td>INC ACC / INC IX</td> </tr> <tr> <td>Unconditional and conditional instructions</td> <td>JPN 100 / JMP 100</td> </tr> <tr> <td>Compare instructions</td> <td>CMP 100</td> </tr> </table>	Instruction Group	Instruction	Data movement	LDM #2	Input and output of data	IN / OUT	Arithmetic Operations	INC ACC / INC IX	Unconditional and conditional instructions	JPN 100 / JMP 100	Compare instructions	CMP 100	4
Instruction Group	Instruction													
Data movement	LDM #2													
Input and output of data	IN / OUT													
Arithmetic Operations	INC ACC / INC IX													
Unconditional and conditional instructions	JPN 100 / JMP 100													
Compare instructions	CMP 100													
8(b)(iii)	1 mark for name, 1 mark for description - Indirect addressing - the address to be used is at the given address - Relative addressing - the address to be used is an offset number of locations away, relative to the address of the current instruction - Indexed addressing - form the address from the given address plus the contents of the index register	2												

44. O/N 21/P13/Q6/a,b

6(a)	1 mark for identification of line and description of error 1 mark for the correct statement <table border="1" data-bbox="288 1312 1305 1608"> <thead> <tr> <th>Line number</th><th>Description of the error</th><th>Correct statement</th></tr> </thead> <tbody> <tr> <td>2</td><td>Program Counter should be incremented, not decremented</td><td>$PC \leftarrow [PC] + 1$</td></tr> <tr> <td>3</td><td>It should be the contents of the address in the MAR</td><td>$MDR \leftarrow [[MAR]]$</td></tr> </tbody> </table>	Line number	Description of the error	Correct statement	2	Program Counter should be incremented, not decremented	$PC \leftarrow [PC] + 1$	3	It should be the contents of the address in the MAR	$MDR \leftarrow [[MAR]]$	4						
Line number	Description of the error	Correct statement															
2	Program Counter should be incremented, not decremented	$PC \leftarrow [PC] + 1$															
3	It should be the contents of the address in the MAR	$MDR \leftarrow [[MAR]]$															
6(b)	1 mark for each correct row <table border="1" data-bbox="288 1720 1305 2040"> <thead> <tr> <th>Current contents of the ACC</th><th>Instruction</th><th>New contents of the ACC</th></tr> </thead> <tbody> <tr> <td>11111111</td><td>OR 101</td><td>11111111</td></tr> <tr> <td>00000000</td><td>XOR #15</td><td>00001111</td></tr> <tr> <td>10101010</td><td>LSR #2</td><td>00101010</td></tr> <tr> <td>01010101</td><td>AND 104</td><td>00000000</td></tr> </tbody> </table>	Current contents of the ACC	Instruction	New contents of the ACC	11111111	OR 101	11111111	00000000	XOR #15	00001111	10101010	LSR #2	00101010	01010101	AND 104	00000000	4
Current contents of the ACC	Instruction	New contents of the ACC															
11111111	OR 101	11111111															
00000000	XOR #15	00001111															
10101010	LSR #2	00101010															
01010101	AND 104	00000000															

45. M/J 21/P11/Q3/a,b,c

3(a)	1 mark for each completed statement The Program Counter holds the address of the next instruction to be loaded. This address is sent to the Memory Address Register. The Memory Data Register holds the data fetched from this address. This data is sent to the Current Instruction Register and the Control Unit decodes the instruction's opcode. The Program Counter is incremented.	5																																																																																																																																																																																																				
3(b)	1 mark for each shaded set of values <table><tr><th rowspan="2">Instruction address</th><th rowspan="2">ACC</th><th colspan="4">Memory address</th><th rowspan="2">IX</th><th rowspan="2">Output</th></tr><tr><th>365</th><th>366</th><th>367</th><th>368</th></tr><tr><td></td><td></td><td>1</td><td>3</td><td>65</td><td>66</td><td>0</td><td></td></tr><tr><td>200</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>201</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>202</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>203</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>204</td><td></td><td>2</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>205</td><td></td><td></td><td></td><td></td><td></td><td>2</td><td></td></tr><tr><td>206</td><td>65</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>207</td><td></td><td></td><td></td><td></td><td></td><td></td><td>A</td></tr><tr><td>208</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>200</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>201</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>202</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>203</td><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>204</td><td></td><td>3</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>205</td><td></td><td></td><td></td><td></td><td></td><td>3</td><td></td></tr><tr><td>206</td><td>66</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>207</td><td></td><td></td><td></td><td></td><td></td><td></td><td>B</td></tr><tr><td>208</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>200</td><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>201</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>202</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>209</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Instruction address	ACC	Memory address				IX	Output	365	366	367	368			1	3	65	66	0		200	1							201								202								203	2							204		2						205						2		206	65							207							A	208								200	2							201								202								203	3							204		3						205						3		206	66							207							B	208								200	3							201								202								209								6
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3(c)(ii)	1 mark for correct answer The number is divided by 8 (and only whole number retained)	1								

46. M/J 21/P12/Q4/a,b(ii,iii)

4(a)	1 mark for each shaded section / bullet point - Load 65 into ACC - Load 100 into ACC, increment and store in 102 - Load 68 into ACC - Load 101 into ACC, decrement and store in 102 <table><tr><th rowspan="2">Instruction address</th><th rowspan="2">ACC</th><th colspan="3">Memory address</th></tr><tr><th>100</th><th>101</th><th>102</th></tr><tr><td></td><td></td><td>68</td><td>65</td><td>100</td></tr><tr><td>70</td><td>65</td><td></td><td></td><td></td></tr><tr><td>71</td><td></td><td></td><td></td><td></td></tr><tr><td>72</td><td></td><td></td><td></td><td></td></tr><tr><td>73</td><td></td><td></td><td></td><td></td></tr><tr><td>74</td><td></td><td></td><td></td><td></td></tr><tr><td>76</td><td>100</td><td></td><td></td><td></td></tr><tr><td>77</td><td>101</td><td></td><td></td><td></td></tr><tr><td>78</td><td></td><td></td><td></td><td>101</td></tr><tr><td>79</td><td></td><td></td><td></td><td></td></tr><tr><td>70</td><td>68</td><td></td><td></td><td></td></tr><tr><td>71</td><td></td><td></td><td></td><td></td></tr><tr><td>72</td><td></td><td></td><td></td><td></td></tr><tr><td>80</td><td>101</td><td></td><td></td><td></td></tr><tr><td>81</td><td>100</td><td></td><td></td><td></td></tr><tr><td>82</td><td></td><td></td><td></td><td>100</td></tr><tr><td>83</td><td></td><td></td><td></td><td></td></tr><tr><td>(70)</td><td></td><td></td><td></td><td></td></tr></table>	Instruction address	ACC	Memory address			100	101	102			68	65	100	70	65				71					72					73					74					76	100				77	101				78				101	79					70	68				71					72					80	101				81	100				82				100	83					(70)					4
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47. M/J 21/P12/Q5/a,b

5(a)	1 mark for each term correctly inserted The control unit/bus transmits the signals to coordinate events based on the pulses of the (system) clock. The data bus carries data to components, while the address bus carries the address where data is being written to or read from. The arithmetic logic unit/ALU performs mathematical operations and logical comparisons.	5
5(b)	1 mark per bullet point to max 3 per factor. max 4 overall. Number of cores: • Each core processes one <u>instruction</u> per clock pulse • More/multiple cores mean that sequences of instructions can be split between them • ... and so more than one <u>instruction</u> is executed per clock pulse // more sequences of instructions can be run at the same time • More cores decreases the time taken to complete task Clock speed: • Each <u>instruction</u> is executed on a clock pulse // one F-E cycle is run on each clock pulse • ... so the clock speed dictates the number of <u>instructions</u> that can be run per second • The faster the clock speed the more <u>instructions</u> can be run per second	4

48. M/J 21/P13/Q3/a,b,c

3(a)	1 mark for each completed statement The Program Counter holds the address of the next instruction to be loaded. This address is sent to the Memory Address Register. The Memory Data Register holds the data fetched from this address. This data is sent to the Current Instruction Register and the Control Unit decodes the instruction's opcode. The Program Counter is incremented.	5
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3(b)	1 mark for each shaded set of values <table><tr><th rowspan="2">Instruction address</th><th rowspan="2">ACC</th><th colspan="4">Memory address</th><th rowspan="2">IX</th><th rowspan="2">Output</th></tr><tr><th>365</th><th>366</th><th>367</th><th>368</th></tr><tr><td></td><td></td><td>1</td><td>3</td><td>65</td><td>66</td><td>0</td><td></td></tr><tr><td>200</td><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>201</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>202</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>203</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>204</td><td></td><td>2</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>205</td><td></td><td></td><td></td><td></td><td></td><td>2</td><td></td></tr><tr><td>206</td><td>65</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>207</td><td></td><td></td><td></td><td></td><td></td><td></td><td>A</td></tr><tr><td>208</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>200</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>201</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>202</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>203</td><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>204</td><td></td><td>3</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>205</td><td></td><td></td><td></td><td></td><td></td><td>3</td><td></td></tr><tr><td>206</td><td>66</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>207</td><td></td><td></td><td></td><td></td><td></td><td></td><td>B</td></tr><tr><td>208</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>200</td><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>201</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>202</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>209</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Instruction address	ACC	Memory address				IX	Output	365	366	367	368			1	3	65	66	0		200	1							201								202								203	2							204		2						205						2		206	65							207							A	208								200	2							201								202								203	3							204		3						205						3		206	66							207							B	208								200	3							201								202								209								6
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Unit 5

SYSTEM SOFTWARE

38 items · 245 marks

Syllabus sections in this unit

5.1 Operating Systems

5.2 Language Translators

Unit 5: System Software

1. M/J 26/P11/Q2/a,b,c

- 2 A computer programmer uses language translators. Two types of translator are an interpreter and a compiler.

(a) Explain why language translators are needed when programming.

[2]

(b) Describe **two** benefits of using an interpreter instead of a compiler when developing a program.

1

2

[4]

(c) Identify **one** other type of translator.

[1]

2. M/J 26/P12/Q7/a,b

7 A student is writing a computer program for a school project.

(a) The student uses a computer that comes with an Operating System (OS) and utility software.

(i) Identify **two** tasks the OS performs as part of process management.

1

2

[2]

(ii) Identify **two** tasks the OS performs as part of memory management.

1

2

[2]

(iii) Explain the reasons why defragmentation software improves the performance of a computer.

[3]

(b) The student is using an Integrated Development Environment (IDE).

(i) Explain how the student can use debugging tools in the IDE.

[4]

(ii) The high-level language the student is using can be partially compiled and partially interpreted.

Explain the reasons why some high-level languages are partially compiled and partially interpreted.

[2]

3. M/J 26/P13/Q5/a,b,c

- 5** A student uses a laptop for school work. The laptop contains system software including an Operating System (OS) and utility software.

(a) Hardware management is one key management task that the OS performs.

- (i)** Identify **two** tasks the OS performs as part of hardware management.

1

2

[2]

- (ii)** Identify **two** other key management tasks that the OS performs.

1

2

[2]

(b) A disk formatter is an example of utility software.

- (i)** State the purpose of utility software in a computer.

[1]

- (ii)** Describe the purpose of a disk formatter.

[2]

(c) The student is writing a computer program in a high-level language.

The student imports routines from program libraries into their program.

- (i)** Define the term program library.

[1]

- (ii)** Describe **one** benefit of the student using a Dynamic Link Library (DLL) instead of a non-Dynamic Link Library.

[2]

4. M/J 26/P21/Q2/b

also Unit 12 (1 mark); this unit 1 mark

- 2** A washing machine contains:

- two trays for the washing powder
- a user interface display
- five sensors.

The washing machine control program includes the following variables:

variable	description
MainsOn	TRUE when the mains power is on
WaterOn	TRUE when the water supply to the machine is turned on
DisplayOn	TRUE when the user interface display is active
Tray1	TRUE when it detects washing powder in tray 1
Tray2	TRUE when it detects washing powder in tray 2

- (b) The user will select from three possible wash options.

The pseudocode for the washing machine program contains a procedure `WashCycle()`

```
PROCEDURE WashCycle()
    // declaration of variables has been omitted
    // TimeSoFar is a global variable which changes
    // as the procedure is executed
    Finished ← FALSE
    TimeSoFar ← 0.0
    OUTPUT "Option (1/2/3) "
    INPUT OptionNumber
    CASE OF OptionNumber:
        // WashTime is the total wash time in minutes
        1 : WashTime ← 60
        2 : WashTime ← 90
        3 : WashTime ← 120
    ENDCASE
    WHILE Finished = FALSE
        TimeRemaining ← WashTime - INT(TimeSoFar)
        
    ENDWHILE
ENDPROCEDURE
```

- (i) The programmer uses an Integrated Development Environment (IDE) to develop and run the washing machine program.

State the IDE feature to display the current value for any variable during the execution of the program.

[1]

- (ii) When option 3 is selected, the programmer interrupts execution when the `TimeSoFar` is 17.6 minutes.

State the current value for `TimeRemaining`

[1]

5. M/J 26/P23/Q2/d

also Unit 12 (1 mark); this unit 1 mark

- 2 A control program is being developed for a robot floor vacuum cleaner. The settings for the robot can be changed and include:

- the strength of the suction (numerical values: 0, 1, 2, 3, 4 or 5)
- the total vacuum cleaning time (a whole number of minutes).

The robot has a number of sensors.

The control program includes the variables:

variable	description
MainsOn	TRUE when the mains power is on
DisplayOn	TRUE when the user display is shown
Suction	TRUE when the program detects that the user has selected a non-zero value for the strength of the suction

(d) The pseudocode for the control program contains the procedure `CleanProgram()`

```

PROCEDURE CleanProgram()
// declaration of variables has been omitted

    Finished ← FALSE
    OUTPUT "Total clean time (minutes) "
    INPUT  TotalCleanTime
    OUTPUT "Suction strength (0,1,2,3,4,5) "
    INPUT  SuctionStrength

    WHILE Finished = FALSE
        // TimeSoFar is a global variable which changes
        // as the procedure is executed
        // TimeSoFar is the time the robot has been cleaning (in seconds)

        TimeRemaining ← TotalCleanTime - INT(TimeSoFar/60)

        }
    ENDWHILE

    }
ENDPROCEDURE

```

- (i) When the program is under development and running, the programmer can temporarily stop the execution at a set point in the program.

Identify this feature of the Integrated Development Environment (IDE).

[1]

- (ii) The programmer runs the program.

- The user is prompted and inputs the value 40 (total clean time in minutes).
- The user is prompted and inputs the value 3 (for the suction strength).
- The programmer interrupts its execution; the `TimeSoFar` is shown in a report window as 125 (the cleaning time so far in seconds).

State the current value of `TimeRemaining`

[1]

6. O/N 25/P11/Q5/a,b,c

5 The Operating System (OS) is responsible for hardware and security management in a computer system.

(a) (i) State **two** tasks that are performed by hardware management.

1

2

[2]

(ii) State **two** tasks that are performed by security management.

1

2

[2]

(b) (i) Describe **one** benefit of using library routines when developing software.

[2]

(ii) Describe **one** drawback of using library routines when developing software.

[2]

(c) Identify **two** presentation features found in a typical Integrated Development Environment (IDE).

1

2

[2]

7. O/N 25/P12/Q8/b(ii)

8 A restaurant accesses software and stores data using cloud computing.

(b) The restaurant uses touchscreens as input devices to allow staff to select items from a menu.

(ii) The image files that are displayed on the touchscreen have been corrupted.

Identify **two** examples of utility software that could be used to recover the images.

1

2

[2]

8. O/N 25/P12/Q10/a

10 A programmer is developing a computer program.

(a) Explain how the programmer can first use an interpreter **and** then a compiler to develop the computer program.

Interpreter

Compiler

[4]

9. O/N 25/P13/Q8/a,b,d

- 8** Compilers and interpreters are used to translate programs written in a high-level language into a low-level language.

(a) State **two** disadvantages of using a compiler compared to an interpreter during program development.

1

2

[2]

(b) Explain how a programmer benefits from using program libraries during program development.

[3]

(d) Memory management is one key management task performed by an OS.

Give **three** other key management tasks that are performed by an OS.

1

2

3

[3]

10. O/N 25/P21/Q1/c

- 1** A program is being developed to meet a particular customer requirement.

(c) During the alpha testing stage, an Integrated Development Environment (IDE) is used to help locate an error that has been identified. The IDE report window feature is used to examine the values assigned to variables.

Explain how **two** other IDE features are used together with the report window feature to help locate the error.

[3]

11. M/J 25/P11/Q6/b

- 6** A programmer is buying a new computer.

(b) Both computers have an Operating System (OS).

Describe the purpose of an OS in a computer.

[5]

12. M/J 25/P13/Q3/a

3 A programmer is writing a program in a high-level language.

(a) Complete the description of compilers and interpreters by writing the missing words.

A compiler checks all the code before attempting to translate the program. If any errors are found, they are all reported at the same time and the program does not translate or run. If there are no errors found, the compiler produces which can run without access to the

An interpreter translates one line of code and then runs it, before moving to the next line of code. If the line of code has an error, the interpreter and displays the error. The programmer can correct the error and then the interpreter continues translating from that point.

[4]

13. O/N 24/P11/Q4/a,c,d

4 A student uses a laptop to write a program that is saved as a text file.

(a) The laptop has utility software and an Operating System (OS).

(i) Describe the file management tasks carried out by an OS.

[2]

(ii) Explain the need for back-up software.

[2]

(c) The student used a program library when writing their program.

Explain the benefits to the student of using library files when writing a program.

[3]

(d) The program code is written using an Integrated Development Environment (IDE).

(i) One presentation feature found in a typical IDE is prettyprint.

Identify **and** describe **one other** presentation feature found in a typical IDE.

Feature

Description

[2]

(ii) One debugging feature found in a typical IDE is single stepping.

Identify **and** describe **one other** debugging feature found in a typical IDE.

Feature

Description

[2]

14. O/N 24/P12/Q3/c

3 A student has a computer.

- (c) The computer has an Operating System (OS). One of the key management tasks of the OS is process management.

Describe the process management tasks performed by an OS.

[4]

15. O/N 24/P12/Q4/a,b

4 A program is written in a high-level language by a team of three programmers using an Integrated Development Environment (IDE).

- (a) Describe how the programmers can use the debugging features of a typical IDE during the development of the program.

[4]

- (b) The programmers created a new program library whilst developing the program.

Describe the benefits to the programmers of creating a program library.

[3]

16. O/N 24/P22/Q1/b,c

1 A program has been developed and released for general use. After a few months of use an error is detected where under certain circumstances it outputs an unexpected value.

- (b) The program contains a function `Lookup()`. After investigation, it is found that this is the function that sometimes returns an incorrect value.

An Integrated Development Environment (IDE) is used to help locate the error.

The IDE features of watch window, single stepping and breakpoint will be used.

Explain these features including the order that they will be used in to locate the error in `Lookup()`.

[3]

- (c) To solve the error a programmer decides to create a new module.

The design of the new module has been completed and the module is being coded.

Identify **two** features of an IDE that will help during the coding of this new module.

1

2

[2]

17. M/J 24/P11/Q3/a,b

3 A software developer is writing a computer program.

- (a) The developer uses an interpreter while writing the program code because it is easier for debugging.

Explain **one** reason why it is easier to debug the program code using an interpreter instead of a compiler.

[2]

- (b) The program is ready to be sold to customers.

The developer uses a compiler because it creates an executable file.

Explain the reasons why the need to create an executable file makes the compiler the appropriate choice when the program is complete.

[3]

18. M/J 24/P12/Q6

6 A computer has an Operating System (OS).

Memory management and process management are two OS tasks.

Explain how memory management **and** process management support multi-tasking.

[4]

19. M/J 24/P12/Q8/a,b,c

8 A programmer uses an Integrated Development Environment (IDE) to write a computer program. The IDE has both a compiler and an interpreter as built-in translators.

- (a) The programmer decides to use the compiler when testing the final program.

Describe the benefits of using the compiler during testing.

[2]

- (b) IDEs have many features other than built-in translators.

Complete the table by identifying **one other** common IDE feature that can be used for each purpose. Describe how each feature helps the user during program development.

Each feature must be different. Do **not** give translator as one of your features.

Purpose	IDE feature	Description
for coding		
for presentation		
for debugging		

[6]

(c) The programmer uses program libraries when developing the program.

Describe **two** benefits to the programmer of using program libraries.

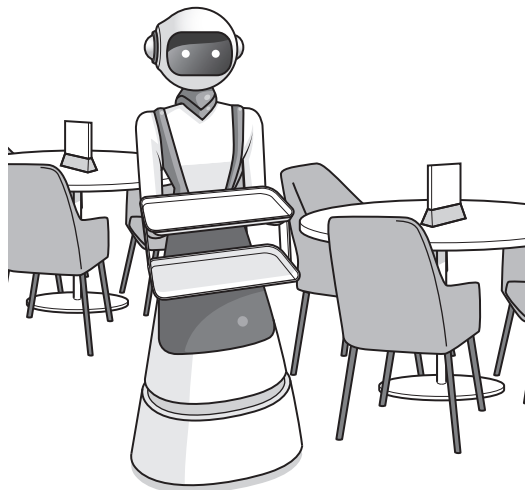
1

2

[2]

20. M/J 24/P13/Q7/e

7 Robots are used to serve food and drink to customers at a restaurant.



(e) Program libraries were used when writing the robot's software.

(i) State what is meant by a **program library**.

[1]

(ii) Some program libraries include Dynamic Link Library (DLL) files.

Describe the benefits of a programmer using a library with DLL files instead of using a library that does not include DLL files.

[4]



21. O/N 23/P11/Q6/a,b,c

- 6** A programmer uses both a compiler and an interpreter to translate a program written in a high-level language.

(a) Describe the advantages of using the interpreter compared to the compiler to translate the program.

[4]

(b) State **one** reason why some high-level languages are partially compiled and partially interpreted.

[1]

(c) (i) Identify **two** features that support the visual presentation of the code in a typical Integrated Development Environment (IDE).

1

2

[2]

(ii) Identify **two** features that support the debugging of the code in a typical IDE.

1

2

[2]

22. O/N 23/P12/Q8/a,b

- 8** A computer has an Operating System (OS).

(a) State **one** purpose of the Operating System.

[1]

(b) The Operating System has utility software.

The purpose of some utility software is to improve security.

Identify **one** example of utility software that is **not** intended to improve security.

Explain why this software is needed.

Utility software

Explanation

[3]

23. O/N 23/P13/Q5/a,b(ii)

- 5 A programmer uses an Integrated Development Environment (IDE) to develop a program that monitors air quality.

(a) Describe the following features of a typical IDE.

Context-sensitive prompts

Single stepping

[4]

(b) The program is distributed by downloading the source code and its library files from a web server.

(ii) The program files are stored on a new hard disk after they have been downloaded.

Describe the reasons why a hard disk formatter is needed for the new hard disk.

[3]

24. M/J 23/P11/Q3/a,b,c

- 3 A computer has an Operating System (OS).

(a) Describe how the Operating System manages the peripheral hardware devices of the computer.

[4]

(b) Hardware management is one key management task carried out by the Operating System.

Identify **two other** key management tasks carried out by the Operating System.

1

2

[2]

(c) The Operating System has utility software including defragmentation software.

Explain how defragmentation can improve the performance of the computer.

[3]

25. M/J 23/P12/Q5/d

- 5 A student has purchased a new laptop.

(d) The laptop has systems software.

(i) Describe how the Operating System (OS) manages processes in the computer.

[5]

(ii) Describe the purpose of utility software in a computer.

[2]

26. M/J 23/P12/Q7/a,b,c

7 A software developer is working in a team writing a program for a client.

(a) The developer is writing a new program library to be used by the other team members.

(i) Define the term **program library**.

[2]

(ii) Explain **two** benefits to the developer of choosing to create a Dynamic Link Library (DLL).

1

2

[4]

(b) The development team needs to use a translator whilst writing the program for the client.

Identify whether an interpreter or a compiler would be more appropriate at this stage of the program development.

Justify your choice.

Translator

Justification

[3]

(c) The development team uses an Integrated Development Environment (IDE).

Complete the table by describing the typical features found in an IDE.

Feature	Description
Breakpoints	
Dynamic syntax checks	
Context-sensitive prompts	
Single stepping	

[4]

27. M/J 23/P13/Q5/a

5 A programmer is developing a computer game in a high-level language to sell to the public.

(a) The programmer uses both an interpreter and a compiler at different stages of the development of the program.

(i) Explain the reasons why the programmer uses an interpreter while writing the program code.

[2]

(ii) Explain the reasons why the programmer uses a compiler when the program has been written.

[3]

28. M/J 23/P21/Q1/a

- 1 A programmer has written a program which includes the function `Calculate()`. When the program is run, the function returns an unexpected value.

(a) Describe how a typical Integrated Development Environment (IDE) could be used to help debug the program to find the errors in the function `Calculate()`.

[4]

29. M/J 23/P22/Q5/c

- 5 A programmer has produced the following pseudocode to output the square root of the numbers from 1 to 10.

Line numbers are for reference only.

```
10 DECLARE Num : REAL
11 Num ← 1.0
...
40 REPEAT
41     CALL DisplaySqrt(Num)
42     Num ← Num + 1.0
43 UNTIL Num > 10
...
50 PROCEDURE DisplaySqrt(BYREF ThisNum : REAL)
51     OUTPUT ThisNum
52     ThisNum ← Sqrt(ThisNum)           // Sqrt returns the square root
53     OUTPUT " has a square root of ", ThisNum
54 ENDPROCEDURE
```

The pseudocode is correctly converted into program code.

Function `Sqrt()` is a library function and contains no errors.

The program code compiles without errors, but the program gives unexpected results. These are caused by a design error.

- (c) Describe how a typical Integrated Development Environment (IDE) could be used to identify this error.

[3]

30. O/N 22/P11/Q7/a,b

7 (a) State **two** benefits to a programmer of using Dynamic Link Library (DLL) files.

1

2

[2]

(b) Memory management is one of the tasks performed by an Operating System (OS).

Describe the ways in which memory management organises and allocates Random Access Memory (RAM).

[2]

31. O/N 22/P12/Q1/a,b

- 1 (a) Draw **one** line from each utility software to its most appropriate purpose.

Utility software	Purpose
virus checker	to reorganise files so they are contiguous
disk formatter	to scan for malicious program code
backup	to decrease the file size
disk repair	to initialise a disk
defragmentation	to create copies of files in case the original is lost
	to check for and fix inconsistencies on a disk

[5]

- (b) Compilers and interpreters translate programs written in a high-level language into a low-level language.

- (i) State **two** drawbacks of using a compiler compared to an interpreter during program development.

1

2

[2]

- (ii) Explain why high-level language programs might be partially compiled and partially interpreted.

[2]

32. O/N 22/P13/Q3

- 3 Draw **one** line from each Operating System (OS) management task to its most appropriate description.

OS Management task	Description
hardware management	dynamically allocates memory to processes
security management	marks unallocated file storage for availability
memory management	installs programs for devices connected to external ports
process management	validates user and process authenticity
	allows processes to transfer data to and from each other

[4]

33. O/N 21/P11/Q4/b(ii),d

4 Francis is starting his first job as a software developer for a multinational company.

(b) Francis is shown the software he will be working on. He is unfamiliar with the Integrated Development Environment (IDE) he is required to use.

(ii) A typical IDE provides debugging tools to support the testing of a program.

Identify **three** other tools or features found in a typical IDE to support the writing of the program.

1

2

3

[3]

(d) Francis's team use language translators.

Complete the descriptions of language translators by writing the missing words.

..... are usually used when a high-level language program is complete. They translate all the code at the same time and then run the program.

They produce files that can be run without the source code.

..... translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line.

Assemblers are used to translate assembly code into

[4]

34. O/N 21/P12/Q7/b,c

7 A computer has system software.

(b) Describe the file management tasks that an Operating System performs.

[4]

(c) Identify **two** utility programs that can be used to improve the performance of a computer **and** state how they improve the performance.

1

2

[4]

35. O/N 21/P13/Q4/b(ii),d

4 Francis is starting his first job as a software developer for a multinational company.

(b) Francis is shown the software he will be working on. He is unfamiliar with the Integrated Development Environment (IDE) he is required to use.

(ii) A typical IDE provides debugging tools to support the testing of a program.

Identify **three** other tools or features found in a typical IDE to support the writing of the program.

1

2

3

[3]

(d) Francis's team use language translators.

Complete the descriptions of language translators by writing the missing words.

..... are usually used when a high-level language program is complete. They translate all the code at the same time and then run the program.

They produce files that can be run without the source code.

..... translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line.

Assemblers are used to translate assembly code into

[4]

36. M/J 21/P11/Q2/a,b

2 Bingwen's computer comes with an Operating System and utility software.

(a) Draw **one** line from each utility software to its correct description.

Utility software	Description
	Scans software for errors and repairs the problems
Disk formatter	Moves parts of files so that each file is contiguous in memory
Defragmentation	Creates a copy of data that is no longer required
Back-up	Sets up a disk so it is ready to store files
Disk repair	Scans for errors in a disk and corrects them
	Creates a copy of data in case the original is lost

[4]

(b) Identify **four** key management tasks that the Operating System will perform.

1

2

3

4

[4]

37. M/J 21/P12/Q7/a,b

7 Jennifer is writing a computer program for her A Level homework.

- (a)** Jennifer uses a program library to help her write her computer program.

Describe how a program library can be used while writing a computer program.

[2]

- (b)** Jennifer uses an Integrated Development Environment (IDE) to write her computer program.

- (i)** The IDE allows Jennifer to use both an interpreter and a compiler while creating her computer program.

Describe the ways in which Jennifer can use **both** a compiler **and** an interpreter while developing the program.

[4]

- (ii)** Identify **two** debugging tools that a typical IDE can provide.

1

2

[2]

38. M/J 21/P13/Q2/a,b

2 Bingwen's computer comes with an Operating System and utility software.

(a) Draw **one** line from each utility software to its correct description.

Utility software	Description
	Scans software for errors and repairs the problems
Disk formatter	Moves parts of files so that each file is contiguous in memory
Defragmentation	Creates a copy of data that is no longer required
Back-up	Sets up a disk so it is ready to store files
Disk repair	Scans for errors in a disk and corrects them
	Creates a copy of data in case the original is lost

[4]

(b) Identify **four** key management tasks that the Operating System will perform.

1

2

3

4

[4]

Answers Section

1. M/J 26/P11/Q2/a,b,c

2(a)	1 mark each to max 2 - To convert a high-level language to a low-level language/assembly/machine code - Computers use machine code and programmers use high-level languages - To allow the computer to process the coded statements	2	Accept human language for HLL in MP2 MP3: Read is NE
2(b)	1 mark for each benefit and 1 mark for an appropriate expansion to max 4 (2 × 2) For example: - Amend the code or debug in real time - and see the changes immediately instead of having to recompile - Stops as soon as an error is encountered - so the position of the error can be easily seen - Only one error is displayed at a time // no dependent errors generated - individual errors are easier to identify/debug/fix - Incomplete code can be tested/run - so the program does not need to be complete to see if it works/see the output	4	
2(c)	1 mark for a correct answer Assembler	1	Correct answer only

2. M/J 26/P12/Q7/a,b

7(a)(i)	1 mark for each correct point to max 2 - Efficient allocation of resources to processes - Moving processes between states - Allocating time slices to processes - Scheduling processes // An example of a scheduling algorithm - Resolution of conflicts when two processes require the same resource - Allows communication between processes	2	Do not accept allows multi-programming – that is a benefit not a task resource management is NE Accept checking priorities as scheduling Take care that answers do not refer to tasks done by the kernel or by memory management.
7(a)(ii)	1 mark for each correct point to max 2 - Dynamically allocates RAM to processes - Frees up memory when the process closes - Makes sure two processes do not access the same memory - Accesses data from RAM when needed	2	Accept software or applications for processes in first mark point Take care that answers do not refer to tasks done by process management or by file management
7(a)(iii)	1 mark for each correct point to max 3 - With continued use files can be split and stored in different locations on the disk so time is spent by the disk head moving between locations to access all parts of file - Defragmentation software moves split files into contiguous blocks // defragmentation software collects all the free space into a contiguous block ... which means that the disk head does not need to move as much to access the next part of the file // ... which means that the disk head does not need to move as much when writing a file - After defragmentation data access/read/write takes less time, so performance is improved	3	Note: The question asks how defragmentation improves computer performance. So just a description of the process is NE

7(b)(i)	1 mark for each correct point to max 4 , mark in pairs • (Single) Stepping • ...the student runs one line of code at a time and looks at the output/results/flow before moving to run the next line • Breakpoints • ...These are inserted in the code to stop the code at a specific point so that the student can check the flow/values / test sub-sections of the code • (Variable) Watch window • ...the student can view how/when the variables change during execution	4	Take care that the answer says how the student uses the tools and doesn't just describe each tool Award a maximum of two marks for names of debugging tools with no further suitable explanation Just check for errors is not enough for an explanation, that is what debugging means
7(b)(ii)	1 mark for each correct point to max 2 • Partially compiled programs can be used on different platforms as they are interpreted when run • The code is optimised for the CPU as machine code is generated at run time • It is more efficient to run as the source code does not need recompiling	2	Note: Candidate responses need to answer the question why?

3. M/J 26/P13/Q5/a,b,c

5(a)(i)	1 mark each to max 2 e.g. • Transfer data between peripheral devices • Translate data from I/O/peripheral devices • Loading and installation of device drivers • Hardware abstraction — hiding hardware differences from applications • Power management — sleep states, CPU throttling, battery control	2	
5(a)(ii)	1 mark each to max 2 e.g. • Memory management • Process management • File management • Security management // managing user accounts • Provide platform for running applications	2	
5(b)(i)	1 mark for: • Perform housekeeping/maintenance tasks // improving/optimising overall performance	1	
5(b)(ii)	1 mark each to max 2 e.g. • Prepared a disk for future use by // initialising • ...changing/setting file system // by example // directory structure // allocation tables • ... naming the disk (volume label) • ... creating a boot sector • ...erasing all data • ...creating partitions	2	
5(c)(i)	1 mark for: • Set of pre-written / pre-compiled / pre-tested routines	1	

5(c)(ii)	1 mark for benefit 1 for expansion e.g. - Maintenance not needed to be done by the programmer ... because the DLL is separate from program // maintained by library creator - The calling program does not need recompilation by the programmer when a DLL file changes ... because the DLL file can be updated independently of the calling program ... updates will apply to all programs that use the DLL file (Executable) file is smaller ... because it does not contain (all) the library routines // code only linked at execution (Main) memory requirements for program is reduced ... because the DLL files are only loaded into memory when required ... one DLL file can be used by several running programs	2	
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4. M/J 26/P21/Q2/b

2(b)(i)	Report window // Watch window	1
2(b)(ii)	103	1

5. M/J 26/P23/Q2/d

2(d)(i)	<u>breakpoint</u>	1	
2(d)(ii)	38 (minutes)	1	

6. O/N 25/P11/Q5/a,b,c

5(a)(i)	1 mark per bullet point, max 2 marks - Installs <u>driver</u> software for devices connected to external ports - Manages communication between devices - Manages <u>hardware</u> interrupts	2
5(a)(ii)	1 mark per bullet point, max 2 marks - Prevents unauthorised access ... by providing authentication // by validating users and processes - Implements access rights and permissions - Makes provision for recovery of lost data - Carries out Operating System security updates as available - Carries out auditing and keeps logs of activity	2

5(b)(i)	One mark for benefit, one mark for appropriate matching expansion e.g. • Less code needs to be developed as code is pre-written • ... thereby reducing the time to write a program • The library routine will be pre-tested and documented • ... reducing the time taken to test a program // giving confidence that the routine works as it should • The code for the library routine can be written in a different programming language • ... which allows the use of special features of that language • The user's source code is more readable • ... as the library routine is simply called in the source code rather than being written out in full • The library could contain the code for complex routines • ... thus allowing the programmer to include modules they might not be able to code	2
5(b)(ii)	1 mark for drawback, 1 mark for appropriate matching expansion e.g. • Compatibility issues • ... the library routine may not work with current program • Not guaranteed to be thoroughly tested • ... there could be unexpected problems (bugs/virus) • The library routine may not match needs exactly // The library routine may give unexpected results • ... so it will need to be edited and tested • If the library routine is later edited or changed (by someone else) • ... there could be unexpected errors or results	2
5(c)	1 mark per bullet point, max 2 marks • Prettyprint • Expanding and collapsing code blocks • <u>Automatic</u> / <u>smart</u> indentation / formatting	2

7. O/N 25/P12/Q8/b(ii)

8(b)(ii)	1 mark per bullet point, max 2 marks • Disk repair / disk contents analysis • Back-up software	2
----------	--	----------

8. O/N 25/P12/Q10/a

10(a)	1 mark per bullet point, max 4 marks Max 2 from each section. Interpreter: • Use an interpreter while writing/coding the program • ... to test / debug the partially completed program • ... because errors can be corrected and processing continue from where the execution stopped // errors can be corrected in real time // errors are identified one at a time Compiler: • Use the compiler after the program is complete • ... to create an executable file so source code not seen • Use the compiler to repeatedly test the same completed section without having to re-interpret every time	4
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9. O/N 25/P13/Q8/a,b,d

8(a)	1 mark per bullet point, max 2 • Large amounts of source code take time to compile • It can be slower to produce the object code than an interpreter • The code must be recompiled when it is changed • The program cannot run if there are errors • It is not possible to correct errors in real-time • One error can cause false reporting of multiple further errors • Sections of code / unfinished code cannot easily be tested	2
8(b)	1 mark per bullet point, max 3 marks • Programming time is saved as code does not have to be written from scratch • Testing time is saved as code is already tested / documented • A library routine is more likely to work, as code is already tested • Library routines automatically update if they are changed / improved • The programmer can use library routines to perform complex functions / procedures that they may not be able to write themselves	3
8(d)	1 mark for each bullet point, max 3 marks • File management • Security management • Hardware / peripheral management // input / output management • Process management	3

10. O/N 25/P21/Q1/c

1(c)	One mark per point: 1 reference to a breakpoint 2 reference to single stepping 3 Correct explanation of both e.g. to stop program execution at specific point / line / statement / instruction and to execute one line / statement / instruction at a time to (locate an/the error) One mark for each bulleted point	3
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11. M/J 25/P11/Q6/b

6(b)	1 mark each to max 5 e.g. • To provide a user interface • so that the user is able to communicate with the hardware • To manage memory • so that data can be stored and accessed • and multitasking is possible • To manage files • allowing the user to create, edit, update and delete files and folders • To manage inputs and outputs from hardware/peripherals • To handle processes • to make sure each process has fair access	5
------	--	----------

12. M/J 25/P13/Q3/a

3(a)	1 mark for each correctly completed term • an executable file/.exe • source/program code • stops • immediately/in real-time A compiler checks all of the code before attempting to translate the program. If any errors are found, they are all reported at the same time and the program does not translate or run. If there are no errors found, the compiler produces an executable file/.exe which can run without access to the source/program code. An interpreter translates one line of code and then runs it, before moving to the next line of code. If the line of code has an error, the interpreter stops and displays the error. The programmer can correct the error immediately/in real-time and then the interpreter continues translating from that point.	4
------	---	----------

13. O/N 24/P11/Q4/a,c,d

4(a)(i)	1 mark for each bullet point (max 2) e.g. • Secondary storage space is divided into file allocation units • Space on secondary storage is allocated to particular files • Maintains/creates directory structures • Specifies the logical method of file storage e.g. FAT or NTFS • Provides file naming conventions • Controls user access // implements access rights //implements password protection // Makes file sharing possible • Controls access to the file by other software • Specifies tasks that can be performed on a file e.g. open, close, delete, copy, create, move, etc • Allows searching for a file	2
4(a)(ii)	1 mark for each bullet point (max 2) • To allow data to be retrieved / restored when lost • To automatically make a duplicate copy of data... • ...so the user does not have to remember to backup data • To make regular duplicate copies of data	2
4(c)	1 mark for each bullet point (max 3) e.g. • It saves development time for the student • ... by using pre-written code • There is less testing to do // the routines are more likely to work correctly • ... because they have already been tested and are error-free • It results in a more robust program • Subroutines can be used that they do not have the skill to write • ... which enables a more complex program to be written	3
4(d)(i)	1 mark for correct name, 1 mark for matching description • Expand/collapse code blocks... • ... sections of source code that are part of the same block can be expanded to see the content or collapsed so that the overall code is seen • Auto-indentation // auto-formatting... • ... automatically indents/formats code as the user types so that the structure is clear // aids readability	2
4(d)(ii)	1 mark for correct name, 1 mark for matching description • Breakpoints ... • ... stops the code running on a set line to view the current status/variable contents/program flow • Report window // variable watch window ... • ... shows the values in variables/data structures and how they change when each line is run	2

14. O/N 24/P12/Q3/c

3(c)	1 mark for each bullet point (max 4) e.g. • It manages the scheduling of processes // Decides which process is to be run next • Allows multi-tasking/multi-processing • Ensures fair access • Handles interrupts • Manages / allocates which resources the processes require • Facilitates the sharing and exchange of data between processes • Prevents interference between processes // conflict resolution	4
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15. O/N 24/P12/Q4/a,b

4(a)	1 mark for each bullet point (max 4) e.g. • Single stepping • Run the program one line at a time • ... and check the variable contents / program flow // show the effect of each line of code • Set breakpoints • ... run the code up to a set line • ... and then check the status • Variable/report watch window • ... view how the data changes as the program is running	4
4(b)	1 mark for each bullet point (max 3) e.g. • Subroutines can be shared / reused • ... between team members who are working independently • ... without having to rewrite/re-test them which saves the programmers' time • A program library provides continuity between programs/programmers • Individual programmers can contribute their specialisms to the library // Individual programmers can use the specialisms of others	3

16. O/N 24/P22/Q1/b,c

1(b)	For example: • Set a breakpoint at the start of / within <code>Lookup</code>, to stop execution at a given statement • then use single stepping to execute one statement / instruction at a time • to display the value of variables using a watch window One mark for each: MP1 Order starting with a breakpoint and an explanation – ‘stop execution at this statement / line’ MP2 Explanation of single stepping – execute ‘line by line’ / statements MP3 Explanation of watch window – displaying the value of <u>variable(s)</u>	3
1(c)	Features include: MP1 Editor MP2 <u>Auto</u>- (syntax) complete / <u>auto</u> correction // identify undeclared variables(s) MP3 Prettyprint / <u>auto</u>-indentation / <u>auto</u> (structure) highlighter MP4 <u>Dynamic</u> syntax checking MP5 Expand / collapse code blocks MP6 Context sensitive prompts Max 2 marks	2

17. M/J 24/P11/Q3/a,b

3(a)	1 mark each to max 2: • The interpreter will stop when an error is found • ... so the error can be corrected in real-time, and the result of changes seen immediately • Only one error is displayed at a time • ... so fewer errors to correct simultaneously and no dependent errors	2
3(b)	1 mark each to max 3: • Program can be distributed without source code • ... so it cannot be edited/stolen/plagiarised • Users do not require the translator to run the program • ... so time is not spent retranslating by user	3

18. M/J 24/P12/Q6

6	1 mark each to max 4 Max 3 marks for each management task: <i>Memory management: Max 3 marks</i> • Stores data from all currently running programs concurrently in RAM • Stops the data from overwriting each other in RAM/primary storage • Decides which processes should be in main memory • Makes efficient use of memory <i>Process management: Max 3 marks</i> • Allows one process to be paused whilst another process can be actioned • Decides which process is to be run next • Switches between processes to allow them to share the use of the processor • Identification/description of scheduling	4
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19. M/J 24/P12/Q8/a,b,c

8(a)	1 mark each to max 2: • Creates an executable file • ... so the code can be tested multiple times without having to recompile • ... so repeated testing takes less time	2
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8(b)	1 mark for identification of each feature and 1 mark for matching description: e.g. <i>For coding:</i> <table><tr><th>IDE feature</th><th>Description</th></tr><tr><td>Context-sensitive prompts</td><td>Gives suggestions for code as the user types instead of having to write/remember the code</td></tr><tr><td>Auto-correct</td><td>Corrects spelling mistakes so that user has fewer errors to correct</td></tr></table> <i>For presentation:</i> <table><tr><th>IDE feature</th><th>Description</th></tr><tr><td>Pretty-printing</td><td>Colour code keywords so the user can identify any errors</td></tr><tr><td>Expand/collapse (code) blocks</td><td>The user can hide code that they are not currently working on</td></tr></table> <i>For debugging:</i> <table><tr><th>IDE feature</th><th>Description</th></tr><tr><td>Single stepping</td><td>Run the code one line at a time // shows the effect of each line of code</td></tr><tr><td>Breakpoints</td><td>Stop the code running at a set point to check the flow/variable contents</td></tr></table>	IDE feature	Description	Context-sensitive prompts	Gives suggestions for code as the user types instead of having to write/remember the code	Auto-correct	Corrects spelling mistakes so that user has fewer errors to correct	IDE feature	Description	Pretty-printing	Colour code keywords so the user can identify any errors	Expand/collapse (code) blocks	The user can hide code that they are not currently working on	IDE feature	Description	Single stepping	Run the code one line at a time // shows the effect of each line of code	Breakpoints	Stop the code running at a set point to check the flow/variable contents	6
IDE feature	Description																			
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Single stepping	Run the code one line at a time // shows the effect of each line of code																			
Breakpoints	Stop the code running at a set point to check the flow/variable contents																			
8(c)	1 mark each to max 2: • Saves programming/testing time as code does not have to be written/re-written from scratch // code does not have to be tested • Code is already tested so it is more robust/likely to work • The programmer does not need to maintain the library // library routines are updated automatically • Can perform complex calculations that the programmer may be unable to do • Makes code more easily readable	2																		

20. M/J 24/P13/Q7/e

7(e)(i)	1 mark for: Pre-written code/functions/routines that can be imported/called in another program	1
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7(e)(ii)	1 mark each to max 4 : • Maintenance not needed to be done by the programmer • ... because the DLL is separate from program • The calling program does not need recompilation by the programmer when a DLL file changes • ... because the DLL file can be updated independently of the calling program • ... updates will apply to all programs that use the DLL file	4
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21. O/N 23/P11/Q6/a,b,c

6(a)	1 mark for each bullet point (max 4) • easier to debug the program • ... because it translates line-by-line and stops when an error is found whereas the compiler translates all the program at the same time • ... only reporting one error at a time • ... which allows the error to be corrected in real time whereas the program would need to be corrected and recompiled • ... and the program can restart at same point when error occurred with a compiler the program needs to be re-run • The effect of any changes made by the programmer can be seen immediately with a compiler the effects can only be seen after re-running • A partially completed program can be translated / tested on its own a compiler cannot translate a partial program	4
6(b)	1 mark for each bullet point (max 1) • Partially compiled programs can be used on different platforms as they are interpreted when run • Code is optimised for the CPU as machine code is generated at run time • Source code does not need recompiling so more efficient to run	1
6(c)(i)	1 mark for each bullet point (max 2) • Prettyprint • Expand/collapse code blocks • Auto indentation / formatting	2
6(c)(ii)	1 mark for each bullet point (max 2) • Single stepping • Breakpoints • Report window • Variable expressions	2

22. O/N 23/P12/Q8/a,b

8(a)	1 mark for each bullet point (max 1) - To hide the complexities of the hardware from the user - To provide a platform for software to run - To provide a user interface	1
8(b)	1 mark for the name of the utility software 2 marks for the explanation - Defragmentation software ... because over time saving and deleting of small files fragments the disk ... the software makes (individual) files contiguous ... so access time to the files is improved ... because head movement is reduced - Disk contents analysis/disk repair software ... to identify and mark bad sectors ... to restore corrupted files ... to recover lost data (due to hardware failure) - File compression ... to reduce the size of files ... which saves storage and memory space // by example ... and reduces transmission time // by example - Disk formatter ... to prepare a disk for use // set up the file system ... to partition the disc ... to delete all the data from the disc	3

23. O/N 23/P13/Q5/a,b(ii)

5(a)	1 mark for each bullet point (max 2 for each feature). Context-sensitive prompts: - As the code is being written ...the options to complete the statement are shown Single stepping: - allows the programmer to execute the program one line at a time ...so that the effects of each statement can be seen	4
5(b)(ii)	1 mark for each bullet point (max 3). - Disk needs to be prepared for initial use - Disk needs to be checked for errors - A new file system needs to be generated on the disk - The file allocation table needs to be set up	3

24. M/J 23/P11/Q3/a,b,c

3(a)	1 mark each to max 4 Examples: • Installs device drivers • ... to allow communication between peripherals and computer • Sends data and receives data to and from peripherals • ... such as to an output device and from an input device/by example • Handles buffers for transfer of data • ... to ensure smooth transfer between devices that transmit and receive at different speeds • Manages interrupts / signals from the device	4
3(b)	1 mark each to max 2 • Memory management • File management • Security management • Process management • Error checking and recovery	2
3(c)	1 mark each to max 3 • Rearranges blocks of individual files (on the HDD) so they are contiguous // moves the free space together • Accessing each file is faster • ...because there is no need to search for the next fragment / block of the file • ...so less head movement is needed	3

25. M/J 23/P12/Q5/d

5(d)(i)	1 mark each to max 5 • Manages the scheduling of processes // decides which order to run processes • Manages which resources the processes require • ... such as allocating memory • Enables processes to share data • Prevents interference between processes // resolution of conflicts • Handles the process queue • It allows multi-tasking / multi-processing • ... by ensuring fair access, handling priorities and handling interrupts	5
5(d)(ii)	1 mark each to max 2 • To help users to set-up / configure / analyse / optimise / maintain the computer ... • ... by for example, making memory allocation more efficient • ... by for example, checking the system for faults	2

26. M/J 23/P12/Q7/a,b,c

7(a)(i)	1 mark each to max 2 • Set of pre-written / pre-compiled / pre-tested subroutines • ... which can be called in other programs • ... by installing/importing the library	2
7(a)(ii)	1 mark for each bullet point. Mark in pairs; 1 mark for a benefit and 1 mark for an appropriate expansion • (main) memory requirements for program is reduced • ... as dynamic link library is loaded only once / when required • the executable file size of the program using the DLL will be smaller • ... because the executable does not contain (all) the library routines • maintenance not needed to be done by the programmer • ... because the DLL is separate from program • no need to recompile the main program when changes are made to DLL • ... because changes / improvements/ error correction to the DLL file code are done independently of the main program • A single DLL file can be made available to several application programs • ... Saving space in memory / easing the pressure on memory	4
7(b)	No mark for choice. 1 mark each to max 3 for justification Interpreter • Allows the developer to make real-time changes • ... so the program can be debugged at each stage • ... the effect of any changes made by the developer can be seen immediately • The developer can test when incomplete • ... so small parts can be tested without having to test the rest of the program • ... if one section does not work others can still be tested • To avoid dependent errors Compiler • The developer can debug multiple errors simultaneously • Produces an executable file • ... so that the developer can test the program multiple times without recompiling	3

7(c)	1 mark each Breakpoints: • Stop the code at a specific line to check the current progress / values Dynamic syntax checks: • Highlight / underline / colour syntax errors as the code is entered Context-sensitive prompts: • Suggest the code to add // automatically complete statements Single stepping: • Run the code one line at a time so the values can be checked	4
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27. M/J 23/P13/Q5/a

5(a)(i)	1 mark each to max 2 • Programmer can test sections of the code without every part working / being written • Programmer can debug in real time • ... so that errors can be fixed and the program continued from that point • The effect of any changes made by the programmer can be seen immediately • To avoid dependent errors	2
5(a)(ii)	1 mark each to max 3 • The compiler produces an executable file • ... so the user cannot access / edit / sell the code • ... and users do not need the translator to run the game • The game can be compiled for different hardware specifications • ... and then used to generate more income for the programmer • The program can be tested multiple times without having to retranslate each time	3

28. M/J 23/P21/Q1/a

1(a)	For example: Could set a <u>breakpoint</u> to stop the program at a particular point / instruction then the value of variables could be checked using a <u>report/watch window</u> while <u>single stepping</u> can be used to execute one statement/line at a time. Marks available as follows: • One mark for each underlined term • One mark for an explanation of each term Note: max 4 marks	4
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29. M/J 23/P22/Q5/c

5(c)	Could use an IDE to: • Set a breakpoint to stop the program at a certain line / statement / point • Step through the program line by line / statement by statement • checking the value of 'num' / a variable using a report / watch window One mark per bullet	3
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30. O/N 22/P11/Q7/a,b

7(a)	1 mark for each benefit (max 2): • (main) memory requirements for program are reduced as dynamic link library is loaded only once / when required • the executable file size is smaller because the executable does not contain all the library routines • maintenance not needed to be done by the programmer because the DLL is separate from program • no need to recompile the main program when changes are made to DLL because changes / improvements/ error correction to the DLL file code are done independently of the main program	2
7(b)	1 mark for each bullet point (max 2): • RAM is assigned into blocks • dynamic allocation of RAM to programs / processes • reclaims unused blocks of RAM • prevents two programs / processes occupying the same area of RAM at the same time • moves data from secondary storage when needed // manages paging, segmentation and virtual memory	2

31. O/N 22/P12/Q1/a,b

1(a)	1 mark for each correct line. <table><thead><tr><th>Utility software</th><th>Purpose</th></tr></thead><tbody><tr><td>virus checker</td><td>to reorganise files so they are contiguous</td></tr><tr><td>disk formatter</td><td>to scan for malicious program code</td></tr><tr><td>backup</td><td>to decrease the file size</td></tr><tr><td>disk repair</td><td>to initialise a disk</td></tr><tr><td>defragmentation</td><td>to create copies of files in case the original is lost</td></tr><tr><td></td><td>to check for and fix inconsistencies on a disk</td></tr></tbody></table>	Utility software	Purpose	virus checker	to reorganise files so they are contiguous	disk formatter	to scan for malicious program code	backup	to decrease the file size	disk repair	to initialise a disk	defragmentation	to create copies of files in case the original is lost		to check for and fix inconsistencies on a disk	5
Utility software	Purpose															
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defragmentation	to create copies of files in case the original is lost															
	to check for and fix inconsistencies on a disk															
1(b)(i)	1 mark for each bullet point (max 2): • larger amounts of source code take time to compile • slower to produce the object code than an interpreter • code cannot be changed without recompilation • the program will not run if there are any errors • errors cannot be corrected in real-time • one error may result in other false errors being reported • cannot easily test specific sections of the source code // cannot easily test unfinished source code	2														
1(b)(ii)	1 mark for each bullet point (max 2): • partially compiled programs can be used on different platforms as they are interpreted when run • code is optimised for the CPU as machine code is generated at run time	2														

32. O/N 22/P13/Q3

3	1 mark for each correct line: <table><tr><th>OS Management task</th><th>Description</th></tr><tr><td>hardware management</td><td>dynamically allocates memory to processes</td></tr><tr><td>security management</td><td>marks unallocated file storage for availability</td></tr><tr><td>memory management</td><td>installs programs for devices connected to external ports</td></tr><tr><td>process management</td><td>validates user and process authenticity</td></tr><tr><td></td><td>allows processes to transfer data to and from each other</td></tr></table>	OS Management task	Description	hardware management	dynamically allocates memory to processes	security management	marks unallocated file storage for availability	memory management	installs programs for devices connected to external ports	process management	validates user and process authenticity		allows processes to transfer data to and from each other	4
OS Management task	Description													
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memory management	installs programs for devices connected to external ports													
process management	validates user and process authenticity													
	allows processes to transfer data to and from each other													

33. O/N 21/P11/Q4/b(ii),d

4(b)(ii)	1 mark for each correct tool e.g. • Colour coding // pretty printing • Auto-complete • Auto-correct • Context sensitive prompts • Expand and collapse code blocks	3
4(d)	1 mark for each correctly completed term Compilers are usually used when a high-level language program is complete. They translate all the code at the same time and then run the program. They produce executable/.exe/object code files that can be run without the source code. Interpreters translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line. Assemblers are used to translate assembly code into binary/machine code.	4

34. O/N 21/P12/Q7/b,c

7(b)	1 mark for each bullet point to max 4 - Storage space is divided into file allocation units - Space is allocated to particular files - Maintains / creates directory structures - Specifies the logical method of file storage (e.g. FAT or NTFS) - Provides file naming conventions - Controls access // implements access rights // implements password protection // Makes file sharing possible - Specifies tasks that can be performed on a file (e.g. open, close, delete, copy, create, move etc.)	4
7(c)	1 mark for identifying program 1 mark for description, max 2 per program e.g. - Defragmentation - Less time is taken to access files because each one is contiguous so there is less head movement - Virus checker - makes more RAM available for programs to run ... because it removes software that might be taking up memory / replicating - Disk repair / Disk contents analysis - preventing bad sectors being used because it identifies / marks them - reduces access times by optimising storage - Disk/system clean up - releases storage by removing unwanted / temporary files	4

35. O/N 21/P13/Q4/b(ii),d

4(b)(ii)	1 mark for each correct tool e.g. - Colour coding // pretty printing - Auto-complete - Auto-correct - Context sensitive prompts - Expand and collapse code blocks	3
4(d)	1 mark for each correctly completed term Compilers are usually used when a high-level language program is complete. They translate all the code at the same time and then run the program. They produce executable/.exe/object code files that can be run without the source code. Interpreters translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line. Assemblers are used to translate assembly code into binary/machine code.	4

36. M/J 21/P11/Q2/a,b

2(a)	1 mark for each correct line <table><thead><tr><th>Utility software</th><th>Description</th></tr></thead><tbody><tr><td>Disk formatter</td><td>Scans software for errors and repairs the problems</td></tr><tr><td>Defragmentation</td><td>Moves parts of files so that each file is contiguous in memory</td></tr><tr><td>Back-up</td><td>Creates a copy of data that is no longer required</td></tr><tr><td>Disk repair</td><td>Sets up a disk so it is ready to store files</td></tr><tr><td></td><td>Scans for errors in a disk and corrects them</td></tr><tr><td></td><td>Creates a copy of data in case the original is lost</td></tr></tbody></table>	Utility software	Description	Disk formatter	Scans software for errors and repairs the problems	Defragmentation	Moves parts of files so that each file is contiguous in memory	Back-up	Creates a copy of data that is no longer required	Disk repair	Sets up a disk so it is ready to store files		Scans for errors in a disk and corrects them		Creates a copy of data in case the original is lost	4
Utility software	Description															
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	Creates a copy of data in case the original is lost															
2(b)	1 mark per bullet point to max 4 • memory management • file management • security management • hardware / device / peripheral / resources management • input/output management • process management • error checking and recovery • provision of a platform for software • provision of a user interface	4														

37. M/J 21/P12/Q7/a,b

7(a)	1 mark per bullet point to max 2 • Program libraries store pre-written functions and routines • The program library can be referenced/imported • the functions/routines can be called in her own program	2
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7(b)(i)	1 mark per bullet point to max 4; max 3 from each section Interpreter: • Use an interpreter while writing the program • ... to test/debug the partially completed program • ... because errors can be corrected and processing continue from where the execution stopped // errors can be corrected in real time // errors are identified one at a time Compiler: • Use the compiler after the program is complete • ... to create an executable file • Use the compiler to repeatedly test the same (completed) section • ... without having to re-interpret every time // compiler not needed at run-time	4
7(b)(ii)	1 mark per correct tool to max 2 e.g. • Breakpoints • Single stepping • Report windows	2

38. M/J 21/P13/Q2/a,b

2(a)	1 mark for each correct line <table><tr><th>Utility software</th><th>Description</th></tr><tr><td>Disk formatter</td><td>Scans software for errors and repairs the problems</td></tr><tr><td>Defragmentation</td><td>Moves parts of files so that each file is contiguous in memory</td></tr><tr><td>Back-up</td><td>Creates a copy of data that is no longer required</td></tr><tr><td>Disk repair</td><td>Sets up a disk so it is ready to store files</td></tr><tr><td></td><td>Scans for errors in a disk and corrects them</td></tr><tr><td></td><td>Creates a copy of data in case the original is lost</td></tr></table>	Utility software	Description	Disk formatter	Scans software for errors and repairs the problems	Defragmentation	Moves parts of files so that each file is contiguous in memory	Back-up	Creates a copy of data that is no longer required	Disk repair	Sets up a disk so it is ready to store files		Scans for errors in a disk and corrects them		Creates a copy of data in case the original is lost	4
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2(b)	1 mark per bullet point to max 4 • memory management • file management • security management • hardware / device / peripheral / resources management • input/output management • process management • error checking and recovery • provision of a platform for software • provision of a user interface	4
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Unit 6

SECURITY, PRIVACY AND DATA INTEGRITY

32 items · 163 marks

Syllabus sections in this unit

6.1 Data Security

6.2 Data Integrity

Unit 6: Security, privacy and data integrity

1. M/J 26/P11/Q4/d

- 4 A shop allows customers to pay to borrow toys. The shop uses a database to record information about the customers and the toys they have borrowed.

CUSTOMER (CustomerID, ContactFirstName, Phone)

TOY (ToyID, ToyName, CostPerDay)

TOY_INSTANCE (UniqueToyID, ToyID, Quality)

BORROW (BorrowID, CustomerID, UniqueToyID, StartDate, NumberDays)

- (d) When data is entered into the database it is validated and verified.

- (i) Explain how validation and verification support data integrity within a database.

[4]

- (ii) Give **one** reason why data may still be incorrect even after it passes validation and verification.

[1]

2. M/J 26/P12/Q2/a,b,c

- 2 Personal data is entered into a computer using a keyboard. The data is validated and verified during data entry.

- (a) Describe **two** methods of data verification that can be used during data entry.

1

2

[4]

- (b) The personal data includes the date of birth, for example 05/07/2010.

Complete the table by identifying **two** methods of validation that can be used on the date of birth.

State how each validation method will be used for the date of birth.

data validation method	use
.....	
.....	

[3]

(c) The data needs to be kept secure whilst on the computer.

(i) A virus is a threat to data.

Complete the table by describing what is meant by a virus.

Identify **and** describe **one** method to restrict the risk posed by a virus.

virus description	
method to restrict risk	
description of method to restrict risk	

[3]

(ii) A hacker is another threat to data.

Complete the table by describing what is meant by a hacker.

Identify **and** describe **one** method to restrict the risk posed by a hacker.

hacker description	
method to restrict risk	
description of method to restrict risk	

[3]

3. M/J 26/P13/Q1

- 1** Validation and verification can be used to protect the integrity of data.

Complete the description of methods of ensuring data integrity by writing the missing terms.

..... is a method of verification where the data is entered by a user and then manually compared to the original data to make sure that they are the same.

A verification method used during data transfer calculates a value from the data which is transferred with the data. The receiver performs the same calculation and compares the result with the value transferred.

This method of verification is

There are several methods of data validation.

check compares the data entered to a list of possible values that can be entered. If the data is not one of these values it is rejected.

..... check can be used to make sure data entered has the correct combination of letters, numbers and symbols. For example, a code may need to be two capital letters, one number followed by one symbol.

..... check is used to make sure the correct quantity of characters has been entered. For example, a name must be more than one character and less than or equal to 100 characters.

[5]

4. O/N 25/P11/Q8/a

- 8** One ethical consideration for a student connecting their personal computer to the school network is the risk of spreading malware on the network.

(a) Viruses and pharming are examples of malware.

Explain what is meant by a virus and pharming.

Virus

Pharming

[2]

5. O/N 25/P12/Q1

- 1 Draw **one** line from each verification method to indicate whether it is used during data transfer or data entry.

Verification Method	Use
Parity byte check	
Checksum	Data transfer
Visual check	
Parity block check	Data entry

[2]

6. O/N 25/P12/Q5/e

- 5 A local area network (LAN) has five computers, one switch and one server.

- (e) Complete the table by identifying **one** threat to computer and data security posed by networks and the internet.

Describe the threat and give a method of prevention.

Threat	Description	Prevention

[3]

7. O/N 25/P12/Q6/c

- (c) A computer system uses even parity. The least significant (rightmost) bit of each byte is the parity bit.

- (i) Complete the byte by writing the missing parity bit:

							parity bit ↓
0	1	0	1	1	1	0	

[1]

- (ii) The computer also uses parity block check. The parity block check uses even parity. Computer A transmits four bytes of data to computer B, followed by a parity byte. Computer B receives the following sequence of bytes.

							parity bit ↓
1	0	1	1	0	1	1	1
0	1	1	1	0	0	0	0
0	0	0	1	1	0	1	1
0	1	1	1	0	1	0	0
parity byte →	1	0	1	0	0	0	0

Following transmission, one of the four bytes of data has an error in one of the bits.

Circle the bit that has been altered during the data transfer.

[1]

8. O/N 25/P12/Q10/c

- 10 A programmer is developing a computer program.

- (c) A user downloads the computer program from the internet.

State what should be included as part of the download to make sure the program is authentic.

[1]

9. O/N 25/P13/Q7/c,e

- 7 A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

- (c) Identify **and** describe **one** method of data verification that can be used when transferring data from the digital video camera to the server.

Method

Description

[3]

- (e) The table contains three algorithms that perform data validation. The third algorithm uses the functions `MID` and `LENGTH`.

The functions `MID` and `LENGTH` are defined as follows:

- `MID(ThisString : STRING, x : INTEGER, y : INTEGER)` RETURNS STRING
returns a string of length `y` starting at position `x` from `ThisString`
- `LENGTH(ThisString : STRING)` RETURNS INTEGER
returns the integer value representing the length of `ThisString`

Complete the table to identify the method of data validation for each algorithm. Each method of data validation must be different.

Algorithm	Method of data validation
<pre> INPUT x IF x < 1 or x > 26 THEN OUTPUT "Invalid" ENDIF </pre>	
<pre> INPUT x IF x <> 'R' AND x <> 'G' AND x <> 'B' THEN OUTPUT "Invalid" ENDIF </pre>	
<pre> INPUT x FLAG ← FALSE FOR INDEX ← 1 TO LENGTH(x) IF MID(x, INDEX, 1) = "@" THEN FLAG ← TRUE ENDIF NEXT INDEX IF NOT FLAG THEN OUTPUT "Invalid" ENDIF </pre>	

10. O/N 25/P21/Q6/a

- 6 Students are learning about a simple check digit method for data validation. In this method, a single check digit is appended to the end of an original number to give a new number.

The students are studying a method which:

- calculates the sum of all the digits in the original number
- uses integer division to calculate the remainder when the sum is divided by 10
- uses the remainder as the check digit
- appends the check digit to the original number, creating the new number.

For example:

original number	4162
sum of all digits	$4 + 1 + 6 + 2 = 13$
remainder when the sum is divided by 10 using integer division	3
new number	41623

The method described can be used to detect single-digit errors. For example, if the new number is incorrectly input as 41633, then the check digit does not match and the input will be rejected.

- (a) An incorrect attempt was made to enter 41623. The **first two** digits were entered incorrectly; the **last three** digits were entered correctly.

When this number was tested, the check digit was found to be correct and the input was accepted.

Identify an example of the incorrect attempt to enter 41623 and explain why this number would **not** be rejected.

Number

Explanation

[2]

11. M/J 25/P11/Q7/a,b,c

7 A banker stores personal data on their work computer.

(a) The banker needs to transfer confidential data across the internet.

Identify and describe **one** method of restricting the risks posed by an unauthorised person intercepting the data whilst it is being transferred across the internet.

Method

Description

[3]

(b) The banker receives confidential data across the internet. The data includes a digital signature.

Explain how a digital signature can make sure the data has **not** been changed during transmission.

[5]

(c) The data that is transferred can also be verified using a checksum.

Explain how data can be verified using a checksum.

[3]

12. M/J 25/P13/Q3/b

3 A programmer is writing a program in a high-level language.

- (b)** The programmer needs to keep the program files secure on their computer and during electronic data transmission over the internet. The files are protected by a password.

Complete the table by identifying **one** other method of keeping the files secure during electronic data transmission **and one** other method of keeping the files secure on the computer.

State how each method protects the data.

The methods must be different.

	Method	How the method protects the data
during transmission		
on the computer		

[4]

13. O/N 24/P12/Q4/c

4 A program is written in a high-level language by a team of three programmers using an Integrated Development Environment (IDE).

- (c)** The file containing the final program code will be sent by email for beta testing.

Identify **one** security method that can be used to protect the program code from unauthorised access during email transfer.

Explain how your chosen method protects the program code.

Security method

Explanation

[3]

14. M/J 24/P11/Q5/b,c(i)

- 5** A bank allows customers to access their accounts using an application that they can download onto a device such as a smartphone.

- (b)** The bank wants to protect the integrity of its data while transferring the data to other banks. Parity check is one example of data verification.

Complete the description of parity check when Computer A is transmitting data to Computer B.

Computer A and Computer B agree on whether to use

parity. Computer A divides the data into groups of The

number of 1s in each group is counted. If the agreed parity is

and the group has an even number of 1s, a parity bit of 1 is appended, otherwise a parity bit of 0 is appended.

In a parity check the bytes are grouped together, for example in a grid. The number of 1s in each column (bit position) is counted. A bit is assigned to each column to make the column match the parity. These parity bits are transmitted with the data as a parity

[5]

- (c)** The bank also needs to keep its customers' data private and secure.

- (i)** The bank's network has a firewall.

Explain how a firewall can help protect the customers' data.

[3]

15. M/J 24/P12/Q3/a

- 3** An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.

(a) The assessment board needs to make sure the data stored on the server is secure.

- (i)** Authentication methods can help to protect the server against hackers.

Identify **one other** security measure that helps to protect the server from hackers.

Describe how the security measure works.

Security measure

Description

[3]

- (ii)** Identify **one** security measure that helps to protect the data when it is being transmitted to its destination. Describe how the security measure works.

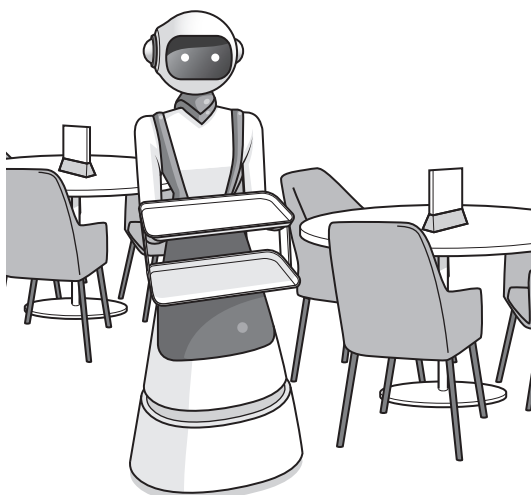
Security measure

Description

[3]

16. M/J 24/P13/Q7/f

- 7** Robots are used to serve food and drink to customers at a restaurant.



- (f)** The data from the robots is transmitted to a central computer using a wireless connection.

- (i)** Complete the table by identifying **and** describing **two** methods of data verification that can be used during data transfer.

	Method	Description
1		
2		

[4]

(ii) Explain how encryption can protect the security of data during transmission.

[2]

17. O/N 23/P12/Q5/a,b,c

5 (a) State the meaning of **privacy of data**.

[1]

(b) State the meaning of **integrity of data**.

[1]

(c) Describe the following threats to a computer system.

Phishing email

Spyware

[4]

18. O/N 23/P13/Q3/a(iii),c

- 3** A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

- (iii)** State how the following security measures can be used to protect computer systems.

Firewall

Encryption

Passwords

[3]

- (b)** The company uses a database, `COURSES`, to store data about the courses and their tutors.

Each course starts at different times of the year and may have a different tutor.

The database has the following structure:

```
COURSE_INFORMATION(CourseID, Description, Cost)
```

```
TUTOR(TutorID, TelephoneNumber, EmailAddress, TutorName)
```

```
COURSE_SCHEDULE(CourseID, DateStarted, TutorID)
```

- (c)** An example of a tutor ID is NK16C6.

An administrative officer enters the tutor ID into the `TUTOR` table.

Explain how data verification can be used when the tutor ID is entered.

[4]

19. O/N 23/P13/Q8/a,b,c

- 8** **(a)** Data verification is one method of protecting the integrity of data.

Describe **one** other method of protecting the integrity of data.

[2]

- (b)** State **one** difference and **one** similarity between pharming and phishing.

Difference

Similarity

[2]

- (c)** Explain how the data security risks of malware can be restricted.

[3]

20. M/J 23/P12/Q2/c(i)

- 2** A horse riding school uses a database, `Lessons`, to store data about lesson bookings.

This database is created and managed using a Database Management System (DBMS).

- (c)** The database `Lessons` has the following tables:

`HORSE(HorseID, Name, Height, Age, HorseLevel)`

`STUDENT(StudentID, FirstName, LastName, RiderLevel, PreferredHorseID)`

`LESSON(LessonID, Date, Time, StudentID, HorseID, LessonContent)`

Dates in this database are stored in the format `#DD/MM/YYYY#`.

The fields `RiderLevel` and `HorseLevel` can only have the values: `Beginner`, `Intermediate` or `Advanced`.

- (i)** Describe **two** methods of validating the field `RiderLevel`.

1

2

[2]

21. M/J 23/P13/Q4/d

- 4** A shop rents cars to customers. The shop uses a relational database to store information about the rentals.

(d) The data is validated and verified when it is entered into the database.

- (i)** The car registration number must be: 1 letter, followed by 3 numbers, followed by 2 letters.

For example, A123AA is valid but A12AA is invalid.

One way that a registration number can be validated is by using a presence check to make sure the registration number has been entered.

Describe **two other** ways that the car registration number can be validated.

1

2

[2]

- (ii)** Describe **two** ways that the car registration number can be verified when it is entered into the database.

1

2

[2]

- (iii)** State why the car registration number might be incorrect even after it has been validated and verified.

[1]

22. M/J 23/P13/Q6/a,b

6 Data needs to be kept secure when stored on a computer and during transmission over a network.

- (a) Explain how a digital signature is used to authenticate a digital document during transmission over a network.

[5]

- (b) Complete the table by identifying **and** describing **two** types of software that can be installed on a computer to prevent threats over a network.

Type of software	Description
.....	
.....	

[2]

23. M/J 23/P23/Q2/b

2 A program stores a date of birth for a student using a variable, `MyDOB`, of type `DATE`.

- (b) A function `CheckDate()` will take three integer parameters representing a day, month and year of a given date.

The function will validate the date of birth for a student that the parameters passed to it represent.

For a date to be valid, a student must be at least 18 in year 2020.

- (i) Two of the parameter values can be checked without reference to the third parameter.

Describe these **two** checks.

Check 1

Check 2

[2]

- (ii) Several values of the parameter representing the day can only be checked completely by referring to the value of **one other** parameter.

Describe this check.

[2]

24. O/N 22/P11/Q2

2 Draw **one** line from each security feature to its most appropriate description.

Security feature	Description
firewall	converts data to an alternative form
pharming	redirects a user to a fake website
anti-virus software	verifies the authenticity of data
encryption	scans files on the hard drive for malicious software
	accepts or rejects incoming and outgoing packets based on criteria

[4]

25. O/N 22/P12/Q4/a,b,c

4 (a) State the difference between **data verification** and **data validation**.

[1]

(b) A checksum can be used to detect errors during data transmission.

Describe how a checksum is used.

[3]

(c) One validation method is a presence check.

Describe **two other** validation methods that can be used to validate non-numeric data.

1

2

[2]

26. O/N 22/P12/Q6/a

6 (a) A student uses a networked laptop computer to send an email to a colleague.

(i) Explain how a digital signature ensures the email is authentic.

[2]

(ii) Describe how a firewall protects the data on the computer.

[3]

27. O/N 21/P11/Q2/a,b,c

2 Xanthe wants to maintain the integrity and security of data stored on her computer.

(a) Explain the difference between data security and data integrity.

[2]

(b) Xanthe uses both data validation and data verification when entering data on her computer.

(i) Describe how data validation helps to protect the integrity of the data. Give an example in your answer.

Description

Example

[2]

(ii) Describe how data verification helps to protect the integrity of the data. Give an example in your answer.

Description

Example

[2]

(c) Two malware threats are spyware and viruses.

Give **two** similarities and **one** difference between spyware and a virus.

Similarity 1

Similarity 2

Difference

[3]

28. O/N 21/P12/Q1

- 1 When designing computer systems, it is important to consider the security, integrity and privacy of the data.

Draw **one** line from each measure to indicate whether it keeps data secure or protects the integrity of data.

Measure

Firewall

Double entry

Presence check

Access rights

Password

Data Security

Data Integrity

[2]

29. O/N 21/P13/Q2/a,b,c

2 Xanthe wants to maintain the integrity and security of data stored on her computer.

(a) Explain the difference between data security and data integrity.

[2]

(b) Xanthe uses both data validation and data verification when entering data on her computer.

(i) Describe how data validation helps to protect the integrity of the data. Give an example in your answer.

Description

Example

[2]

(ii) Describe how data verification helps to protect the integrity of the data. Give an example in your answer.

Description

Example

[2]

(c) Two malware threats are spyware and viruses.

Give **two** similarities and **one** difference between spyware and a virus.

Similarity 1

Similarity 2

Difference

[3]

30. M/J 21/P11/Q6/a,b,c

6 Each of the following algorithms performs data validation.

State the type of validation check that each of the algorithms performs.

(a)

```
INPUT x  
  
IF x < 0 OR x > 10 THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

(b)

```
INPUT x  
  
IF x = "" THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

(c)

```
INPUT x  
  
IF NOT(x = "Red" OR x = "Yellow" OR x = "Blue") THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

31. M/J 21/P12/Q8/a,b,c

- 8** A company has several security measures in place to prevent unauthorised access to the data on its computers.

(a) Describe the difference between the security and privacy of data.

[2]

- (b)** Each employee has a username and password to allow them to log onto a computer. An employee's access rights to the data on the computers is set to either read-only, or read and write.

Identify **one** other software-based measure that could be used to restrict the access to the data on the computers.

[1]

- (c)** The company is also concerned about threats posed by networks and the internet.

Identify **two** threats to the data that are posed by networks and the internet.

Threat 1

Threat 2

[2]

32. M/J 21/P13/Q6/a,b,c

6 Each of the following algorithms performs data validation.

State the type of validation check that each of the algorithms performs.

(a)

```
INPUT x  
  
IF x < 0 OR x > 10 THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

(b)

```
INPUT x  
  
IF x = "" THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

(c)

```
INPUT x  
  
IF NOT(x = "Red" OR x = "Yellow" OR x = "Blue") THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

Answers Section

1. M/J 26/P11/Q4/d

4(d)(i)	1 mark for each correct point to max 4 - Validation only ensures that data meets certain criteria - Validation prevents the input of inappropriate data - Verification makes sure that the data that is entered matches the original data / the data that was intended to be entered - Verification prevents data being accidentally changed from the original - Any example of validation/verification in practice e.g. entering a password twice	4	Accept a type of validation for MP1 BOD or a type of verification for MP3 BOD but not for both
4(d)(ii)	1 mark for a correct answer The original data was incorrect // the same wrong data was entered twice, and the wrong data was also valid // by example	1	User error is TV

2. M/J 26/P12/Q2/a,b,c

2(a)	1 mark for method 1 mark for correct matching description (* 2) - Visual check - The entered data is compared with the original by a person - Double entry - Enter the data twice and the computer automatically compares if they match	4	Note: Question asks for methods of data verification not validation Accept examples of visual checks, such as printing a copy and comparing the printout to the original Data as intended is NE. It is the comparison to the original that is important
2(b)	1 mark for two correct names of methods 1 mark for each correct matching use to max 2 - Presence check - The Date of Birth (DoB) field cannot be left blank - Range check - DoB must be after one date and before another date // between specific dates – must include both ends - Limit check - DoB must be after or before a specific date – must be only one end - Format check - DoB must be dd/mm/yyyy - Length check - DoB must be exactly 10 characters - Type check - The data type must be DATE or DATE/TIME	3	Note: Question asks for methods of data validation not verification Mark the first answer in each box The use must match the example given in the question 05/07/2010 Do not accept range and limit as two different checks Take care that answers do not just repeat the method, must be applied in some way. The use must match the method If no methods are given read the uses which must be applied to the DoB. For format check accept mm/dd/yyyy
2(c)(i)	1 mark for correct description of a virus Description: malicious software that replicates on the computer and can fill storage or delete/corrupt/change data - cause harm is NE 1 mark each for a correct method and matching description, 2 marks Method: Anti-virus software // anti-malware software Description: The software scans files/downloads/device and compares against database of known viruses, and quarantines/deletes/alerts the user if a match found	3	The description of the method must match the method chosen

2(c)(ii)	1 mark for correct description of a hacker Description: person who gains unauthorised access to a computer system 1 mark each for a correct method and matching description, 2 marks For example: Method: Firewall Description: The firewall monitors all incoming and outgoing data compares it to stored criteria and blocks transmissions that are not allowed Method: Multi-factor authentication Description: A second form of verification such as biometrics or use of an authenticator app is required for logging in Method: <u>Strong</u> passwords Description: - The password contains a mixture of different types of characters and is at least a set number of characters and is not something connected to the individual such as a date of birth Method: Encryption Description: Scramble/encode the data/turn into ciphertext so that even if it is accessed it will not be understood/useable without the decryption key	3 The description of the method must match the method chosen Just malicious intent is NE for unauthorised access Accept two-factor for multi-factor If firewall is the candidate's answer for question 2(c)(i) ensure it is not repeated here Just authentication is NE for multi-factor authentication Map biometrics/use of biometric password as the method to strong password Description: Uses features of an individual that are unique and which are almost impossible to duplicate
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3. M/J 26/P13/Q1

1	1 mark for each correct term Visual check is a method of verification where the data is entered by a user and then manually compared to the original data to make sure that they are the same. A verification method used during data transfer calculates a value from the data which is transferred with the data. The receiver performs the same calculation and compares the result with the value transferred. This method of verification is checksum. There are several methods of data validation. Existence check compares the data entered to a list of possible values that can be entered. If the data is not one of these values, it is rejected. Format check can be used to make sure data entered has the correct combination of letters, numbers and symbols. For example, a code may need to be two capital letters, one number followed by one symbol. Length check is used to make sure the correct quantity of characters has been entered. For example, a name must be more than one character and less than or equal to 100 characters.	5
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4. O/N 25/P11/Q8/a

8(a)	1 mark for each correct explanation, max 2 marks Virus: Malicious program/software that replicates/copies itself and deletes/alters files/data stored on a computer Pharming: Malicious code/software installed on a computer which redirects user to a fake website to obtain personal data	2
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5. O/N 25/P12/Q1

1	1 mark for 2 or 3 correctly connected verification method boxes, 2 marks for all 4 correct Verification Method Parity byte check Checksum Visual check Parity block check Use Data transfer Data entry	2
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6. O/N 25/P12/Q5/e

5(e)	1 mark for correct threat 1 mark corresponding description 1 mark for correct method of prevention <table border="1"> <thead> <tr> <th>Threat</th><th>Description</th><th>Prevention</th></tr> </thead> <tbody> <tr> <td>Malware / Virus</td><td>Malicious code that can alter/delete files</td><td>Anti-virus // Anti-malware // Firewall</td></tr> <tr> <td>Spyware</td><td>Records keystrokes which are sent to third party</td><td>Anti-spyware // Firewall // Anti-malware</td></tr> <tr> <td>Hacking</td><td>Gaining <u>unauthorised</u> access to a computer network / device</td><td>Authentication // Firewall</td></tr> <tr> <td>Phishing</td><td><u>Emails</u> supposedly from reputable companies are sent to trick people into revealing personal information</td><td>Use a spam filter // Do not open emails from unknown sources</td></tr> <tr> <td>Pharming</td><td>Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information</td><td>Use a VPN // Anti-malware// Do not open links or download attachments</td></tr> </tbody> </table>	Threat	Description	Prevention	Malware / Virus	Malicious code that can alter/delete files	Anti-virus // Anti-malware // Firewall	Spyware	Records keystrokes which are sent to third party	Anti-spyware // Firewall // Anti-malware	Hacking	Gaining <u>unauthorised</u> access to a computer network / device	Authentication // Firewall	Phishing	<u>Emails</u> supposedly from reputable companies are sent to trick people into revealing personal information	Use a spam filter // Do not open emails from unknown sources	Pharming	Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information	Use a VPN // Anti-malware// Do not open links or download attachments	3
Threat	Description	Prevention																		
Malware / Virus	Malicious code that can alter/delete files	Anti-virus // Anti-malware // Firewall																		
Spyware	Records keystrokes which are sent to third party	Anti-spyware // Firewall // Anti-malware																		
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7. O/N 25/P12/Q6/c

6(c)(i)	1 mark for correct answer parity bit ↓ <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table>	0	1	0	1	1	1	0	0	1
0	1	0	1	1	1	0	0			

6(c)(ii)	1 mark for correct answer parity bit ↓ <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td></td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td></td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td></td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td></td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>parity byte→</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>		1	0	1	1	0	1	1	1		0	1	1	1	0	0	0	0		0	0	0	1	1	0	1	1		0	1	1	1	0	1	0	0	parity byte→	1	0	1	0	0	0	0	0	[1] 1
	1	0	1	1	0	1	1	1																																							
	0	1	1	1	0	0	0	0																																							
	0	0	0	1	1	0	1	1																																							
	0	1	1	1	0	1	0	0																																							
parity byte→	1	0	1	0	0	0	0	0																																							

8. O/N 25/P12/Q10/c

10(c)	1 mark for a correct answer • Digital Signature	1
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9. O/N 25/P13/Q7/c,e

7(c)	1 mark for the name 2 marks for the matching description, max 3 marks - Parity Byte Check - a parity bit is added to each byte to make the number of 1s match the parity, odd or even - each byte can be checked on receipt and request to be resent if the byte does not match parity - Parity Block Check - a bit is added to each byte, but a parity byte is also set for each block - the location of an error can be found using vertical and horizontal parity - Checksum - A calculation is made from the data and transmitted with the data - The receiver performs the same calculation and compares with received checksum to see if they match	3
7(e)	1 mark for each validation check correctly identified, max 3 marks Range Check Existence Check Format Check	3

10. O/N 25/P21/Q6/a

6(a)	Number: Any number where the first two digits sum to 5 / 15 and the last three digits are 623 Explanation: The remainder is the same / 3 // The sum of the (first four) digits is the same / 13	2
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11. M/J 25/P11/Q7/a,b,c

7(a)	1 mark for a correct method: 1 mark each to max 2 for a corresponding description Method: Encryption Description: - Data is encoded/scrambled using a key to create cipher text - if intercepted it cannot be understood - without being decrypted using a key	3
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7(b)	1 mark each to max 5 • The sender hashes the document / message • to produce a digest • The sender encrypts the digest to create the digital signature • The message and the signature are sent to the banker / receiver • The receiver decrypts the signature to reproduce the digest • The receiver uses the same hashing algorithm on the document received to produce a second digest • The receiver compares this digest with the one from the digital signature • If both of the receiver's digests are the same the document has not changed	5
7(c)	1 mark each to max 3 • The data is put through an algorithm to create a checksum value • The data and checksum are sent to the receiver • The receiver performs the same algorithm on the data • if both checksums match the data is verified	3

12. M/J 25/P13/Q3/b

3(b)	1 mark for method and 1 mark for corresponding description During transmission e.g. • Encryption // by example such as VPN • Jumble / encode data so it cannot be decrypted/understood without the key On computer e.g. • Firewall / proxy • Filter incoming transmissions and stop any that could be attempting unauthorised access • Anti-malware • Find and delete or quarantine any malware that could delete the data / files • Encryption • Jumble / encode data so it cannot be decrypted / understood without the key • Physical method // by example • For example, the computer storing the data cannot be accessed without the key to the room	4
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13. O/N 24/P12/Q4/c

4(c)	1 mark for the security method. 2 marks for explanation Security method: Encryption Explanation • File contents are converted to cipher text • If intercepted the data cannot be understood without the decryption key	3
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14. M/J 24/P11/Q5/b,c(i)

5(b)	1 mark for each correctly completed term: • odd or even • 7-bits • odd • block • byte Computer A and Computer B agree on whether to use odd or even parity. Computer A divides the data into groups of 7-bits. The number of 1s in each group is counted. If the agreed parity is odd and the group has an even number of 1s a parity bit of 1 is appended, otherwise a parity bit of 0 is appended. In a parity block check the bytes are grouped together, for example in a grid. The number of 1s in each column (bit position) is counted. A bit is assigned to each column to make the column match the parity. These parity bits are transmitted with the data as a parity byte.	5
5(c)(i)	1 mark each to max 3: • Compares all incoming and outgoing transmissions • ... against set criteria/whitelist/blacklist • Blocks all transmissions that do not meet rules • Blocks data entering from specific ports • Blocks unauthorised/unknown internal software transmitting data	3

15. M/J 24/P12/Q3/a

3(a)(i)	1 mark for security measure 1 mark each to max 2 for how the chosen measure works: • Firewall • Checks incoming connections • ... against criteria • Blocks data from entering specific ports • Blocks data that does not meet whitelist that meets blacklist • Proxy server • Prevents devices accessing the web server directly • Intercepts any requests • Forwards the request using its own IP address • Screens returning data before sending it to the user	3
3(a)(ii)	1 mark for security measure 1 mark each to max 2 for description of the chosen measure: • Encryption • Encodes/scrambles data • ... so if it is intercepted it cannot be understood • Algorithm/key is required to decode the data	3

16. M/J 24/P13/Q7/f

7(f)(i)	1 mark for each correct method and 1 mark for corresponding description to max 4 : <table><tr><th>Method</th><th>Description</th></tr><tr><td>Parity byte</td><td>An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.</td></tr><tr><td>Parity block</td><td>Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.</td></tr><tr><td>Checksum</td><td>A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.</td></tr></table>	Method	Description	Parity byte	An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.	Parity block	Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.	Checksum	A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.	4
Method	Description									
Parity byte	An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.									
Parity block	Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.									
Checksum	A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.									
7(f)(ii)	1 mark each to max 2 : • Encodes/scrambles data • ... so if it is intercepted it cannot be understood • Algorithm/key is required to decode the data	2								

17. O/N 23/P12/Q5/a,b,c

5(a)	1 mark for: Either - Ensuring data can only be accessed by / disclosed to authorised persons Or - Ensuring data cannot be accessed by / disclosed to unauthorised persons	1
5(b)	1 mark for each bullet point (max 1) - Ensuring the accuracy / completeness / consistency of data (during / after processing) - Ensuring the data is up to date	1
5(c)	1 mark for each bullet point. Phishing email (max 2) - The email pretends to be from an official body ... persuading individuals to disclose private information // by example such as bank details ... or requesting authentication by redirecting to an unofficial/unauthorised website // inviting a user to click a link Spyware (max 2) - Malware downloaded without the user's knowledge ... which secretly records the user's actions / keystrokes on the computer ... and sends logs of the actions to a third party	4

18. O/N 23/P13/Q3/a(iii),c

3(a)(iii)	1 mark each for firewall, encryption and passwords. Firewall: - Monitors incoming and outgoing traffic and rejects any traffic that does not meet the set rules Encryption: - Ensures that if data is intercepted / obtained it cannot be understood without the decryption key Passwords: - Ensures only users with the correct password can access the resources // prevents unauthorised access	3
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3(c)	1 mark for each bullet point. - The administrator completes a visual check / checks by eye ...that the tutor identifier input matches the tutor identifier on the original document - Double entry check // the administrator (or a second person) enters the number a second time ...and the system compares it with the first entry	4
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19. O/N 23/P13/Q8/a,b,c

8(a)	1 mark for each bullet point (max 2) - Validation // a validation method named or described ...protects the data by ensuring that the data is reasonable / sensible and within specified bounds	2
8(b)	1 mark for difference 1 mark for similarity Difference: - Pharming is malicious code that redirects to a fake website. Phishing uses an email to prompt user action. - Pharming is automatic. Phishing requires user action. Similarity: - Both try to obtain financial or personal information - Both are a false representation of an official organisation, e.g. a bank - Both make use of fake websites	2
8(c)	1 mark for each bullet point (max 3). - Download programs from reputable websites / sources ...as these are less likely to contain malware - Backup / archive computer systems ...so they can be restored in case of data loss from malware program installation Install and run anti-malware program ...so that regular scans can be made for known malware ...and if malware is found it can be quarantined / removed ...and computer's anti-malware definitions are regularly updated - Using a firewall to block unused ports ...so that malware cannot enter the computer system - Deny administrator privileges to everyday users ...so that malware cannot be downloaded by everyday users - Avoid the use of / access to removable devices ...so that malware cannot be installed from these devices	3

20. M/J 23/P12/Q2/c(i)

2(c)(i)	1 mark each to max 2 • Presence check to make sure that the (rider level) is entered • Look-up / Existence check to make sure the rider level is only Beginner, Intermediate or Advanced • Length check to make sure the rider level entered is either 8 or 12 characters • Type check to make sure the rider level is alphanumeric	2
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21. M/J 23/P13/Q4/d

4(d)(i)	1 mark each to max 2 • Length check: the registration number must be 6 characters long • Format check: the registration number must be in the format letter-digit-digit-digit-letter-letter • Type check: the registration number must be alphanumeric	2
4(d)(ii)	1 mark each • Visual check: Manually compare the registration number entered with the source document • Double entry: Enter the registration number twice and the computer compares to check they are the same	2
4(d)(iii)	The registration number on the original document might be in the correct format but may be the incorrect registration number for that car.	1

22. M/J 23/P13/Q6/a,b

6(a)	1 mark each to max 5 • The sender hashes the document • ... to produce a <u>digest</u> • The sender <u>encrypts</u> the digest to create the digital signature • The message and the signature are sent to the receiver • The receiver <u>decrypts</u> the signature to reproduce the digest • The receiver uses the <u>same</u> hashing algorithm on the document received to produce a second digest • The receiver compares this digest with the one from the digital signature • If both of the receiver's digests are the same the document is authentic	5
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6(b)	1 mark each for identification and appropriate description of 2 pieces of software, max 2 <table><tr><th>Type of software</th><th>Description</th></tr><tr><td>Antivirus</td><td> - scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing </td></tr><tr><td>Antispyware</td><td> - scans the computer for spyware and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known spyware and prevents the download continuing. </td></tr><tr><td>Firewall</td><td> - monitors incoming and outgoing traffic and compares it to criteria that are set by the user such as through a whitelist/blacklist/identifying allowed / blocked IP addresses - compares incoming and outgoing traffic to criteria blocks those that do not match criteria </td></tr><tr><td>Antimalware</td><td> - scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing </td></tr></table>	Type of software	Description	Antivirus	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing	Antispyware	- scans the computer for spyware and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known spyware and prevents the download continuing.	Firewall	- monitors incoming and outgoing traffic and compares it to criteria that are set by the user such as through a whitelist/blacklist/identifying allowed / blocked IP addresses - compares incoming and outgoing traffic to criteria blocks those that do not match criteria	Antimalware	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing	2
Type of software	Description											
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23. M/J 23/P23/Q2/b

2(b)(i)	MP1 Value for <u>month</u> is between 1 and 12 (inclusive) MP2 Value of <u>year</u> is <= 2002	2
2(b)(ii)	MP1 Reference to <u>month</u> and <u>day</u> MP2 Clear description for a <u>check</u> that the day number matches with a relevant month (Either day matches with month // month matches with day)	2

24. O/N 22/P11/Q2

2	1 mark for each correct line. Security feature firewall pharming anti-virus software encryption converts data to an alternative form redirects a user to a false website verifies the authenticity of data scans files on the hard drive for malicious software accepts or rejects incoming and outgoing packets based on criteria	4
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25. O/N 22/P12/Q4/a,b,c

4(a)	data verification is checking if input data is the same as the original whereas data validation is checking that the data is reasonable / sensible	1
4(b)	1 mark for each bullet point (max 3): - checksum value is calculated from the data before transmission // correct description of a checksum algorithm ... this calculated value is transmitted with the data receiving computer recalculates the checksum from the received data - if the checksum received and calculated match, no error has occurred // if the checksum received and calculated do not match, an error has occurred	3
4(c)	1 mark for each bullet point (max 2). For example: - to make sure data is in the required format // only expected characters allowed - to make sure the data is already present in the system - to make sure the data contains the correct number of characters - to ensure that non-numeric data is entered	2

26. O/N 22/P12/Q6/a

6(a)(i)	1 mark for each method of ensuring authenticity (max 2): • (email) message put through hashing algorithm to produce a digest • Digest encrypted with <u>sender's private</u> key (to create the digital signature) • the (digital) signature can only be decrypted with matching <u>sender's public</u> key	2
6(a)(ii)	1 mark for each bullet point: • monitors <u>incoming</u> and <u>outgoing</u> packets / traffic • checks against an allow list / deny list of IP addresses // checks against a set of rules for acceptable data / ports etc. • blocks transmissions that do not meet criteria / rules // allows through is satisfies the criteria /rules	3

27. O/N 21/P11/Q2/a,b,c

2(a)	1 mark per bullet point • security is protecting data from loss / corruption • integrity is ensuring the consistency / accuracy of the data	2
2(b)(i)	1 mark per bullet point • validation checks that data is reasonable / sensible • example e.g. checking data is the right number / type of characters	2
2(b)(ii)	1 mark per bullet point • verification checks that data is the same as the original • by example e.g. double entry	2
2(c)	1 mark per similarity to max 2 • Both are pieces of malicious software • Both are downloaded / installed/run without the user's knowledge • Both can pretend to be / are embedded in other legitimate software when downloaded // both try to avoid the firewall • Both run in the background 1 mark for difference • Virus can damage computer data; spyware only records / accesses data • Virus does not send data out of the computer; spyware sends recorded data to third party • Virus replicates itself; spyware does not replicate itself	3

28. O/N 21/P12/Q1

1	1 mark for 3 correct lines only from Data Security 1 mark for 2 correct lines only from Data Integrity <pre> graph LR Firewall --> DS[Data Security] DoubleEntry[Double entry] --> DS PresenceCheck[Presence check] --> DS AccessRights[Access rights] --> DI[Data Integrity] Password --> DI </pre>	2
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29. O/N 21/P13/Q2/a,b,c

2(a)	1 mark per bullet point • security is protecting data from loss / corruption • integrity is ensuring the consistency / accuracy of the data	2
2(b)(i)	1 mark per bullet point • validation checks that data is reasonable / sensible • example e.g. checking data is the right number / type of characters	2
2(b)(ii)	1 mark per bullet point • verification checks that data is the same as the original • by example e.g. double entry	2
2(c)	1 mark per similarity to max 2 • Both are pieces of malicious software • Both are downloaded / installed/run without the user's knowledge • Both can pretend to be / are embedded in other legitimate software when downloaded // both try to avoid the firewall • Both run in the background 1 mark for difference • Virus can damage computer data; spyware only records / accesses data • Virus does not send data out of the computer; spyware sends recorded data to third party • Virus replicates itself; spyware does not replicate itself	3

30. M/J 21/P11/Q6/a,b,c

6(a)	Range (check)	1
6(b)	Presence (check)	1
6(c)	Existence (check)	1

31. M/J 21/P12/Q8/a,b,c

8(a)	1 mark per bullet point • Security protects data against loss • Privacy protects data against unauthorised access	2
8(b)	1 mark for a correct answer • Two factor authentication • Biometric passwords • Key Card Access • Firewall	1
8(c)	1 mark per correct answer to max 2 • Malware // viruses // spyware // by example • Hacking • Phishing • Pharming	2

32. M/J 21/P13/Q6/a,b,c

6(a)	Range (check)	1
6(b)	Presence (check)	1
6(c)	Existence (check)	1

Unit 7

ETHICS AND OWNERSHIP

27 items · 125 marks

Syllabus sections in this unit

7.1 Ethics and Ownership

Unit 7: Ethics and Ownership

1. M/J 26/P11/Q2/d

- 2 A computer programmer uses language translators. Two types of translator are an interpreter and a compiler.

(d) The completed computer program is distributed under a licence.

Complete the table by writing the name of each software licence that is described.

software licence	description
	The software is released on a trial period. After the trial period ends, a fee may be required to continue using the software.
	The software is purchased for a fee. There is no access to the source code.
	The software is released with the source code. The software can be changed and distributed with the changes under the same licence restrictions.

[3]

2. O/N 25/P11/Q8/b,c

- 8 One ethical consideration for a student connecting their personal computer to the school network is the risk of spreading malware on the network.

(b) Give **three** other ethical considerations for a student using their personal computer to connect to the school network.

1

2

3

[3]

(c) Describe **two** social impacts of students using Artificial Intelligence (AI) to complete their homework.

1

2

[4]

3. O/N 25/P12/Q2/a,b

2 (a) Complete the table by describing each term.

Term	Description
Copyright	
Open Source (Initiative)	
Shareware	
Software Licence	

[4]

(b) Describe the purpose of a code of conduct.

[2]

4. O/N 25/P12/Q10/b

10 A programmer is developing a computer program.

(b) The programmer releases the program as Free Software.

Describe what is meant by Free Software.

[2]

5. O/N 25/P13/Q7/a

7 A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

(a) Describe **one** ethical impact of this use of AI in the classroom.

[2]

6. O/N 25/P13/Q8/c

- 8** Compilers and interpreters are used to translate programs written in a high-level language into a low-level language.

(c) A programmer is developing an Operating System (OS).

Identify **two** types of software licence the programmer can use to allow other people to edit and redistribute the OS.

1

2

[2]

7. M/J 25/P11/Q4/a

- 4** A shop installs a new system that allows users to purchase items without going through a manual checkout.

The new system:

- identifies customers when they enter the shop and matches them to their account
- prevents a customer from walking through the automatic barriers if they do not have an account
- automatically detects the items that a customer has taken from a shelf and charges these to the customer's account.

(a) The new system uses digital cameras and Artificial Intelligence (AI) to identify the customers.

Explain how the new system uses AI to identify each customer.

[4]

8. M/J 25/P12/Q3/a,b,c

- 3** A computer program uses a digital camera to read the words on an item.

The program can read the words that are written on the item, translate the words to a chosen language and then output the words as audio. For example, when used in a supermarket, the program can output the words written on the labels on products.

The program uses Artificial Intelligence (AI).

- (a)** Explain how AI is used in the computer program described.

[4]

- (b)** State **two** social benefits of the use of AI in the computer program described.

1

2

[2]

- (c)** The computer program is released under a commercial software licence.

- (i)** Describe the features of a commercial software licence.

[3]

- (ii)** Explain the reasons why an open source software licence might **not** be appropriate for the computer program described.

[3]

9. M/J 25/P13/Q3/d

- 3** A programmer is writing a program in a high-level language.

- (d)** Describe the possible consequences of the programmer **not** joining a professional ethical body.

[3]

10. O/N 24/P11/Q6

- 6** A car park system uses a camera to record the registration number of each car as it enters and leaves the car park.

Explain how artificial intelligence is used in the car park system to identify the car's registration number.

[4]

11. O/N 24/P11/Q7/a,b

7 Software is distributed with a licence.

(a) Give **two** benefits of distributing software using a shareware software licence.

1

2

[2]

(b) Give **two** benefits of distributing software using a commercial software licence.

1

2

[2]

12. O/N 24/P12/Q5/a,b

5 A computer programmer is learning about the importance of acting ethically at work.

(a) Explain the reasons why the computer programmer needs to act ethically towards colleagues and the public.

Colleagues

The public

[4]

(b) The programmer writes a program to be used in a business. The source code of the program can be modified by the business. The business pays the programmer for the maintenance and security updates.

(i) Identify **one** suitable type of software licence for the program **and** describe the reasons for your choice.

Type of software licence

Reasons for choice

[3]

(ii) Explain the reasons why the programmer should copyright their program.

[2]

13. O/N 24/P13/Q5/a,b

5 A computer programmer writes programs that are distributed with a software licence.

(a) Complete the table by writing the type of software licence each statement describes.

Statement	Software licence
A fee is charged for the software. The source code cannot be accessed. Users do not get a free trial.	
Users can try the software before buying it. Users may not be able to access all features during the trial.	
Software is usually free of charge. Users can access the source code and alter the program to their needs.	

[3]

(b) Explain the reasons why it is important for the computer programmer to join a professional ethical body.

[3]

14. M/J 24/P11/Q5/c(ii)

5 A bank allows customers to access their accounts using an application that they can download onto a device such as a smartphone.

(c) The bank also needs to keep its customers' data private and secure.

(ii) Customers need to use biometric authentication to access their accounts.
One biometric authentication method is facial recognition.

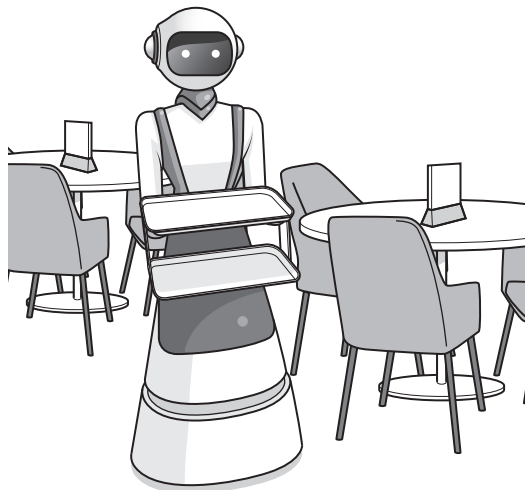
Facial recognition uses Artificial Intelligence (AI).

Describe how AI is used in facial recognition.

[4]

15. M/J 24/P13/Q7/b

- 7 Robots are used to serve food and drink to customers at a restaurant.



- (b) The robot uses Artificial Intelligence (AI) to communicate with the customers. The customers speak to the robot to order their food and drinks.

Explain how AI will be used in this part of the robot.

[3]

**16. O/N 23/P13/Q6/a,b**

- 6 (a) State **two** benefits to a programmer of distributing a program using a shareware licence.

1

2

[2]

- (b) Explain why it is important for a programmer to join a professional ethical body.

[4]

17. M/J 23/P11/Q4/c

- 4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

- (c) The CCTV system uses Artificial Intelligence (AI) to identify the presence of a person in the house and to track their movements.

Describe how AI is used in this system.

[3]

18. M/J 23/P12/Q7/d

7 A software developer is working in a team writing a program for a client.

(d) One section of the program being developed will convert user's speech into commands.

Explain how Artificial Intelligence (AI) can be used in this program.

[3]

19. M/J 23/P13/Q3/b

3 A mobile telephone is used to record a video.

(b) The mobile telephone uses a built-in digital camera to record the video.

The digital camera automatically focuses on the faces of people.

Explain how Artificial Intelligence (AI) is used by the camera to automatically focus on the faces of people.

[3]

20. M/J 23/P13/Q5/b

5 A programmer is developing a computer game in a high-level language to sell to the public.

(b) The programmer needs to publish the game under a software licence so that it can be sold to the public.

Identify the **most appropriate** type of software licence for the game **and** justify your choice.

Licence

Justification

[4]

21. O/N 22/P12/Q9

9 One use of Artificial Intelligence (AI) is for facial recognition software.

Describe the social impact of using facial recognition software to identify individuals in an airport.

[2]

22. O/N 22/P13/Q8/a,b

- 8 (a) (i) Explain why some programs are distributed under an open source licence.

[2]

- (ii) Explain how a programmer benefits from distributing a program under a commercial licence.

[2]

- (b) A commercial program for a vehicle repair garage includes an Artificial Intelligence (AI) module that can diagnose faults and suggest repairs.

Describe **one** economic impact the AI module may have on the garage.

[2]

23. O/N 21/P11/Q4/a,b(i),c

- 4 Francis is starting his first job as a software developer for a multinational company.

- (a) Francis has been advised to join a professional ethical body.

Describe the benefits to Francis of joining a professional ethical body.

[3]

- (b) Francis is shown the software he will be working on. He is unfamiliar with the Integrated Development Environment (IDE) he is required to use.

- (i) Describe the ways in which Francis can act ethically in this situation.

[2]

- (c) Francis is part of a team writing a program. He finds an error in part of the program that has already been tested. He decides not to tell anyone because he is worried about the consequences.

Explain the reasons why Francis acted unethically in this situation.

[2]

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Explain the reasons why Francis acted unethically in this situation.

[2]

25. M/J 21/P11/Q1/d

1 Anya scans an image into her computer for a school project.

(d) Anya made sure that the image was not subject to any copyright before scanning it.

Describe what is meant by **copyright**.

[2]

26. M/J 21/P12/Q2/a,b,c

2 Aisha manages a team of software developers.

- (a) Explain the reasons why it is important that Aisha acts ethically in relation to her team members.

[2]

- (b) The team are developing a computer game where the user plays a board game (such as chess) against the computer.

Describe how the computer would use Artificial Intelligence (AI) to play the board game.

[3]

- (c) The final game will be released under a licence.

Tick (✓) **one or more** boxes in each row to identify the licence(s) each statement describes.

Statement	Free Software Foundation	Open Source Initiative	Shareware	Commercial Software
The user can edit the source code				
The user must always pay before being able to use the software				
The user can redistribute the software				
The user always gets a trial period				

[4]

27. M/J 21/P13/Q1/d

1 Anya scans an image into her computer for a school project.

- (d) Anya made sure that the image was not subject to any copyright before scanning it.

Describe what is meant by **copyright**.

[2]

Answers Section

1. M/J 26/P11/Q2/d

2(d)	1 mark for each correct licence		3	Do not accept copyleft for open source
	Software licence	Description		Do not accept freeware for open source
	Shareware	The software is released on a trial period. After the trial period ends, a fee may be required to continue using the software.		
	Commercial // proprietary	The software is purchased for a fee. There is no access to the source code.		
	Open source initiative // free software	The software is released with the source code. The software can be changed and distributed with the changes under the same licence restrictions.		

2. O/N 25/P11/Q8/b,c

8(b)	1 mark per bullet point, max 3 marks e. g. • Students should consider privacy issues • Students should not make inappropriate use of social networking // Students should not engage in cyberbullying • Students should not copy work that is not their own • Students should abide by the school rules/guidelines for accessing the network • Students should only use the school network for legitimate purposes • Students should not hack / crack other computers	3
8(c)	1 mark per bullet point, max 4 marks e. g. Negative Impacts • If students use AI to shortcut learning, they may miss out on developing reasoning and problem-solving skills // Students may not be learning anything // It could lead to a decline in specific skills • It could have negative impacts on student communication // heavy reliance on AI may limit opportunities for collaboration, teamwork, and face-to-face communication • Emphasises the digital divide// some students may not have access to the technology • The AI answer might not be correct leading to mis-information • Increased screen time and isolation could contribute to anxiety or loneliness Positive Impacts • Students given additional support may do better • Pupils who struggle in traditional settings may gain self-esteem through AI-assisted learning at their own pace Long-Term Impacts • Protection of intellectual property // ethical issues and considerations • Schools may need to redesign curricula and/or assessment to integrate AI	4

3. O/N 25/P12/Q2/a,b

2(a)	1 mark for each correct row, max 4 marks	4										
<table><tr><th>Term</th><th>Description</th></tr><tr><td>Copyright</td><td>Gives the holder the legal right of ownership of intellectual property</td></tr><tr><td>Open Source (Initiative)</td><td>Software can be freely copied, distributed or adapted</td></tr><tr><td>Shareware</td><td>Gives the user a trial period / limited functionality before a payment/sign up may be required</td></tr><tr><td>Software Licence</td><td>A (legal) agreement that defines the rights and terms of usage for the software</td></tr></table>			Term	Description	Copyright	Gives the holder the legal right of ownership of intellectual property	Open Source (Initiative)	Software can be freely copied, distributed or adapted	Shareware	Gives the user a trial period / limited functionality before a payment/sign up may be required	Software Licence	A (legal) agreement that defines the rights and terms of usage for the software
Term	Description											
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Software Licence	A (legal) agreement that defines the rights and terms of usage for the software											
2(b)	1 mark per bullet point, max 2 marks e.g. • to create a safe, respectable and professional working environment for all employees • to ensure that employees understand the expectations • ... and the consequences of their actions • to provide a clear outline of what is considered good practice • so there are clear rules to follow in certain situations • to protect the reputation of the organisation // maintain public trust	2										

4. O/N 25/P12/Q10/b

10(b)	1 mark per bullet point, max 2 • The software is free of restrictions • The software is not necessarily free of charge • The <u>source code</u> comes with the program // A user can edit the <u>source code</u> to add functionality / suit their needs • Users can share the software with others	2
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5. O/N 25/P13/Q7/a

7(a)	1 mark per point, max 2 marks e.g. • The learning experience could be improved • ... by identifying students who are struggling • and employing more personalised learning • There could be early Intervention • ... to determine which students require extra support • Teachers and students could understand the challenges better • ...and identify optimal times for learning • There could be privacy concerns • students / teachers could be uncomfortable with constant surveillance • ... students might not want their data / actions passed to third parties • There could be Mental Health concerns • due to constant monitoring and pressure • contributing to student stress and anxiety	2
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6. O/N 25/P13/Q8/c

8(c)	1 mark per bullet point, max 2 marks • Open Source (Initiative) • Free Software (Foundation)	2
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7. M/J 25/P11/Q4/a

4(a)	1 mark each to max 4 e.g. • Uses image recognition / facial recognition • Measures the distance between facial features of the customer • and stores the values in a database of user information • An image is captured using the digital camera • the AI identifies that the image is a face • by analysing the pixels to find patterns • the distance between features is calculated and compared to those in database	4
------	---	----------

8. M/J 25/P12/Q3/a,b,c

3(a)	1 mark each to max 4 e.g. • Uses image recognition // optical character recognition • to analyse a photograph/pixels to identify the location of characters within an image • in order to convert the pattern of pixels into individual characters • Uses natural language • to combine the characters into words • and compare the words to database to identify a match in the language of the writing • it may also predict what is missing / incomplete / from what has been read • then compare the words to a database of the user's chosen language to find a match and translate • Uses text-to-speech • to generate the audio waveforms corresponding to the words	4
3(b)	1 mark each to max 2 e.g. • It can help people with visual impairment to identify which products to purchase • It can help people who cannot understand the language • It can help to overcome learning / reading difficulties	2
3(c)(i)	1 mark each to max 3 • The program is usually sold for a fee • Users have no access to source code • It protects the intellectual rights of the programmer / developer • Users cannot legally alter or share the program • There are limitations on the use of the software e.g. number of times it can be installed • Support and maintenance is usually provided e.g. updates, bug fixes etc.	3
3(c)(ii)	1 mark each to max 3 e.g. • The source code would be accessible therefore it could be changed • which might lead to the program outputting an incorrect response • so people could purchase the wrong item • which might have serious consequences // by example e.g. allergies	3

9. M/J 25/P13/Q3/d

3(d)	1 mark each to max 3 e.g. • There is no / limited access to legal advice • in case action is taken against them • There are fewer networking opportunities • so they could miss out on contacts / jobs • There would be less access to training • There would be no clear laid out ethical guidelines • and / or people to discuss potential ethical problems with • possibly leading to inappropriate / unethical actions • which might lead to legal proceedings / recourse	3
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10. O/N 24/P11/Q6

6	1 mark for each bullet point (max 4) e.g. • It uses image recognition • The pixels of each image from the camera are stored • ... and matched to the expected shape/size/colour of registration number • It uses optical character recognition • The pixels within the registration number area identified are analysed • ... and compared to expected characters/letters/numbers	4
---	--	----------

11. O/N 24/P11/Q7/a,b

7(a)	1 mark for each bullet point (max 2) • The user can test/trial the program before purchasing to see whether it meets needs • There is no initial cost • Users who try the software may spread the word so the developer gets more custom • The developer can get feedback to improve the program	2
7(b)	1 mark for each bullet point (max 2) • The owner/developer can earn a fee • Comprehensive support is usually offered so problems can be easily solved • The software is usually well tested so few bugs	2

12. O/N 24/P12/Q5/a,b

5(a)	1 mark for each bullet point max 2 for public max 2 for colleagues e.g. Colleagues: The programmer should: • Treat colleagues fairly and avoid acts of discrimination • Be prepared to accept / offer critique • ... so that technical work can be improved • Credit the contributions of colleagues • ... so that they feel valued / respected • Help / train colleagues etc. e.g. The public: The programmer should: • Maintain the health, safety and welfare of the public • ... so as not to endanger anyone • Be honest / realistic in making claims about the software • ... so that public trust is maintained • Maintain the security of the public's data etc.	4
5(b)(i)	Type of software licence 1 mark for • Open Source Initiative // Free Software Foundation Reason for choice: 1 mark for each bullet point (max 2) • Allows program source code to be examined • ... so that the business can adapt/modify the program to meet their needs • Allows the programmer to earn money for the software and updates	3
5(b)(ii)	1 mark for each bullet point (max 2) • To identify themselves as the owner/author // To gain formal recognition of ownership • To allow for legal consequences if anyone steals/copies it • To restrict competition	2

13. O/N 24/P13/Q5/a,b

5(a)	1 mark for each correct licence		3
	Statement	Software Licence	
	A fee is charged for the software. The source code cannot be accessed. Users do not get a free trial.	Commercial	
	Users can try the software before buying it. Users may not be able to access all features during the trial.	Shareware	
	Software is usually free of charge. Users can access the source code and alter the program to their needs.	Open Source // Free software	
5(b)	1 mark for each bullet point (max 3) e.g. • The professional ethical body has ethical guidelines to follow • ... so clients/other staff know the standards being applied • ... so the computer programmer does not have to decide what is ethical it's written down • Clients/staff know the computer programmer is reputable • ... there may be a test/requirement for entry • ... giving recognition of the computer programmer's skills/knowledge • The professional ethical body provides help and support • ... for example, if the computer programmer needs legal advice • The professional ethical body runs training courses • ... to keep the computer programmer's skills up to date		3

14. M/J 24/P11/Q5/c(ii)

5(c)(ii)	1 mark each to max 4 : e.g. • Captures an image of the face • Uses image recognition • Trained to identify the features of a face in an image • ... using a large number of images • Analyse images for facial features • Uses the probability of a match	4
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15. M/J 24/P13/Q7/b

7(b)	1 mark each to max 3 : e.g. • Voice/speech recognition is used • ... to identify if someone speaking • The sound is recorded and analysed • The audio recordings are compared to a database of words/sound waves • ... to identify the word that has the highest probability of being said • Natural language recognition is used • Words are combined and compared to known sentences • ... programmed action(s) for matching sentence(s) are performed	3
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16. O/N 23/P13/Q6/a,b

6(a)	1 mark for each bullet point (max 2). • Users are able to trial the program and may offer the programmer constructive feedback • More people might try the program because the trial is free of charge • Allows bugs to be found and corrected on a wide range of computer system configurations improving the programmer's original program • Users that find the trial useful will buy the program so programmer gets income • Allows the program to be copyrighted and so protects the programmer's intellectual property rights // no illegal modification allowed	2
6(b)	1 mark for each bullet point (max 4). • The programmer has ethical guidelines to follow • ...so purchasers of programs know the standards being applied • ...so programmer does not have to decide what is ethical or not • Clients know programmer is reputable • ...recognition of programmer's skills/knowledge • There may be an entry requirement / exam • ... so clients know that the programmer is competent • Professional ethical body provides help and support • Such as legal advice // by appropriate example • Enables programmer to attend the ethical body's training courses • ...to keep skills up to date // to develop skills	4

17. M/J 23/P11/Q4/c

4(c)	1 mark each to max 3 Examples: • Uses image recognition • Monitors every image taken to identify matching images/shapes/features to a 'person' ... • ... starts recording to secondary storage/permanently when a person is identified • System identifies direction of movement of person and uses this to decide where/how to move the camera/record • System identifies other cameras to start recording based on direction of movement	3
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18. M/J 23/P12/Q7/d

7(d)	1 mark each to max 3 • Uses speech recognition • ... which identifies key phrases / words spoken • ... and matches these to a database • ... and generates the most likely sentence / command / word	3
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19. M/J 23/P13/Q3/b

3(b)	1 mark each to max 3 Examples: • Scans the scene in real time • Identifies if there are faces in the image • Uses facial recognition • ... uses image recognition • ... takes each frame individually • ... analyses the pixels • ... stores pattern for a face • ... looks for patterns that match/come close to the pattern for a face • Camera focuses on the pattern identified	3
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20. M/J 23/P13/Q5/b

5(b)	1 mark for appropriate licence; 1 mark for each point to max 3 - Commercial software licence - User has to pay for the product so the programmer can gain an income - Enables the program to be copyrighted ... so the user cannot legally edit the program // the programmer retains control over product ... and can take legal action against people who attempt to illegally copy it /sell it on - Shareware licence - Enables the program to be copyrighted - The user cannot legally edit the program so the developer retains control over product - User can try the program for free and then pay for the full game which allows the programmer to gain an income - so more people can experience it and therefore be more likely to buy it	4
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21. O/N 22/P12/Q9

9	1 mark for each bullet point (max 2). For example: - incorrect recognition of faces leads to mistakes such as ... access to facilities / systems may be denied - privacy issues / people do not like data being stored - individuals will feel safer ... there might be a reduction in crime - faster boarding - catching criminals	2
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22. O/N 22/P13/Q8/a,b

8(a)(i)	1 mark for each bullet point (max 2): - to allow users to customise the code - to allow errors to be reported / identified / fixed by users - to allow additional features to be added to the code - to allow for collaboration	2
8(a)(ii)	1 mark for each correct point (max 2) Example: - enables the program to be copyrighted - prevents illegal changes to the program / protects the source code - prevents illegal copies of the program being made - a fee can be charged for the program	2

8(b)	1 mark for a correct economic impact and 1 mark for corresponding description Example: • reduce costs to the garage • ... because less time taken for diagnosis • increase profits for the garage • ... as technicians spend more time repairing, so completing more jobs in a day • decrease costs passed to customer • ... so garage may gain customers • profit margins can be reduced • ... because program may be expensive to buy / maintain / update	2
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23. O/N 21/P11/Q4/a,b(i),c

4(a)	1 mark per bullet point to max 3 e.g. • He has ethical guidelines to follow • ... so clients/other staff know the standards being applied • ... so he does not have to decide what is ethical it's written down • Clients / staff know he is reputable • ... recognition of his skills / knowledge • ... there may be a test / requirements for entry • They provide help and support • ... for example if he needs legal advice • They run training courses • ... to keep his skills up-to-date	3
4(b)(i)	1 mark per bullet point to max 2 e.g. • He can tell the manager he has not used it • ... and how he will get up-to-date • He can perform his own research on how to use it • He can explain to the manager that he needs additional training • He can(ask the manager to book on a training course • He can ask for a mentor / to shadow someone • He can practice at home before starting	2

4(c)	1 mark per bullet point to max 2 e.g. • He didn't act in best interest of product • ... because the product might fail because he didn't report the error • He didn't act in best interest of client • ... because if the product does not work then they have been let down because he didn't report the error • He didn't act in the best interest of the profession • ... he is letting his profession down because he didn't report the error • He didn't act in best interest of the company • ... not correcting the error early could lead to later problems	2
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24. O/N 21/P13/Q4/a,b(i),c

4(a)	1 mark per bullet point to max 3 e.g. • He has ethical guidelines to follow • ... so clients/other staff know the standards being applied • ... so he does not have to decide what is ethical it's written down • Clients / staff know he is reputable • ... recognition of his skills / knowledge • ... there may be a test / requirements for entry • They provide help and support • ... for example if he needs legal advice • They run training courses • ... to keep his skills up-to-date	3
4(b)(i)	1 mark per bullet point to max 2 e.g. • He can tell the manager he has not used it • ... and how he will get up-to-date • He can perform his own research on how to use it • He can explain to the manager that he needs additional training • He can(ask the manager to book on a training course • He can ask for a mentor / to shadow someone • He can practice at home before starting	2

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25. M/J 21/P11/Q1/d

1(d)	1 mark per bullet point to max 2 • The formal and legal rights to ownership // intellectual property rights • Protects against unauthorised reproduction of work • Provides for legal right of redress	2
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26. M/J 21/P12/Q2/a,b,c

2(a)	1 mark per bullet point to max 2 • To make sure the team members feel valued • To get the best work out of the team • To enable them to work well together • To enable them to create the best product for the client	2
2(b)	1 mark per bullet point to max 3 • The rules / past moves / decision making algorithms of the game will be stored • The AI program is trained, by playing many times • AI will look (ahead) at possible moves • ... and/or analyse the pattern of past choices • ... and choose the move most likely to be successful • Computer could learn how to improve // learn from previous mistakes • ... by storing the positive/negative result of choices • ... and changing its future choices	3

2(c)

1 mark for each correct column

4

Statement	Free Software Foundation	Open Source Initiative	Shareware	Commercial Software
The user can edit the source code	✓	✓		
The user must always pay before being able to use the software				✓
The user can redistribute the software	✓	✓	✓	
The user always gets a trial period			✓	

27. M/J 21/P13/Q1/d

1(d)	1 mark per bullet point to max 2 • The formal and legal rights to ownership // intellectual property rights • Protects against unauthorised reproduction of work • Provides for legal right of redress	2
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Unit 8

DATABASES

30 items · 401 marks

Syllabus sections in this unit

8.1 Database Concepts

8.2 Database Management Systems (DBMS)

8.3 Data Definition Language (DDL) and Data Manipulation Language (DML)

Unit 8: Databases

1. M/J 26/P11/Q4/a,b,c

- 4 A shop allows customers to pay to borrow toys. The shop uses a database to record information about the customers and the toys they have borrowed.

CUSTOMER (CustomerID, ContactFirstName, Phone)

TOY (ToyID, ToyName, CostPerDay)

TOY_INSTANCE (UniqueToyID, ToyID, Quality)

BORROW (BorrowID, CustomerID, UniqueToyID, StartDate, NumberDays)

- (a) Complete the table by describing the relationship between each pair of tables. Include the primary key(s) and foreign key(s) and how these create each relationship.

pair of tables	description of relationship
CUSTOMER and BORROW	
TOY_INSTANCE and TOY	

[4]

- (b) Write a Structured Query Language (SQL) script to return the total number of days of all toys borrowed by each customer. The customer ID and first name for each customer also need to be returned.

The result needs to include a suitable field name for the total number of days.

[4]

- (c) The table BORROW needs amending to include an additional field.

- (i) The field will store whether the toy has been returned and will store either Yes or No.

Write an SQL script to include the additional field in the table BORROW.

[2]

- (ii) The toy that was borrowed with the borrow ID A2226 has been returned.

Write an SQL script to amend the record and store that the toy has been returned.

[3]

2. M/J 26/P12/Q3/a,b,c

- 3** A computer game developer stores data about a computer game in a database. The database has three entities:

- the users, who are called players
- the characters that users can create
- the groups that users can create and allow characters to join.

(a) Describe the relationships between the **three** entities.

[2]

(b) The database is created with **four** tables:

PLAYER(PlayerID, PlayerName, DateJoined)

CHARACTER(CharacterID, CharacterName, DateCreated, Level, NumberOfPoints, PlayerID)

GROUP(GroupID, GroupName, Category)

CHARACTER_GROUP(CharacterGroupID, CharacterID, GroupID, DateJoinedGroup)

Sample data for the table CHARACTER is given:

CharacterID	CharacterName	DateCreated	Level	NumberOfPoints	PlayerID
00123	Crystal	05/05/2025	5	35687	85AD
00124	Diamond	06/06/2025	4	21055	69CD
00125	Ruby	07/10/2025	1	1002	85AD

- (i) The given Structured Query Language (SQL) script is written to define the table CHARACTER.

```

01  CREATE TABLE CHARACTER(
02      CharacterID INT,
03      CharacterName VARCHAR,
04      DateCreated DATE,
05      TheLevel INT,
06      NumberOfPoints INT,
07      PlayerID VARCHAR,
08      PRIMARY KEY CharacterID,
09      FOREIGN KEY (PlayerID)
10  );

```

The SQL script contains **four** errors.

Give the line number of each error and the corrected line.

error line 1

corrected line

error line 2

corrected line

error line 3

corrected line

error line 4

corrected line

[4]

- (ii) Write an SQL script to return each player's name and the number of characters each player has created.

The result needs to include a suitable field name for the number of characters.

Order the results in descending order of player ID.

[5]

- (c) Complete the description of database key terms by writing the missing words.

The stores data about the data in the database. The table names, field names and relationships will be stored. Two other examples of data that will be stored are and

When designing a database, the schema is the design of the database tables, fields and relationships independent of any Database Management System (DBMS).

A DBMS provides security features to protect the data.

are used to give user accounts permission to read data, some to read and write data and others may be unable to read the data.

When creating a database, a provides tools to create visual components for example forms and reports.

[5]

3. M/J 26/P13/Q4/a,b,c

- 4 A database needs designing for a school library.

The library stores data about books including the author, title, category and number of pages.

The library will also store data about the students and the books they have borrowed. The data needs to include the student's name and student's class.

Books are borrowed for 14 days. A book is late if it has not been returned within the 14 days. The database needs to store whether the book is late, or not late.

- (a) Create a 3-table design for the library database normalised to Third Normal Form (3NF).

Give your table design in the format:

TableName (Field1, Field2, Field3, ...)

A primary key is underlined with a solid line.

A foreign key is underlined with a broken line.

table 1

table 2

table 3

[6]

(b) Explain the reasons why the library database should enforce referential integrity.

[2]

(c) The school is also creating a database to store data about the departments, the staff and classes in the school. The database is called `SCHOOLDATA`. The database has four tables:

`STAFF(StaffID, FirstName, FamilyName, DepartmentID)`

`DEPARTMENT(DepartmentID, DepartmentName, DepartmentLeader)`

`CLASS(ClassID, YearGroup, SubjectCode)`

`SUBJECT(SubjectCode, SubjectName, DepartmentID)`

(i) The table shows some sample data for `SUBJECT`.

SubjectCode	SubjectName	DepartmentID
CS1	Computer Science IGCSE	ITCS
CS2	Computer Science IAS-level	ITCS
MA3	Mathematics IGCSE	MTS
ME4	Media IGCSE	ITCS

The Structured Query Language (SQL) scripts to create the database and the table `SUBJECT` are shown.

```

01  CREATE DATA BASE SCHOOLDATA;
02  CREATE TABLE SUBJECT(
03      SubjectCode VARCHAR NONULL,
04      SubjectName VARCHAR
05      DepartmentID VARCHAR,
06      PRIMARY KEY (SubjectCode),
07      FOREIGN KEY (DepartmentID) REFERENCES (DepartmentID)
08  );

```

The SQL scripts contain **four** errors.

Give the line number of each error and the corrected line.

error line 1

corrected line

error line 2

corrected line

error line 3

corrected line

error line 4

corrected line

[4]

- (ii) Write an SQL script to return each department's name and the number of subject codes for that department.

The result needs to include a suitable field name for the number of subject codes.

Order the results in ascending order of department name.

[5]

4. O/N 25/P11/Q2/a,b,c,d

- 2 A relational database, `WORKEXP`, stores data about students and the companies where they complete work experience.

Students can complete multiple work experience placements but can only complete one placement at a time.

Students can complete more than one placement at the same company.

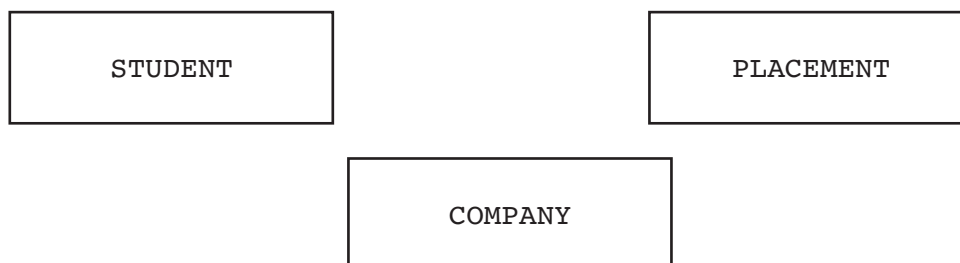
Part of the database is shown:

`STUDENT`(`StudentID`, `FirstName`, `TelephoneNumber`, `UniversityName`)

`PLACEMENT`(`PlacementID`, `StudentID`, `CompanyID`, `StartDate`, `EndDate`, `Complete`)

`COMPANY`(`CompanyID`, `CompanyName`, `MaxStudentsPerPlacement`)

- (a) Complete the entity-relationship (E-R) diagram for the database `WORKEXP`.



[2]

- (b) Explain why the database `WORKEXP` is in Third Normal Form (3NF).

[2]

- (c) Some example data from the `PLACEMENT` table is shown:

<code>PlacementID</code>	<code>StudentID</code>	<code>CompanyID</code>	<code>StartDate</code>	<code>EndDate</code>	<code>Complete</code>
PC001	LDEA01	MSCM	01/01/2023	12/07/2023	TRUE
PC002	LDEA01	MSCM	01/04/2024	08/05/2024	FALSE
PC003	ALAU02	NEAM	09/03/2020	10/03/2021	FALSE
PC004	LOLI75	GOUZ	07/06/2018	11/09/2018	TRUE

- (i) Write a Structured Query Language (SQL) script to delete all placements that have been completed.

[2]

- (ii) Write an SQL script to return the total number of placements completed by the student with ID LDEA01 at the company with ID NEAM. The total should be given an appropriate name.

[4]

- (d) The database tables are repeated here:

STUDENT(StudentID, FirstName, TelephoneNumber, UniversityName)

PLACEMENT(PlacementID, StudentID, CompanyID, StartDate, EndDate,
Complete)

COMPANY(CompanyID, CompanyName, MaxStudentsPerPlacement)

Explain what is meant by referential integrity, and how it applies to this database.

[3]

5. O/N 25/P12/Q4/a,b,c,d

- 4 A relational database, SHIPPING, stores data about the ships in a company and the containers that are carried on the ships.

The database has the following tables:

CONTAINER(ContainerID, Type, Weight, OwnerName, ShipID)

SHIP(ShipID, Type, Capacity, ShipName)

- (a) Describe the relationship between the two tables. Refer to the primary and foreign keys in your answer.

[2]

- (b) The table CONTAINER needs an additional field to store the data for the last inspection date.

Write a Structured Query Language (SQL) script to add **one** field to the table CONTAINER to store the date of last inspection of the container, for example 08/07/2019.

[2]

- (c) Write an SQL script to return the number of containers stored in the database for the ship with the name Caledonia.

[4]

- (d) Describe the purpose of a developer interface in a Database Management System (DBMS).

[2]

6. O/N 25/P13/Q5/a,b,c,d

5 A company creates a relational database to store data about its customers.

The database, `REVIEWS`, stores data about the customers, products, complaints and staff.

The database has four tables:

`CUSTOMER`(`CustomerID`, `CustomerFirstName`, `CustomerLastName`, `CustomerEmail`)

`PRODUCT`(`ProductID`, `ProductName`, `ProductDetail`, `Price`, `Rating`)

`COMPLAINT`(`ComplaintID`, `ProductID`, `CustomerID`, `ComplaintDetails`, `StaffID`)

`STAFF`(`StaffID`, `StaffFirstName`, `StaffLastName`, `Department`, `RemoteWorker`)

(a) Complete the entity-relationship (E-R) diagram for the database `REVIEWS`.



[3]

(b) Some example data from the `STAFF` table is shown.

<code>StaffID</code>	<code>StaffFirstName</code>	<code>StaffLastName</code>	<code>Department</code>	<code>RemoteWorker</code>
1	Ralph	Jura	E	Yes
2	Luca	Emcee	A	Yes
3	Darwin	Acula	F	No

Write a Structured Query Language (SQL) script to define the table `STAFF`.

[4]

(c) Products are given a rating between 1 and 10 inclusive.

Write an SQL script to return only the `ProductID`, `ProductName` and `ComplaintDetails` for all products with a rating of 5 or less. The results need to be displayed in descending order of rating.

[5]

(d) A Database Management System (DBMS) is used to maintain and manage the database.

Describe **two** ways in which the DBMS can be used to ensure the security of the customer data.

1

2

[4]

7. M/J 25/P11/Q5/a,b,c,d,e

- 5** A company builds and sells furniture to customers. The company stores data about customers, their payment cards and their furniture orders in a database.

The database, FURNITURE, has the following tables:

CUSTOMER(CustomerID, Name, Phone)

CUSTOMER_CARD_DATA(CardID, CustomerID, CardType, CardNumber, EndDate)

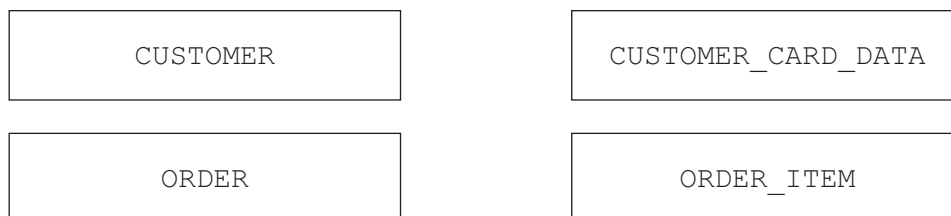
ORDER(OrderID, CustomerID, TotalCost, Paid, OrderDate, Complete)

ORDER_ITEM(OrderItemID, OrderID, Type, Height, Width, Depth, Details)

The primary keys are underlined in each table.

The attribute `Complete` in the table `ORDER` stores the Boolean value `TRUE` if the order has been built and `FALSE` if the order has not been built.

- (a) Complete the entity-relationship (E-R) diagram for the database.



[3]

- (b) Identify **one** attribute in the table `CUSTOMER_CARD_DATA` that could be a candidate key.

[1]

- (c) Identify **two** tables in the database that contain one or more foreign keys.
Give **one** attribute that is a foreign key in each table.

	Table	Foreign key
1		
2		

[2]

- (d) Explain the reasons why the data in the table `ORDER_ITEM` cannot be stored in the table `ORDER`

[3]

- (e) Write an Structured Query Language (SQL) script to output the customer ID, the customer's name and the total cost of the customer's orders that have **not** been paid.

The output of the total cost must have an appropriate title.

[4]

8. M/J 25/P12/Q5/a,b,c,d

- 5 An online game has a database that stores data about users and the characters each user creates in the game. Each user can create multiple characters and purchase multiple items for each character.

The normalised database has the following design:

USER (Username, Password, DateOfBirth)

CHARACTER (CharacterName, CharacterID, Username, Level, Money)

ITEM (ItemName, MinimumLevel, Cost)

CHARACTER_ITEM (CharacterID, ItemName)

- (a) Explain the purpose of the table CHARACTER_ITEM in the database.

[2]

- (b) Underline the attribute, or attributes, that form the primary key in each of the tables.

USER (Username, Password, DateOfBirth)

CHARACTER (CharacterName, CharacterID, Username, Level, Money)

ITEM (ItemName, MinimumLevel, Cost)

CHARACTER_ITEM (CharacterID, ItemName)

[2]

- (c) A Database Management System (DBMS) provides data security.

- (i) Identify **two** methods the DBMS can use to protect the data in the table USER from unauthorised access.

Explain how each method protects the data.

Method 1

Explanation

Method 2

Explanation

[4]

- (ii) The DBMS also supports data integrity.

Give **two** ways that a DBMS can support data integrity.

1

2

[2]

- (d) (i) Write a Structured Query Language (SQL) script to count the number of items purchased by the user with the username "KAT123".

[3]

- (ii) The following changes need to be made to the character with the ID "0002":

- level changed to 3
- money changed to 10000.00

Write an SQL script to change the character's data.

[3]

9. M/J 25/P13/Q6/a,b,c,d

- 6** A shop sells pens to customers. Customers place an order with the shop and collect the items the next day. The shop uses a database to store the information about the orders.

The database contains the following tables:

`CUSTOMER(CustomerID, CustomerName, Email)`

`ORDER(OrderID, CustomerID, Date, Collected)`

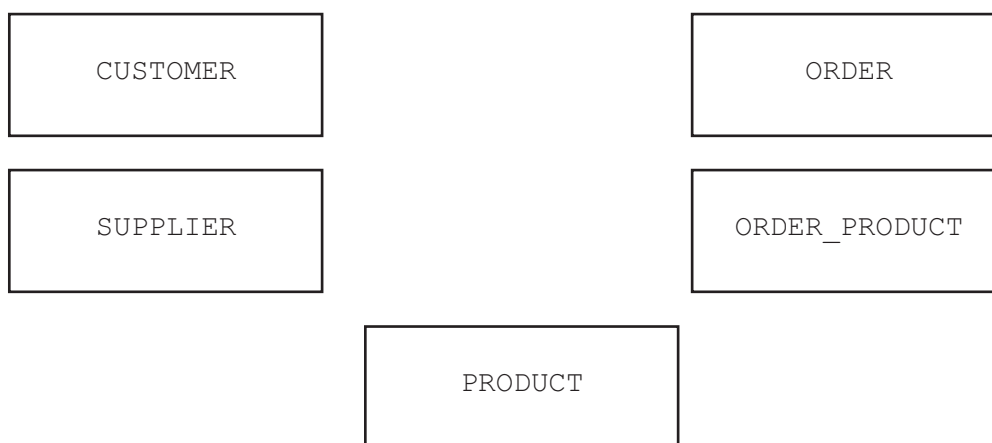
`ORDER_PRODUCT(OrderID, ProductID, Quantity)`

`PRODUCT(ProductID, ProductName, QuantityInBox, Cost, SupplierID)`

`SUPPLIER(SupplierID, SupplierName, SupplierEmail)`

The primary keys are underlined.

- (a)** Complete the entity-relationship (E-R) diagram for the database.



[4]

- (b)** A new product needs to be entered into the database. The product has the ID 002323, the product name 'Blue ball point 2mm', there are 50 in a box, the product costs \$5.00 and the supplier has the ID SFX223.

Write a Structured Query Language (SQL) script to enter the new product into the table `PRODUCT`.

[4]

- (c)** The attribute `Collected` in the table `ORDER` stores the Boolean value `TRUE` if the order has been collected and `FALSE` if the order has not been collected.

Write an SQL script to return the customer name for each customer that has orders they have not collected. Include the number of orders each customer has not collected with an appropriate title.

An example output might be:

CustomerName	NotCollected
Jack Wright	2
Lin Cho	1
Santaya Yui	1

[4]

(d) The shop uses a Database Management System (DBMS).

(i) One feature provided by a DBMS is a data dictionary.

The data dictionary stores the attribute names, table names, foreign keys and primary keys in the database.

Identify **three** other items stored in a data dictionary.

1

2

3

[3]

(ii) The DBMS provides a developer interface.

Explain how a database designer can make use of the developer interface.

[3]

10. O/N 24/P11/Q2/a,b,c

2 A shop repairs electronic devices, for example mobile phones and tablet computers. The shop owner stores the data about the repairs using a file-based approach.

(a) Give **one** limitation of using a file-based approach to store the data **and** explain how a relational database addresses this limitation.

Limitation

Explanation

[3]

(b) The shop owner creates a relational database called `FIXIT`.

The database stores data about the customers and the devices for repair.

Some devices need new parts that are ordered from suppliers.

The database `FIXIT` is designed to include the following tables:

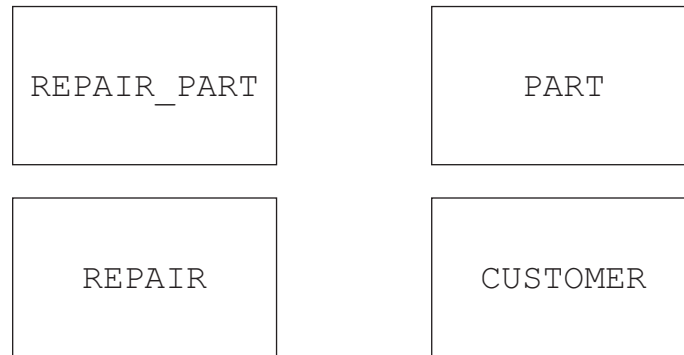
`PART(PartID, Description, Price, SupplierID)`

`CUSTOMER(CustomerID, FirstName, LastName, ContactNumber)`

`REPAIR(RepairNumber, StartDate, EndDate, CustomerID, Device)`

`REPAIR_PART(PartID, RepairNumber, Quantity)`

- (i) Complete the entity-relationship (E-R) diagram for the given tables.



[3]

- (ii) The table shows sample data for the table `REPAIR_PART`.

PartID	RepairNumber	Quantity
ACD128SA	0022	3
PPOR543DWW	0022	1
TR453	0023	1
PPOR543DWW	0023	2
WED5	0024	5

Write a Structured Query Language (SQL) script to define the table `REPAIR_PART`.

Include constraints (restrictions) on the data that can be entered into each field where appropriate.

[5]

- (iii) Suppliers send invoices to the company for the parts that are used. A new table, `INVOICE`, stores the data about each invoice and whether it has been paid or not.

The design for the table `INVOICE` is shown:

```
INVOICE(InvoiceID, SupplierID, AmountDue, Paid, DatePaid)
```

The table shows sample data for the table `INVOICE`.

InvoiceID	SupplierID	AmountDue	Paid	DatePaid
000001	JK675	22.50	TRUE	01/01/2024
000002	WR443	358.99	FALSE	
000003	JK675	10.21	FALSE	

Write an SQL script to return the total amount due to the supplier with the ID of JK675 for all the invoices that have **not** currently been paid.

[3]

(c) Complete the table by writing a definition for each of the database terms.

Term	Definition
Referential integrity	
Candidate key	
Tuple	

[3]

11. O/N 24/P12/Q6/a,b

- 6** A company uses a relational database to store data about its customers, employees and the individual repair jobs that customers have booked.

(a) Explain the benefits of using a relational database instead of a file-based approach.

[3]

(b) The company decides which employees will work on each repair job. An employee can log into the database to access information about their repair jobs.

The database is normalised and includes these tables:

- CUSTOMER stores personal data about each customer
- EMPLOYEE stores personal data about each employee
- LOGIN_DATA stores the username and password for each employee
- JOB stores the data about each repair job
- JOB_EMPLOYEE stores the employees that are working on each repair job.

(i) Identify each relationship between the database tables **and** explain how each relationship can be implemented in the normalised database.

[6]

(ii) The database also has the table INVOICE that stores data about each invoice that is sent to a customer.

Example data from the table INVOICE is given.

InvoiceID	DateSent	Amount	Paid	JobID
29262	12/12/2023	105.20	Y	221
26765	11/11/2023	200.00	Y	315
13290	02/01/2024	50.00	Y	315
34090	05/02/2024	25.95	N	569

Write a Structured Query Language (SQL) script to return the total amount of all the invoices sent in the year 2023 that have been paid.

[3]

12. O/N 24/P13/Q4/a,b,c,d,e

- 4** A company makes ice cream and sells it to shops.

The ice cream is made in batches: a large quantity of one type and flavour of ice cream that is then split into smaller quantities for sale.

The company's owner has designed a relational database, ICECREAM, to store data about their ice cream and customers.

Some of the tables in the database are given. The database is not normalised.

BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)

CUSTOMER(CustomerID, CompanyName, EmailAddress, TelephoneNumber)

SALE(SaleID, BatchID, CustomerID, Quantity, Date)

(a) Identify **two** foreign keys in the table SALE **and** the table that each foreign key references.

Foreign key 1

Table name 1

Foreign key 2

Table name 2

[2]

- (b) Write an SQL script to return the total quantity of ice cream sold to the customer with the ID of 0034E in the year 2023.

[3]

- (c) The table definition for `BATCH` is repeated here:

`BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)`

Sample data for the table `BATCH` is given:

BatchID	Type	Flavour	Size	SellingPrice	EndDate
KIV12	Plain	Vanilla	1	2.20	12/12/2024
RIC14	Plain	Chocolate	0.5	1.80	12/12/2024
TYL1	Non-dairy	Lemon	0.5	2.10	01/02/2025
FYV2	Non-dairy	Vanilla	0.25	1.50	02/02/2025
BIV13	Plain	Vanilla	1	2.20	02/02/2024

- (i) Write an SQL script to define the table `BATCH`.

Include constraints (restrictions) on the data that can be entered into each field where appropriate.

[5]

- (ii) The table `BATCH` is not normalised.

Normalise the database table `BATCH`.

Write the table definitions for your new tables.

Identify any primary and foreign keys in your tables.

Do **not** change or include the tables `CUSTOMER` and `SALE`.

[4]

(d) Complete the following table by defining each database term.

Database term	Definition
Entity	
Attribute	

[2]

(e) A Database Management System (DBMS) supports data integrity.

Explain how a DBMS supports data integrity.

[3]

13. M/J 24/P11/Q6/a,b,c

- 6 A company is developing a website that will allow users to create an account and then play a quiz every day. The data about the users and the quizzes are stored in a database.

A user must select a unique username and enter a valid email address to create an account. All users must be over the age of 16. A new quiz is given to the users every day. Each quiz is stored in its own text file.

The database stores the filename of each quiz and the date it can be played. The user gets a score for each quiz they complete, which is stored in the database. The scores are used to give each user a rating, for example Gold.

(a) Create a 3-table design for this database normalised to Third Normal Form (3NF).

Give your table design in the format:

TableName (PrimaryKey, Field1, Field2, ...)

[6]

(b) The company is using a Database Management System (DBMS) to set up the database.

Describe what is meant by the following DBMS features:

Data dictionary

Logical schema

[4]

(c) The company has another database, FARMING, for a different game.

The database FARMING has a table named EVENT which is shown with some sample data.

PlayerID	EventID	Category	Points
000123	3	Build	100
000124	1	Grow	36
000123	4	Grow	22
000123	7	Create	158
000125	3	Grow	85
000125	4	Build	69

- (i) The database `FARMING` has a second table created named `PLAYER` that has the primary key `PlayerID`.

The field `PlayerID` in `EVENT` needs to be set up as a foreign key to link to `PlayerID` in `PLAYER`.

Write a Structured Query Language (SQL) script to change the table definition for `EVENT` to link the foreign key to `PLAYER`.

[2]

- (ii) Write an SQL script to return the number of events that each player has completed.

[3]

14. M/J 24/P12/Q4/a,b,c,d

- 4** An assessment board wants to store the marks students achieved in exams in a database named RECORDS.

Part of the database design includes these two tables:

EXAM(ExamID, Subject, Level, TotalMarks)

EXAM_QUESTION(ExamQuestionID, ExamID, QuestionNumber, Question, MaxMark)

- (a)** Identify the relationship between EXAM and EXAM_QUESTION.

[1]

- (b)** Sample data for the table EXAM is shown:

ExamID	Subject	Level	TotalMarks
00956124	Computer Science	2	75
00956125	Computer Science	3	120
00956126	Mathematics	2	100
00956127	Mathematics	3	150
00956128	Physics	2	70
00956129	Physics	3	80

Write a Structured Query Language (SQL) script to define the table EXAM.

[3]

- (c)** The table EXAM_QUESTION has been created but the foreign key has not been linked.

Write an SQL script to update EXAM_QUESTION **and** link the foreign key to EXAM.

[2]

- (d)** The database also needs to store data about the students, the exams the students have taken and the marks the students achieved in each question of each exam.

Describe the additional tables that will need to be included in the database **and** explain how all the tables in the database will be linked.

[5]

15. M/J 24/P13/Q4/a,b,c,d

- 4** A theatre wants to use a database to store data about the shows that are scheduled, their customers and the seats that the customers have booked.

In the theatre:

- Each show can take place on multiple dates.
- Some dates can have more than one performance.
- There are multiple rows of seats.
- Each seat can be individually booked by its row letter and seat number, for example row E seat 2.

Part of the database design includes these tables:

SHOW(ShowID, Title, Duration)

SEAT(SeatID, RowLetter, SeatNumber)

PERFORMANCE(PerformanceID, ShowID, ShowDate, StartTime)

- (a)** Identify the relationship between the tables PERFORMANCE and SHOW.

[1]

- (b)** Sample data for the table PERFORMANCE is shown:

PerformanceID	ShowID	ShowDate	StartTime
0001	MK12	5/5/2025	13:00
0002	MK12	5/5/2025	19:30
0003	MK12	6/5/2025	19:00
0004	OP3	7/5/2025	18:30
0005	OP3	8/5/2025	18:30
0006	OP3	9/5/2025	13:00

Write a Structured Query Language (SQL) script to define the table PERFORMANCE.

[4]



- (c)** Write an SQL script to return the number of times each show is scheduled. For example, in the sample data in part **(b)**, the show MK12 is scheduled three times.

The result needs to include the show name and a suitable field name for the number of times it is scheduled.

[4]

- (d)** Customers give their first name, last name and email address when they are making a booking. One booking can include multiple seats.

Describe the additional tables that will need to be included in the database **and** explain how these tables will be linked within the database.

[5]



16. O/N 23/P11/Q3/a,b,c

- 3** A shop manager has designed a relational database to store customer orders.

The database will have the following tables:

CUSTOMER(CustomerID, FirstName, LastName, Town)

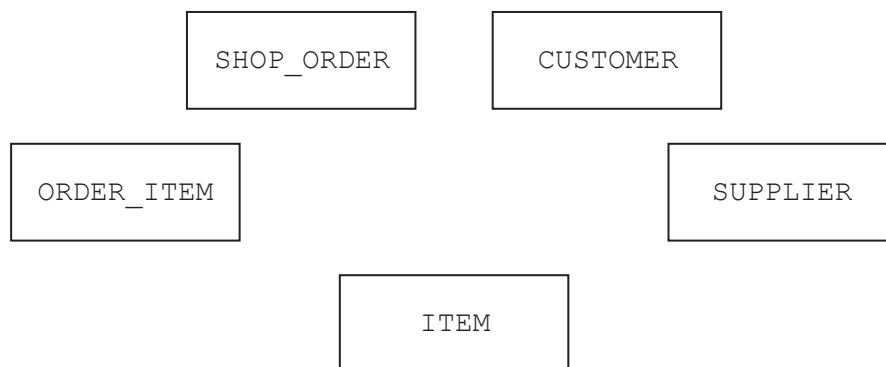
SHOP_ORDER(OrderNo, CustomerID, OrderDate)

SUPPLIER(SupplierID, EmailAddress, TelephoneNumber)

ITEM(ItemNumber, SupplierID, Description, Price)

ORDER_ITEM(ItemNumber, OrderNo, Quantity)

- (a)** Complete the entity-relationship (E-R) diagram for the relational database.



[3]

- (b)** Identify **three** advantages of a relational database compared to a file-based approach.

1

2

3

[3]

- (c) (i)** Write a Structured Query Language (SQL) script to define the database called *SHOP*.

[1]

- (ii)** Write the SQL script to return the total quantity of items that the customer with the ID of HJ231 has ordered.

[4]

17. O/N 23/P12/Q2/a,b,c

- 2 (a)** State what is meant by the following terms in a relational database model.

Entity

Primary key

Referential integrity

[3]

- (b)** Authentication is one method a Database Management System (DBMS) can use to improve the security of a database.

Describe **other** methods that a DBMS can use to improve the security of a database.

[4]

- (c)** The following database table is not normalised.

StudentName	DateOfBirth	TutorGroup	Subject	SubjectCode
Yuwei Chen	01/09/2004	SMH	English, Maths, Computer Science	EN, MA, CS
Claudia Raj	23/02/2005	JMB	Maths, Physics, Art	MA, PY, AR
Aamil Akram	24/01/2005	KMB	Art, Design, English language	AR, DE, EN
Areeba Faraz	21/12/2004	SMH	English language, Chemistry, Design	EN, CH, DE

Explain how to modify the table to put it into First Normal Form (1NF).

[4]

18. O/N 23/P13/Q3/b

- 3** A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

- (b)** The company uses a database, `COURSES`, to store data about the courses and their tutors.

Each course starts at different times of the year and may have a different tutor.

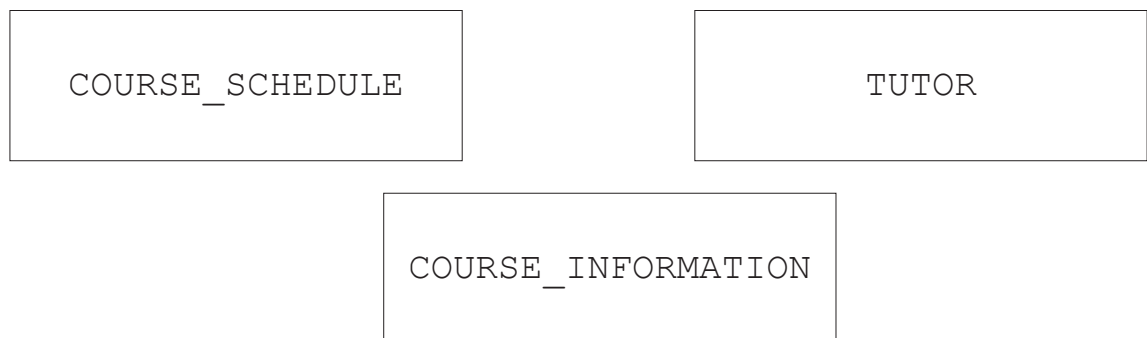
The database has the following structure:

```
COURSE_INFORMATION(CourseID, Description, Cost)
```

```
TUTOR(TutorID, TelephoneNumber, EmailAddress, TutorName)
```

```
COURSE_SCHEDULE(CourseID, DateStarted, TutorID)
```

- (i)** Complete the entity-relationship (E-R) diagram for the database `COURSES`.



[1]

- (ii)** Write the Structured Query Language (SQL) script to return the total number of courses that have started after 9 September 2023.

The value returned must have an appropriate field name.

[4]

19. M/J 23/P11/Q2/a,b

- 2** An organisation uses a database to store data about the types of bird that people have seen.

- (a)** The database is managed using a Database Management System (DBMS).

- (i)** State what is meant by a data dictionary **and** give **one** example of an item typically found in a data dictionary.

Definition

Example

[2]

- (ii)** State what is meant by data integrity **and** give **one** example of how this is implemented in a database.

Definition

Example

[2]

- (b) The database, `Birds`, stores information about the types of bird and the people who have seen them.

Data about each bird seen is stored with its location and data about the person who saw the bird.

Database `Birds` has the following tables:

```
BIRD_TYPE(BirdID, Name, Size)
```

```
BIRD_SEEN(SeenID, BirdID, Date, Location, PersonID)
```

```
PERSON(PersonID, FirstName, LastName, EmailAddress)
```

- (i) Complete the table by identifying **two** foreign keys and the database table where each is found.

Foreign key	Database table

[2]

- (ii) The database `Birds` has been normalised.

Draw **one** line from each Normal Form to the most appropriate definition.

Normal Form

Definition

First Normal Form (1NF)

All fields are fully dependent on the primary key.

Second Normal Form (2NF)

There are no repeating groups of attributes.

Third Normal Form (3NF)

There are no partial dependencies.

[1]

- (iii) Part of the database table `BIRD_TYPE` is shown:

BirdID	Name	Size
0123	Blackbird	Medium
0035	Jay	Large
0004	Raven	Large
0085	Robin	Small

The database only supports these data types:

- character
- varchar
- Boolean
- integer
- real
- date
- time

Write a Structured Query Language (SQL) script to define the table `Bird_Type`.

[4]

(iv) The database tables are repeated here for reference:

```
BIRD_TYPE(BirdID, Name, Size)
```

```
BIRD_SEEN(SeenID, BirdID, Date, Location, PersonID)
```

```
PERSON(PersonID, FirstName, LastName, EmailAddress)
```

Complete the SQL script to return the number of birds of each size seen by the person with the ID of J_123.

```
SELECT BIRD_TYPE.Size, ..... (BIRD_TYPE.BirdID)

      AS NumberOfBirds

FROM BIRD_TYPE, .....

WHERE ..... = "J_123"

AND BIRD_TYPE.BirdID = .....

..... BIRD_TYPE.Size;
```

[5]

20. M/J 23/P12/Q2/a,b,c(ii,iii)

2 A horse riding school uses a database, `Lessons`, to store data about lesson bookings.

This database is created and managed using a Database Management System (DBMS).

(a) The table contains names and descriptions of DBMS features and tools.

Complete the table by writing down the missing names and descriptions.

Name	Description
Data dictionary	
Query processor	
.....	A model of a database that is not specific to one DBMS.
.....	A software tool that allows the user to create items such as tables, forms and reports.

[4]

(b) Explain the reasons why referential integrity is important in a database.

[3]

(c) The database `Lessons` has the following tables:

```
HORSE(HorseID, Name, Height, Age, HorseLevel)
```

```
STUDENT(StudentID, FirstName, LastName, RiderLevel, PreferredHorseID)
```

```
LESSON(LessonID, Date, Time, StudentID, HorseID, LessonContent)
```

Dates in this database are stored in the format `#DD/MM/YYYY#`.

The fields `RiderLevel` and `HorseLevel` can only have the values: `Beginner`, `Intermediate` or `Advanced`.

(ii) Write a Structured Query Language (SQL) script to return the names of all the horses that have the horse level intermediate or beginner.

[4]

(iii) The following SQL script should return the number of riders that have the rider level beginner and have a lesson booked on 09/09/2023.

```
SELECT SUM(STUDENT.RiderLevel) AS NumberOfRiders  
  
FROM STUDENT, LESSON  
  
WHERE StudentID = StudentID  
  
OR Date = #09/09/2023#  
  
AND STUDENT.RiderLevel = Beginner;
```

There are **four** errors in the script.

Identify **and** correct each error.

1

2

3

4

[4]

21. M/J 23/P13/Q4/a,b,c

- 4 A shop rents cars to customers. The shop uses a relational database to store information about the rentals.

(a) Describe **two** ways in which a relational database addresses the limitations of a file-based approach.

1

2

[4]

(b) Complete the table by writing the missing term or description for each database feature.

Term	Description
.....	An object that data is stored about.
Tuple	
Secondary key	
.....	A field in one table that is linked to a primary key in another table.

[4]

(c) The car rental database is not normalised. The current database design is:

```
BOOKING (CarRegistration, StartDate, EndDate,
         CarModel, CarColour, CustomerFirstName)
```

```
CUSTOMER (CustomerFirstName, CustomerLastName, EmailAddress,
          TelephoneNumber)
```

Write a normalised database design for this database.

All tables must be in Third Normal Form (3NF).

Use the field names given **and** underline the primary key fields.

[4]

22. O/N 22/P11/Q4/a,b,c,d

- 4 A photographer creates a relational database to store data about photographs taken at birthday parties.

The database, PHOTOGRAPHS, stores details of the customer, the party, the photographs taken and the cameras used.

The photographer has several cameras that are used for taking the photographs at the parties.

Each camera has a specific lens type (for example, XY32Z) and lighting type (for example, F1672).

Data about each photograph is stored in the database including the party at which it was taken, the time it was taken and the camera used.

The database has these four tables:

CUSTOMER(CustomerID, FirstName, LastName, Telephone)

PARTY(PartyID, CustomerID, PartyDate, StartTime)

PHOTO_DATA(PhotoID, PartyID, TimeTaken, CameraID)

CAMERA_DATA(CameraID, LensType, LightingType)

(a) Complete the entity-relationship (E-R) diagram for the database PHOTOGRAPHS.



[3]

(b) The database is normalised and is in Third Normal Form (3NF).

Describe the characteristics of a database that is in Third Normal Form (3NF).

[3]

(c) The table shows some sample data for the table PHOTO_DATA.

PhotoID	PartyID	TimeTaken	CameraID
ST23-56	BD987	08:34	NIK-02
ST23-57	BD987	08:55	NIK-02
ST23-60	BC08	09:01	CAN-01
ST23-61	BC08	10:23	CAN-12
ST23-62	BC08	10:56	NIK-01

(i) State what is meant by a **tuple**. Give an example of a tuple from PHOTO_DATA.

Tuple

Example

[2]

(ii) Complete the Structured Query Language (SQL) script to display the total number of photographs that have been taken using a camera with a camera ID starting with CAN.

```

SELECT ..... (.....)

FROM

WHERE CameraID LIKE ;

```

[4]

- (d) Write an SQL script to include two new fields in CAMERA_DATA to store the number of photographs currently on the camera **and** the date the camera was last used.

[3]

23. O/N 22/P12/Q5/a,b,c,d

- 5 A relational database, GARDEN, has the following tables:

OWNER(OwnerID, FirstName, TelephoneNo, TreeID, TreePosition)

TREE(TreeID, ScientificName, MaxHeight, FastGrowing)

- (a) The database is **not** in Third Normal Form (3NF).

Explain how the database can be normalised to 3NF.

[3]

- (b) Write the Structured Query Language (SQL) script to add a new record in the table TREE to store the following data:

Attribute	Value
TreeID	LOW_1276
ScientificName	Salix_Alba
MaxHeight	30.00
FastGrowing	TRUE

[3]

- (c) State what is meant by a **candidate key** in a relational database.

[1]

- (d) (i) Describe, using an example, what is meant by a **data dictionary**.

[2]

- (ii) Describe what is meant by a **logical schema**.

[2]

24. O/N 22/P13/Q2/a,b,c,d,f

- 2** The relational database `ASTRONOMY` is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

`COMPANY (TelephoneNumber, CompanyID, CompanyName)`

`PHOTOGRAPH (PhotoID, TelescopeID, DateTaken, TimeTaken, Elevation)`

`TELESCOPE (TelescopeID, CompanyID, SerialNumber)`

- (a)** Complete the following table by writing the correct answer for each item.

Item	Answer
a suitable field for the primary key in <code>COMPANY</code>	
a candidate key in <code>TELESCOPE</code>	
the degree of relationship between <code>TELESCOPE</code> and <code>PHOTOGRAPH</code>	

[3]

- (b)** A Database Management System (DBMS) has several features.

Identify the feature that describes the relationship between data and its structure.

[1]

- (c)** Complete the SQL script to return the total number of telescopes owned by the company whose ID begins with `HW`.

```
SELECT ..... (.....)
FROM TELESCOPE
WHERE ..... LIKE .....
```

[4]

- (d)** Write the SQL script to add **one** field to the table `PHOTOGRAPH` to store the resolution of the photograph, e.g. `1920 × 1068`.

[2]

- (f)** Describe the purpose of a query processor in a DBMS.

[2]

25. O/N 21/P11/Q5/a,b,c

- 5** Javier owns many shops that sell cars. He employs several managers who are each in charge of one or more shops. He uses the relational database `CARS` to store the data about his business.

Part of the database is shown:

`SHOP(ShopID, ManagerID, Address, Town, TelephoneNumber)`

`MANAGER(ManagerID, FirstName, LastName, DateOfBirth, Wage)`

`CAR(RegistrationNumber, Make, Model, NumberOfMiles, ShopID)`

- (a)** Tick (✓) **one** box in each row to identify whether each field is a primary key or a foreign key.

Table	Field name	Primary key	Foreign key
MANAGER	ManagerID		
SHOP	ManagerID		
CAR	RegistrationNumber		
CAR	ShopID		

[2]

- (b)** Describe the ways in which access rights can be used to protect the data in Javier's database from unauthorised access.

[3]

- (c)** Javier uses Data Definition Language (DDL) and Data Manipulation Language (DML) statements in his database.

- (i)** Complete the following DML statements to return the number of cars for sale in each shop.

`SELECT COUNT (.....)`

`FROM .....`

`..... ShopID`

[3]

- (ii)** Complete the DML statement to include the following car in the table `CAR`.

Field	Data
RegistrationNumber	123AA
Make	Tiger
Model	Lioness
NumberOfMiles	10500
ShopID	12BSTREET

CAR

..... ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")

[2]

26. O/N 21/P12/Q6/a,b,c

- 6 A shop sells plants to customers. The shop manager has a relational database to keep track of the sales.

The database, PLANTSALLES, has the following structure:

PLANT(PlantName, QuantityInStock, Cost)

CUSTOMER(CustomerID, FirstName, LastName, Address, Email)

PURCHASE(PurchaseID, CustomerID)

PURCHASE_ITEM(PurchaseID, PlantName, Quantity)

(a) The database is normalised.

(i) The table lists the following three stages of normalisation:

- The first stage is from a database that is not normalised (0NF) to First Normal Form (1NF).
- The second stage is from 1NF to Second Normal Form (2NF).
- The third stage is from 2NF to Third Normal Form (3NF).

Tick (✓) **one** box in each row to identify the appropriate stage for each task.

Task	Normalisation stage		
	0NF to 1NF	1NF to 2NF	2NF to 3NF
Remove any partial key dependencies			
Remove any repeating groups of attributes			
Remove any non-key dependencies			

[2]

(ii) Draw an entity-relationship (E-R) diagram for the database PLANTSALLES.



[3]

(b) The shop manager uses a Database Management System (DBMS).

Describe the purpose **and** contents of the data dictionary in the DBMS.

[3]

(c) The shop manager uses both Data Definition Language (DDL) and Data Manipulation Language (DML) statements to create and search the database.

(i) Complete the DML statements to return the total number of items purchased with the purchase ID of 3011A.

SELECT SUM (.....)

FROM

WHERE =;

[4]

(ii) Write DDL statements to include a field in the table `PURCHASE` to store the date of the order.

[3]

27. O/N 21/P13/Q5/a,b,c

5 Javier owns many shops that sell cars. He employs several managers who are each in charge of one or more shops. He uses the relational database `CARS` to store the data about his business.

Part of the database is shown:

`SHOP(ShopID, ManagerID, Address, Town, TelephoneNumber)`

`MANAGER(ManagerID, FirstName, LastName, DateOfBirth, Wage)`

`CAR(RegistrationNumber, Make, Model, NumberOfMiles, ShopID)`

(a) Tick (✓) **one** box in each row to identify whether each field is a primary key or a foreign key.

Table	Field name	Primary key	Foreign key
MANAGER	ManagerID		
SHOP	ManagerID		
CAR	RegistrationNumber		
CAR	ShopID		

[2]

(b) Describe the ways in which access rights can be used to protect the data in Javier's database from unauthorised access.

[3]

(c) Javier uses Data Definition Language (DDL) and Data Manipulation Language (DML) statements in his database.

- (i) Complete the following DML statements to return the number of cars for sale in each shop.

SELECT COUNT (.....)

FROM

..... ShopID

[3]

- (ii) Complete the DML statement to include the following car in the table CAR.

Field	Data
RegistrationNumber	123AA
Make	Tiger
Model	Lioness
NumberOfMiles	10500
ShopID	12BSTREET

CAR

..... ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")

[2]

28. M/J 21/P11/Q7/a,b,c

7 Bobby and Kim are discussing databases.

- (a)** Bobby tells Kim that a file-based approach is usually better than a relational database.

Explain why Bobby is incorrect.

[3]

- (b)** Bobby has a shop that sells products to customers. His database will store data about his customers, their payment details, orders and the products he sells. Customers will have login details to access their accounts. The database will update customers' payment and login details without keeping any historical records.

- (i)** Give **one** example of each of the following relationships from Bobby's database.

one-to-one

one-to-many

many-to-many

[3]

- (ii)** Tick (✓) **one** box to identify the relationship that cannot be directly implemented in a normalised relational database.

Relationship	Tick (✓)
one-to-one	
one-to-many	
many-to-many	

[1]

- (iii)** Bobby wants to name his database `SHOPORDERS`.

Write a Data Definition Language (DDL) statement to define a new database with the name `SHOPORDERS`.

[1]

- (c)** A database has a data dictionary.

Give **three** items that are stored in a data dictionary.

1

2

3

[3]

29. M/J 21/P12/Q1/a,b,c

- 1 Raj owns houses that other people rent from him. He has a database that stores details about the people who rent houses, and the houses they rent. The database, `HOUSE_RENTALS`, has the following structure:

`CUSTOMER`(`CustomerID`, `FirstName`, `LastName`, `DateOfBirth`, `Email`)

`HOUSE`(`HouseID`, `HouseNumber`, `Road`, `Town`, `Bedrooms`, `Bathrooms`)

`RENTAL`(`RentalID`, `CustomerID`, `HouseID`, `MonthlyCost`, `DepositPaid`)

- (a) Give the definition of the following database terms, using an example from the database `HOUSE_RENTALS` for each definition.

Term	Definition and example
Field	
Entity	
Foreign key	

[6]

- (b) Tick (✓) **one** box to identify whether the database `HOUSE_RENTALS` is in Third Normal Form (3NF) or not in 3NF.

Justify your choice using one or more examples from the database `HOUSE_RENTALS`.

In 3NF	
Not in 3NF	

Justification

[2]

- (c) Example data from the table `RENTAL` are given:

RentalID	CustomerID	HouseID	MonthlyCost	DepositPaid
1	22	15B5L	1000.00	Yes
2	13	3F	687.00	No
3	1	12AB	550.00	Yes
4	3	37	444.50	Yes

- (i) Complete the following Data Definition Language (DDL) statement to define the table RENTAL.

```
CREATE ..... (
    RentalID INTEGER NOT NULL,
    CustomerID INTEGER NOT NULL,
    HouseID ..... (5) NOT NULL,
    MonthlyCost ..... NOT NULL,
    DepositPaid BOOLEAN NOT NULL,
    ..... (RentalID)
);
```

[4]

- (ii) Write a Data Manipulation Language (DML) script to return the first name and last name of all customers who have **not** paid their deposit.

[4]

30. M/J 21/P13/Q7/a,b,c

7 Bobby and Kim are discussing databases.

- (a)** Bobby tells Kim that a file-based approach is usually better than a relational database.

Explain why Bobby is incorrect.

[3]

- (b)** Bobby has a shop that sells products to customers. His database will store data about his customers, their payment details, orders and the products he sells. Customers will have login details to access their accounts. The database will update customers' payment and login details without keeping any historical records.

- (i)** Give **one** example of each of the following relationships from Bobby's database.

one-to-one

one-to-many

many-to-many

[3]

- (ii)** Tick (✓) **one** box to identify the relationship that cannot be directly implemented in a normalised relational database.

Relationship	Tick (✓)
one-to-one	
one-to-many	
many-to-many	

[1]

- (iii)** Bobby wants to name his database `SHOPORDERS`.

Write a Data Definition Language (DDL) statement to define a new database with the name `SHOPORDERS`.

[1]

- (c)** A database has a data dictionary.

Give **three** items that are stored in a data dictionary.

1

2

3

[3]

Answers Section

1. M/J 26/P11/Q4/a,b,c

4(a)	1 mark for each correct point - CUSTOMER BORROW is a one-to-many relationship // 1:M - TOY_INSTANCE to TOY is a many-to-one relationship // M:1 - Identifying the correct Primary Key and matching Foreign Key in both relationships - Stating how the keys form the relationship	4	
4(b)	1 mark for each correct point - Selecting the contact name, customer ID and totalling the number of days - needs all 3 - Using an appropriate name - Linking or joining the correct tables - Grouping the results Example 1: <pre>SELECT CUSTOMER.ContactFirstName, CUSTOMER.CustomerID, SUM(NumberDays) AS TotalDays FROM BORROW, CUSTOMER WHERE CUSTOMER.CustomerID = BORROW.CustomerID GROUP BY CUSTOMER.CustomerID;</pre> MP1 MP2 MP3 MP4 Example 2: <pre>SELECT CUSTOMER.ContactFirstName, CUSTOMER.CustomerID, SUM(NumberDays) AS TotalDays FROM BORROW, INNER JOIN CUSTOMER ON CUSTOMER.CustomerID = BORROW.CustomerID GROUP BY CUSTOMER.CustomerID;</pre> MP1 MP2 MP3 MP4	4	CustomerID needs the table reference Accept any correct method of joining the tables, but no mix and match between methods
4(c)(i)	1 mark for each correct statement - Altering the correct table - Adding an appropriately named field with an appropriate data type <pre>ALTER TABLE BORROW ADD Returned BOOLEAN;</pre> accept VARCHAR/CHAR(3)/BOOL	2	Award just 1 mark if statements are in the wrong order or if there are any additional statements Accept any sensible field name
4(c)(ii)	1 mark for each correct statement - Updating the correct table - Setting the additional field to the correct data - Where clause data type must match 4(c)(i) Example: <pre>UPDATE BORROW SET Returned = "Yes" WHERE BorrowID = "A2226";</pre>	3	Award a maximum of 2 marks if statements are in the wrong order or if there are any additional statements

2. M/J 26/P12/Q3/a,b,c

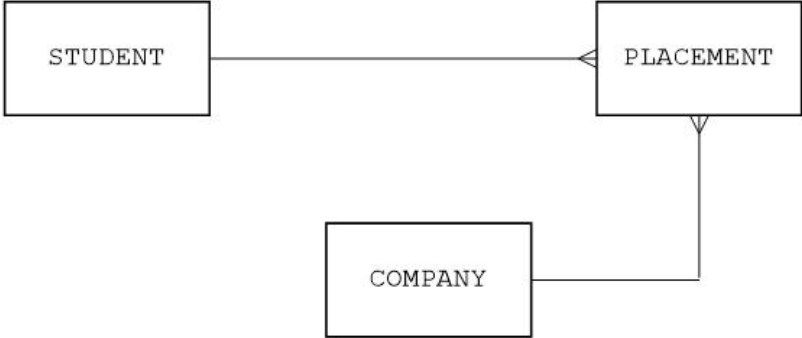
3(a)	1 mark for each correct relationship - The relationship between PLAYER and CHARACTER is one to many /1:M // one player can have many characters, but each character can only be created by one player - The relationship between GROUP and CHARACTER is many to many / M:M // one group can have many characters, and one character can be in many groups	2	The second option in each case needs both 'halves' Accept a m:m relationship between PLAYER and GROUP Accept user for player Mark the writing first, but accept accurate E-R diagram(s), if needed for clarification
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3(b)(i)	1 mark for each correct line number and the appropriate correction - Line 02 - CharacterID VARCHAR, // CharacterID VARCHAR (5), - Line 05 Level INT, - Line 08 - PRIMARY KEY (CharacterID) - Line 09 - FOREIGN KEY (PlayerID) REFERENCES PLAYER (PlayerID)	4	Ignore extra NOT NULL on line 02 but not on line 05 or line 08
3(b)(ii)	1 mark for each correct point - Selecting the player's name and counting a field - Using an appropriate name - Linking or joining the correct tables - Grouping the results - Ordering in descending order by player ID Example 1: <pre>SELECT PlayerName, COUNT(CharacterID) MP1 AS TotalCharacters MP2 FROM CHARACTER, PLAYER WHERE CHARACTER.PlayerID = PLAYER.PlayerID MP3 GROUP BY PlayerName MP4 ORDER BY PLAYER.PlayerID DESC; MP5</pre> Example 2: <pre>SELECT PlayerName, COUNT(CharacterID) MP1 AS TotalCharacters MP2 FROM CHARACTER INNER JOIN PLAYER ON CHARACTER.PlayerID = PLAYER.PlayerID MP3 GROUP BY PlayerName MP4 ORDER BY PLAYER.PlayerID DESC; MP5</pre>	5	Accept any correct method of joining the tables, but no mix and match between methods The only attribute that needs a table reference is the PlayerID. Ignore other table references as long as the correct table is used Take care with the DESC
3(c)	1 mark for two data dictionary examples For example: - data types of individual fields/attributes - type of validation - primary key - foreign key 1 mark for each remaining term (4 marks) - data dictionary - logical - access rights - developer interface The data dictionary stores data about the data in the database. The table names, field names and relationships will be stored. Two other examples that will be stored are example 1 and example 2 When designing a database, the logical schema is the design of the database tables, fields and relationships independent of any Database Management System (DBMS). A DBMS provides security features to protect the data. Access rights are used to give user accounts permission to read data, some to read and write data and others may be unable to read the data. When creating a database a developer interface provides tools to create visual components for example forms and reports.	5	Mark the first answer in each space Do not accept table names or field names or relationships, they are given in the question There are many answers for the examples, the list is suggestions only Do not accept metadata as an example or instead of data dictionary Just 'key' is NE for primary or foreign keys Just data type is TV for the second example

3. M/J 26/P13/Q4/a,b,c

4(a)	1 mark each 3 tables with suitable names e.g. Student, Book, Borrow - All tables given have a suitable PK - Book has fields: author, title, category, number of pages - Student has fields: FName LName, class - Borrow has fields: book PK, student PK, date borrowed, field to identify book is late - FKs in borrow are clearly identified as the Book PK and the Student PK e.g. Book(<u>BID</u>, Author, Title, Category, Pages) Student(<u>SID</u>, FName, LName, Class) Borrow(<u>BwlD</u>, <u>StudentID</u>, <u>BookID</u>, DateBorrowed, Late)	6	Allow composite primary key for Borrow table. Borrow table must contain sensible linking related fields Do not accept Title or Name as PKs. Allow alternatives to late e.g. DateBorrowed, DateDueBack, DateReturned.
4(b)	1 mark each to max 2 - Ensure consistency of data // up-to-date // not redundant data - Ensure relationships between data remain valid and prevent orphaned records - If a change is made in one place it is made everywhere else // cascading updates/deletes - Every foreign key has a corresponding primary key	2	
4(c)(i)	1 mark for each error line and correction - Line 01 CREATE DATABASE SCHOOLDATA; - Line 03 SubjectCode VARCHAR NOT NULL, - Line 04 SubjectName VARCHAR, - Line 07 FOREIGN KEY (DepartmentID) REFERENCES DEPARTMENT (DepartmentID)	4	Do not award mark for Line 04 if extra Not Null has been added
4(c)(ii)	1 mark for each correct point - Selecting department name and counting subject code ... - Using suitable field name - From statement where with join / other join method - Grouping by department name / department ID - Ordering by department name in ascending order Example 1: <pre> SELECT DepartmentName, COUNT(SubjectCode) (MP1) AS "NumberSubjectCodes" (MP2) FROM DEPARTMENT, SUBJECT WHERE DEPARTMENT.DepartmentID = SUBJECT.DepartmentID (MP3) GROUP BY (DepartmentID) (MP4) ORDER BY DepartmentName; (MP5) </pre> Example 2: <pre> SELECT DepartmentName, Count(Subject.SubjectCode) (MP1) AS CountOfSubjectCode (MP2) FROM DEPARTMENT (INNER) JOIN SUBJECT ON DEPARTMENT.DepartmentID = SUBJECT.DepartmentID (MP3) GROUP BY DEPARTMENT.DepartmentName (MP4) ORDER BY (DEPARTMENT.DepartmentName) ASC; (MP5) </pre> Please note that these two lines are equivalent to MP1 and MP2 <pre> SELECT DepartmentName, NumberSubjectCodes = Count(SUBJECT.subjectcode) </pre> MP1: SELECT DepartmentName, and a correct COUNT function MP2: CountOfSubject =	5	Accept == for = Accept any correct method of joining the tables, but no mix and match between methods In example 1 the join mark needs correct FROM and WHERE clauses In example 2 the join mark needs correct FROM... JOIN and ON clauses

4. O/N 25/P11/Q2/a,b,c,d

2(a)	1 mark for each correct relationship, max 2 marks 	2
2(b)	1 mark per bullet point, max 2 marks • There are no repeating groups of attributes • There are no many-to-many relationships • There are no partial key dependencies // no non-key dependencies // no transitive dependencies	2
2(c)(i)	1 mark per bullet point, max 2 marks • DELETE FROM and correct table • Correct condition Example answer: DELETE FROM PLACEMENT WHERE Complete = TRUE;	2
2(c)(ii)	1 mark per bullet point, max 4 marks • SELECT COUNT any appropriate field • AS and an appropriate name • FROM and correct table and WHERE and one correct condition • Two AND clauses with the other two correct conditions Example answer: SELECT COUNT (CompanyID) AS TotalPlacements FROM PLACEMENT WHERE CompanyID = "NEAM" AND StudentID = "LDEA01" AND Complete = TRUE;	4

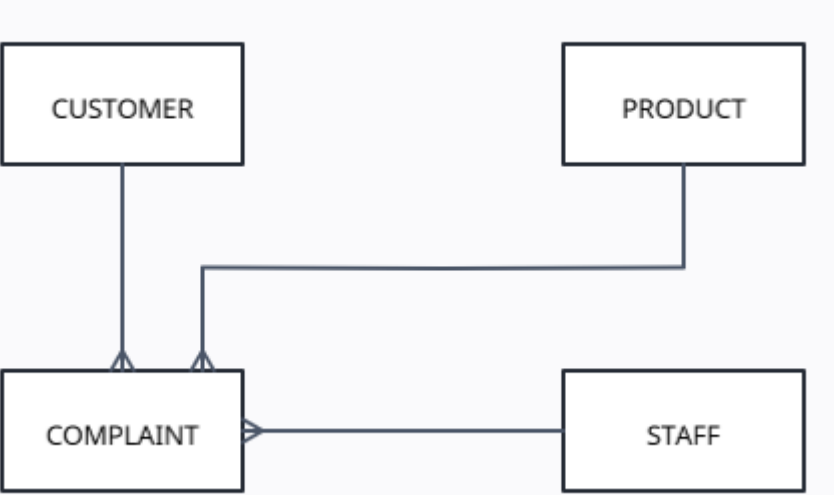
2(d)	1 mark per bullet point, max 3 marks Generic mark points (max 2) • Referential integrity ensures that related data is consistent • Referential integrity ensures that every foreign key has a corresponding primary key • Referential integrity provides for cascading update / delete • Referential integrity ensures that if a primary key is deleted or modified all linked records in foreign table will be deleted or modified // Referential integrity stops "orphaned records"—records in a table that point to an entry in another table that no longer exists Specific mark points (max 2) for good examples e.g. • CompanyID is a foreign key in PLACEMENT table and is dependent on the primary key CompanyID in COMPANY table • If a record is deleted from the STUDENT table, then all records with the same StudentID will be deleted from the PLACEMENT table • If a CompanyID is modified in the company table, all records with that CompanyID in the placement table will also be modified	3
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5. O/N 25/P12/Q4/a,b,c,d

4(a)	1 mark per bullet point, max 2 marks • The relationship between SHIP and CONTAINER is one-to-many (1:M) • The primary key <u>ShipID</u> in the SHIP table is linked to the foreign key ShipID in the CONTAINER table	2
4(b)	1 mark per bullet point, max 2 marks • ALTER TABLE statement • Including the additional field with a suitable field name and suitable field type Example answer <pre>ALTER TABLE CONTAINER ADD InspectionDate DATE;</pre>	2

4(c)	1 mark per bullet point, max 4 marks • SELECT <u>COUNT</u> statement • Using the correct tables • Joining the tables • The condition for the name Example Answer 1: <pre>SELECT COUNT(ContainerID) FROM CONTAINER, SHIP WHERE CONTAINER.ShipID = SHIP.ShipID AND ShipName = "Caledonia";</pre> Example Answer 2: <pre>SELECT COUNT(ContainerID) FROM CONTAINER INNER JOIN SHIP ON CONTAINER.ShipID = SHIP.ShipID WHERE ShipName = "Caledonia";</pre>	4
4(d)	1 mark per bullet point, max 2 marks • To allow the user to create / modify / delete tables // maintain the database • To allow the user to set up / modify relationships • To allow the user to create a form for data input • To allow a user to add tools to a form // for example, drop-down boxes / buttons etc. • To allow a user to design a report (to show the output in an organised manner) • To allow a user to add a menu to enable a choice of different actions • To allow a user to inspect the database contents and metadata • ... by creating and/or running different SQL queries	2

6. O/N 25/P13/Q5/a,b,c,d

5(a)	1 mark for each correct relationship, max 3 marks  <pre> graph TD CUSTOMER --> COMPLAINT PRODUCT --> COMPLAINT COMPLAINT --> STAFF </pre>	3
------	--	----------

5(b)	1 mark per bullet point, max 4 marks • Create table with opening and closing brackets and commas separating the attributes • Appropriate data types for StaffFirstName, StaffLastName and Department • Appropriate data types for StaffID and RemoteWorker • Primary key correctly defined Example Answer 1: <pre>CREATE TABLE STAFF(StaffID INTEGER, StaffFirstName VARCHAR, StaffLastName VARCHAR, Department CHAR, RemoteWorker BOOLEAN, PRIMARY KEY (StaffID));</pre> Example Answer 2: <pre>CREATE TABLE STAFF(StaffID INTEGER NOT NULL PRIMARY KEY, StaffFirstName VARCHAR, StaffLastName VARCHAR, Department CHAR, RemoteWorker BOOLEAN);</pre>	4
5(c)	1 mark per bullet point, max 5 marks • SELECT and the correct attributes • FROM and the correct tables • Tables joined correctly • Correct condition for the rating • Correct ORDER BY clause Example Answer 1 <pre>SELECT PRODUCT.ProductID, ProductName, ComplaintDetails FROM PRODUCT, COMPLAINT WHERE PRODUCT.ProductID = COMPLAINT.ProductID AND Rating <=5 ORDER BY Rating DESC;</pre> Example Answer 2 <pre>SELECT PRODUCT.ProductID, ProductName, ComplaintDetails FROM PRODUCT INNER JOIN COMPLAINT ON PRODUCT.ProductID = COMPLAINT.ProductID WHERE Rating <=5 ORDER BY Rating DESC;</pre>	5

5(d)	One mark for identification of method one mark for corresponding description max 4 marks • Authentication methods / passwords / biometrics / 2-factor authentication can be implemented • ... which prevents unauthorised access to the customer's data • Access rights / privileges can be set • ... so that only those with correct permissions can read / edit customer's data • Regular backups can be scheduled • ... so that a second copy of the customer's data is available in case of loss/damage • The data can be encrypted • ... so that the customer's data cannot be understood by anyone who gains unauthorized access • Different views can be created • ... so that not everyone can see the customer's data	4
------	---	----------

7. M/J 25/P11/Q5/a,b,c,d,e

5(a)	1 mark for each relationship. Max 2 if any extra <pre>graph TD; CUSTOMER --> CUSTOMER_CARD_DATA; CUSTOMER --> ORDER; ORDER --> ORDER_ITEM;</pre>	3								
5(b)	1 mark for: CardNumber	1								
5(c)	1 mark per row to max 2 <table><thead><tr><th>Table</th><th>Foreign key</th></tr></thead><tbody><tr><td>ORDER_ITEM</td><td>OrderID</td></tr><tr><td>ORDER</td><td>CustomerID</td></tr><tr><td>CUSTOMER_CARD_DATA</td><td>CustomerID</td></tr></tbody></table>	Table	Foreign key	ORDER_ITEM	OrderID	ORDER	CustomerID	CUSTOMER_CARD_DATA	CustomerID	2
Table	Foreign key									
ORDER_ITEM	OrderID									
ORDER	CustomerID									
CUSTOMER_CARD_DATA	CustomerID									

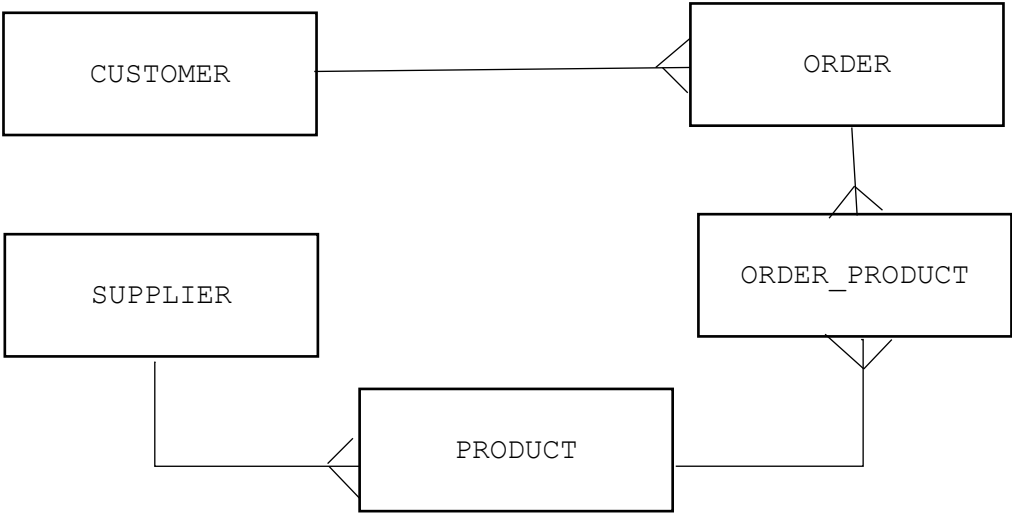
5(d)	1 mark each to max 3 e.g. - Each order would only be able to have one item - or the database would not be normalised - it would not be in 1NF - due to repeated groups of attributes - in the ORDER table	3
5(e)	1 mark each - Selecting the customer ID, customer name and sum of TotalCost with appropriate identifier - FROM clause with suitable join of tables (ON or WHERE) - ORDER.Paid = FALSE condition (with correct key word) - GROUP BY condition e.g. SELECT CUSTOMER.CustomerID, CUSTOMER.Name, Sum(ORDER.TotalCost) AS TotalOwed FROM CUSTOMER INNER JOIN ORDER ON CUSTOMER.CustomerID = ORDER.CustomerID WHERE ORDER.Paid = FALSE GROUP BY CUSTOMER.CustomerID; Alternative JOIN Statement SELECT CUSTOMER.CustomerID, CUSTOMER.Name, Sum(ORDER.TotalCost) AS TotalOwed FROM CUSTOMER,ORDER WHERE CUSTOMER.CustomerID = ORDER.CustomerID AND ORDER.Paid = FALSE GROUP BY CUSTOMER.CustomerID;	4

8. M/J 25/P12/Q5/a,b,c,d

5(a)	1 mark each to max 2 - To remove the many-to-many relationship - between CHARACTER and ITEM tables - To allow each character to have many items // to allow each item to be purchased for many characters - by creating a linking table - between the characters and the items purchased for each one	2
5(b)	1 mark for the three single Primary Keys 1 mark for the composite key USER(<u>Username</u> , Password, DateOfBirth) CHARACTER(<u>CharacterName</u> , <u>CharacterID</u> , Username, Level, Money) ITEM(<u>ItemName</u> , MinimumLevel, Cost) CHARACTER_ITEM(<u>CharacterID</u> , <u>ItemName</u>)	2

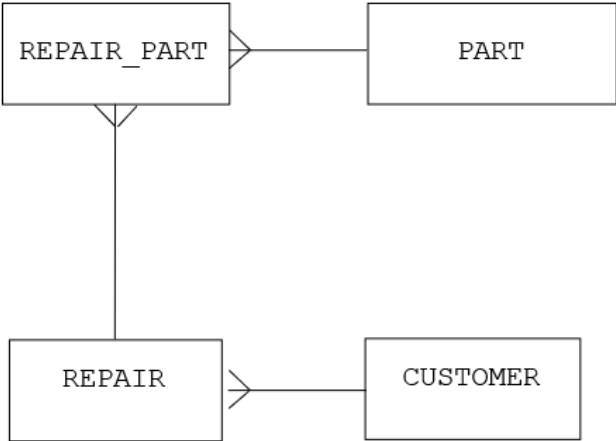
5(c)(i)	1 mark for method 1 mark for matching explanation e.g. • Access rights • appropriate permissions for the table USER are needed to read or edit the data • A password for the database or the USER table • prevents users without the password from accessing the data • Encrypting the database • stops users without the decryption key from decoding / understanding the data • Views • users can be given a view to the database that does not include the data in the table USER	4
5(c)(ii)	1 mark each to max 2 e.g. • Validation • Enforce referential integrity • Cascade update / delete • Ensuring the database is normalised	2
5(d)(i)	1 mark for each point • Select COUNT • FROM (using correct tables) and joining • WHERE (or AND) condition for correct username Example 1: SELECT COUNT(ItemName) FROM CHARACTER_ITEM INNER JOIN CHARACTER ON CHARACTER.CharacterID = CHARACTER_ITEM.CharacterID WHERE CHARACTER.Username = "KAT123"; Example 2: SELECT COUNT(ItemName) FROM CHARACTER_ITEM, CHARACTER WHERE CHARACTER.CharacterID = CHARACTER_ITEM.CharacterID AND CHARACTER.Username = "KAT123";	3
5(d)(ii)	1 mark for each point • UPDATE character • SET level and money • WHERE condition e.g. UPDATE CHARACTER SET Level = 3, Money = 10000.00 WHERE CharacterID = "0002";	3

9. M/J 25/P13/Q6/a,b,c,d

6(a)	1 mark for each correct relationship  <pre> graph LR CUSTOMER --> ORDER ORDER --> ORDER_PRODUCT SUPPLIER --> PRODUCT ORDER_PRODUCT --> PRODUCT </pre>	4
6(b)	1 mark each • INSERT INTO PRODUCT • VALUES with opening and closing brackets • Inserting ProductID, ProductName and SupplierID correctly as strings including quotation marks into correct fields • Inserting quantity and cost correctly into correct fields Example 1: INSERT INTO PRODUCT VALUES ("002323", "Blue ball point 2 mm", 50, 5.00, "SFX223"); Example 2: INSERT INTO PRODUCT (ProductID, ProductName, QuantityInBox, Cost, SupplierID) VALUES ("002323", "Blue ball point 2 mm", 50, 5.00, "SFX223");	4

6(c)	1 mark each • Selection of customer name and counting any field from ORDER as an appropriate identifier • Joining tables ORDER and CUSTOMER • AND (or WHERE) clause: Collected = FALSE • Grouping by customer ID or customer name Example 1: SELECT CustomerName, COUNT(OrderID) AS NotCollected FROM ORDER, CUSTOMER WHERE ORDER.CustomerID = CUSTOMER.CustomerID AND Collected = FALSE GROUP BY CUSTOMER.CustomerID Example 2: SELECT CustomerName, COUNT(OrderID) AS NotCollected FROM ORDER INNER JOIN CUSTOMER ON ORDER.CustomerID = CUSTOMER.CustomerID WHERE Collected = FALSE GROUP BY CUSTOMER.CustomerID	4
6(d)(i)	1 mark each to max 3 e.g. • Relationships • Views • Data types • Validation rules	3
6(d)(ii)	1 mark each to max 3 e.g. • To create / modify / delete database objects • To create a form for data input • To add tools to a form • for example, drop-down boxes / buttons etc. • To design a report to show the output in an organised manner • To add a menu to enable users to choose different actions / run different queries	3

10. O/N 24/P11/Q2/a,b,c

2(a)	1 mark for limitation and 2 marks for corresponding explanation e.g. Data redundancy // data duplication • Separate linked tables are used • Data items are stored once so reduces data duplication / data redundancy Data inconsistency / Poor data integrity • Data changed once in one place will automatically update elsewhere • Linked data cannot be entered differently in two tables • Referential integrity can be enforced The data structure used depends on the application • Changes to the data structure are managed by the DBMS • ... and queries are not dependent on the structure of the data • ... and changes to the data do not require programs to be re-written	3
2(b)(i)	1 mark for each correct relation 	3
2(b)(ii)	1 mark for each bullet point • Create table REPAIR_PART with opening and closing brackets, all statements inside the brackets • PartID and RepairNumber as varchar (or equivalent) • Quantity as integer • At least 1 appropriate constraint • Dual primary key For example: <pre>CREATE TABLE REPAIR_PART(PartID VARCHAR(20) NOT NULL, RepairNumber VARCHAR(4) NOT NULL, Quantity INT NOT NULL, PRIMARY KEY (PartID, RepairNumber));</pre>	5

2(b)(iii)	1 mark for each bullet point • Select SUM of AmountDue • From the correct table and one correct condition • Second correct condition <pre>SELECT SUM(AmountDue) FROM INVOICE WHERE SupplierID = "JK675" AND Paid = FALSE;</pre>	3								
2(c)	1 mark for each correct definition <table><tr><th>Database Term</th><th>Definition</th></tr><tr><td>Referential Integrity</td><td>All duplicate entries of data between tables are consistent // all foreign keys are matched to an appropriate primary key</td></tr><tr><td>Candidate Key</td><td>A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value</td></tr><tr><td>Tuple</td><td>A row / record in a table // one instance of an entity in a table</td></tr></table>	Database Term	Definition	Referential Integrity	All duplicate entries of data between tables are consistent // all foreign keys are matched to an appropriate primary key	Candidate Key	A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value	Tuple	A row / record in a table // one instance of an entity in a table	3
Database Term	Definition									
Referential Integrity	All duplicate entries of data between tables are consistent // all foreign keys are matched to an appropriate primary key									
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Tuple	A row / record in a table // one instance of an entity in a table									

11. O/N 24/P12/Q6/a,b

6(a)	1 mark each to max 3 e.g. - There is reduced data redundancy // less repeated data ... because each item of data is only stored once - Data consistency is maintained // Data integrity is improved ... changes in one table will automatically update in another ... linked data cannot be entered differently in two tables - Program-data independence is ensured ... changes to the data do not require programs to be re-written // queries are not dependent on the structure of the data - Complex queries are easier to run - Different views can be provided so users can only see specific aspects of the database - Multiple concurrent access is possible ... through record locking	3
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6(b)(i)	1 mark each to max 6 - CUSTOMER to JOB is 1 to many ... implemented by Primary Key in CUSTOMER is Foreign Key in JOB - EMPLOYEE to LOGIN_DATA is 1 to 1 ... implemented by Primary Key in EMPLOYEE is Foreign Key in LOGIN_DATA - JOB to JOB_EMPLOYEE is 1 to many ... implemented by Primary Key in JOB is Foreign Key in JOB_EMPLOYEE - EMPLOYEE to JOB_EMPLOYEE is 1 to many ... implemented by Primary Key in EMPLOYEE is Foreign Key in JOB_EMPLOYEE	6
6(b)(ii)	1 mark each - Select SUM of Amount - From the correct table and one correct condition - Remaining correct condition Example: <pre>SELECT SUM(Amount) FROM INVOICE WHERE Paid = "Y" AND DateSent >= #01/01/2023# AND DateSent <= #31/12/2023#</pre>	3

12. O/N 24/P13/Q4/a,b,c,d,e

4(a)	1 mark each: - Foreign key: BatchID, table BATCH - Foreign key: CustomerID, table CUSTOMER	2
4(b)	1 mark each: - SELECT SUM(Quantity) - FROM SALE WHERE and one correct condition - AND with remainder correct conditions e.g. <pre>SELECT SUM(Quantity) FROM SALE WHERE CustomerID = "0034E" AND Date >= #01/01/2023# AND Date <= #31/12/2023#;</pre>	3

4(c)(i)	1 mark for each bullet point • Create table BATCH with opening and closing brackets, all statements within brackets • BatchID, Type and Flavour as varchar or equivalent • ...with suitable constraint(s) • Size as decimal, SellingPrice as currency, EndDate as date (or equivalent) • Primary key identified as BatchID e.g. <pre>CREATE TABLE BATCH (BatchID VARCHAR(6) NOT NULL, Type VARCHAR (20) NOT NULL, Flavour VARCHAR (20) NOT NULL, Size FLOAT, SellingPrice CURRENCY, EndDate DATE, PRIMARY KEY (BatchID));</pre>	5
4(c)(ii)	1 mark for each bullet point each (max 4) e.g. • Ice cream table with an appropriate name • ... containing type, flavour, size, selling price • ... with suitable primary key • ... foreign key identified in BATCH that links to the primary key in ICECREAM Example table definitions – not example answer: <pre>BATCH(BatchID, IceCreamID, EndDate) ICE_CREAM(IceCreamID, Type, Flavour, Size, SellingPrice)</pre>	4
4(d)	1 mark for each definition Entity: • A real-life object that is represented as a table Attribute: • An item of data about an entity	2
4(e)	1 mark for each bullet point each (max 3) • Referential integrity is enforced • ... such as cascade update/delete // if the data is changed in one place it is updated in every other place • ... and ensures each foreign key has a corresponding primary key	3

13. M/J 24/P11/Q6/a,b,c

6(a)	1 mark each: • User table with the username as the Primary Key • ... containing at least email address, date of birth / age and rating • Quiz table with Quiz ID or date or file name as the Primary Key. • ... containing at least the other field(s) not used as the PK • A joining table with an appropriate name including at least fields for user identification, quiz identification and score • ... with an appropriate Primary Key • ... and Foreign Keys matching the Primary Keys of the other two tables USER(<u>Username</u>, Email, DateOfBirth, Rating) QUIZ(<u>QuizID</u>, Date, Filename) USER QUIZ(<u>Username</u>, <u>QuizID</u>, Score)	6
6(b)	1 mark each to max 2 for data dictionary and max 2 for logical schema: Data dictionary: • Data about the data in the database // metadata • Identifies the characteristics of the data that will be stored • Appropriate example e.g. field names, table name, validation rules, data types, primary / foreign keys, relationships etc. Logical schema: • Conceptual design • Platform/database independent overview of the database • Is used to design the physical structure • Appropriate example e.g. Design of entities / E-R diagram / views	4
6(c)(i)	1 mark for each correct clause: • Alter table EVENT • Adding foreign key as PlayerID referencing correct table ALTER TABLE EVENT ADD FOREIGN KEY(PlayerID) REFERENCES PLAYER(PlayerID);	2
6(c)(ii)	1 mark each: • Selecting PlayerID from EVENT • Counting EventID • Grouping by the PlayerID Example: SELECT PlayerID, COUNT(EventID) FROM EVENT GROUP BY PlayerID;	3

14. M/J 24/P12/Q4/a,b,c,d

4(a)	1 mark for: 1-to-many	1
4(b)	1 mark each: • Creating table EXAM with opening and closing brackets • All fields with appropriate data types and commas at end of lines • ExamID as primary key Example: <pre>CREATE TABLE EXAM(ExamID varchar NOT NULL, Subject varchar, Level int, TotalMarks int, PRIMARY KEY (ExamID));</pre>	3
4(c)	1 mark each: • Altering table EXAM_QUESTION • Linking ExamID to ExamID in EXAM Example. <pre>ALTER TABLE EXAM_QUESTION ADD FOREIGN KEY (ExamID) REFERENCES EXAM (ExamID);</pre>	2
4(d)	1 mark each to max 5: • STUDENT table identified with suitable Primary Key • A linking table between STUDENT and EXAM with suitable Primary Key and appropriate name • ... that includes the Primary Key of the STUDENT table as a Foreign Key to join with STUDENT • ... and includes the Primary Key of the EXAM table as a Foreign Key to join with EXAM • A linking table between STUDENT and EXAM_QUESTION with suitable Primary Key and appropriate name • ... that includes the Primary Key of Table 2 as a Foreign Key to join with Table 2 • ... that stores the ExamQuestionID and the mark for that question	5

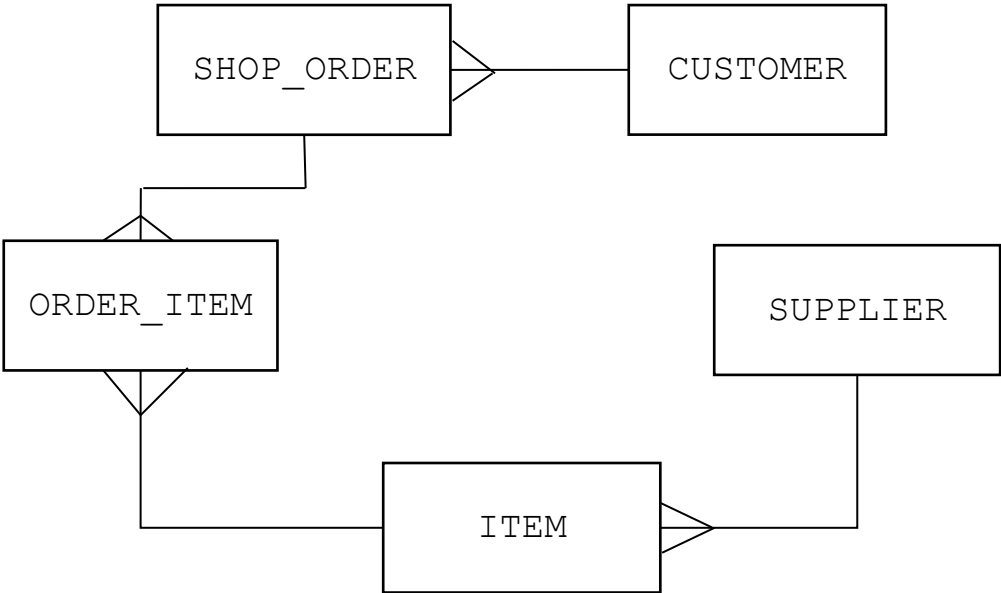
15. M/J 24/P13/Q4/a,b,c,d

4(a)	1 mark for: many to 1 // there are many performances of each show	1
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4(b)	1 mark each for: • Creating table <code>PERFORMANCE</code> with opening and closing brackets • Setting all four attributes with appropriate data types • Setting <code>PerformanceID</code> as primary key • Setting <code>ShowID</code> as foreign key referencing <code>SHOW</code> table Example: <pre>CREATE TABLE PERFORMANCE(PerformanceID varchar NOT NULL, ShowID varchar, ShowDate Date, StartTime Time, PRIMARY KEY (PerformanceID), FOREIGN KEY (ShowID) REFERENCES SHOW (ShowID));</pre>	4
4(c)	1 mark each for: • Selecting <code>COUNT</code> of an attribute in <code>PERFORMANCE</code> table with suitable name • <code>FROM</code> clause • Joining tables • Grouping by the title and selecting the title Example 1: <pre>SELECT SHOW.Title, Count (PERFORMANCE.PerformanceID) AS NumberOfShowings FROM PERFORMANCE, SHOW WHERE PERFORMANCE.ShowID = SHOW.ShowID GROUP BY SHOW.Title;</pre> Example 2: <pre>SELECT SHOW.Title, Count (PERFORMANCE.PerformanceID) AS NumberOfShowings FROM PERFORMANCE INNER JOIN SHOW ON PERFORMANCE.ShowID = SHOW.ShowID GROUP BY SHOW.Title;</pre>	4

4(d)	1 mark each to max 5: e.g. • CUSTOMER table identified with suitable Primary Key and appropriate name • ... and other suitable fields including name and email • BOOKING TABLE identified with suitable Primary Key and appropriate name • ... that stores the Primary Key of the CUSTOMER table as a Foreign Key to join with CUSTOMER table • ... and stores the Primary Key of the PERFORMANCE table as a Foreign Key to join with PERFORMANCE table • A linking table between Table 2 and SEAT with suitable Primary Key and appropriate name • ... that includes the Primary Key of Table 2 as a Foreign Key to join with Table 2 • ... that stores the SeatID.	5
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16. O/N 23/P11/Q3/a,b,c

3(a)	1 mark for each correct relationship or relationships • 1:M between CUSTOMER and SHOP_ORDER • 1:M between SUPPLIER and ITEM • 1:M between SHOP_ORDER and ORDER_ITEM and M:1 between ORDER_ITEM and ITEM  <pre> graph TD SHOP_ORDER[SHOP_ORDER] -- "1:M" --> CUSTOMER[CUSTOMER] SHOP_ORDER -- "1:M" --> ORDER_ITEM[ORDER_ITEM] ORDER_ITEM -- "M:1" --> ITEM[ITEM] SUPPLIER[SUPPLIER] -- "1:M" --> ITEM </pre>	3
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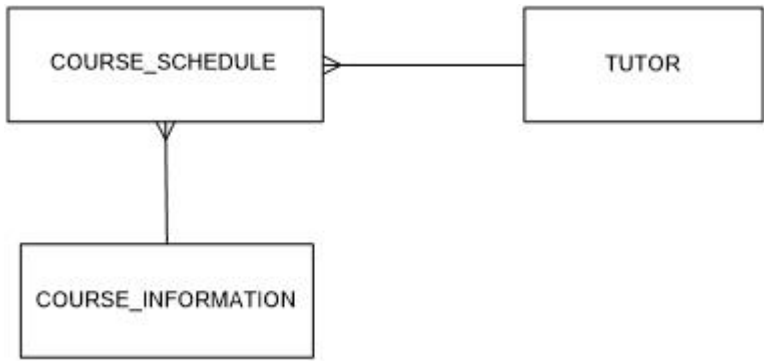
3(b)	1 mark for each bullet point (max 3) - Reduced data redundancy - Improved data integrity / consistency / referential integrity - Allows for views / improved privacy - Allows for program-data independence - Complex queries can be executed	3
3(c)(i)	1 mark for CREATE DATABASE SHOP;	1
3(c)(ii)	1 mark for each line (max 4) Either: SELECT SUM(Quantity) FROM ORDER_ITEM, SHOP_ORDER WHERE ORDER_ITEM.OrderNo = SHOP_ORDER.OrderNo AND SHOP_ORDER.CustomerID = 'HJ231'; OR SELECT SUM(Quantity) FROM ORDER_ITEM (INNER) JOIN SHOP_ORDER ON ORDER_ITEM.OrderNo = SHOP_ORDER.OrderNo WHERE SHOP_ORDER.CustomerID = 'HJ231';	4

17. O/N 23/P12/Q2/a,b,c

2(a)	1 mark for each term (max 3) Entity: - An object about which data can be stored Primary key: - The unique attribute / combination of attributes used to identify the record / tuple Referential integrity: - Makes sure that if data is changed in one place the change is reflected in all related records - cascading update/delete - Makes sure that data that does not exist cannot be referenced - Ensures that every foreign key has a corresponding primary key // A logical dependency of a foreign key on a primary key - Ensures that the data in the database is consistent / up to date - Prevents records from being added/deleted/modified incorrectly - Makes sure any queries return accurate and complete results	3
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2(b)	1 mark for each bullet point (max 4) Max 2 if no descriptions • Backup / recovery procedures • ... automatically takes copies of the database and store off site on a regular basis / weekly, etc. • ... so that the data can be recovered if lost • Use of access rights • ... some users are given different access permissions to different tables • ... read/write, read only, full access, etc. • Views • ... different users are able to see different parts of the database • ... only see what users need to see // by example • Record and table locking • ... prevents simultaneous access to data • ... so updates are not lost // data is not overwritten • Encryption • ... the data is turned into ciphertext • ... so it cannot be understood without a decryption key	4
2(c)	1 mark for each bullet point (max 4) • Identify repeating groups of attributes ... • ... Subject and SubjectCode • Ensure each field is atomic • ... StudentName should be split into e.g. FirstName and LastName • Identify the primary key for the table	4

18. O/N 23/P13/Q3/b

3(b)(i)	1 mark for both 1:M relationships as follows:  <pre> graph TD CS[COURSE_SCHEDULE] --> T[TUTOR] CS --> CI[COURSE_INFORMATION] </pre>	1
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3(b)(ii)	1 mark for each bullet point. • <code>SELECT Count (CourseID)</code> • <code>AS NumOfCourses</code> • <code>FROM COURSE_SCHEDULE</code> • <code>WHERE DateStarted > "09/09/23";</code> <code>SELECT Count (CourseID) AS NumOfCourses FROM COURSE_SCHEDULE WHERE DateStarted > "09/09/23";</code>	4
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19. M/J 23/P11/Q2/a,b

2(a)(i)	1 mark for definition: • Data about the data in the database // data about the structure of the database // metadata for a database 1 mark for a suitable example Examples: • table names • data types • field names	2						
2(a)(ii)	1 mark for definition • Methods of making sure the data is consistent 1 mark for example Examples: • Enforcing referential integrity • If data in one table is deleted/edited all tables are updated // cascading update/delete • Validation/verification rules	2						
2(b)(i)	1 mark for each field name and table <table><tr><th>Foreign key</th><th>Database table</th></tr><tr><td>BirdID</td><td>BIRD_SEEN</td></tr><tr><td>PersonID</td><td>BIRD_SEEN</td></tr></table>	Foreign key	Database table	BirdID	BIRD_SEEN	PersonID	BIRD_SEEN	2
Foreign key	Database table							
BirdID	BIRD_SEEN							
PersonID	BIRD_SEEN							

2(b)(ii)	1 mark for all 3 correct lines Normal Form First Normal Form (1NF) Second Normal Form (2NF) Third Normal Form (3NF) Definition All fields are fully dependent on the primary key. There are no repeating groups of attributes. There are no partial dependencies.	1
2(b)(iii)	1 mark each • CREATE TABLE start and end bracket • Bird ID as CHAR/VARCHAR • Name and size as VARCHAR/CHAR • Bird ID as primary key Example answer: <pre>CREATE TABLE BIRD_TYPE (BirdID CHAR(4) NOT NULL, Name VARCHAR(9), Size VARCHAR(6), PRIMARY KEY (BirdID));</pre>	4
2(b)(iv)	1 mark for each correctly completed space <pre>SELECT BIRD_TYPE.Size, COUNT(BIRD_TYPE.BirdID) AS NumberOfBirds FROM BIRD_TYPE, BIRD_SEEN WHERE BIRD_SEEN.PersonID = "J_123" AND BIRD_TYPE.BirdID = BIRD_SEEN.BirdID GROUP BY BIRD_TYPE.Size;</pre>	5

20. M/J 23/P12/Q2/a,b,c(ii,iii)

2(a)	1 mark for each correct feature or description <table><tr><th>Feature</th><th>Description</th></tr><tr><td>Data dictionary</td><td>Data about the data in the database // data about the structure of the database // metadata for a database</td></tr><tr><td>Query processor</td><td>Software that allows the user to enter criteria, then finds and returns the appropriate result // software that processes and executes queries written in SQL</td></tr><tr><td><u>Logical schema</u></td><td>A model of a database that is not specific to one DBMS</td></tr><tr><td><u>Developer interface</u></td><td>A software tool that allows the user to create items such as tables, forms and reports</td></tr></table>	Feature	Description	Data dictionary	Data about the data in the database // data about the structure of the database // metadata for a database	Query processor	Software that allows the user to enter criteria , then finds and returns the appropriate result // software that processes and executes queries written in SQL	<u>Logical schema</u>	A model of a database that is not specific to one DBMS	<u>Developer interface</u>	A software tool that allows the user to create items such as tables, forms and reports	4
Feature	Description											
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<u>Logical schema</u>	A model of a database that is not specific to one DBMS											
<u>Developer interface</u>	A software tool that allows the user to create items such as tables, forms and reports											
2(b)	1 mark each to max 3 • Referential Integrity makes sure data is consistent • Referential Integrity makes sure all data is up-to-date • Referential integrity ensures that every foreign key has a corresponding primary key • Referential Integrity prevents records from being added / deleted / modified incorrectly • Referential Integrity makes sure that if data is changed in one place the change is reflected in all related records • Referential Integrity makes sure any queries return accurate and complete results	3										
2(c)(ii)	1 mark each • SELECT field Name • FROM table HORSE • WHERE with Intermediate / Beginner • OR with Beginner / Intermediate Example answer: SELECT Name FROM HORSE WHERE HorseLevel = "Intermediate" OR HorseLevel = "Beginner";	4										

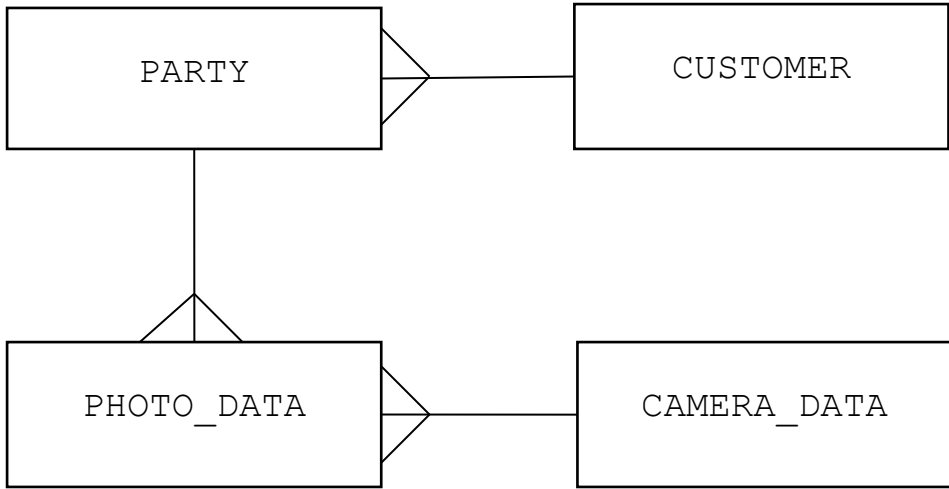
2(c)(iii)	1 mark each - SUM should be COUNT // SELECT COUNT(STUDENT.RiderLevel) - The WHERE statement needs the table names before each field name // WHERE STUDENT.StudentID = LESSON.StudentID - The OR should be AND // AND Date = #09/09/2023# - Beginner is missing the speech marks // STUDENT.RiderLevel = "Beginner";	4
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21. M/J 23/P13/Q4/a,b,c

4(a)	1 mark for each bullet point. Mark in pairs. Max 2 for each description. • Reduces data redundancy • ... because linked tables mean that each data item is stored only once • Reduces program-data dependency • ... because the data is separate from the software so changes to the data do not require programs to be re-written • Reduces data inconsistency // improves data integrity • ... because by only storing data once it only needs to be updated once // changes in one table will automatically update in another // linked data cannot be entered differently in two tables • Complex queries are easier to run • Can provide different views •so users can only see specific aspects of the database	4										
4(b)	1 mark each <table><tr><th>Term</th><th>Description</th></tr><tr><td>Entity</td><td>An object that data is stored about.</td></tr><tr><td>Tuple</td><td>A row of data in a table about one instance of an object.</td></tr><tr><td>Secondary key</td><td>An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.</td></tr><tr><td>Foreign key</td><td>A field in one table that is linked to a primary key in another table.</td></tr></table>	Term	Description	Entity	An object that data is stored about.	Tuple	A row of data in a table about one instance of an object.	Secondary key	An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.	Foreign key	A field in one table that is linked to a primary key in another table.	4
Term	Description											
Entity	An object that data is stored about.											
Tuple	A row of data in a table about one instance of an object.											
Secondary key	An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.											
Foreign key	A field in one table that is linked to a primary key in another table.											

4(c)	1 mark each - Only 3 tables with appropriate identifiers (i.e. one table for customer, one for booking and one for car) - Appropriate Primary key in each table underlined - Booking table includes Primary key from car and Primary key from customer as Foreign keys - All original fields are in correct tables Example answer: BOOKING(<u>BookingID</u>, CarRegistration, CustomerID, StartDate, EndDate) CAR(<u>CarRegistration</u>, CarModel, CarColour) CUSTOMER(<u>CustomerID</u>, CustomerFirstName, CustomerLastName, EmailAddress, TelephoneNumber)	4
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22. O/N 22/P11/Q4/a,b,c,d

4(a)	1 mark for each correct relationship: 1:M between CUSTOMER and PARTY 1:M between PARTY and PHOTO_DATA 1:M between CAMERA_DATA and PHOTO_DATA 	3
4(b)	1 mark for each bullet point (max 3): - no repeating groups of attributes // data is atomic - no partial key dependencies - no non-key dependencies // no transitive dependencies	3
4(c)(i)	1 mark for the definition, 1 mark for the example: - definition: a single row in a table - example: from the PHOTO_DATA table	2

4(c)(ii)	1 mark for each correctly completed empty space: • COUNT • PhotoID • PHOTO_DATA • 'CAN*' // 'CAN%' <pre>SELECT COUNT(PhotoID) FROM PHOTO_DATA WHERE CameraID LIKE 'CAN*'; // WHERE CameraID LIKE 'CAN%';</pre>	4
4(d)	1 mark for each bullet point: • ALTER TABLE CAMERA_DATA • ADD NumberStored INTEGER • , LastUsed DATE; <pre>ALTER TABLE CAMERA_DATA ADD NumberStored INTEGER, LastUsed DATE;</pre>	3

23. O/N 22/P12/Q5/a,b,c,d

5(a)	1 mark for each bullet point (max 3): Solution 1: • removing the many-to-many relationship between Owner and Tree • ... by removing TreeID and TreePosition from the Owner table • ... and creating a linking table between Owner and Tree • ... that contains OwnerID, TreeID and TreePosition • ... (composite) primary key of the linking table should be OwnerID and TreeID // insert a named new primary key in the linking table Solution 2: • removing the many-to-many relationship between Owner and Tree • move TreePosition into TREE table • ... put OwnerID into TREE table • create a new table with suitable name (for the species of tree) • ... containing ScientificName, MaxHeight and FastGrowing • ... with ScientificName as primary key // or another suitable primary key	3
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5(b)	1 mark for each bullet point: • INSERT INTO TREE • VALUES () and correct values • Values in correct order Option 1: INSERT INTO TREE(TreeID, ScientificName, MaxHeight, FastGrowing) VALUES('LOW_1276', 'Salix_Alba', 30.00, TRUE); Option 2: INSERT INTO TREE VALUES('LOW_1276', 'Salix_Alba', 30.00, TRUE);	3
5(c)	1 mark for: An attribute / field (or set of attributes / fields) that could be a primary key	1
5(d)(i)	1 mark for description • stores metadata about the database 1 mark for a correct example For example: • field / attribute names • table name • validation rules • data types • primary keys // foreign keys • relationships	2
5(d)(ii)	1 mark for each bullet point (max 2): • the overview of a database structure • models the problem / situation • ... by using methods such as an ER diagram • independent of any particular DBMS	2

24. O/N 22/P13/Q2/a,b,c,d,f

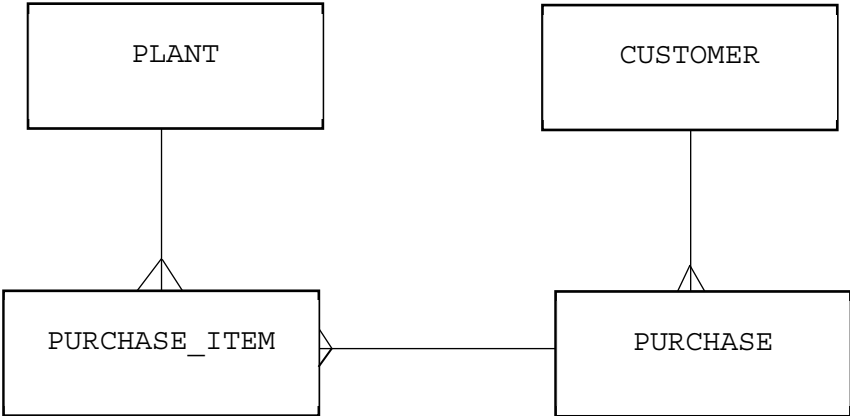
2(a)	1 mark for each correct answer. <table><thead><tr><th>Item</th><th>Answer</th></tr></thead><tbody><tr><td>a suitable field for the primary key in COMPANY</td><td>CompanyID</td></tr><tr><td>a candidate key in TELESCOPE</td><td>SerialNumber // TelescopeID</td></tr><tr><td>the degree of relationship between TELESCOPE and PHOTOGRAPH</td><td>1:M / 1 to many</td></tr></tbody></table>	Item	Answer	a suitable field for the primary key in COMPANY	CompanyID	a candidate key in TELESCOPE	SerialNumber // TelescopeID	the degree of relationship between TELESCOPE and PHOTOGRAPH	1:M / 1 to many	3
Item	Answer									
a suitable field for the primary key in COMPANY	CompanyID									
a candidate key in TELESCOPE	SerialNumber // TelescopeID									
the degree of relationship between TELESCOPE and PHOTOGRAPH	1:M / 1 to many									
2(b)	Logical schema	1								

2(c)	1 mark for each correctly completed missing part: SELECT COUNT (TelescopeID) FROM TELESCOPE WHERE CompanyID LIKE ' HW% ';	4
2(d)	1 mark for each bullet point: - ALTER TABLE PHOTOGRAPH - ADD Resolution TEXT; ALTER TABLE PHOTOGRAPH ADD Resolution TEXT / VARCHAR(11);	2
2(f)	1 mark for each bullet point (max 2): - allows the user to enter criteria - searches for the data that meets the entered criteria - organises the results to be displayed to the user	2

25. O/N 21/P11/Q5/a,b,c

5(a)	1 mark for 2 or 3 correct ticks, 2 marks for 4 correct ticks	2																				
	<table><tr><th>Table</th><th>Field name</th><th>Primary Key (PK)</th><th>Foreign Key (FK)</th></tr><tr><td>MANAGER</td><td>ManagerID</td><td>✓</td><td></td></tr><tr><td>SHOP</td><td>ManagerID</td><td></td><td>✓</td></tr><tr><td>CAR</td><td>RegistrationNumber</td><td>✓</td><td></td></tr><tr><td>CAR</td><td>ShopID</td><td></td><td>✓</td></tr></table>	Table	Field name	Primary Key (PK)	Foreign Key (FK)	MANAGER	ManagerID	✓		SHOP	ManagerID		✓	CAR	RegistrationNumber	✓		CAR	ShopID		✓	
Table	Field name	Primary Key (PK)	Foreign Key (FK)																			
MANAGER	ManagerID	✓																				
SHOP	ManagerID		✓																			
CAR	RegistrationNumber	✓																				
CAR	ShopID		✓																			
5(b)	1 mark per bullet point - Access rights give managers / himself access to different elements ... by having different accounts / logins ... which have different access rights e.g. read only // no access / read / write - Specific <u>views</u> can be assigned to himself and to the managers ... e.g. managers can only see the data for their own shop(s)	3																				
5(c)(i)	1 mark per correctly completed statement SELECT COUNT(RegistrationNumber) FROM CAR GROUP BY ShopID	3																				
5(c)(ii)	1 mark for each correct statement INSERT INTO CAR VALUES ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")	2																				

26. O/N 21/P12/Q6/a,b,c

6(a)(i)	1 mark for 1 tick in the correct place 2 marks for all 3 ticks correct <table><thead><tr><th rowspan="2">Task</th><th colspan="3">Normalisation stage</th></tr><tr><th>0NF to 1NF</th><th>1NF to 2NF</th><th>2NF to 3NF</th></tr></thead><tbody><tr><td>Remove any partial key dependencies</td><td></td><td>✓</td><td></td></tr><tr><td>Remove any repeating groups of attributes</td><td>✓</td><td></td><td></td></tr><tr><td>Remove any non-key dependencies</td><td></td><td></td><td>✓</td></tr></tbody></table>	Task	Normalisation stage			0NF to 1NF	1NF to 2NF	2NF to 3NF	Remove any partial key dependencies		✓		Remove any repeating groups of attributes	✓			Remove any non-key dependencies			✓	2
Task	Normalisation stage																				
	0NF to 1NF	1NF to 2NF	2NF to 3NF																		
Remove any partial key dependencies		✓																			
Remove any repeating groups of attributes	✓																				
Remove any non-key dependencies			✓																		
6(a)(ii)	1 mark for each correct relationship 	3																			
6(b)	1 mark for description of purpose - Stores metadata about the database 1 mark for each example of contents to max 2 e.g. - field / attribute names - table name - validation rules - data types - primary keys // foreign keys - relationships	3																			
6(c)(i)	1 mark for each correctly completed space <pre>SELECT SUM(Quantity) FROM PURCHASE_ITEM WHERE PurchaseID = "3011A";</pre>	4																			

6(c)(ii)	1 mark per bullet point - ALTER TABLE PURCHASE - ADD OrderDate - Suitable data type, e.g. DATE ALTER TABLE PURCHASE ADD OrderDate DATE;	3
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27. O/N 21/P13/Q5/a,b,c

5(a)	1 mark for 2 or 3 correct ticks, 2 marks for 4 correct ticks	2																				
	<table><tr><th>Table</th><th>Field name</th><th>Primary Key (PK)</th><th>Foreign Key (FK)</th></tr><tr><td>MANAGER</td><td>ManagerID</td><td>✓</td><td></td></tr><tr><td>SHOP</td><td>ManagerID</td><td></td><td>✓</td></tr><tr><td>CAR</td><td>RegistrationNumber</td><td>✓</td><td></td></tr><tr><td>CAR</td><td>ShopID</td><td></td><td>✓</td></tr></table>	Table	Field name	Primary Key (PK)	Foreign Key (FK)	MANAGER	ManagerID	✓		SHOP	ManagerID		✓	CAR	RegistrationNumber	✓		CAR	ShopID		✓	
Table	Field name	Primary Key (PK)	Foreign Key (FK)																			
MANAGER	ManagerID	✓																				
SHOP	ManagerID		✓																			
CAR	RegistrationNumber	✓																				
CAR	ShopID		✓																			
5(b)	1 mark per bullet point - Access rights give managers / himself access to different elements ... by having different accounts / logins ... which have different access rights e.g. read only // no access / read / write - Specific <u>views</u> can be assigned to himself and to the managers ... e.g. managers can only see the data for their own shop(s)	3																				
5(c)(i)	1 mark per correctly completed statement SELECT COUNT(RegistrationNumber) FROM CAR GROUP BY ShopID	3																				
5(c)(ii)	1 mark for each correct statement INSERT INTO CAR VALUES ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")	2																				

28. M/J 21/P11/Q7/a,b,c

7(a)	1 mark per bullet point to max 3 - Flat-file has more data redundancy ... because the same data is stored many times // data is stored in different tables which are linked - There is program-data dependence with flat-files ... because any changes to the structure of the data means the programs that access that data have to be re-written - Flat-file has more data inconsistency // worse data integrity ... because duplicated data might be stored differently //...because when data is updated in one place, it is not updated everywhere - It is not easy to perform complex searches /queries ... because a new program has to be written each time - Flat files could have a lack of privacy ... as user views cannot easily be implemented	3								
7(b)(i)	1 mark for each correct example one-to-one - e.g. customer to payment details // customer to login details one-to-many - e.g. customer to order many-to-many - e.g. order to product // customer to product	3								
7(b)(ii)	1 mark <table><tr><th>Relationship</th><th>Tick (✓)</th></tr><tr><td>one-to-one</td><td></td></tr><tr><td>one-to-many</td><td></td></tr><tr><td>many-to-many</td><td>✓</td></tr></table>	Relationship	Tick (✓)	one-to-one		one-to-many		many-to-many	✓	1
Relationship	Tick (✓)									
one-to-one										
one-to-many										
many-to-many	✓									
7(b)(iii)	1 mark CREATE DATABASE SHOPORDERS;	1								
7(c)	1 mark per item to max 3 - table name - field name // attribute - data type - type of validation - Primary Key - Foreign Key - relationships	3								

29. M/J 21/P12/Q1/a,b,c

1(a)	1 mark for definition, 1 mark for appropriate example in each <table><tr><td>Term</td><td>Definition and example</td></tr><tr><td>Field</td><td>A column/attribute in a table e.g. CustomerID in the table CUSTOMER</td></tr><tr><td>Entity</td><td>Anything that data can be stored about e.g. A customer or a house</td></tr><tr><td>Foreign Key</td><td>A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u></td></tr></table>	Term	Definition and example	Field	A column/attribute in a table e.g. CustomerID in the table CUSTOMER	Entity	Anything that data can be stored about e.g. A customer or a house	Foreign Key	A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u>	6
Term	Definition and example									
Field	A column/attribute in a table e.g. CustomerID in the table CUSTOMER									
Entity	Anything that data can be stored about e.g. A customer or a house									
Foreign Key	A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u>									
1(b)	1 mark per bullet point to max 2 - All fields in all tables are dependant fully on the PK and on no other fields - for example all fields in Customer table are fully dependent on CustomerID	2								
1(c)(i)	1 mark for each correctly completed line <pre>CREATE TABLE RENTAL (RentalID INTEGER NOT NULL, CustomerID INTEGER NOT NULL, HouseID VARCHAR (5) NOT NULL, MonthlyCost REAL/CURRENCY NOT NULL, DepositPaid BOOLEAN NOT NULL, PRIMARY KEY (RentalID));</pre>	4								
1(c)(ii)	1 mark per bullet point - Select FirstName and LastName - From both tables - Where DepositPaid = No - Joining tables (either AND, or INNER JOIN) Example script: SELECT FirstName, LastName FROM CUSTOMER, RENTAL WHERE DepositPaid = No AND RENTAL.CustomerID = CUSTOMER.CustomerID;	4								

30. M/J 21/P13/Q7/a,b,c

7(a)	1 mark per bullet point to max 3 • Flat-file has more data redundancy • ... because the same data is stored many times // data is stored in different tables which are linked • There is program-data dependence with flat-files • ... because any changes to the structure of the data means the programs that access that data have to be re-written • Flat-file has more data inconsistency // worse data integrity • ... because duplicated data might be stored differently //...because when data is updated in one place, it is not updated everywhere • It is not easy to perform complex searches /queries • ... because a new program has to be written each time • Flat files could have a lack of privacy • ... as user views cannot easily be implemented	3								
7(b)(i)	1 mark for each correct example one-to-one • e.g. customer to payment details // customer to login details one-to-many • e.g. customer to order many-to-many • e.g. order to product // customer to product	3								
7(b)(ii)	1 mark <table border="1"><thead><tr><th>Relationship</th><th>Tick (✓)</th></tr></thead><tbody><tr><td>one-to-one</td><td></td></tr><tr><td>one-to-many</td><td></td></tr><tr><td>many-to-many</td><td>✓</td></tr></tbody></table>	Relationship	Tick (✓)	one-to-one		one-to-many		many-to-many	✓	1
Relationship	Tick (✓)									
one-to-one										
one-to-many										
many-to-many	✓									
7(b)(iii)	1 mark CREATE DATABASE SHOPORDERS ;	1								
7(c)	1 mark per item to max 3 • table name • field name // attribute • data type • type of validation • Primary Key • Foreign Key • relationships	3								

Topic index: where each part was filed

Reference	Unit	Marks	Page	Also / context
M/J 26/P11/Q1/a,b,c,d	1	12	4	
M/J 26/P11/Q2/a,b,c	5	7	336	
M/J 26/P11/Q2/d	7	3	425	
M/J 26/P11/Q3/a,b,c	4	9	218	
M/J 26/P11/Q4/a,b,c	8	13	451	
M/J 26/P11/Q4/d	6	5	385	
M/J 26/P11/Q5/a,b,c	1	9	5	
M/J 26/P11/Q6/a,b	3	4	133	
M/J 26/P11/Q7/a,b,c	2	13	85	
M/J 26/P12/Q1/a,b,c,d	1	8	6	
M/J 26/P12/Q2/a,b,c	6	13	385	
M/J 26/P12/Q3/a,b,c	8	16	452	
M/J 26/P12/Q4/a,b	1	4	7	
M/J 26/P12/Q5/a,b	1	7	7	
M/J 26/P12/Q6/a,b,c	3	7	134	
M/J 26/P12/Q6/d	2	4	85	
M/J 26/P12/Q7/a,b	5	13	337	
M/J 26/P12/Q8/a,b	3	3	134	
M/J 26/P13/Q1	6	5	387	
M/J 26/P13/Q2/a,b	3	3	136	
M/J 26/P13/Q3/a,b,c(i)	1	10	8	
M/J 26/P13/Q3/c(ii,iii)	2	3	86	
M/J 26/P13/Q4/a,b,c	8	17	453	
M/J 26/P13/Q5/a,b,c	5	10	338	
M/J 26/P13/Q6/a,b,c,d	1	8	9	
M/J 26/P13/Q7/a,b,c	4	9	221	
M/J 26/P13/Q8/a,b,c	3	10	137	
M/J 26/P21/Q2/b	5	2	338	also U12:1
M/J 26/P23/Q2/d	5	2	339	also U12:1
O/N 25/P11/Q1/a,b	1	7	10	
O/N 25/P11/Q2/a,b,c,d	8	13	455	
O/N 25/P11/Q3/a,b	3	4	138	
O/N 25/P11/Q4/a,b,c,d	4	4	221	
O/N 25/P11/Q5/a,b,c	5	10	341	
O/N 25/P11/Q6/a,b	4	4	224	
O/N 25/P11/Q6/c	3	3	138	
O/N 25/P11/Q7/a,b,c	2	8	86	
O/N 25/P11/Q7/d	1	6	10	
O/N 25/P11/Q8/a	6	2	387	
O/N 25/P11/Q8/b,c	7	7	425	
O/N 25/P11/Q9	3	3	139	
O/N 25/P11/Q10/a,b	3	4	139	
O/N 25/P12/Q1	6	2	388	
O/N 25/P12/Q2/a,b	7	6	426	
O/N 25/P12/Q3/a,b,c	4	10	224	
O/N 25/P12/Q4/a,b,c,d	8	10	456	
O/N 25/P12/Q5/a,b,c,d	2	10	87	
O/N 25/P12/Q5/e	6	3	388	
O/N 25/P12/Q6/a,b,d	1	4	11	
O/N 25/P12/Q6/c	6	2	389	
O/N 25/P12/Q7/a,b,c	1	4	12	
O/N 25/P12/Q8/a	2	4	87	
O/N 25/P12/Q8/b(i)	3	4	139	
O/N 25/P12/Q8/b(ii)	5	2	341	
O/N 25/P12/Q9/a,b	3	4	140	
O/N 25/P12/Q10/a	5	4	341	
O/N 25/P12/Q10/b	7	2	426	
O/N 25/P12/Q10/c	6	1	389	
O/N 25/P12/Q11	3	3	141	
O/N 25/P13/Q1/a,b,c	1	7	13	
O/N 25/P13/Q2/a,b,d	1	5	14	
O/N 25/P13/Q2/c	4	1	226	

Reference	Unit	Marks	Page	Also / context
O/N 25/P13/Q3/a,b	3	4	142	
O/N 25/P13/Q4/a,b	2	4	88	
O/N 25/P13/Q4/c	3	3	143	
O/N 25/P13/Q5/a,b,c,d	8	16	457	
O/N 25/P13/Q6/a,b,c	4	11	226	
O/N 25/P13/Q7/a	7	2	426	
O/N 25/P13/Q7/b	1	2	14	
O/N 25/P13/Q7/c,e	6	6	389	
O/N 25/P13/Q7/d	3	4	143	
O/N 25/P13/Q8/a,b,d	5	8	342	
O/N 25/P13/Q8/c	7	2	427	
O/N 25/P21/Q1/c	5	3	342	
O/N 25/P21/Q6/a	6	2	391	
M/J 25/P11/Q1/a,b	3	4	143	
M/J 25/P11/Q2/a,c	2	6	88	
M/J 25/P11/Q2/b	1	7	15	
M/J 25/P11/Q3/a,b	1	7	16	
M/J 25/P11/Q4/a	7	4	427	
M/J 25/P11/Q4/b	3	2	145	
M/J 25/P11/Q5/a,b,c,d,e	8	13	458	
M/J 25/P11/Q6/a(i)	3	4	145	
M/J 25/P11/Q6/a(ii,iii)	4	5	228	
M/J 25/P11/Q6/b	5	5	342	
M/J 25/P11/Q7/a,b,c	6	11	392	
M/J 25/P11/Q8/a,b	4	7	228	
M/J 25/P12/Q1/a,b,c	4	9	231	
M/J 25/P12/Q2/a,b,c	1	7	17	
M/J 25/P12/Q3/a,b,c	7	12	428	
M/J 25/P12/Q4/a,b	3	4	146	
M/J 25/P12/Q5/a,b,c,d	8	16	459	
M/J 25/P12/Q6/a,b,c,d,e	2	19	89	
M/J 25/P13/Q1/a,b,c	1	14	17	
M/J 25/P13/Q2/a,b,c	4	10	233	
M/J 25/P13/Q3/a	5	4	343	
M/J 25/P13/Q3/b	6	4	393	
M/J 25/P13/Q3/c	2	4	89	
M/J 25/P13/Q3/d	7	3	428	
M/J 25/P13/Q4/a,b	3	4	147	
M/J 25/P13/Q5/a,b	4	9	233	
M/J 25/P13/Q6/a,b,c,d	8	18	460	
M/J 25/P13/Q7/a	4	2	237	
M/J 25/P13/Q7/b,c	1	3	18	
O/N 24/P11/Q1/a,b,c,d	1	7	19	
O/N 24/P11/Q2/a,b,c	8	17	461	
O/N 24/P11/Q3/a,b	4	7	237	
O/N 24/P11/Q3/c,d	3	9	148	
O/N 24/P11/Q4/a,c,d	5	11	343	
O/N 24/P11/Q4/b	1	6	19	
O/N 24/P11/Q5/a,b,c	3	6	148	
O/N 24/P11/Q6	7	4	428	
O/N 24/P11/Q7/a,b	7	4	429	
O/N 24/P11/Q8/a,b	4	4	238	
O/N 24/P12/Q1/a,b	3	4	150	
O/N 24/P12/Q2/a,b,c	3	6	151	
O/N 24/P12/Q3/a,b	4	8	239	
O/N 24/P12/Q3/c	5	4	344	
O/N 24/P12/Q4/a,b	5	7	344	
O/N 24/P12/Q4/c	6	3	393	
O/N 24/P12/Q5/a,b	7	9	429	
O/N 24/P12/Q6/a,b	8	12	464	
O/N 24/P12/Q7/a,b	1	7	20	
O/N 24/P12/Q8/a,b	4	11	239	
O/N 24/P12/Q9/a,b	3	4	151	
O/N 24/P13/Q1/a,b,c	3	5	152	
O/N 24/P13/Q2/a,b	3	6	153	
O/N 24/P13/Q3	3	3	153	

Reference	Unit	Marks	Page	Also / context
O/N 24/P13/Q4/a,b,c,d,e	8	19	464	
O/N 24/P13/Q5/a,b	7	6	430	
O/N 24/P13/Q6/a,b,c	1	10	21	
O/N 24/P13/Q7/a,b	4	6	242	
O/N 24/P13/Q8/a,b	1	4	21	
O/N 24/P13/Q8/c	4	2	245	
O/N 24/P13/Q9/a,b,c,d	2	14	90	
O/N 24/P22/Q1/b,c	5	5	344	
M/J 24/P11/Q1/a,b	3	3	153	
M/J 24/P11/Q2/a,b,c	3	12	154	
M/J 24/P11/Q2/d	1	3	22	
M/J 24/P11/Q2/e	2	3	91	
M/J 24/P11/Q3/a,b	5	5	345	
M/J 24/P11/Q4/a,b	4	7	245	
M/J 24/P11/Q5/a	2	4	91	
M/J 24/P11/Q5/b,c(i)	6	8	394	
M/J 24/P11/Q5/c(ii)	7	4	430	
M/J 24/P11/Q6/a,b,c	8	15	466	
M/J 24/P11/Q7	1	3	22	
M/J 24/P11/Q8/a,b	2	8	92	
M/J 24/P12/Q1/a,b	3	6	156	
M/J 24/P12/Q2/a,b,c	3	10	157	
M/J 24/P12/Q2/d	1	7	23	
M/J 24/P12/Q3/a	6	6	395	
M/J 24/P12/Q3/b,c	2	8	93	
M/J 24/P12/Q4/a,b,c,d	8	11	468	
M/J 24/P12/Q5/a,b	4	7	249	
M/J 24/P12/Q6	5	4	345	
M/J 24/P12/Q7/a,b,c	1	6	23	
M/J 24/P12/Q8/a,b,c	5	10	345	
M/J 24/P13/Q1/a,b,c,d	1	9	24	
M/J 24/P13/Q2/a,b,c	1	9	25	
M/J 24/P13/Q3/a,b	4	7	252	
M/J 24/P13/Q4/a,b,c,d	8	14	469	
M/J 24/P13/Q5/a,b,c,d	2	11	94	
M/J 24/P13/Q6	3	3	158	
M/J 24/P13/Q7/a,c,d	3	8	159	
M/J 24/P13/Q7/b	7	3	431	
M/J 24/P13/Q7/e	5	5	347	
M/J 24/P13/Q7/f	6	6	395	
O/N 23/P11/Q1/a,b,c	1	7	26	
O/N 23/P11/Q2/a,b,c,d	2	10	95	
O/N 23/P11/Q3/a,b,c	8	11	470	
O/N 23/P11/Q4/a,b	3	4	160	
O/N 23/P11/Q5/a,b,c	4	12	255	
O/N 23/P11/Q6/a,b,c	5	9	348	
O/N 23/P11/Q7/a,b,c	3	7	161	
O/N 23/P11/Q8/a,b,c	4	10	255	
O/N 23/P11/Q9/a,b	3	5	161	
O/N 23/P12/Q1/a,b,c	3	6	162	
O/N 23/P12/Q2/a,b,c	8	11	471	
O/N 23/P12/Q3/a,b,c	1	6	27	
O/N 23/P12/Q4/a,b	3	5	163	
O/N 23/P12/Q5/a,b,c	6	6	396	
O/N 23/P12/Q6/a,b,c	1	8	27	
O/N 23/P12/Q7/a,b,c,d	2	17	96	
O/N 23/P12/Q8/a,b	5	4	348	
O/N 23/P12/Q8/c	4	4	258	
O/N 23/P12/Q9/a,b	4	8	258	
O/N 23/P13/Q1/a,b	1	3	28	
O/N 23/P13/Q2/a,b	1	4	28	
O/N 23/P13/Q3/a(i,ii)	2	6	97	
O/N 23/P13/Q3/a(iii),c	6	7	397	ctx (b) intro
O/N 23/P13/Q3/b	8	5	472	
O/N 23/P13/Q4/a,b	3	4	163	
O/N 23/P13/Q5/a,b(ii)	5	7	349	

Reference	Unit	Marks	Page	Also / context
O/N 23/P13/Q5/b(i)	1	2	28	
O/N 23/P13/Q6/a,b	7	6	431	
O/N 23/P13/Q7/a,b,c	3	8	165	
O/N 23/P13/Q7/d	4	3	261	
O/N 23/P13/Q8/a,b,c	6	7	397	
O/N 23/P13/Q9/a,b,c,d	4	13	261	
M/J 23/P11/Q1/a,b,c	1	13	29	
M/J 23/P11/Q2/a,b	8	16	472	
M/J 23/P11/Q3/a,b,c	5	9	349	
M/J 23/P11/Q3/d(i,ii,iii,iv,v)	1	7	30	
M/J 23/P11/Q3/d(vi)	4	1	263	
M/J 23/P11/Q4/a,b	3	8	165	
M/J 23/P11/Q4/c	7	3	431	
M/J 23/P11/Q4/d	2	9	98	
M/J 23/P11/Q5/a,b	3	4	166	
M/J 23/P11/Q6	4	5	263	
M/J 23/P12/Q1/a,b,c,d,e	2	11	99	
M/J 23/P12/Q2/a,b,c(ii,iii)	8	15	474	
M/J 23/P12/Q2/c(i)	6	2	398	
M/J 23/P12/Q3/a,b	4	6	264	
M/J 23/P12/Q4/a,b,c	1	5	31	
M/J 23/P12/Q4/d	4	1	264	
M/J 23/P12/Q5/a,b,c	4	8	265	
M/J 23/P12/Q5/d	5	7	349	
M/J 23/P12/Q6/a,b	3	4	167	
M/J 23/P12/Q7/a,b,c	5	13	350	
M/J 23/P12/Q7/d	7	3	432	
M/J 23/P13/Q1/a,b	3	3	168	
M/J 23/P13/Q2/a,b,c,d,e	2	15	100	
M/J 23/P13/Q3/a	3	5	169	
M/J 23/P13/Q3/b	7	3	432	
M/J 23/P13/Q3/c	1	7	31	
M/J 23/P13/Q4/a,b,c	8	12	476	
M/J 23/P13/Q4/d	6	5	399	
M/J 23/P13/Q5/a	5	5	351	
M/J 23/P13/Q5/b	7	4	432	
M/J 23/P13/Q6/a,b	6	7	400	
M/J 23/P13/Q7/a	1	3	32	
M/J 23/P13/Q7/b,c	4	6	266	
M/J 23/P21/Q1/a	5	4	352	
M/J 23/P22/Q5/c	5	3	352	
M/J 23/P23/Q2/b	6	4	400	
O/N 22/P11/Q1/a,b,c,d	1	11	33	
O/N 22/P11/Q2	6	4	401	
O/N 22/P11/Q3/a,b	3	4	170	
O/N 22/P11/Q4/a,b,c,d	8	15	477	
O/N 22/P11/Q5/a,b,c,d	4	11	267	
O/N 22/P11/Q6/a,b,c	4	11	267	
O/N 22/P11/Q7/a,b	5	4	353	
O/N 22/P11/Q7/c	1	3	33	
O/N 22/P11/Q7/d,e	4	4	272	
O/N 22/P11/Q8	2	4	100	
O/N 22/P11/Q9/a,b	3	4	171	
O/N 22/P12/Q1/a,b	5	9	354	
O/N 22/P12/Q2/a,b	1	5	34	
O/N 22/P12/Q3/a,b,c	3	7	171	
O/N 22/P12/Q4/a,b,c	6	6	401	
O/N 22/P12/Q5/a,b,c,d	8	11	479	
O/N 22/P12/Q6/a	6	5	402	
O/N 22/P12/Q6/b	1	4	34	ctx (a) intro
O/N 22/P12/Q7/a,b,c	4	12	272	
O/N 22/P12/Q8/a,b,c	1	8	34	
O/N 22/P12/Q9	7	2	432	
O/N 22/P12/Q10/a,b	2	6	101	
O/N 22/P13/Q1/a,b	1	4	36	
O/N 22/P13/Q1/c	3	3	173	

Reference	Unit	Marks	Page	Also / context
O/N 22/P13/Q2/a,b,c,d,f	8	12	480	
O/N 22/P13/Q2/e	1	2	37	
O/N 22/P13/Q3	5	4	355	
O/N 22/P13/Q4/a,b,c	4	7	276	
O/N 22/P13/Q5/a,b	3	5	174	
O/N 22/P13/Q6/a,b,c	4	10	276	
O/N 22/P13/Q7/a,b,c	2	10	101	
O/N 22/P13/Q8/a,b	7	6	433	
O/N 22/P13/Q9/a,b	1	4	37	
O/N 22/P13/Q10/a,b	3	8	175	
O/N 21/P11/Q1/a,b,c	1	9	38	
O/N 21/P11/Q2/a,b,c	6	9	402	
O/N 21/P11/Q3/a,b,c	3	8	175	
O/N 21/P11/Q4/a,b(i),c	7	7	433	
O/N 21/P11/Q4/b(ii),d	5	7	356	
O/N 21/P11/Q5/a,b,c	8	10	481	
O/N 21/P11/Q6/a,b	4	8	280	
O/N 21/P11/Q7/a,b	1	8	39	
O/N 21/P11/Q8/a,b,c	2	9	103	
O/N 21/P12/Q1	6	2	403	
O/N 21/P12/Q2/a,b	3	4	177	
O/N 21/P12/Q3/a	3	3	178	
O/N 21/P12/Q3/b	2	5	103	
O/N 21/P12/Q4/a,b,c,d,e	1	8	40	
O/N 21/P12/Q5/a,b	1	12	40	
O/N 21/P12/Q6/a,b,c	8	15	482	
O/N 21/P12/Q7/a	4	3	282	
O/N 21/P12/Q7/b,c	5	8	356	
O/N 21/P12/Q8/a,b	4	15	282	
O/N 21/P13/Q1/a,b,c	1	9	42	
O/N 21/P13/Q2/a,b,c	6	9	404	
O/N 21/P13/Q3/a,b,c	3	8	178	
O/N 21/P13/Q4/a,b(i),c	7	7	434	
O/N 21/P13/Q4/b(ii),d	5	7	357	
O/N 21/P13/Q5/a,b,c	8	10	483	
O/N 21/P13/Q6/a,b	4	8	284	
O/N 21/P13/Q7/a,b	1	8	43	
O/N 21/P13/Q8/a,b,c	2	9	104	
M/J 21/P11/Q1/a,b,c	1	15	43	
M/J 21/P11/Q1/d	7	2	434	
M/J 21/P11/Q2/a,b	5	8	358	
M/J 21/P11/Q3/a,b,c	4	13	287	
M/J 21/P11/Q4/a,b,c,d	2	14	105	
M/J 21/P11/Q5/a,b,c	3	6	179	
M/J 21/P11/Q6/a,b,c	6	3	405	
M/J 21/P11/Q7/a,b,c	8	11	485	
M/J 21/P11/Q8	3	3	179	
M/J 21/P12/Q1/a,b,c	8	16	486	
M/J 21/P12/Q2/a,b,c	7	9	435	
M/J 21/P12/Q3/a,b	3	6	180	
M/J 21/P12/Q4/a,b(ii,iii)	4	7	291	
M/J 21/P12/Q4/b(i)	1	1	44	
M/J 21/P12/Q5/a,b	4	9	295	
M/J 21/P12/Q5/c,d	2	8	106	
M/J 21/P12/Q6/a,b,c	1	6	46	
M/J 21/P12/Q7/a,b	5	8	359	
M/J 21/P12/Q8/a,b,c	6	5	406	
M/J 21/P13/Q1/a,b,c	1	15	47	
M/J 21/P13/Q1/d	7	2	435	
M/J 21/P13/Q2/a,b	5	8	360	
M/J 21/P13/Q3/a,b,c	4	13	295	
M/J 21/P13/Q4/a,b,c,d	2	14	107	
M/J 21/P13/Q5/a,b,c	3	6	181	
M/J 21/P13/Q6/a,b,c	6	3	407	
M/J 21/P13/Q7/a,b,c	8	11	488	
M/J 21/P13/Q8	3	3	181	